**FA8075-18-D-0004**

# Project ADRIAN
## An Analysis of Logistics Data Exchange Requirements at the Tactical Edge

**Abstract:** This study is the culmination of a comprehensive examination of the data prioritization and exchanges required to enable logistics support to tactical, geographically disparate Marine forces operating in command and control (C2) denied or degraded environments (C2D2E). This analysis identifies baseline logistics data exchange capabilities required by Stand-in Forces (SIF) conducting Expeditionary Advanced Base Operations (EABO) against near-peer adversaries across the competition continuum.

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Approved: 10 October 2023

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**Published by**
National Information Standards Organization (NISO)
3600 Clipper Mill Road
Suite 302
Baltimore, MD 21211
www.niso.org

## Abstract

This report is the culmination of a comprehensive examination of the data exchange requirements and corresponding prioritization to enable logistics support to tactical, geographically disparate Marine forces operating in command and control (C2) denied or degraded environments (C2D2E). This analysis identifies baseline logistics request capabilities required by Stand-in Forces (SIF) conducting Expeditionary Advanced Base Operations (EABO) against a peer or near-peer adversary, and across the competition continuum.

Additionally, this analysis supports the risk assessment and decision-making processes for logisticians to develop recommendations optimizing combat readiness and meet demands (e.g., survivability and lethality) in the context of specific threat capabilities against U.S. forces in the USINDOPACOM area of operations (AO). The analysis recommends improvements to the SIF through new technology, innovative solutions, doctrine, C2, warfare concepts, military operations center procedures, future concepts, and has enterprise-wide application across the range of military operations.

This initiative was conducted under the cognizance of Logistics Combat Element (LCE) Integration Division (LCE ID), Capabilities Development Directorate (CDD), Deputy Commandant for Combat Development and Integration (DC CD&I), in support of the Logistics Information Technology (IT) Capabilities Based Assessment (CBA).

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# Preface

This report details the minimum data requirements from the perspective of demand initiation, primarily from the smallest, most forward deployed tactical user in a communications degraded/denied environment. This report is directed primarily to those in the position to map and create the data architecture necessary to pass the minimum demand and request information, either vertically or horizontally, in the fulfillment of the demand. This report applies to those who are tasked with developing data exchanges across multiple types of networks and communication nodes either via secure or non-secure means and across security enclaves.

This work represents many hours of research on what data is, what it means, where it comes from, how data is used, how data is developed, doctrinal understanding, and candid conversations between ground truth reality and perceived assumptions. The report is part of the greater work to develop a single, holistic approach to five of the six doctrinal functions of Marine Corps logistics and their applicable subfunctions as it stands today without depending on future material capabilities yet to be matured and employed.[1]

While this report is voluminous—perhaps dauntingly so—it is necessary to clearly articulate the environment, context, rationale, and urgency of a unified, holistic logistics data solution for the survivability of our most valuable resource—the Marine. There exists a moral and an ethical imperative to do this the right way, and to protect our people accordingly. Failure to address these findings carries potential mortal consequences.

Lastly, this report identifies and discusses doctrinal and capability gaps discovered through the various working groups and breakout sessions. What is unique about this report is the discussion on the rationale, impacts, and then the granular recommendations which then will enable future capabilities to be integrated into the Marine Corps Installations and Logistics Enterprise (MCILE).

[1] The sixth doctrinal function of Marine Corps Logistics—Services—is omitted from this report as it is not applicable to tactical-level logistics operations in a warfighting environment.

# Section 1: Executive Summary

The Global War on Terror, in many ways, spurred a revolution in military technology and accelerated an evolution in tactics, techniques, and procedures. In other areas, however, we as a military have become complacent. With uncontested freedom of movement throughout the battlespace, sustaining our combat forces was easy, and the competencies of our military logistics apparatus severely atrophied. In Iraq and Afghanistan, we were primarily engaged in counterinsurgency-type warfare against adversaries with comparatively primitive communications systems. Self-assured of our technological superiority, our logistics information systems experienced little advancement or improvement during those twenty years of conflict in the Middle East.

However, the great power competition has seen the rise of new threats and a different class of opponent entirely. Ongoing geopolitical tension with both China and Russia has signaled a refocus on what conflict might look like with a near-peer adversary. We can no longer afford to take for granted that our adversary's most sophisticated electronics are cell phones and walkie-talkies, or that our forces will not have to operate in an environment where *our* ability to communicate may be degraded or denied entirely.

The Marine Corps' logistics information technology (IT) portfolio currently has two major weaknesses. First, there exist 77 unique, stand-alone software systems meant to enable specific, tasks. There is effectively no interoperability or compatibility between any of these disparate applications; data cannot be exchanged between them without manual transcription from human operators. The inability to automate processes that require data input from different sources not only takes up valuable man-hours that Marines could devote to other mission-critical tasks, it also greatly increases the odds of human error. Moreover, the time that it currently takes for leaders to assemble and process this information lends itself to delays in decision-making that could have catastrophic real-world implications. The second issue is even more alarming. At present, every one of those 77 logistics applications—and nearly *all* logistics data—reside on unclassified networks and servers. This poses an enormous risk to both mission and force. Our adversaries possess significant cyber and electronic warfare capabilities. Non-secure communications are especially susceptible to interception and exploitation. Our unsecured logistics data feeds the enemy's 'kill chain' and can be used to target our forces. The very real, tangible consequence of continuing to operate under this paradigm is loss of life.

The **Project ADRIAN** initiative was born out of moral imperative to address these critical vulnerabilities. Our mission is to deliver an IT-enabled logistics command and control (C2) capability to the operating forces, one that ensures the integrity and availability of logistics data and provides for its secure and reliable exchange between the warfighter and the relevant combat support agency. That logistics C2 functionality must be holistically integrated into *the* Common Operational Picture (COP). This study is just the first phase of Project ADRIAN—define information exchange requirements. This detailed analysis deconstructs the functions of Marine Corps logistics to identify core tasks and determine all the information and data that a future-state IT system would need to accommodate. Additionally, the *critical data analysis* contained herein identifies the minimum data exchanges required to enable logistics operations and effectively sustain our forces in communications denied or degraded environments (C2D2). Publication of this document signifies a transition to development and engineering of a software solution that will strengthen and streamline our logistics processes and enhance our ability to face new and emerging threats.

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# Section 2: Introduction

## 2.1 Purpose

This section provides the operational context in which logistics operations will occur. Logistics information technology (IT) systems have a critical dependency on communications network pathways. A logical assumption is the data pathways and information exchange requirements need to be predicated on the operating conditions. If the system is designed around austere conditions, enemy threat capabilities, and operational mission requirements, then rationally, it should work well in an uncontested environment. Therefore, the purpose of this analysis is driven by the operating environment, near-peer threat, the Stand-in Force (SIF) mission, and the resulting logistical challenges these changes present.

### 2.1.1 Operating Environment

The United States (U.S.) Indo-Pacific Command (USINDOPACOM) area of responsibility (AOR) provides the operational context from which this analysis is derived. The AOR encompasses about half the earth's surface, stretching from the waters off the west coast of the U.S. to the western border of India, and from Antarctica to the North Pole. Within the AOR boundaries, 83% of the surface area is water, with only 17% of the surface comprised of land mass.

This region is unparalleled in its diversity—culturally, socially, economically, and geo-politically—and is home to more than 50% of the world's population. The AOR comprises 36 national identities, 3,000 unique languages, several of the world's largest militaries, five nations allied with the U.S. through mutual defense treaties (Australia, Japan, the Republic of Korea (ROK), the Philippines, and Thailand), and two nations opposed or confrontational to the U.S. (People's Republic of China (PRC) and Democratic People's Republic of Korea (DPRK)). The region encompasses over 25,000 inhabited and uninhabited islands in varying size and shape.

### 2.1.2 Near-Peer Threat

The following assessment provides the context within the great power competition against potential near-peer adversaries (i.e., China) in contested operating environments. This presents compounding challenges for freedom of movement and domain dominance that logistics communication, coordination, and execution must consider.

#### 2.1.2.1 Diplomatic Efforts

China is the preeminent near-peer competitor to the U.S. and is challenging free and open shipping lanes by claiming sovereignty over maritime regions and territories. Diplomatically, China has established formal relations with Malaysia, Thailand, the Philippines, Bangladesh, Maldives and recently Fiji and Papua New Guinea to expand its influence to gain additional footholds and bolster its maritime claims.

#### 2.1.2.2 Technological Capabilities

As a near-peer threat, China possesses technological capabilities that are equivalent—or in some cases superior—to those of the U.S. The following are some of the anti-access, area denial (A2AD) capabilities representing a direct threat to U.S. forces across the range of military operations in the USINDOPACOM AOR.

1) Satellite surveillance. According to a U.S. Department of Defense (DoD) report to Congress in November 2022, China nearly doubled the number of intelligence, surveillance, and reconnaissance (ISR) satellite platforms in the past four years, and its total satellite fleet, numbering 260 platforms, is second only to that of the United States.

This provides China with the ability to employ a persistent and consistent observation capability over the operating area. It is assumed these satellites are capable of visible, imagery at a resolution of around 10 centimeters, coupled with additional multispectral, hyperspectral, and synthetic aperture radar capabilities on par with U.S. systems. [1]

2) Electromagnetic (EM) detection and deception. Electromagnetic Spectrum Superiority is firstly about people and is a twofold capability. On the detection side, the People's Liberation Army (PLA) is using drones, satellites, and cyber monitoring to find electromagnetic "heat patterns" to confirm the presence of military equipment and personnel. Every electronic and electro-mechanical device emits an electromagnetic signal and is detectible within a given range. Additionally, deception targets human decision-makers. Notionally, the PLA could target U.S. satellite uplinks and downlinks supporting ISR, communications, early warning, and navigation systems. The consequences could have a profound negative impact on a commander's planning, decision-making, and execution cycle, and complicate effective air, ground, and naval operations. To achieve the above objectives, the PLA emphasizes the deception strategy of "hide the real and inject the false," effecting signals and information to mislead U.S. operators and decision-makers. They also highlight the importance of surprise — as the perceived weaker side in a competition, employing asymmetric means to defeat the stronger rival via a fast electromagnetic spectrum attack against a critical vulnerability when least expected. [2]

3) Cyber detection and service interruption. For years, China has tested its cyber capabilities by targeting the personal data of Americans, including the theft of personnel records from the U.S. Office of Personnel Management (OPM), the intrusion into Marriott hotels, Anthem health insurance company, and now the wholesale theft of credit and other information from Equifax. Additionally, China has practiced its ability to interrupt and deny cyber services as a pressure tactic against its neighbors. [2]

4) Radio signal detection and jamming. Since 2020, Chinese military's ground-based jammers have seen a severalfold increase in numbers, potentially targeting the most distant and powerful U.S. communications satellites. It is a development that poses an expanded threat to a critical enabler of modern military warfare to include the global positioning systems (GPS) utility that provides users with positioning, navigation, and timing (PNT) services which almost every major piece of equipment uses. [1] [3]

5) Cellular monitoring and service interruption. China has and continues to exploit vulnerabilities in the global telecommunications network to target, track and intercept phone communications of U.S. phone subscribers traveling abroad as they connect to foreign cell towers. In 2018, China carried out the highest number of apparent surveillance actions targeting U.S. mobile phone users over 3G and 4G networks through Chinese state-owned Unicom. Tens of thousands of phone users were affected by the attacks between 2018 and 2020. This mass surveillance incident was intended primarily for intelligence collection. Such attacks pinpoint the location of U.S. based cell phones operating abroad which indicate a mass of U.S. forces. Additionally, such methods that are used for monitoring can also be used for service denial. A simple command can shut down all cellular service for U.S. based cellphones operating on a foreign cellular tower. [4]

6) Financial monitoring and service interruption. China has proven its ability to intrude and monitor financial transactions through data gateways. While the actual transaction data may not be compromised, the header transaction data is readable allowing monitoring of the country origin of the financial instrument (e.g., credit cards, debit cards, etc.) and the

location of the point of sale. Through this type of monitoring China can ascertain U.S. financial activity within a particular foreign country exposing vendors and U.S. forces to hostilities and coordinated cyber-attacks. [5]

While the technological capabilities are impressive, it is impossible for our near-peer competitor to shut down all communications pathways indefinitely. The collective energy output and coverage required to do so is impractical and unsustainable. Therefore, the more logical approach is that these capabilities will be applied to targeted areas and with limited duration against key U.S. forces and capabilities.

#### 2.1.2.3 Military Strategy and Capabilities

China's investments in ISR and its precision-strike complex have enabled operational reach to the U.S. mainland. However, these capabilities are more likely applicable within the USINDOPACOM region. For at least the past three decades, China has studied U.S. concepts for joint force power projection and maneuver warfare. In response, China developed strategies and capabilities to counter these concepts, aiming to keep the joint force at "arm's length," disrupt U.S. planning, and complicate U.S. decision-making.

While China does not make its military capabilities transparent, it is known that it has expended significant resources on research, development, and procurement of ship mounted, long range missile technologies. The table below represents the current estimate of China's shipborne missile capabilities. [6]

| Type | Range | Speed | Launch | Target | Quantity |
|---|---|---|---|---|---|
| YJ-12 | 400 Km | Mach 2.5 | Air/Ship | Sea/Land | >800 |
| YJ-91/A | 400 Km | Mach 2.5 | Air/Ship | Sea/Land | >1,000 |
| FL-2 | 40 Km | Mach 1.7 | Air/Ship | Sea/Land | >1,000 |
| HY-3 | 130 Km | Mach 2.5 | Ship | Sea | >1,200 |
| SS-N-22 | 220 Km | Mach 3.0 | Ship | Sea | >4,900 |
| HQ-7 | 12 Km | UNK | Ship | Air | >2,000 |
| SA-N-7 | 30 Km | Mach 3.0 | Ship | Air | >1,000 |
| SA-N-6 | 90 Km | Mach 4.0 | Ship | Air | >1,000 |
| Approximate Total of Ship Mounted Missiles | | | | | >12,900 |

**Table 1: China's Shipborne Missile Capability**

Based on the presented data, it is reasonable to assume the Chinese military strategy is one of "detect, triangulate, and engage" from mobile platforms to provide significant stand-off from U.S. engagement capabilities. China's intent is to avoid direct engagement with large, land-based forces, in favor of an asymmetric strategy.

### 2.1.3 The Stand-in Force Concept of Operations

To defeat the Chinese strategy of "detect, triangulate, and engage" at distance, the SIF mission and strategy is adapted from large land-based maneuver warfare competencies (easily detectable, easily engaged) to small, lethal, low signature, and highly mobile elements (not easily detectable, not easily engaged) to deny the enemy maritime mobility and key terrain.

The main ideas behind the SIF concept are simple:

1) Find. Detect and/or discover a potential adversary's people and things (such as weapon systems, sensor systems, submarines, etc.) in an assigned operating area and then track them at a level that facilitates targeting by fleet or joint weapons until they leave that area (i.e., kill chain concept). This finding and tracking effort starts as soon as the

possible target is identified and continues at every point along the competition continuum. [7]

2) Strike. If armed conflict begins, the SIF will use knowledge of the adversary to help the fleet or other elements of the joint force attack quickly and effectively, blind the adversary, and disrupt the enemy's plans to force a move into other places where SIF and the fleet have an advantage. [7]

3) Deny. In the event of armed conflict, SIF will conduct sea denial in specific areas, disrupting adversarial plans. Sea denial is nested within the fleet and joint force campaign plans. For example, denying a specific area can present a forward edge behind which the U.S. fleet can maneuver. Area denial can also "herd" an adversary into areas in which U.S. forces have a relative advantage, or it can sanitize key straits or chokepoints. [7]

4) Counter. The forward SIF element must make itself difficult to detect by a potential adversary by maintaining a low signature, moving frequently and unpredictably, and using deception. This imposes costs on potential adversaries, as they are forced to expend resources to find and engage friendly forces. Winning the counter-reconnaissance fight in the context of a naval campaign also means that Marines in the littorals will screen for the fleet as a measure of force protection, and thereby increase the fleet commander's freedom of action/movement/decision options. This enables fighting inside a contested area and disrupts an adversary's plans. [7]

### 2.1.4 The Logistics Challenge

While the SIF operating construct is relatively simple in design, maintaining and sustaining these elements operating in geographically disparate locations across the spectrum of competition introduces complex challenges for logistical support. Logistically speaking, this is a new challenge to plan, communicate, and execute. Adding to the challenge, the SIF may include elements from the Army, Marine Corps, Navy, Coast Guard, Air Force, Space Force, special operations forces (SOF), interagency organizations, as well as allies and partners. These factors introduce a completely new dimension not previously exercised as a core competency. Therefore, logistics—as a whole—is then challenged to collect the *right* data at the appropriate level, and to exchange that data between the relevant agencies to enable effective decision-making by commanders at both supporting and supported units. [7]

### 2.1.5 Purpose Summary

The driving force behind this analysis is the collective logistics challenges presented by the operating environment, the SIF and EABO operating concepts, and the threat posed by near-peer adversaries across the competition continuum. When accounting for these factors, a new imperative emerges to determine the minimum data exchange requirements. Therefore, the focus is only on the most critical data elements that must be transmitted.

## 2.2 Objective

### 2.2.1 Marine Corps Enterprise Objectives

In response to the factors of the operating environment, enemy threat, and new operating missions, logistics data must be prioritized the same as — and fall under the umbrella of — what is considered "kill chain" traffic. To transform and prioritize logistics data as kill chain traffic, the Marine Corps' objectives include, but are not limited to:

1) Define the current logistics operating environment (as-is state).
2) Outline the capabilities and systems that enable the current state.
3) Identify gaps and overlaps.
4) Define requirements at the tactical level (to-be state).
5) Outline future capabilities to achieve the to-be state.
6) Reconcile gaps and overlaps between the as-is and to-be state.
7) Generate a capabilities requirement document—or series thereof—to address gaps and operational risks identified in the Log IT CBA, and present recommendations for material and procedural solutions.

These objectives enable planning and execution of logistics operations to nest with and support the SIF. Logistics is the most data intensive warfighting function, dependent upon multiple sources and streams at various levels to enable critical decision making and execution.

Just as maneuver warfare concepts require force redesign and optimization to counter the enemy threat, logistics too must parse the data inputs and streams down to the most critical elements to initiate action and register demand.

### 2.2.2 Analysis Objective

The objective of this analysis is to support the overall Marine Corps' objectives. However, not all of the previous objectives will be met in this analysis, but they will support parallel and ongoing analyses and initiatives headed by other elements of the enterprise. As such, this analysis is to apply scientific based methods to identify the minimum data exchange requirements at the lowest tactical level. Identifying the critical data inputs from the tactical level flowing upward to initiate logistics decision/action is the foundation upon which to build out further logistics capabilities.

## 2.3 Scope

The scope of this analysis is limited to ground logistics at the lowest tactical level communicating upward and the minimal essential data elements for five out of six functions of logistics and their corresponding sub-functions. The omitted doctrinal function of Marine Corps logistics is 'Services', which is mainly an internal command function with minimal external support requirements.

Aviation logistics is beyond the scope of this analysis. However, the findings contained herein may be relevant in applying a similar methodology to identify the minimum data exchange requirements and pathways in these related fields.

Additionally, this study provides a comparative analysis incorporating modern industry standards/concepts and then operationalizing them to fit into the current military data exchange requirements where appropriate and logical. While focusing on the threat environment, it strives to present a holistic approach in all aspects of logistics information management regardless of operating conditions.

This analysis supports Contract FA8075-18-D-0004, Task 3.4.7. in analyzing the risk assessment and decision analysis for logistics to develop recommendations optimizing combat readiness and meet demands (e.g., survivability and lethality) in the context of specific threat capabilities against III MEF forces in the first island chain. The analysis recommends improvements to the Stand-in Force through new technology, innovative solutions, C2, warfare

concepts, military operation center processes, and future concepts and has enterprise-wide application.

## 2.4 Methodology, Assumptions, and Procedures

### 2.4.1 Problem Framing

The Logistics IT portfolio—as it stands today—is not capable of supporting Stand-in Forces operating in austere, contested, communications denied/degraded, and distributed maritime environments. The lack of a holistic logistics C2 capability prevents the secure and timely exchange of data required to enable combat support to the tactical edge, thereby presenting an enormous risk to mission and force. The following sections provide the analytical basis of the problem frame assessment.

#### 2.4.1.1 Quantity of Logistics Information Technology Systems

As of this analysis, there are currently 77 unique logistics IT systems in Marine Corps inventory.[2] That number averages to over 12 systems for each of the six doctrinal functions of logistics. The sheer volume of logistics information systems operating on existing network architectures is unsustainable and incapable of supporting warfighter needs at the tactical edge, especially given the defined operating environment.

#### 2.4.1.2 Operating Enclaves

All logistics information systems currently in inventory are needlessly complex, cumbersome and resource intensive (data center storage, processing power, bandwidth, and throughput requirements), and reside primarily and/or exclusively on the Non-Secure Internet Protocol Router (NIPR) Network (NIPRNet). With these systems existing exclusively on NIPR, the ability to exchange data between security enclaves is severely restricted—impractical in most cases, if not outright impossible.[3] Logistics data is routed through or to many joint force military and/or interagency organizations, and may ultimately be routed through or to hundreds of potential contracted vendors. Unsecure data traversing unsecure networks is susceptible to enemy interception and exploitation and may ultimately feed the enemy's kill chain. A simple resupply request, for instance, can indicate composition, disposition, strength, and operational capability of friendly forces at a given, precise location. Given the projected operating conditions in the USINDOPACOM AOR, unclassified networks will be unavailable and there is a high likelihood of limited or severely restricted access to the Secure Internet Protocol Router (SIPR) Network (SIPRNet). Unsecured logistics data constitutes targeting information that feeds the enemy's "kill chain" and represents an enormous force protection risk.

#### 2.4.1.3 Quality of Service

1) Gaps in Service. While there are approximately 12 information systems per function of logistics, it is noted these systems do not cover the full logistical process requirement. Out of 1,800 logistics processes described in doctrine, policy, and directives, 0% originate digitally and are not holistically supported by information technology. 8% of those processes have some support, while 92% have no support. Initiating, supporting, and terminating records do not start digitally, remain digital, and/or are stored digitally within the logistics information system(s). The preponderance of records are paper based and stored in a file apart from the supporting system. Records that were found to

[2] 20 November 2022 USMC Logistics Systems Inventory

[3] Out of 77 logistics information systems, only two are authorized on SIPR and CENTRIX: TCPT and CLC2S but these two systems are not cross-domain solutions (feeding data across enclaves).

be digital were scanned from paper sources and did not originate as a digital record organically within the logistics system. Paper and digital records are combined into a self-competing system of filing and maintenance which do little to complement each other. At no point could a digital thread be holistically traced from the point of demand to fulfilment to completion without some kind of "middleware" or paper-based catalyst. Decision-making tools (i.e., email, Microsoft Excel spreadsheets, PDFs, etc.) are disaggregated and do not support automated workflows.

2) Database vice Automated Application. Logistics IT systems currently in inventory are inherently complicated, lack automation, and are not connected (interfaced vice integrated). Essentially, all these systems are a collection of siloed databases with an abundance of free text fields not capitalizing on business process workflows, digital signatures, or automated task behaviors. To align these multiple databases requires human reconciliation and manual "swivel chair" methods to transcribe data between systems.

3) Unfederated Data. Due to the multitude of IT systems and varying data schemas, all the logistics data remains unfederated making it impossible to query data from different sources into a single virtual environment. Additionally, as data is not within a shared environment, data is copied repetitively from one data source to another data source to another data source. This results in the generation, storage, and transmission of duplicative and redundant data, thereby significantly increasing costs (i.e., infrastructure to house that data) and demanding increased network resources—which may be untenable in the aforementioned operating environment.

4) Limited Analytical Tools. None of the IT systems have a robust analytical framework to assist in forecasting, identifying trends, prioritizing demands, consolidating similar demands, capturing and sharing decision rationales, or supporting data-driven decision making.

5) Needlessly Susceptible to Human Error. The manual nature of these IT systems introduces a vast propensity for human error. These systems rarely have robust data correction, validation, or business behavior rules applied which creates an unusually permissive ability to input/ingest incorrect data.

#### 2.4.1.4 As-is State of Logistics IT: Mission Impacts, Shortfalls, and Gaps

1) Inefficient use of Time. 80% of logistics administrative work time is spent manually collecting data from multiple systems and/or filling in data gaps, 19% of work time is spent analyzing or correcting data via "middleware" tools (including manual human intervention), and 1% of work time is spent on decision making and/or action.[4]

2) Negates Commanders' Decision Advantage. Logistics is the "pacing function"; operations move at the speed of logistics. The slower the logistics, the slower the commander's decision-making cycle, and the slower decision-making cycle results in decreased warfighter support and ultimately lethality.

3) Unacceptable Risk. These impacts, shortfalls, and gaps result in a very real risk to mission accomplishment and human life, which can both be avoided. There is nothing preventing the Marine Corps from solving these issues other than the will to do so and unifying similar, disparate efforts across the enterprise.

[4] Assessment of the LCE during MAGTF Warrior-17.

4) No Automated Data Inputs. The information systems are 100% reliant on human input for identification, reporting, and updates. This is especially true in determining asset existence and daily physical tracking. For example, on a quarterly basis, all owners and responsible officers of assigned equipment must physically reconcile what is in their possession with the automated report generated by Global Combat Support System – Marine Corps (GCSS-MC). This is done by paper and pencil. Discrepancies are then researched retroactively. Such reconciliations can be minimized or completely eliminated through the implementation of technologies such as passive radio frequency identification (pRFID), allowing for real time to near-real time tracking of assets thereby providing a "digital truth."

#### 2.4.1.5 Study Initiation: Inherent Biases

This section provides transparency to the analysts' cognitive biases upon initiation of this study and throughout the course of this analysis. Recognizing this, these issues are not bound to training, knowledge, small-unit leader, or command involvement constructs. Rather, this problem space is defined by the methodologies supported—or not supported—by existing logistics information systems. The plethora of disaggregated and isolated IT systems owned and operated across the MCILE, at present, prevents the enterprise from fully capitalizing on these investments.

The authors of this study hypothesized the following key problems to Marine Corps' logistics information and data management:[5]

1) Paper-based processes are the primary data capture tool.
2) When logistics processes are paper based, they are susceptible to human error and incur significant time delay when transcribing/converting data into digital records, which adversely impacts commanders' decision-making advantage.
3) These logistics processes and their supporting IT systems are not integrated because the enterprise does not have unity of effort in developing and/or deploying IT solutions.
4) The enterprise—as a whole—does not understand what data needs to be collected, for what purpose, and at what level to initiate action in a contested environment. The enterprise assumes the current collection of data is acceptable and relevant. This observation is not an indictment on anyone or department but is a fact based on the current state of logistics processes.
5) Neither the logistics processes or their IT systems are hierarchical in architecture, nor are they holistic in design.

#### 2.4.1.6 Assuming the Problem of Conditional Performance

The analysis assumes that shortfalls in current logistics information systems are compounded by conditional performance. Conditional performance, in this context, is defined as the employment of different processes and/or IT systems for analogous situations, due only to a change in materiel, environment, or circumstance. For example, a logistics process in a deployed environment for requesting fuel is significantly different than it is in garrison.

It is assumed the Marine Corps experiences conditional performance in two significant areas:

1) Application of different logistics processes for each of the different services or materiel classifications. Some logistics processes are supported by IT, and some are not. Those

[5] Based on the problems identified by the logisticians at various leadership roles the analysts attempt to formulate a basic reason why these conditions exist

logistics processes that are supported by IT are not holistically integrated or capable of supporting automation. A single unified logistics information system is absent.

2) The Marine Corps utilizes different systems in different operating conditions (i.e., deployed vs. garrison). When units deploy, they must adhere to a new set of logistics processes designed specifically for the deployed environment. Garrison processes and procedures do not migrate, on the whole, to the deployed environment. This was especially true during Operations Enduring Freedom (OEF) and Iraqi Freedom (OIF) where food, water, parts, and administrative supplies that were cataloged with an active valid National Stock Number (NSN) were collectively ordered on a unified single courier transaction. However, while in garrison, a separate process, system, and supply chain exists for food, water, parts (consumable and secondary reparable), and administrative supplies.

The myriad logistics processes and their associated information systems, coupled with the conditional performance factor described above, illustrate challenges to current IT solution priorities. The MCILE must refocus its attention on the operating environment.

### 2.4.2 Approach and Methodology

#### 2.4.2.1 Approach Design: Applying DCOVA

Current operating environments are complex, especially so when considering the proliferation of technology, IT systems and applications, methodologies, and other required capabilities. These complexities demand an objective, scientific, systematic approach to address the problems identified herein.

Establishing a business objective always precedes analysis, application of science, and statistical modeling to logistics decision making. Completion of the problem framing phase, triggers the application of the Define, Collect, Organize, Visualize, and Analyze (DCOVA) framework. [8]

The DCOVA framework ensures the problem framing is systematic, identifies key relevant data sources, and reduces complexity to the basic elements. DCOVA is comprised of five tasks: [8]

1) **D**EFINE the variables that require study to solve the problem or meet the objective.
2) **C**OLLECT the data for the defined variables from authoritative data sources.
3) **O**RGANIZE the data collected by table, relevance, and relation.
4) **V**ISUALIZE the organized data into charts, graphs, and diagrams to observe the effects, contribution, and relation thereof to the problem.
5) **A**NALYZE the data collected by applying science, industry standards, and applicable research methodologies to reach conclusions, make recommendations, and present results.

Using this method, depicted in Figure 1, establishes quantitative and qualitative relevance to the problem defined and can connect the logistics business decision to performance, implementation, insight, and innovation.

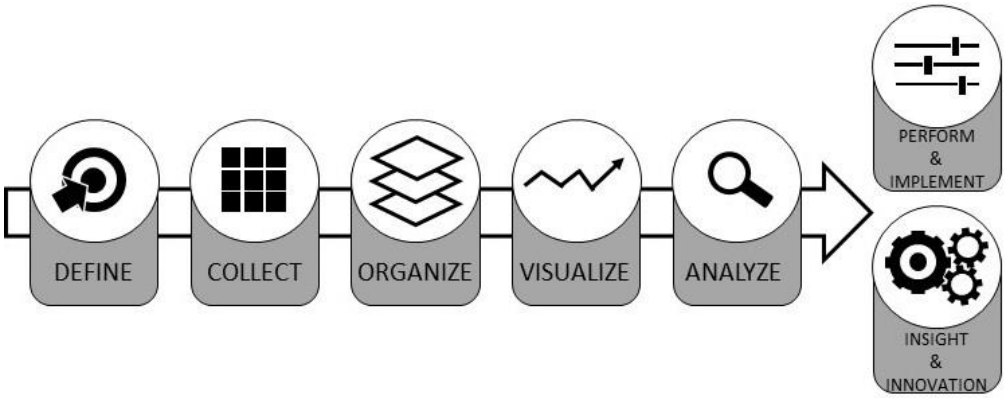


**Figure 1: DCOVA Framework**

#### 2.4.2.2 Approach Questions

The following are questions that need to be answered or identify a path forward in determining logistics data exchange requirements.

1) How do we provide for the integrity and availability of logistics information, and ensure its secure and reliable exchange between the warfighter and the relevant combat service support agency in a timely manner?
2) What does the current logistics operating paradigm look like?
3) What does the logistics concept of support NEED to look like to support future operational scenarios? Has it, or will it change?
4) What does the current logistics IT system architecture look like? Does anybody know? Is it different for each application?
5) What does that IT system architecture NEED to look like to enable logistics concepts of support?
6) Can we leverage existing architectures?
7) Does a new solution need to be engineered entirely?
8) Can Information (or Data) Exchange Requirements (IER) be precisely defined?
9) What information needs to be communicated to whom and where—higher, adjacent, subordinate, supporting, joint, coalition, contracted vendors?

#### 2.4.2.3 Approach Methodology

1) **D**efine. Five of the six functions of logistics were decomposed into their sub-functions—as defined in doctrine—as the initial structure of this analysis. The variables are comprised of the data elements in each of those categories.
2) **C**ollect. Data points were collected for each function/sub-function. The system name, purpose, and type were not relevant to the collection process and were omitted. However, each data point was evaluated in its origin, echelon value, schema, authority, and type.
3) **O**rganize. The collected data points were organized by function/sub-function category. The anticipated command/organizational echelon was then specified for each of those data points.

4) **V**isualize. Once the collected data points were organized, a pivot table was created to visualize how each data point aligned to its corresponding function/sub-function. Due to the sheer volume of data points (approximately 1,000 in total), the full table is not displayed below in Figure 2, but is included in Appendices B - F.

| Functions to Data Elements | Functions to Data Elements | Functions to Data Elements | Functions to Data Elements | Functions to Data Elements |
|---|---|---|---|---|
| ⊟ Supply | ⊟ Transportation | ⊟ Maintenance | ⊟ General Engineering | ⊟ Health Services |
| ⊞ A. Requirements | ⊞ A. Embarkation | ⊞ A. Inspection | ⊞ A. Engineer Reconnaissance (ER) | ⊞ A. Medical Command and Control |
| ⊞ B. Distribution | ⊞ B. Landing Support | ⊞ B. Classification | ⊞ B. Obstacle Removal | ⊞ B. Casualty Management |
| ⊞ C. Procurement | ⊞ C. Motor Transport | ⊞ C. Evacuation | ⊞ C. Explosive Ordnance Disposal | ⊞ C. Dental |
| ⊞ D. Storage | ⊞ D. Aerial Delivery | ⊞ D. Calibration | ⊞ D. Power Generation/Distribution | ⊞ D. Medical Stability Operations |
| ⊞ E. Salvage | ⊞ E. Material Handling Equipment | ⊞ E. Servicing | ⊞ E. Bulk Liquid Support | |
| ⊞ F. Disposal | ⊞ F. Freight Transportation | ⊞ F. Repair | ⊞ F. Demolition | |
| | ⊞ G. Passenger Transportation | ⊞ G. Modification | ⊞ G. Facilities Maintenance | |
| | ⊞ H. Port and Terminal Operations | ⊞ H. Reclamation | ⊞ H. Horizontal Construction | |
| | | ⊞ I. Recovery | ⊞ I. Vertical Construction | |
| | | ⊞ J. Overhaul | ⊞ J. Bridging | |
| | | ⊞ K. Rebuild | ⊞ K. Water Production/Storage | |

**Figure 2: Functions of Logistics Visualization**

5) **A**nalyze. The analysis was tested and validated through a tabletop wargame. The participants of the wargame were provided a realistic, current state scenario, intelligence brief, and operation order, and given the task to identify the minimum data that must be communicated to trigger required actions and facilitate the requested combat support.[6]

### 2.4.3 Structure of the Analysis

The structure of this analysis provides the operating context demanding a paradigm shift, defines and scopes the problem, and identifies the preliminary findings and data commonalities.

Key terms and definitions are provided to ensure uniform, common understanding and their contextual use throughout the analysis. Data exchange requirements for each function of logistics (minus ‘Services’) is then presented. Lastly, critical capability gaps and recommended changes to doctrine are proposed.

### 2.4.4 Reusability

This analysis is scalable and can be applied to multiple settings. The process and methodology used herein offer an overall “how-to” structure and procedure. To reuse this analysis, the analyst would:

1) Collect, read, and extract the required processes in the six functions of logistics, or any project requiring data, from the library of directives, orders, and regulations as well as observation of work in the six functions of logistics at each echelon. This collection is the baseline of what is functionally required.

2) Extract and list current data requirements for each function and sub-function of logistics. This process will reveal overlaps, redundancies, duplications, gaps, and help determine the point of data creation/origin and subsequent curation into useable information.

3) Lastly, determine the echelon value, priority, and criticality of each data element in terms of who, what, when, where, why, and how the data is curated and leveraged. Determine the minimum data required for action/decision makers to begin their cycle/process for support.

[6] This wargame was reviewed by the Marine Corps Warfighting Laboratory (MCWL) and approved for unclassified use.

# Section 3: Tactical Logistics Data Analysis

## 3.1 Preliminary Findings Common to All Functions

### 3.1.1 Shared Data Elements

The participants of the Project ADRIAN wargame identified key data elements common to every aspect of logistics, regardless of the function or sub-function. Traditionally, these common data elements have varying degrees of granularity depending on the use.

#### 3.1.1.1 Shared Observation of Data Relevance by Echelon

As data is developed upwardly, the granularity becomes less relevant. For instance, the serial number of a truck identified for embark by the unit becomes less relevant as it moves upward in the supporting chain as it is consolidated with other units' embarkation requests. However, the quantity, weight, and size are the enduring relevant data elements. The relevance condition applies to all functions and sub-functions of logistics, without exception.

#### 3.1.1.2 Differing Data Variations of "Who, What, When, Where, Why, and How"

Through the analytical process, it was determined that the basic data requirements were common to each task but comprised of differing formats of the "five W's and H".

For instance, in terms of "who", the Marine Corps captures this data for a unit/element as a callsign, brevity code, Department of Defense Activity Address Code (DoDAAC), Unit Identification Code (UIC), Monitored Command Code (MCC), organization name, Reporting Unit Code (RUC), and in some cases Routing Identifier Code (RIC). Some of these elements are hierarchical or nest into larger organizations but all primarily identify the element as "who" within a system. Of note, some of these data variations are more akin to accounts, ledgers, and balance sheets than truly element/unit identification, mission, purpose, and capability.

Each logistics IT system uses different forms of identification, location, and date/time formats resulting in disparities in what should be commonly understood. These disparities produce data gaps and forces data reinterpretation, thereby delaying decision cycle time and follow-on action.

#### 3.1.1.3 Identical Data Replicated System to System

Since none of the data is federated from and to the highest authoritative source(s), current logistics IT systems are not integrated, interoperable, or limited in system-to-system interface. Basic data is generated, replicated, and copied repetitiously from authoritative sources to each system independently.

Due to the lack of integration and interoperability, data becomes stale and unreliable as it traverses between systems. To align data between systems, there is a systematic reliance on human capital acting as system interfacing agents, consuming an enormous number of man-hours to reconcile, validate, and transcribe data from one system to another. Additionally, the Marine Corps has created and codified a culture of acceptance of human-based reconciliation/validation through mandates, directives, and orders by making these actions inspectable provisions that require proof via paper-based records. This creates conditions in which Marines—as human capital—are working *for* the IT system(s) rather than the system working for Marines.

This culture of acceptance places the burden of correcting data impurities and disparities on the backs of the Marines, vice demanding system-wide changes to eliminate the need for data reconciliations/validations in the first place. Systems lag in these improvements due to

perceived fiscal constraints (internally or externally imposed) in developing or correcting information system flaws up-front.

#### 3.1.1.4 Logistics Data Capture Approach

The analysis revealed that all logistics data is generated by analog sources. If data is captured digitally, that data is transcribed from analog sources or back into analog sources. Logistics processes exhibit a systematic reliance on paper correspondence to initiate action and assign/respond to inquiries/requests, rather than leveraging information technology in the form of business workflows. In tactical environments, data may be generated via chat applications (i.e., mIRC, Mako, Transverse, Matrix), email, and verbally via voice-over-radio. In most instances, if data is generated in a digital environment, it is then migrated to an analog intermediary reporting source, processed via human “middleware”, and then transcribed into another digital reporting system for decision/action. Data is transcribed from analog to digital form by individual Marines—human beings—who manually enter data into specific fields of a database or database-like system from paper sources of documentation (i.e., printed letters, reports, forms, spreadsheets, etc.).

This human-dependent paradigm is inherently susceptible to data gaps and errors. Inaccurate, erroneous, or irrelevant data is processed into misleading information with inconsistent variable extrapolation. The Marine Corps, as an enterprise, does not have a clear and concise plan to generate and capture data originating at the lowest echelons.

Despite the proliferation of new technology in recent years, the enterprise has yet to establish a unified data collection plan. Lack of unity between the various logistics agencies, organizations, and departments has only compounded this critical shortfall.

The result of not having a data capture plan, is that the enterprise does not consider or enable where data originates at the most forward tactical edge, where the preponderance of actual work and reporting is taking place.

For example, none of the maintenance lots, bays, stations, and benches in 3rd Marine Logistics Group (MLG) have network access extended beyond their office spaces. The technology gap results in lane non-commissioned officers (NCOs), technicians, operators, and mechanics to develop analog stop-gap solutions. Marines on the tactical edge must typically transcribe reporting data into an analog, organizationally unique form, and send it to a keying clerk for manual input into the relevant logistics IT system. This problem is amplified in exercise/deployed settings, causing the enduring reliance on paper logbooks as “true” data systems of record.

If the Marine Corps logistics data capture plan does not consider the connectedness of devices and/or connected/disconnected synchronization, every data-driven capture technology, no matter how innovative, is degraded in its utility.

## 3.2 Key Terms and Definitions

In scientific and academic documentation, key terms and definitions are typically provided in a glossary. However, organizations across the enterprise have used these terms inconsistently and often deviate from their proper definitions. This section provides terms and definitions to provide the appropriate context and establish common understanding throughout this analysis.

### 3.2.1 Defining ‘Logistics’

This section provides context to the term “logistics,” specifically, *military* logistics. While similar to industry, military logistics encompasses greater scope. (i.e., human resources, engineering,

postal, etc.). The term “logistics” considers and recognizes the six doctrinal functions and multitude of sub-functions, even though only five of the six are subject to this analysis.

When the term “logistics” is used, the analysis means this:

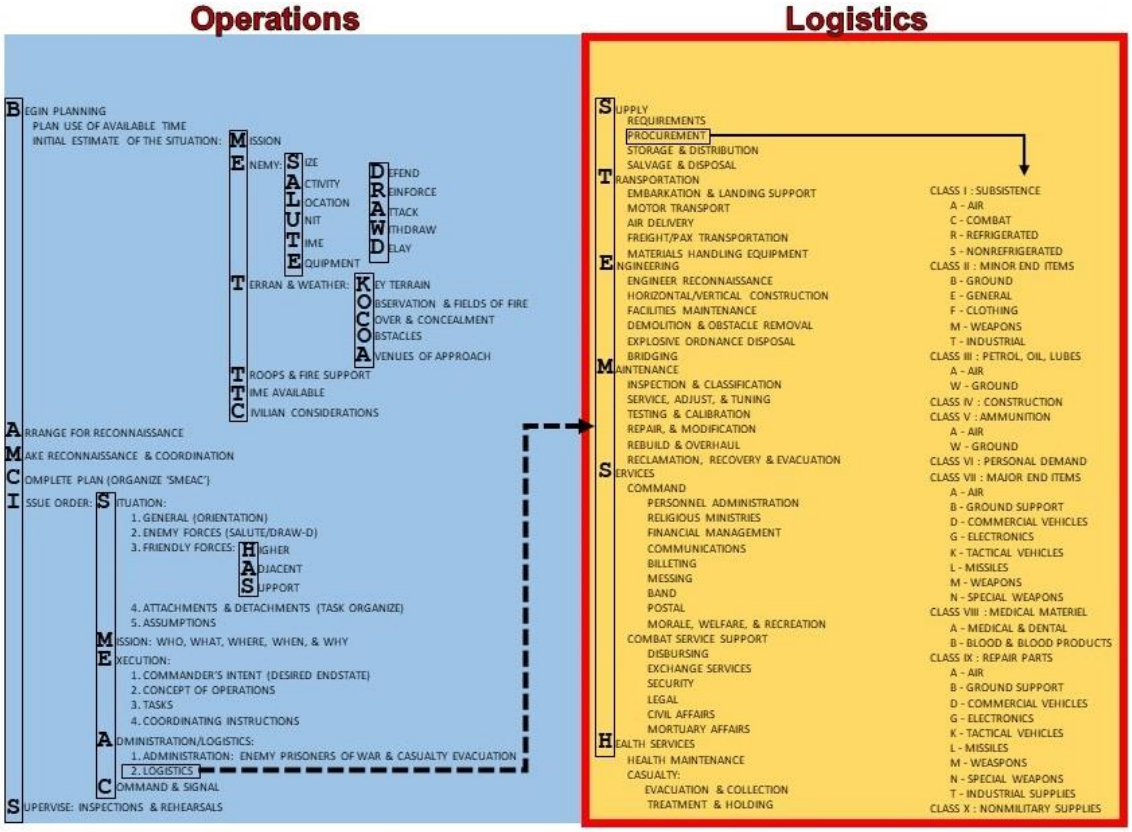


**Figure 3: Operations to Logistics Relationship**

To overcome the problem space defined in this analysis fill capability gaps, the logistics data solution must holistically encompass *all* tasks associated within the five functions of logistics. The solution must be homogeneous in terms of processes, data availability and accessibility, and data exchange across the MCILE.

The Marine Corps has specified and directed how these functions will support and sustain the force across the full spectrum of military operations on an operational and tactical level. It should be noted, while *Joint Publication 4-0—Joint Logistics* (JP 4-0) and *NATO Logistics Handbook* recognize seven functions, the Marine Corps only recognizes six (omission of Operational Contract Support - OCS). [9] The continuance of this analysis will follow the Marine Corps’ doctrine and omit the OCS function and relating tasks.

### 3.2.2 Defining ‘Contested’

For our military context, this means any action engaged in *competition with* or *opposition to* a position of strength, power, or influence.

Great power competitors are now—at this time—engaged in a variety of actions to oppose U.S. positions of strength, power, or influence in a variety of contexts, including operations, communications, information, logistics, culture, unity of effort, or intelligence.

The term “contested logistics” invokes ideas of maritime shipping lanes and sustainment within the first island chain of the western Pacific region once open conflict occurs. However, the predominant near-peer competitor is already engaged in actions to challenge supply chain power, position, and influence, not only regionally, but globally and *within* the U.S.

### 3.2.3 Defining ‘System’

The analysis requires a clear definition and context on how a system is defined and understood. The following is a top-to-bottom summary of the components that comprise a ‘system’.

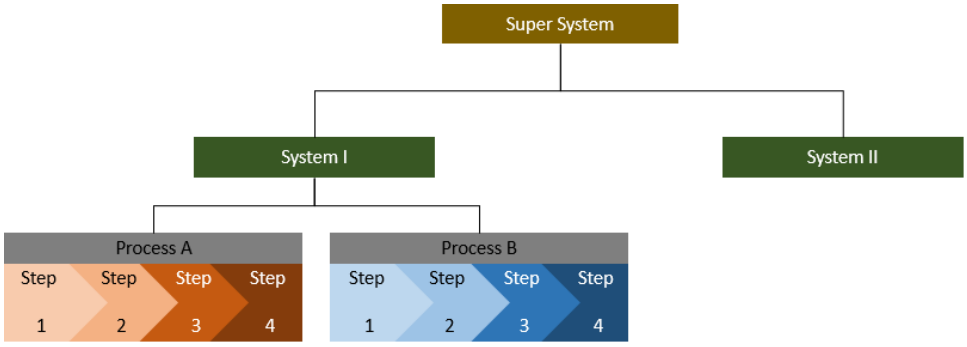


**Figure 4: Step to Super System Hierarchy**

1) Super System, or System-of-Systems: The assembly of multiple, independent systems in context as part of a larger, more complex system (i.e., Logistics as a function is a super system).
2) System: The collection of interconnecting processes organized within an operating framework (i.e., the sub-function of ‘Supply’ is a system).
3) Process: A group of procedures taken in sequence to achieve a specific outcome.
4) Procedure: When a series of tasks/steps are conducted in a certain order.
5) Step or Task: The singular piece of work to be performed.

For this analysis, the super system is the warfighting function ‘logistics’, the sub-systems are the six functions of logistics, and the processes are the collection of the series of actions or steps taken to achieve the results, product, or end state desired within those logistics sub-functions. The analysis attempts to avoid focusing on particular steps, as this type of detail detracts from the context of the process of logistics at the functional level.

### 3.2.4 Defining ‘Capability’

For this analysis a ‘capability’ is defined as the quality, power, or ability in a present state to efficiently/effectively accomplish something; the ability to *do something*.

To quantify a capability, however, three conditions must be met:

1) Alignment: A logistics capability must be specifically aligned to a function, and if applicable, the appropriate subfunction; its purpose.
2) Enabling Factor: The logistics capability must enable a specific process, procedure, or task.
3) Level: The logistics capability must be aligned to the echelon at which it is expected to be performed or useful.

Below is an example of a quantifiable logistics capability.

**Figure 5: Properly Defined Capability Example**

In this example the alignment is “materiel”, which specifically falls under the function of supply. The enabling factor is “request”, and the level at which the capability is to be performed is “the point of need or lowest tactical level”.

### 3.2.5 Defining ‘Kill Chain’

In the abstract, a ‘kill chain’ is defined as the signal pathway from “sensor” to “shooter” on the battlefield.[7] A ‘kill web’ is comprised of multiple kill chains linked together through shared physical or logical signal paths.

Not every sensor needs to communicate with every shooter all the time. That model would create congestion in the communications architecture and present an enormous force protection risk when the adversary has significant electronic warfare (EW) capabilities.

“Shortening the kill chain” requires optimizing the signal pathway from “sensor” to “shooter” that will enable effective action (kinetic or non-kinetic) in the quickest, most effective manner possible. The *right* sensor communicates with the *right* shooter at the *right* time to achieve the desired end state, and it must do so within a communications denied/degraded environment. Therefore, a *critical data analysis* is required—that is, the minimum data exchange requirements must be identified to facilitate the desired decision and/or action in each function of logistics. Logistics IT must adopt this “kill chain/web” framework to enable effective decision making and combat support in operational environments.

Applying this concept to logistics means the right “sensor”—human or technology device—sending the right data to the right logistician at precisely the right time to initiate the required decision and/or action.

### 3.2.6 Defining ‘Data’

Put simply, ‘data’ refers to individual facts and statistics collected for reference or analysis. In a more granular sense, data represents the quantities, characters, or symbols on which operations are performed by a computer, stored, and transmitted in the form of electrical signals and recorded on magnetic, optical, or mechanical recording media.

For the purposes of this analysis data is not “information.” Data is the basic building block for information. This analysis is focused on the “basement” level of data creation, and the purpose for which it was created. Data at the lowest level allows all other capabilities to build upward.

[7] The term “shooter” is not used exclusively to refer to kinetic fires assets. In the abstract, “shooter” encompasses any system, node, or even human being that executes a task associated with the ingestion of sensor data.

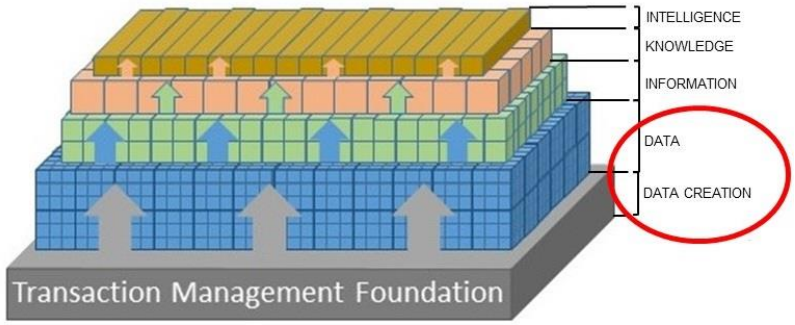


**Figure 6: Data Foundation to Wisdom Hierarchy**

### 3.2.6.1 Data Types

For this analysis, the taxonomy of data types is defined as follows:

1) Data Field (Element): A container for the storage of a recorded fact, character, or symbol. By and large, a data field/element is usually unusable in this basic state. A collection of unrelated data fields is considered a “lake” of data.

**Figure 7: Data Field (Element) Example**

2) Data Block (Object): A logical, topic-based grouping of data fields (elements). This data structure links individual data fields/elements into something more useful.

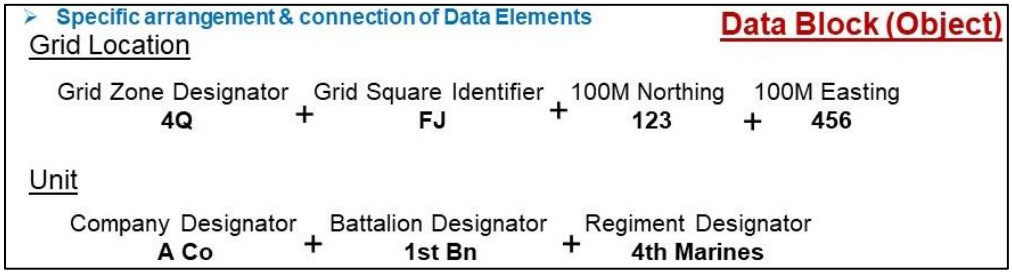


**Figure 8: Data Block (Object) Example**

3) Data File: A collection of data blocks (objects) on which operations, analysis, and/or calculations are performed at a higher level. This is where information begins to take shape.

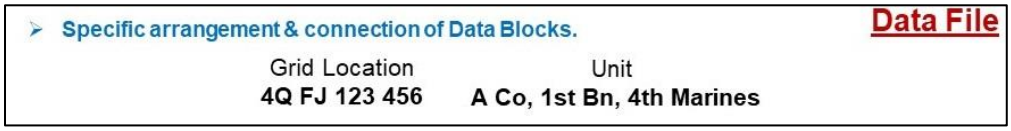


**Figure 9: Data File Example**

#### 3.2.6.2 Data Significance

The majority of data collected in the Marine Corps today is input/created at the data transactional layer, also known as the transaction management foundation. Data creators/inputters are, more often than not, 18 – 22-year-old human beings—the Private First Class (PFC) and Corporal (Cpl) population at the beginning of their military careers.

Anyone above the data transactional layer is a "data consumer". Data consumers are divorced of data creation. Additionally, there is currently no means by which to ensure that data—manually input by a human being—is correct, accurate, timely, and verified upon its entry into the system.

***Strategic decisions at the highest echelons are therefore based on data generated at the lowest echelons and manually input into permissive logistics IT systems by the most junior population of the Marine Corps' workforce.***

### 3.2.7 Defining 'Knowledge Management'

Data is the nucleus of knowledge. Data must be collected, processed, and developed for it to have value. This section explains the importance and impacts of this science as it pertains to logistics data, how it is exchanged, and what is required for it to enable action/decision.

#### 3.2.7.1 Military Application of Knowledge Management

Knowledge management (KM) in theoretical, scientific, and academic circles has only recently emerged as a body of science. While not a new concept, the KM framework is relatively immature in the DoD, with only 20 years compared to more than 200 years of other military disciplines. [10]

KM must not be conflated with information management (IM) and should be addressed. In many instances these terms are often used interchangeably, however, they differ in both meaning and content. The following sources provide a basic point of reference that illustrate that disparity of definitions between the service departments: [10]

1) *Marine Corps Doctrinal Publication (MCDP) 2 – Intelligence*. Describes information as "unevaluated material of any kind… [used as] raw material from which intelligence is derived."[8] From this perspective, information is data that can be processed and put into an understandable form. It is the basic building block of knowledge and understanding and is essential to intelligence. [11]
2) *MCDP 6 – Command and Control*. Describes information as "…all manner of descriptions or representations from raw signals on the one hand, to knowledge and understanding on the other."[9] In this context, information categories range from data—raw or unprocessed signals—to more highly developed forms evaluated/integrated into meaningful knowledge and understanding. [12]
3) *Department of the Navy Knowledge Management Strategy*. The DoN Chief Information Officer (CIO) defines KM as "the integration of people and processes, enabled by technology, to facilitate the exchange of operationally relevant information and expertise to increase organizational performance." [13]

[8] Page 1-7
[9] Page 66

For the purposes of this analysis, 'knowledge' management is defined as follows:

***The process and ability of an organization to capture, develop, share, and effectively use knowledge as it pertains to the organization's mission, purpose and/or goal.*** [14][10]

This definition is the foundation upon which the efficacy/efficiency of data supporting logistics processes and measures of effectiveness thereof are determined. This may be a concept to which the Marine Corps is not accustomed, and is outside the normative institutional emphasis on training, leadership, and command influence to resolve KM issues.

*Knowledge requires data that is timely, accurate, and factually sound. That data must be thoroughly examined in the context of the 'kill chain' framework for it to be useful, efficient, and effective.*

#### 3.2.7.2 Data Development in the 'Data, Information, Knowledge, Wisdom (DIKW)' Framework

Developing basic elements of data into something useful requires two conditions to be met: *connectedness* and *understanding*.

1) Connectedness indicates the level of accessibility and reliability of the data as it is transformed through *understanding* (i.e., a shared environment or integrated system), commonly facilitated through systems interconnected with one another via communications network pathways (i.e., internet, intranet, cloud platforms, etc.).
2) Understanding is the application of cognitive and analytical processes to synthesize new knowledge from previously held beliefs. [15] Understanding applies to both tacit *and* explicit knowledge.[11] [12]

The chart below depicts data development within the framework of *connectedness* and *understanding*.

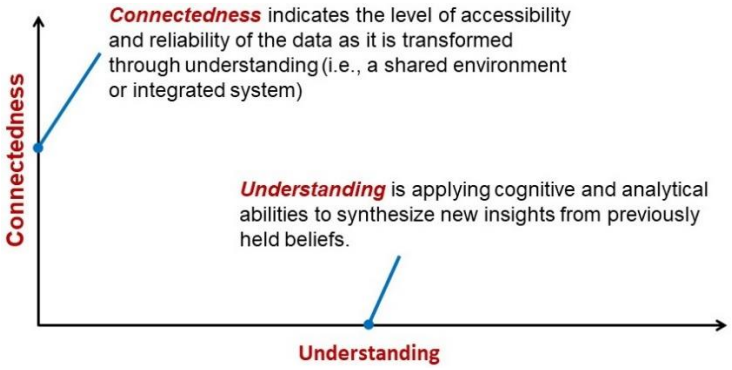


**Figure 10: Data Development Framework**

Applying the data development framework reveals the necessary relationships between *connectedness* and *understanding* required to mature data and transition it through the hierarchy. This framework provides fundamental insights into what it takes to turn raw data into something useful.

[10] Original definition contextualized to fit military applications.
[11] Tacit knowledge is a kind of knowledge that is difficult to transfer via written or verbal form. It must be gained and accumulated primarily through experience. [17]
[12] Explicit knowledge is the kind of knowledge that can be transferred through written or verbal forms. It can be gained through codification or formalization as well as academic education. [17]

1) Data. Data is the foundation of the DIKW framework, and usually exists as a transaction or entry. [16] Data is a discrete element; it is distinctly individual and unconnected. Connectedness must be in a shared data environment. Paper-based or multiple stove-piped information systems will always impede data movement through the hierarchy. An understanding of relationships is required for data to be processed and synthesized into information. ***Data quality*** at this level is imperative and has an upstream effect on all levels of this hierarchy. In a shared data environment, bad data leads to bad decisions—"garbage in, garbage out". This is why many organizations invest in linear transactional standardization with little area to deviate. In industry, deviations from transactional data inputs usually require a supervisor to override in special circumstances and must be justified.

2) Information. When data points are contextualized and compared, then data has transitioned to the information level in the hierarchy. [16] Usually, information comes in the form of reports or listings in which separate data points can be compared within a certain context. Connectedness contributes through interfaced or integration of multiple systems so data points can be arranged to provide meaning. In order to move from information to knowledge, an understanding of patterns or trends must be established. Information usually looks to the past, relative to the present.

3) Knowledge. Information arranged to show/visualize a trend or pattern has transitioned to knowledge. [16] Knowledge commonly manifests as dashboards, graphs, and charts, which are employed to indicate a basis by which to measure performance. Connectedness influences knowledge by presenting information from different-but-related data sources. To move from knowledge to wisdom, an understanding of principles and/or theories must be applied. Knowledge usually looks to real-time or near-real-time in relation to the present.

4) Wisdom. Formulated knowledge that is analytical in nature can be used to forecast future events or situations. [16] Wisdom enables decision within time of opportunity, creating a "decision advantage." Thus, wisdom becomes time-sensitive; as the lower hierarchal levels change, so does the wisdom affecting the decision of opportunity. Connectedness supports wisdom by providing real-time or near-real-time updates across multiple but related data points. Wisdom looks to determine future outcomes based on current developments and are continually changing as updates occur.

Data becomes more strategic in nature the higher in the DIKW hierarchy it moves. Individual transactional data detail becomes less relevant as knowledge becomes more directional in nature (i.e., serial number of an asset becomes less relevant compared to quantity of assets the further the data is developed). The graph below depicts the hierarchies and the type of understanding required to transition from one stage to the next.

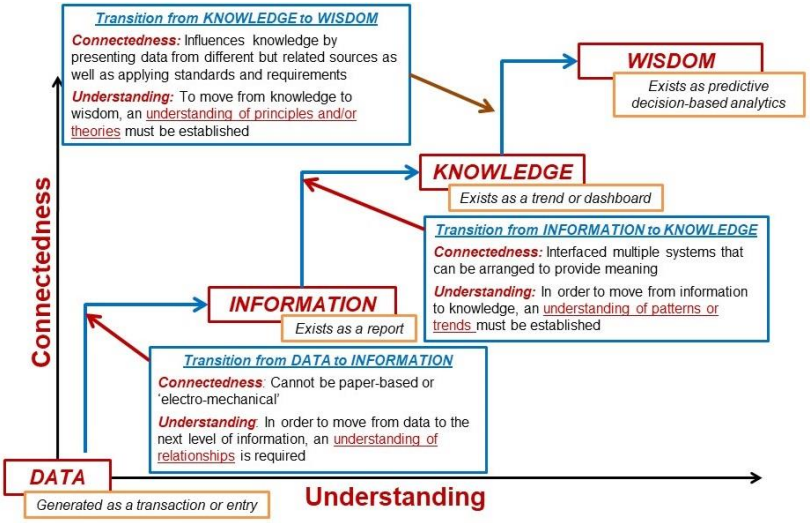


**Figure 11: Data Development Hierarchy**

The process of knowledge management is essential to apply the science to logistics information systems as the process frames the general goal of development of data in these systems. When comparing the components of knowledge management, an "intelligent" (either machine assisted or artificially enhanced) based system requires a unified system which incorporates the subject's theory to provide an analytically based forecast on future outcomes. The table below shows the knowledge management requirements at each level in order to achieve the ideal state of each.

| | Data | Information | Knowledge | Wisdom |
|---|---|---|---|---|
| **Form** | Transaction | Report | Dashboard | Analysis/Forecast |
| **Time Relation** | Discrete | Past | Present | Future |
| **Connectedness** | None | Interfaced | Integrated | Unified |
| **Understanding** | None | Relationships | Pattern/Trend | Theory |

**Table 2: Data Development Relationships**

### 3.2.7.3 Logistics Information Technology Drivers

Industry has identified drivers that possess relevant roles in logistics information technology and data exchange requirements. These drivers are identified in this analysis as they are pertinent in governing the modernization in logistics information systems. The identified technology drivers have been modified to fit into a military context.

| Technology Driver | Definition | Role of Technology |
|---|---|---|
| **Optimization** | The alignment of global supply and logistics chain resources, both tangible and intangible, to facilitate the success of organizational mission | Optimization tools analyzes all possible options to find the best solution to a logistics problem, at all levels – horizontally as well as vertically |
| **Synchronization** | The ability to coordinate, organize and manage end-to-end logistics flows (materiel, services, and information) in such a way the supply and logistics chain functions as a single entity throughout the enterprise | Technology facilitates real-time data availability and sharing between organizations. The information provides common insight and facilitates collaborative decision making between the supporting and supported organizations (i.e., common operational picture) |
| **Agility** | The degree to which an organization can change practices, processes, and/or structures of systems and networks in response to unexpected events, their effects, or impacts | Tools which generate actionable information by collecting data regarding demand, inventory flows, and requests; filter the data; and presenting it in a form that can be readily used. That is, technology tools allow decision makers to quickly adjust to changes in supply and logistics chains |
| **Velocity** | The speed at which end-to-end flows occur in the supply and logistics chain. It encompasses speed-to-battlefield when conditions are rapidly changing | Properly implemented technologies help organizations rapidly respond to customer requirements for faster, more consistent flows of materiel, services, and information. Event trigger management tools have the capability to address problems dynamically, provide recommendations, and automatically resolve issues |
| **Resourcing** | The result of being good stewards of public funds throughout the supply and logistics chain. Asset performance, depreciation, cost-to-value, time, and people are some of the critical components that require alignment to mission need in a global environment | Technology tools help organizations manage inventory, transportation, and other key logistics functions more effectively than could be accomplished through manual means. The outcomes are greater control of costs and enhanced effectiveness with greater efficiency |
| **Resilience** | The ability to transition logistics from a garrison state to the field and back seamlessly with little to no interruption in service | Logistics systems assist in the transitional nature of the logistics network by scaling and scoping resources within the available resource model. Through redirection and reapplication, the logistics chain can anticipate and transform itself as the situation requires and back again. The ability to transform requires a clear signal pathway on a stable communications platform. |

**Table 3: Technology Drivers**

#### 3.2.7.4 Knowledge Management as a Demand for Logisticians

Previous sections have detailed the evolution and drivers of information technology in the logistics sector. For military applications, what has changed in the knowledge management context? As in industry, the military is now beginning to see the comparative difference in 'knowledge work' and more traditional 'manual work'. Unlike repetitive tasks generally associated with manual work, knowledge work involves the cognitive abilities of the human brain.

| Manual Work Is: | | Knowledge Work Is: |
|---|---|---|
| Highly visible | → | Largely invisible |
| Highly specialized | → | Holistic |
| Stable | → | Ever-changing |
| Fixated on the right answers | → | Fixated on the right questions |
| Focused on structure with few decisions | → | Focused on less structure with more decisions |

**Table 4: Manual Work vs. Knowledge Work**

KM shares many of the same characteristics as what the Marine Corps would call "smart staff work", the fundamentals of staff action which have existed as long as our Armed Forces. In March 2014, the DoN CIO reissued an updated KM strategy outlining a vision to "… to create, capture, share, and reuse knowledge to enable effective and agile decision-making, increase the efficiency of task accomplishment, and improve mission effectiveness." [17] This DoN KM Strategy outlines four primary goals:

1) Goal 1: Continue to expand Departmental awareness that KM applied to operational and business processes will enable significant improvements in mission accomplishment.
2) Goal 2: Instill KM principles and methodologies into the DoN culture.
3) Goal 3: Maximize the utility of existing DoN KM experience and resources.
4) Goal 4: Drive towards a more centrally supported universally available and executed DoN KM.

Overall, the DoN has identified KM opportunities to adopt and mature communications, knowledge sharing, and collaboration to promote an increase staff productivity by 20 to 25 percent. [18][13]

<u>Since a large percentage of military staff work is knowledge-based, especially that of logisticians, it is essential to increase the productivity of the logistician as a primary means of responding to increased mission demands and decreased personnel/financial resources. In other words, technology must be leveraged in a way that gives time and money back to the unit where velocity in the logistics chain is based on data flow; or, alternatively, leveraging IT-enabled automation to fill the resource gap</u>.

In order for the benefits of the above strategy to be realized, it is necessary to understand the four basic information characteristics which facilitate productivity through data-based logistics technology. These characteristics drawn from industry do in fact apply to government/military applications [19]:

[13] "…the McKinsey Global Institute (MGI) finds that twice as much potential value lies in using social tools to enhance communications, knowledge sharing, and collaboration within and across enterprises. MGI's estimates suggest that by fully implementing social technologies, companies have an opportunity to raise the productivity of interaction workers—high-skill knowledge workers, including managers and professionals—by 20 to 25 percent."

1) Information accuracy. Good decisions are difficult or nearly impossible without information that gives a true picture of the state of logistics. Information does not need to be 100% precise all the time, but rather the data available must paint a picture that is directionally accurate. The more automated the system, the more accurate information becomes.
2) Information accessibility. Decision makers need data at the right time, in the right place, and it must be accessible. This is the application of the “kill chain” model. Most times accurate data may exist but by the time it becomes available (by transcribing from analog to digital or interfacing) it is either out of date or is not accessible in the operating environment. The more integrated the system (vice interfacing) the more accessible the system becomes.
3) Information type. Organizations typically capture vast amount of data. However, the quantity of data captured does not ensure good decision making. Decision makers require information that is relevant to the decision they are making. Therefore, enterprises must think about the data that is recorded, how it is recorded, and if it should be recorded at all so that valuable resources are not wasted collecting meaningless data while important data is not being recorded. This is the essential component of federated data.[14]
4) Information sharing. Logistics can only be effective if all organizations are sharing a common view of the information used to make decisions. Different information with different sub-organizations results in misaligned action plans that hamper logistic chain performance. Data sharing must occur horizontally (lateral) as well as vertically (hierarchical). It should be noted paper-based systems do not lend themselves to information sharing in any form and at any level. Paper-based methods must be used as a last resort in the direst of circumstances and only temporarily.

Therefore, we can conclude information, as represented as knowledge management, is the lifeline of the logistics chain process which drives effective decisions and actions on data exchange requirements [20]. The criticality of logistics information to the logistician is essential as their direct line of sight to logistics processes is very limited – this is especially true for the SIF concept which operates in an austere distributed operating environment.

## 3.3 Logistics Data Elements and Interdependencies

Through analysis and the LOG IT Wargame, it became evident there exists essential data requirements common to all functions, agnostic to the type of request (material, information, or service) or if the request is push or pull or echelon level of origination. These essential data elements provide context and must endure through the lifecycle of the request providing trace back.

The perspective all the following data requirements is from the lowest tactical user leveraging the demand.

[14] A federated database is virtual and does not physically store any data. Instead, it provides an interface where users can query information across multiple data sources and coordinates information across non-congruent systems.

### 3.3.1 Solution to Voice-over-Radio Communications

#### 3.3.1.1 Legacy Radio Communications

Pre-formatted military reports are designed and geared towards voice-over-radio communications. These pre-formatted reports have changed little since field radios were first used by the United States Army in the Spanish American War of 1898 and the Philippine Insurrection of 1899–1902. The standardization of the reports ensures the recipient has the right data for the purpose of the communication. Even so, transmitting voice-over-radio reports and requests require military phonetic spelling of key data elements with the repetitive transmitting and confirmation repeating of data. Voice-over-radio transmissions exposes forces in the following ways:

1) Near-peer competitors have the ability to scan, detect, and triangulate EM transmissions, including radio frequency transmissions. Every time the handset is repetitively keyed, the radio transmits exposing forces to detection by pattern.
2) The time it takes to transmit, even the simplest of reports/demands, via voice-over-radio because of the phonetic spelling of each data element elongates the electromagnetic detection exposure risk. Even through the method of spelling, reiteration, and confirmation the message is subject to human error and mis-transliteration.
3) Voice-over-radio transmissions are verbally passed data elements. These elements are transcribed onto various paper forms, logbooks, etc. for processing. Again, paper-based forms of data capture are the least reliable form of data capture.

Voice-over-radio in this competition continuum must be the very last option for communication.

#### 3.3.1.2 Using High Frequency (HF) for Data Transmission

HF Radio is slow, with bandwidth ranging from 75 to 9600 bits per second, with a typical rate of 1200 bits per second. However, such HF radios can achieve worldwide radio coverage when propagation conditions are right by bouncing signals off the ionosphere. If the analysis assumes the farthest forward, smallest tactical unit has access to organic Marine Corps Falcon III AN/PRC-160 radio systems, they will have the reach, but the limitation of data speed transport still remains. Therefore, data exchange requirements need a complete overhaul to scale down the data transmission size. Data no longer needs to be in the legacy voice-over-radio template.

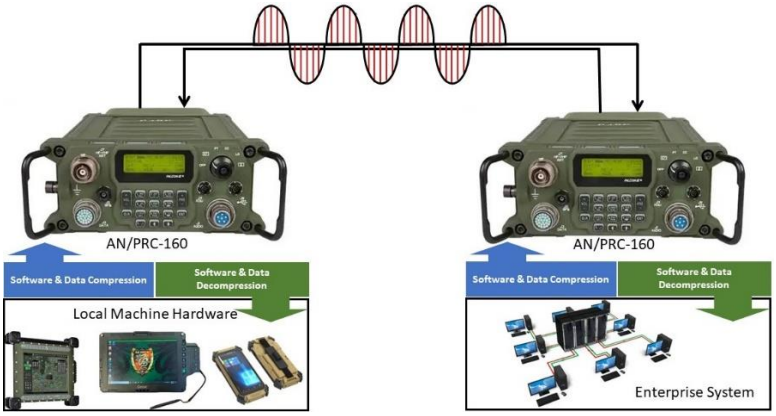


**Figure 12: HF to HF Data Transmission**

Data processing with a data burst capability across the available communication networks must be the primary method. Data burst is a small compact transmission limited only by the radio carrier wavelength. The quality checking by the hardware/software combination ensures

trustworthy connectedness through data "handshake" and checks. With a shorter transmit time and quality data assurance, the risk to forces is minimized without feeding the enemy's kill chain but the data passed must be the absolute critical elements.

### 3.3.2 Data Focus: User Input and Request Data

The following is focusing on the forward tactical user in a communication constrained (intermittent) environment. This is the basis of providing the critical data elements to generate a request across the five of six functions of logistics and its sub-functions. The following will challenge previous held positions on data requirements, flow, and accessibility/retrieval.

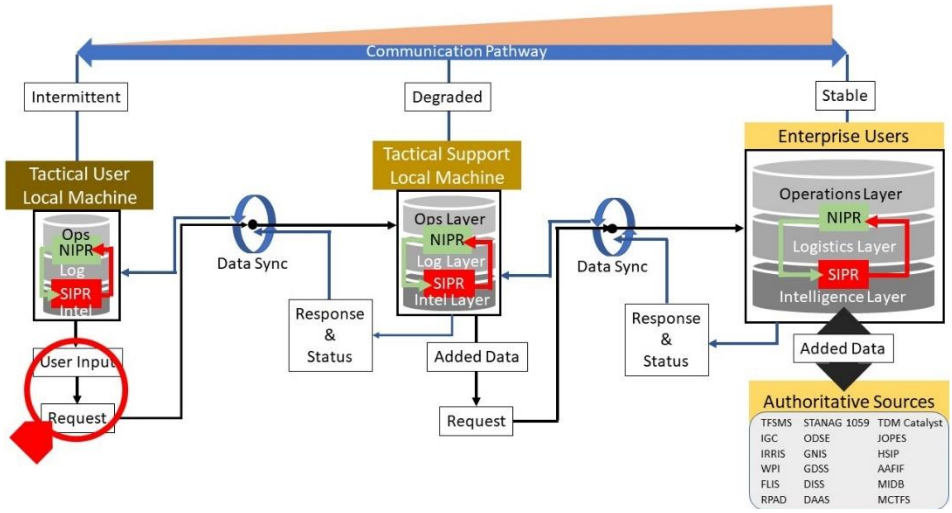


**Figure 13: Data Focus: User Input to a Request**

### 3.3.3 Request Header Concept

The request header concept is the embedded data populated at the time of request creation with little to no user interaction. Header data is an integral part of the data exchange requirements especially for disconnected resynchronization and/or limited bandwidth. The header precedes the actual request information and initiates the demand process as well as providing a common context for follow-on information gathering, status, and response from supporting activities.

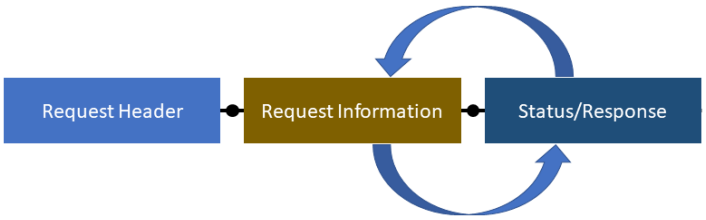


**Figure 14: Data Cycle for Requests**

#### 3.3.3.1 Request Header Data Elements

The following is the required data elements to be embedded into the header of request initiation and are not input by the user. The user has no direct interaction or control over the header data elements.

1) System Date Format – Local. Date expressed as CCYYMMMDD.

2) System Time Format – Local. System time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99). Used to ensure non-duplication of requests. (Note: Decimal seconds are required for the header in order to prevent duplicate time stamps on separate transactions.)

3) Universal Time Coordinate (UTC) Time (Calculated-Data not Passed). In accordance with International Standards Organization (ISO) standard 8601, time can be specified by a + or - and an indication in hours in relation to Universal Time Coordinate (UTC) time; local time is calculated as UTC (plus or minus hours/minutes). Since + is a restricted character, + and – can be substituted by P (plus) and M (minus).

4) Military Date/Time Group (DTG) (Calculated-Data not Passed). DTG expressed as UTC Time/Date: DD HHMM (SS) Z MMM YY based and calculated off the system date and time.

5) System Synchronized Geospatial Reference. Expressed as last system synchronized reference in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical location does not require delivery to this location but serves as a reference in data collection and origination history which can correlate to unit/element use.

6) Local Key Place Holder. Unique system temporarily assigned local key place holder until request is connected, acknowledged, and synchronized to the enterprise system. The device will generate a unique 10-digit alpha-numeric place holder for transmission. Once synchronized, this key is replaced.

7) Request Tracking Identification. Enterprise system unique generated. The enterprise assigned tracking number can be assumed to be an alpha-numeric randomly assigned 12-digit ID. A 12-digit alpha-numeric ID with 24 alpha characters (removing “O” and “I”) and ten numeric characters (zero through nine) offers 2,704,222,032,852,670,000 unique combinations (over 2 quintillion combinations) therefore, the chance of a duplicated request ID is statistically impossible.

8) Request Priority. Expressed as a two-digit numerical input. Priority assignment throughout the logistics system is based on the Uniform Materiel Movement and Issue Priority System (UMMIPS) on the unit/element Force Activity Designator (F/AD) based on the Operational Need Goals (ONG) delineated by mission requirements. Priority government is correlated from the assigned F/AD.

| Force/Activity Designator (F/AD) | | Urgency of Need/Priority | | |
|---|---|---|---|---|
| | | A | B | C |
| | | Critical | Urgent | Routine |
| I | Active Combat/Contingency | 01 | 04 | 11 |
| II | Immediate Combat Ready | 02 | 05 | 12 |
| III | Deployable Force in <30 | 03 | 06 | 13 |
| IV | Deployable Force in <90 | 07 | 09 | 14 |
| V | Non-deployable or deploy >90 | 08 | 10 | 15 |

**Table 5: UMMIPS Priority Table**

Priorities are pre-loaded based on the unit/element's F/AD. The data element is populated by the segment's priority. If the request has multiple segments with differing priorities, the highest priority has precedence.

9) Element/Unit Identification/Call Sign. Expressed as a four-digit alpha-numeric reference key which aligns to the clear text naming convention. "Clear text" is written in normal English, as visible to the user, but the information is stored and transmitted using a unique code. An example in this case, is "1st Platoon, Transportation Services Company, Combat Logistics Battalion 4" is displayed to the user, but the information system transmits the unit identifier data as "G8AQ". The clear text naming convention does not need to be passed, only the unique index key which was randomly assigned by system. The origination of callsigns/brevity codes for units/elements is with the operations function. This stresses the importance of operation planning and execution occurring in a digital environment.

   Alpha-numeric combination of a 4-digit key could be a combination as follows:

   a. 4 alpha characters, 0 numeric characters (4 to 0)

   b. 3 alpha characters, 1 numeric character (3 to 1)

   c. 2 alpha characters, 2 numeric characters (2 to 2)

   d. 1 alpha character, 3 numeric characters (1 to 3)

   e. 0 alpha characters, 4 numeric characters (0 to 4)

   The alpha characters are not case sensitive. While there are 26 letters in the alphabet, the letters "O" and "I" are removed from the available combinations as not to be confused with "zero" and "one" leaving 24 letters for the unique key.

   Numeric characters are zero through nine providing 10 optional characters in the key.

   Available alpha-numeric unit identification/call sign combinations provide 1,080,016 unique keys.

| Ratio Combination | Formula Calculation | Possible Combinations |
|---|---|---|
| 4 alpha characters, 0 numeric characters (4 to 0) | $\binom{4}{0} 24^4 10^0$ | 331,776 |
| 3 alpha characters, 1 numeric character (3 to 1) | $\binom{3}{1} 24^3 10^1$ | 138,240 |
| 2 alpha characters, 2 numeric characters (2 to 2) | $\binom{2}{2} 24^2 10^2$ | 576,000 |
| 1 alpha character, 3 numeric characters (1 to 3) | $\binom{1}{3} 24^1 10^3$ | 24,000 |
| 0 alpha characters, 4 numeric characters (0 to 4) | $\binom{0}{4} 24^0 10^4$ | 10,000 |
| Total possible 4-digit alpha-numeric key combinations: | | 1,080,016 |

**Table 6: Possible 4-Digit Combinations**

10) Request Segments. Expressed as a single numeric digit from one to nine. This means the maximum allowable requests is limited to nine but must be greater than zero. This does not restrict the number of lines within the request segment itself.

   The request segment is used for data verification to ensure the full data file has been passed and corresponds to children requests within the main request container.

#### 3.3.3.2 Request Header Data Exchange Requirements

The request header acts as a container for the follow on and related children or sub-requests. As the header is automated, it does not require human interaction. The following table provides the data segments generated by the logistics application, the use, if the data is repeated, the minimum/maximum of the segment size, the segment usage, and if the segment requires exchange over a communications pathway.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date Request Created - Local | 1 | No | 9/9 | Must Use | Pass |
| 2. Time Request Created - Local | 1 | No | 8/8 | Must Use | Pass |
| 3. Time Request Created - UTC | 1 | No | 11/11 | Optional | Not Passed |
| 4. Date Time Group - UTC | 1 | No | 14/14 | Optional | Not Passed |
| 5. Physical Location of Requestor | 1 | No | 14/14 | Must Use | Pass |
| 6. Request Unique Identification - Local | 1 | No | 10/10 | Must Use | Pass |
| 7. Request Unique Identification - Sync | 1 | No | 12/12 | Must Use | Sync Response |
| 8. Request Priority | 1 | No | 2/2 | Must Use | Pass |
| 9. Element/Unit Identification/Call Sign | 1 | No | 4/4 | Must Use | Pass |
| 10. Request Segments | 1 | No | 1/1 | Must Use | Pass |

**Table 7: Header Data Segments**

#### 3.3.3.3 Request Header Schema

The following is the data schema of the exchanged data segments which would be passed part of the initial communication data burst.

| |
|---|
| Header data schema format: |
| CCYYMMMDD-HHMMSSDD-XXXX123456-XXXXXXXXXX-00-XXXX-00 |
| Header data schema example: |
| 2027OCT13-15470352-4QFJ123456-3838JBNM5-02-KL9K-01 |

**Table 8: Header Data Schema and Example**

The example header portrays a request generated on 13 October 2027 at 15:47 for a single request which is priority 02 (critical) for a unit corresponding to unit/activity ID KL9K. The last synchronized physical location is embedded along with the temporary request ID. The header text size is equivalent to 50 bits.

Upon synchronization the temporary request ID is replaced, and the local machine notes the date/time of synchronization.

| |
|---|
| Synchronized response data: |
| 3838JBNM5-KL9K15474QFJ |

**Table 9: Header Synchronized Data**

The data received from the enterprise replaces the local machine ten-digit tracking ID (3838JBNM5) with the enterprise assigned 12-digit tracking ID (KL9K15474QFJ).

## 3.4 Transportation Data Requirements

The transportation data requirements presented in this section are based on the future doctrinal changes proposed in Section 4 and will not be discussed here. While the header and title of this

section refers to the legacy term for this function as ‘Transportation’, throughout this section the function will be referred to as ‘Mobility’.

Each of the following data requirements are the request segments which are attached to the request header. Request header information will not be duplicated within the request segments.

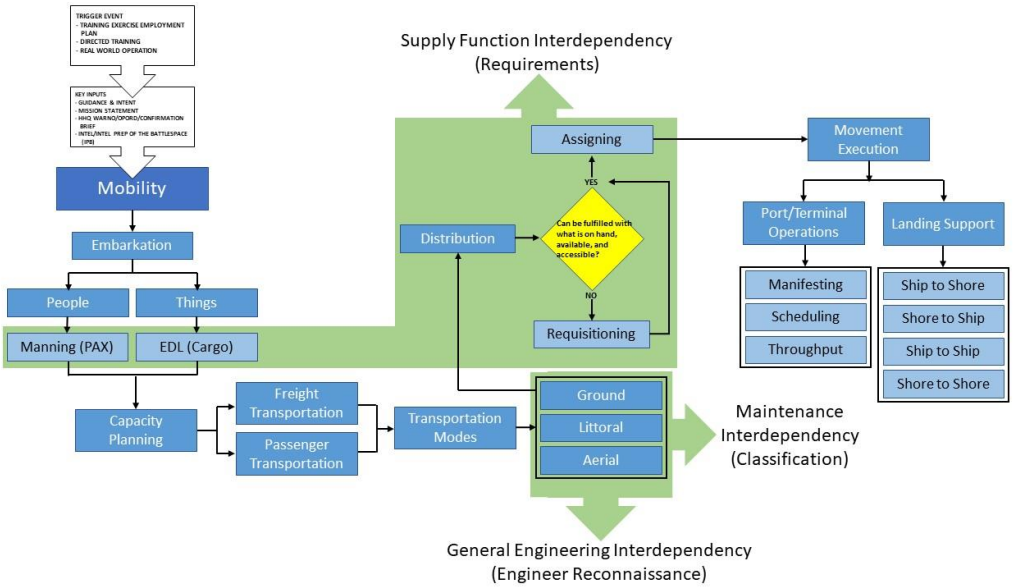


**Table 10: Mobility Cycle**

## 3.4.1 Functional Interdependencies

### 3.4.1.1 Supply Function

The data requirements for the embarkation of people and things (cargo) are interdependent with the ‘Requirements’ sub-function of Supply, which dictate the generation of equipment density lists (EDL) and manning/staffing documents. Any variation in the element’s staffing and equipment level compared to its established structure becomes a staffing document and an EDL.

Output data files of the staffing and EDL hold the critical unit or element data. It is not anticipated forward deployed units would have the entirety of these files on the local machine but only the data slice which pertains to the unit/element directly.

Data slices ensure the minimum retrievable data is encapsulated and stored locally, reducing potential surface exploitation vectors. Regardless, the embark data is wholly dependent on data anchored in the staffing and EDL, found in the ‘Requirements’ sub-function of ‘Supply’.

The user making the request will need to have access to their specific data on the local machine. This data slice should be apportioned and scaled to the appropriate level. For example, the EDL is apportioned from the original structure found in Total Force Structure Management for allowances and the supply system for on hand quantities, the EDL would be apportioned into a consolidated memorandum receipt (CMR), which would be apportioned further into mission equipment or gear lists and/or individual issue lists.

### 3.4.1.2 Maintenance Function

‘Mobility’ is interdependent on the ‘Maintenance’ function for the operational classification of its assets to perform its combat mission. The role of maintenance is to determine what conditions meet the standards of the primary role the asset is to perform (i.e., shoot, move, or

communicate). Mobility as owner/operators is limited in the capacity to effect critical restorative repairs and is reliant of the maintenance function to provide the current classification (operational, degraded, or deadline), but also to correct the defects to return the asset to an operational state.

#### 3.4.1.3 General Engineering

Mobility requires the services from General Engineering in terms of providing area, zone, and route reconnaissance for all modes of transportation. General Engineering provides surveys and measurements from roads, bridges, bypasses, bi-ways, highways, surf, shore, beach, landing zones, runways, etc., without which Mobility cannot be successfully executed.

### 3.4.2 Processing Cycle at the Functional Level

Once the mobility request is received, the request is processed for additional decision or action. The data requirements for action/decision after the request has been initiated are not within scope of this analysis. However, the general process is worth noting for future development.

#### 3.4.2.1 Capacity Planning

The next step in demand fulfillment for movement of PAX and/or cargo is capacity planning. The data provided on the request is evaluated against available and accessible platforms according to time, weight, and destination/origination constraints. Capacity planning occurs at all echelons and only changes in scale and scope. Capacity planning has a dependency on at least two distinct data sets: what is available, and what is requested.

#### 3.4.2.2 Transportation Mode Selection

Once capacity planning has compared the platform capacity/availability data with the demand data, the best mode of transportation is selected. The original request may spawn a related request or be consolidated with similar requests. Once the mode of transportation is selected it is then assigned to the movement execution activity.

#### 3.4.2.3 Movement Execution

Movement execution will respond to the capacity planning assignment through manifesting, dispatching, monitoring, and tracking the physical movement. Additionally, the movement execution will assign landing support as required depending on the conditions of the mission (permissive vs. non-permissive). Movement execution will append additional data (funding data, MHE requests, etc.) to the original/supplemental request(s) as required until the mission is complete, at which time the request is closed.

### 3.4.3 Data Exchange Requirements

The embark data requirements presented here are based on the lowest level tactical unit requesting movement. This is a departure from the traditional paradigm, wherein embarkation is associated with the movement of mass formations to support an operation or exercise. In that legacy model, smaller movement requests and their corresponding level 4 (nested) and level 6 (mobile/loaded items) data are classified as “transportation requests” or “logistics requests”, thereby bypassing the core competencies of the unified execution of embark, distribution, and transportation mode selection.

This analysis revealed that there is little variation in required data between smaller and larger embarkation movements. Because frequent, smaller movement requests are generated in analog forms, that data is frequently lost.

The minimum data requirements for the embarkation of PAX/cargo are ubiquitous regardless of scale and scope of movement (i.e., from the fire team level to Marine Expeditionary Brigade (MEB)).

#### 3.4.3.1 PAX Request Data

The PAX segment identifies the people requiring movement from one place to another. It is agnostic to transportation mode as this occurs during capacity planning.

1) Segment Number: System assigned, as the request is created under the header. A single digit indicator; up to nine independent request types can be added under the header but does not restrict the number of individual lines contained in the request.
2) Request Type: Expressed as a two-digit background code when the user selects “PAX Movement”. In this case PM is the code for a PAX movement.
3) Request priority – Expressed as two-digit code based on the urgency aligned to the element/unit’s F/AD.
4) ZAP/EDI-PI: Expressed as either 8 to 9-digit ZAP or 10-digit EDI-PI. ZAP data element is a pre-formatted and system-assigned abbreviation for USMC personnel. EDI-PI is DoD-assigned for all DoD personnel (active, reserve, government service (GS), and contractor). User should not necessarily need to type in these identifiers but select from a preloaded list on the local machine or scan a government issued identification card if the individual is not in the list.

| Position 1 | Position 2 | Position 3-6 | Position 7-8 | Position 9 |
|---|---|---|---|---|
| Last Name Initial | First Name Initial | Last 4 digits of EDI-PI | Blood Group | Blood RhD |

**Table 11: ZAP Number Data Schema**

| **Individual Marine Data** | | | | |
|---|---|---|---|---|
| Last Name | First Name | EDI-PI | Blood Group | Blood RhD |
| Smith-Hernandez | Chris | 1010919789 | AB | Positive |
| **ZAP Conversion** | | | | |
| Position 1 | Position 2 | Position 3-6 | Position 7-8 | Position 9 |
| Last Name Initial | First Name Initial | Last 4 digits of EDI-PI | Blood Group | Blood RhD |
| S | C | 9789 | AB | + |
| **ZAP Number for Individual Marine** | | | | |
| SC9789AB+ | | | | |

**Table 12: ZAP Number Conversion Example**

5) Earliest Departure Date – Local. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day. The first full day available day for departure.
6) Latest Departure Date – Local. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day. The last full day available day for departure.
7) Departure Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.
8) Destination Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-

meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., MCAS Miramar, etc.) for ease of use.

9) Total Estimated Baggage Weight. Expressed in pounds to the nearest whole number of hand carried/passenger compartment residing baggage.

10) Hazardous Material Type. Required response expressed as a single digit alpha or numeric code when the clear text response is selected. A – Ammunition; E – Explosives; P – Power Source (Lithium or Acid based); C – Chemical/Biological/Radiological Sample; Z – Other; X – None. While hazardous material does not negate movement, it is used for capacity planning and transportation mode selection. Certain precautions must be taken depending on the response of hazardous material being hand carried or expected to be in the passenger area of the supporting platform.

11) Person Count: Calculated. Expressed as the count of unique ZAP/EDI-PI identified in the request segment. As this is a calculated field, the data does not need to be passed to/from across the communication pathway.

12) Total Baggage Weight: Calculated. Expressed as the sum of all baggage weight identified in the request segment. As this is a calculated field, the data does not need to be passed to/from across the communication pathway.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Segment Number | 1 | No | 1/1 | Must Use | Pass |
| 2. Request Type | 1 | No | 2/2 | Must Use | Pass |
| 3. Request Priority | 1 | No | 2/2 | Must Use | Pass |
| 4. ZAP/EDI-PI | 1 | No | 8/10 | Must Use | Pass |
| 5. Earliest Departure Date - Local | 1 | No | 9/9 | Must Use | Pass |
| 6. Latest Departure Date - Local | 1 | No | 9/9 | Must Use | Pass |
| 7. Departure Location | 1 | No | 14/14 | Must Use | Pass |
| 8. Destination Location | 1 | No | 14/14 | Must Use | Pass |
| 9. Total Estimated Baggage Weight (lbs.) | 1 | No | 3/3 | Must Use | Pass |
| 10. Hazardous Material Type | 5 | No | 1/1 | Must Use | Pass |
| 11. Person Count - Calculated | 1 | No | 1/5 | Optional | Not Passed |
| 12. Total Baggage Weight - Calculated | 1 | No | 1/5 | Optional | Not Passed |

**Table 13: PAX Request Segment**

The following is the data schema of the exchanged data segments which would be passed part of the request portion. In this example the header is included as part of the entire data burst.

| |
|---|
| Header data schema format: |
| CCYYMMMDD-HHMMSSDD-XXXX123456-XXXXXXXXXX-00-XXXX-00 |
| Header data schema example: |
| 2027OCT13-15470352-4QFJ123456-3838JBNM5-02-KL9K-01 |
| PAX request data schema format: |
| 0-XX-00-XXXXXXXXXX-CCYYMMMDD-CCYYMMMDD-XXXX123456-XXXX123456-000-X |
| PAX request data schema example for (1) PAX: |
| 1-PM-02-1010919789-2027OCT15-2027OCT20-4QFJ123456-4QFJ456789-075-X |

**Table 14: PAX Request Schema**

#### 3.4.3.2 Cargo Request Data

Cargo requests are used for the movement of military materiel. A Cargo request is not primarily used for personal baggage, while it could be, it is not the intent.

1) Segment Number. System assigned as the request is created under the header. A single digit indicator: up to nine independent request types can be added under the header.
2) Request Type. Expressed as a two-digit background code when the user selects “Cargo Movement”. In this case CM is the code for a PAX movement.
3) Request Priority. Expressed as two-digit code based on the urgency aligned to the element/unit’s F/AD.
4) Item by NIIN. When the user selects the serial number or the clear text of the item(s) to be shipped, the system will populate the North Atlantic Treaty Organization (NATO) Item Identification Number (NIIN) into the request segment. The NIIN is a code in and of itself.
5) Item Quantity. If the cargo does not have serialization, the quantity of item(s) can be input.
6) Serial Number. User selects from the pre-populated list from the gear list and/or EDL. If the serial number is selected, the NIIN will auto populate. Serial numbers are counted as (1) per item.
7) Gross Weight. The total weight, in pounds, of the pallet, box, or end item being shipped.
8) Actual Height. The actual measured height in whole inches.
9) Actual Width. The actual measured width in whole inches.
10) Actual Length. The actual measured length in whole inches.
11) Hazardous Material Indicator Code (HMIC). A non-user inputted single (1) character alphabetic code used to identify all known or suspected hazardous items as defined by safety, health, transportation and/or environmental regulations to serve as a flag that an activity must acquire a Material Safety Data Sheet (MSDS) to inform Federal Catalog System users when more detailed information on an item is available in the Hazardous Material Information System. The HMIC is related to the NIIN, once the NIIN is accepted into the enterprise connected system, the HMIC will populate. D - There is no information in the Hazardous Materials Information Reporting System (HMIRS); however, the National Stock Number (NSN) is in a Federal Stock Class (FSC) in Table I of Federal Standard 313 and an MSDS should be available to the user. N - There is no data in the HMIRS and the NSN is in an FSC not generally suspected of containing hazardous materials. P - There is no information in the HMIRS; however, the NSN is in an FSC in Table II of Federal Standard 313 and an MSDS may be required by the user. The requirement for an MSDS is dependent on a hazard determination of the supplier or the intended end use of the product. Y - Information is found in HMIRS.
12) Handling. Expressed as a single alphabetic code from the user’s clear text selection to indicate special handling at the point of destination: C – Crane or Hoist; M – MHE; T – Tow; R – Ramp; X – None
13) Earliest Departure Date - Local. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day. The first full day available day for departure.
14) Latest Departure Date - Local. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day. The last full day available day for departure.

15) Departure Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

16) Destination Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., MCAS Miramar, etc.) for ease of use.

17) Item Count – Calculated. Expressed as the sum of items identified in the request segment. As this is a calculated field, the data does not need to be passed to/from across the communication pathway.

18) Total Weight – Calculated. Expressed as the sum of the weight identified in the request segment. As this is a calculated field, the data does not need to be passed to/from across the communication pathway.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Segment Number | 1 | No | 1/1 | Must Use | Pass |
| 2. Request Type | 1 | No | 2/2 | Must Use | Pass |
| 3. Request Priority | 1 | No | 2/2 | Must Use | Pass |
| 4. Item by NIIN | 1 | No | 9/9 | Must Use | Pass |
| 5. Item Quantity (if not serialized) | 1 | No | 1/5 | Must Use | Pass |
| 6. Serial Number | 1 | No | 1/10 | Must Use | Pass |
| 7. Gross Weight (lbs.) | 1 | No | 5/5 | Must Use | Pass |
| 8. Actual Height (inches) | 1 | No | 3/3 | Must Use | Pass |
| 9. Actual Width (inches) | 1 | No | 3/3 | Must Use | Pass |
| 10. Actual Length (inches) | 1 | No | 3/3 | Must Use | Pass |
| 11. HMIC | 5 | No | 1/1 | Must Use | Pass |
| 12. Handling | 1 | No | 1/1 | Must Use | Pass |
| 13. Earliest Departure Date - Local | 1 | No | 9/9 | Must Use | Pass |
| 14. Latest Departure Date - Local | 1 | No | 9/9 | Must Use | Pass |
| 15. Departure Location | 1 | No | 14/14 | Must Use | Pass |
| 16. Destination Location | 1 | No | 14/14 | Must Use | Pass |
| 17. Item Count - Calculated | 1 | No | 1/5 | Optional | Not Passed |
| 18. Total Weight - Calculated | 1 | No | 1/5 | Optional | Not Passed |

**Table 15: Cargo Request Segment**

The following is the data schema of the exchanged data segments which would be passed part of the request portion. In this example the header is included as part of the entire data burst.

| |
|---|
| Header data schema format: |
| CCYYMMMDD-HHMMSSDD-XXXX123456-XXXXXXXXXX-00-XXXX-00 |
| Header data schema example: |
| 2027OCT13-15470352-4QFJ123456-3838JBNM5-02-KL9K-01 |
| Cargo request data schema format: |
| 0-XX-00-000000000-00000-XXXXXXXXXX-00000-000-000-000-X-X-CCYYMMMDD-CCYYMMMDD- XXXX123456-XXXX123456 |
| Cargo request data schema example for (1) MTVR Truck, Serial Number 598742: |
| 1-CM-02-015519434-1-598742-28000-126-100-315-D-R-2027OCT15-2027OCT20-4QFJ123456-4QFJ456789 |

**Table 16: Cargo Request Schema**

## 3.5 Supply Data Requirements

The analysis determined, the function of supply is a multi-faceted operation relying heavily on data being communicated routinely to establish demand signals, response by status, and reporting of progress. Furthermore, due to the close proximity of the mobility function to successfully execute supply fulfillment, this has led to the misconception of “distribution” being “transportation” and thereby, supply being wholly responsible for the physical movement of supplies and materiel in the transportation network/pipeline. Therefore, this analysis aims to retain the functional boundary between mobility and distribution within the supply function and emphasize the importance of data sharing, availability, and accessibility in a unified, singular logistics information system.

Additionally, the supply function has been greatly compartmentalized around classification of supplies. Each of the ten classes of supply have established one or more distinct, separate, and specialized methods of requesting, record keeping, accounting, access requirements, logistics IT support (if any), and independent supply chains. These factors demand both the customer and the supporting organization supply personnel to be somewhat proficient in navigating these artificially manufactured and self-imposed processes and systems. This analysis presents the data exchange requirements from the perspective of the warfighter. The analysis determined the warfighter should not be required to navigate – or be subjected to – the current established complexities for communication of materiel requests even if the enterprise continues to hold on to these self-imposed structures.

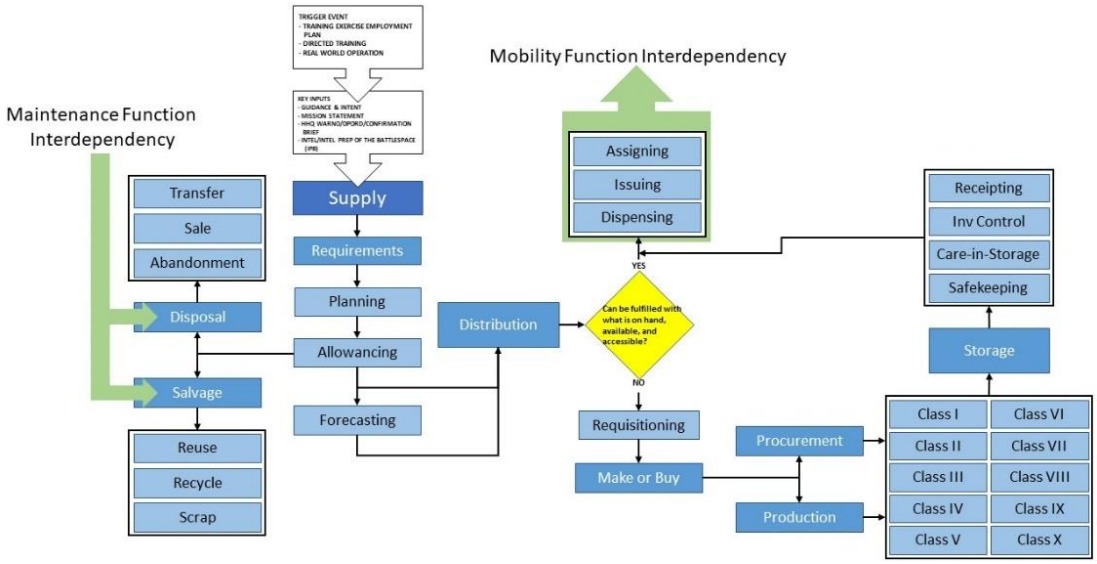


**Figure 15: Supply Cycle**

## 3.5.1 Functional Interdependencies

### 3.5.1.1 Mobility

There exists a partnership between the two functions of supply and mobility to ensure the customer's requirement is fulfilled. This partnership requires coordination, data sharing, and mutual accountability since requisitioning is a two-part process: 1) fulfillment 2) transportation to customer.

The supply function is responsible for the first part of the requisitioning process, which is determining fulfilment method, inventory management, and stocking. Once the demand has been fulfilled the second part of requisitioning is transportation which is the function of mobility.

### 3.5.1.2 Maintenance

Supply activities rely on the doctrinal subfunction of maintenance in classification of repair as well as maintenance related physical preparatory work in the area of disposal/salvage operations. Maintenance may classify equipment as unrepairable due to cost, lack of resources, obsolesce or extended maintenance cycle time. Supply's functional role is to administratively communicate the item identification information, request higher directed disposition/actions, identify unique demilitarization requirements, and report the completion of actions. However, Marines performing supply actions do not perform the physical preparatory work and any demilitarization actions on the item itself, this separation of actions requires unique information to be shared for successful completion.

## 3.5.2 Processing Cycle at the Functional Level

Supply is the most visible link between the operations warfighting function and the logistics warfighting function. The direct inject of operational data (desired capabilities, intent, etc.) become the sourcing requirements in the supply function.

### 3.5.2.1 Requirements Determination

Once the operational order is issued, the supply function begins the requirements determination. There are no logistics IT systems dedicated to this sub-function. Requirements determination is comprised of three subordinate functions:

1) Planning. Determining what the operational objective requires both in terms of equipment and personnel for the duration of the objective.
2) Allowancing. Sets the required allowance of equipment to meet the operational objective. Allowance pertains primarily to equipment but also has an application to people as well commonly referred to as ‘staffing’ or ‘manning’.
3) Forecasting. The sustainment based on the allowancing factor. Forecasting is determining future requirements and is accomplished through the application of various scientific principles by adjusting various variable sets that account for atmospherics, mobility, environment, etc. Forecasting can range from rudimentary to complex. Even with the best intentions, forecasting cannot foresee and anticipate every situation and therefore, requires the customer ability to request materiel to fill the forecast gap.

#### 3.5.2.2 Distribution

Distribution executes the ability to identify, assign, and dispense the materiel, equipment, and/or supplies as well as assign military personnel to identified activities, units, and/or billets. This function must determine if the assets required to the fill the established allowances are currently on hand (within the inventory), if what is on hand is available (operational), and if these assets are accessible (can be physically obtained and employed). Depending on the answers to these variables sets, the next set of actions by the distribution action are as follows:

1) If the assets are on hand, available, and accessible, then such assets can be allocated from inventory by assigning, issuing, or dispensing.
2) If the assets do not meet the criteria, then distribution would create a requisition. Requestions is the act of establishing a demand or signaling a requirement. Requestioning by its nature can come in two forms:
   a. Forecast signal. Where the data of consumption rate, capacity, and order/ship time determines the availability of current inventory compared to future demand can generate a requisition.
   b. Customer direct demand. Where the customer has a direct need beyond what is currently available, forecasted, and/or determined and is usually based on unique mission parameters or circumstances previously unforeseeable.

   In terms of routing, the routing of the requisition depends on the authority to release assets according to the next level’s distribution cycle.

#### 3.5.2.3 Fulfillment: Make or Buy

The action of satisfying the demand leads to the decision to make or buy. These two actions fall under two sub-factors agnostic of echelon, scale, and/or scope:

a. Make (internally source): The decision to make is based on the data indicating if it is more effective, efficient, or supportive to make, internally source, or provide the service, information, and/or materiel to satisfy what is requested. This will lead to a production action.
b. Buy (externally source): The decision to buy means to purchase/obtain a service, information, and or materiel outside the organization. This will lead to a procurement action, usually attributed to materiel, but can apply to services and information.

#### 3.5.2.4 Storage

Storage operations is the act of receiving, holding, protecting, and preparing something until it is to be issued, assigned, or dispensed. While this sub-function pertains mainly to materiel, it certainly has immediate application to both services and information. Storage, as a cyclic function, is comprised of four subordinate functions:

1) Receipting. The act of accepting inventory into the activity to make it available. The action also provides an audit trail of where the inventory originated, who accepted it into the inventory, when it entered the inventory, and where it is being kept.
2) Inventory Control. Inventory control ensures the accurate accountability of assets between what is the real-world physical quantity and the recorded data medium. Additionally, it ensures the inventory is moved to the assigning, dispensing, and issuing function only by authorized individuals acting under the appropriate authority.
3) Care-in-Storage: This subfunction is primarily concerned with the appropriate rotation of stock, ensures serviceability, and preserves usability.
4) Safekeeping: The protection from the elements, theft/pilferage, and in some cases limiting observation/accessibility (as a force protection measure) are all part of the physical safekeeping of inventory.

#### 3.5.2.5 Disposal and Salvage

The end of the lifecycle or usefulness of inventory culminates in either disposal or salvage. Supply is rarely the physical executor of disposal or salvage operations, rather it is the administrative provider reporting the action. The result of either action relieves the government from the responsibilities of accounting and maintaining the inventory.

1) Disposal is the act of removing the inventory physically and administratively from active use, availability, and accessibility according to its original function. Disposal consists of three subordinate functional methods:
    a. Transfer. The physical and administrative movement of the inventory to another Department of Defense (DoD) activity.
    b. Sale. The exchange of money/funds, when authorized, to civilians; foreign governments, and/or departments/agencies outside of DoD.
    c. Abandonment. Leaving the inventory, usually on the field of battle, when recovery or other methods of disposal are not possible or viable. This method is the least desirable of all other disposal methods as it leaves the possibility of salvage or reclamation to adversarial forces. Every effort should be made to ensure the inventory is totally disabled and proves no value in terms of functional use and/or reverse engineering.
2) Salvage is the act of separating the components of the asset into usable inventory outside of its original function. Salvage consists of three subordinate functional methods:
    a. Reuse. To use the inventory for another purpose other than it was originally intended. Usually converting the inventory to a garrison or administrative role rather than its original combat/war-time purpose. This action requires the facilitation of accountability from combat inventory roles to garrison/administrative inventory roles.
    b. Recycle. Breaking down the original inventory into its useful subcomponents that can be used in other like items or satisfy open demands for such subcomponents.

c. Scrap. Reducing the inventory into its primary elemental components as raw material.

### 3.5.3 Data Exchange Requirements

The data exchange requirements presented here remove the customer's burden of being acquainted with the complexities in supply requisitioning procedures. Additionally, this methodology streamlines requests into a single form where any class of supply (and sub-class) can be requested. Therefore, anything from paper, pens, MREs, ammunition, bulk fuel, medical supplies, etc. are all ordered on this form and would be separated out to the different departments for fulfilment. These separate departments would then add any necessary additional data, a process extraneous to the requisitioner.

1) Segment Number. System assigned, as the request is created under the header. A single digit indicator; up to nine independent request types can be added under the header but does not restrict the number of individual lines contained in the request.

2) Request Type. Expressed as a two-digit background code when the user selects "Supply Request". In this case, SR is the code for a Supply Request.

3) Request Priority. Expressed as two-digit code based on the urgency aligned to the element/unit's F/AD. User selects from a clear text list of Critical, Urgent, or Routine.

4) Item by NIIN. When the user selects the clear text name of the item(s) to be requested, the system will populate the NIIN into the request segment. The user is not expected to know the NIIN but could select an item based on the NIIN. Selection lists can be preloaded with the most common items regardless of Class. The supply request is agnostic to classifications; therefore, ammunition, meal ready to eat (MRE), fuel, etc. are requested on a singular form and is parsed out at the supporting activity for action by commodity group. If the NIIN or item name is unknown, the user can input a description in the "Narrative" box. Supporting activity can update the NIIN field once technical research is complete in identifying the materiel item(s) required.

5) Item Quantity. Expressed as a five-digit numeric quantity. User inputs the quantity desired. Unit of issue (U/I) or Unit of Measure (UOM) are background enterprise data elements which does not need to be passed. These terms should be displayed but the data is irrelevant to the data exchange requirement.

6) Required Delivery Date. Local. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day. User inputs desired date the supplies are to be delivered.

7) Delivery Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical locations can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

8) Attachment. Expressed as an embedded object of a video, picture, or voice recording of the of the item(s) being requested. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

9) Narrative. Free text for additional information up to 100 characters to describe the materiel demand when a NIIN does not exist (i.e., bulk water) or additional coordination.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Segment Number | 1 | No | 1/1 | Must Use | Pass |
| 2. Request Type | 1 | No | 2/2 | Must Use | Pass |
| 3. Request Priority | 1 | No | 2/2 | Must Use | Pass |
| 4. Item by NIIN | 1 | No | 9/9 | Optional | Pass |
| 5. Item Quantity | 1 | No | 1/5 | Must Use | Pass |
| 6. Required Date - Local | 1 | No | 9/9 | Must Use | Pass |
| 7. Delivery Location | 1 | No | 14/14 | Must Use | Pass |
| 8. Attachment | 1 | Yes | 1/1 | Optional | Pass |
| 9. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 17: Supply Data Exchange Requirements**

## 3.6 Maintenance Data Requirements

Maintenance is active restoration of equipment to functionally perform its combat mission (i.e., shoot, move, and/or communicate). Even with the application of predictive analytics and condition-based monitoring, the maintenance mission is primarily reactive to the outputs of the equipment's operating efficiency. Within the context of the operating environment, condition-based monitoring and predictive maintenance may be severely limited due to the availability of communication pathways. Therefore, the human owner/operator of military equipment is the first and most essential maintenance "sensor" in the logistics kill chain to report and identify equipment maintenance issues. Requesting maintenance will remain a crucial component for ensuring operational availability to support of mission execution. It should be noted the maintenance function is not responsible for materiel "readiness reporting" but the actions of maintenance produces 'readiness' as a byproduct. However, readiness should not be the litmus test for maintenance execution and efficiency.

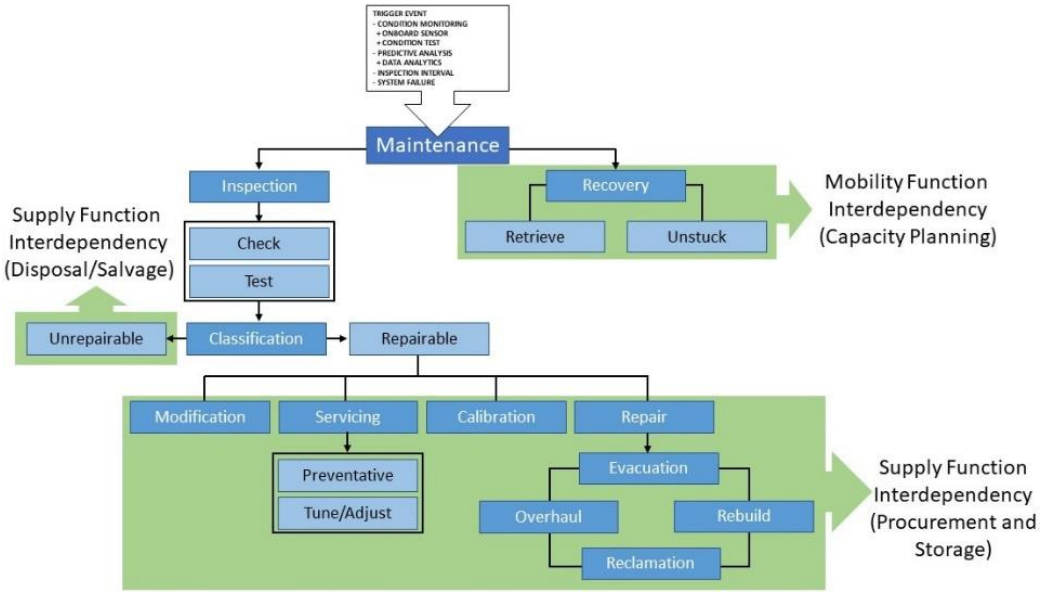


**Figure 16: Maintenance Function Cycle**

## 3.6.1 Functional Interdependencies

### 3.6.1.1 Supply Function

The ability for maintenance to execute its mission relies on two areas in the supply:

1) The ability to procure materiel requests in support of the maintenance function, specifically in Class III and IX.
2) The ability to request or confirm unrepairable government equipment for further disposition in terms of disposal or salvage.

### 3.6.1.2 Mobility Function

Maintenance, and the ability to recover or free from immobility, requires the service provider execute tasking, dispatching, and capacity planning sub-functions of mobility. Functionally, this can be performed at the appropriate echelon regarding scale/scope, but it *does* necessitate the requirement for access to the required data to perform.

## 3.6.2 Processing Cycle at the Functional Level

### 3.6.2.1 Recovery

1) Retrieve. The asset is not able to move or operate on its own power and must be brought back to a predetermined location for follow on maintenance classification. This type of recovery does not necessarily apply only to self-propelled assets but can apply to any military equipment.
2) Free from Immobility (Unstuck). Once the asset is free from the condition of immobility, it is able to operate under its own power.

### 3.6.2.2 Inspection

When a defect or maintenance issue is suspected or identified, the maintenance activity will check and/or test the condition before entering the maintenance cycle. The check/test is not diagnostics but confirmation of the condition which warrants entry into the maintenance cycle.

1) Check. A check is usually performed in response to fault indicator, notification, or sensor reading or some other condition monitoring method. As automated notifications rarely indicate the actual cause, a mechanic/technician must investigate the area/place/system of fault to confirm the condition.
2) Test. Usually done in response to a condition found by the owner/operator. The mechanic/technician recreates or exposes the defect through a standardized process—or through a similar condition the fault was discovered—in order to confirm maintenance is required.

### 3.6.2.3 Classification: Unrepairable Conditions

Unrepairable conditions are governed by cost, obsolesce, service life, and/or maintenance limitations. The maintenance activity could classify an asset based on any one or combination of these conditions. Based on the identified conditions, the maintenance activity will request disposition instructions from higher headquarters. Typically, requests for disposition will be routed to the appropriate program manager, who has final decisional authority and will provide instruction for follow-on action.

#### 3.6.2.4 Classification: Repairable Conditions

Repairable conditions are deemed to be within the capacity, scope, authority, cost, skill, and capabilities of the maintenance activity either organically or by next higher maintenance activity. Repairable conditions will bring the asset to its combat functional use. The repairable conditions are in the following categories:

1) Modifications. An authorized change in the original asset's form, fit, or function. The management of modifications of an asset is not a maintenance activity responsibility. However, the application/installation of the modification is a maintenance function. Owners/operators of assets requiring modifications will request maintenance support for directed modifications per the modification instruction.
2) Servicing. Cyclic and routine preventative maintenance (usually annual) outside owner/operator responsibility identified in the asset's technical manual. Minor adjustments and tuning which an owner/operator would not have the tools, facilities, training, and/or authority to perform are a servicing action conducted by the maintenance function.
3) Calibration. Periodic adjustments for precision equipment/tools. Calibration requires specialized tools, testing equipment, and training to ensure the established operating standards are met. Owner/operators of precision equipment/tools request calibration to the supporting maintenance activity at the appropriate interval per the technical manual.
4) Repair. Most commonly referred to as 'corrective maintenance'. This action brings an asset from a degraded or deadline condition to operational condition focusing a specific defect or set of defects. If the defect is repairable but out of the current maintenance activity's capability to perform, the asset will be evacuated to the next higher maintenance activity. Within the repair action, a subset of more in-depth repair can occur but is usually performed at the depot level maintenance activity:
    a. Overhaul. The restoration of an asset's systems and/or major components. An overhaul is primarily concerned and focused on the replacing or restoring of primary components.
    b. Rebuild. The systematic disassembly on an asset down to its core/basic components and then reconstructed to a new or near new like condition. Every element of the asset from internal wiring to paint is replaced and/or repaired.
    c. Reclamation. Both overhaul and rebuild result in reclamation. Reclamation is to repair an asset and return it to the supply inventory system in usable condition. Reclamation is rarely conducted by organizational maintenance activities and only by direct authorization from a higher-level authority.

### 3.6.3 Data Exchange Requirements

The following data entries are constructed around the precept the requestor is not a mechanic/technician by specialty but more likely is the owner/operator of the equipment at the lowest tactical level. It also must be assumed the owner/operator has exhausted all available maintenance tasks authorized at their level. Lastly, it must be assumed additional data needed by the mechanic/technician is federated and available (i.e., maintenance history, predictive/condition monitoring, etc.). Therefore, the maintenance request is not centered around 'parts ordering' but maintenance as a service which may conclude or result in parts being ordered. The form can also be used to evacuate equipment from one maintenance activity to another. The intent is for data to create the maintenance record once synchronized with the

enterprise logistics IT system and serves as the beginning of an ongoing maintenance conversation.

1) Segment Number. System assigned as the request is created under the header. A single digit indicator: up to nine independent request types can be added under the header.
2) Request Type. Expressed as a two-digit background code when the user selects “Maintenance Request”. In this case CM is the code for data exchange.
3) Request Priority. Expressed as two-digit code based on the urgency aligned to the element/unit’s F/AD. Based on the “Equipment Operational Condition” Routing - Operational: A, Routine - Minor (can perform most combat functions); B, Urgent - Degraded (can perform some combat functions; C, Critical - Deadline (cannot perform any combat functions – cannot shoot, move, or communicate).
4) Serial Number. Optional if not serialized. User selects from available selection within their data slice or EDL. The serial number acts as a header for report data. This data segment is optional since not all reparable equipment is serialized. If the item is not serialized or found in the data slice, then the NIIN and/or name of item for repair must be added. If the serial number is longer than 15 digits, only the last 15 digits (to include special characters) will be transmitted.
5) National Item Identification Number (NIIN). Auto populated if the serial number is selected. Expressed as a 9-position alpha-numeric code (XXXXXXXXX).
6) Model of Equipment. Auto populated based on the NIIN and/or serial number. If neither are selected the user can activate the text box to input the common model number up to 20 characters. If the serial number is selected, this information is not passed.
7) Item Nomenclature. Auto populated based on the NIIN, serial number, and/or model number. If neither are selected the user can activate the text box to input the common item nomenclature up to 20 characters. If the serial number or NIIN is selected, this information is not passed.
8) Number of Pieces Requiring Maintenance Support. Auto calculated based on the number of serial numbers inputted. If the equipment is not serialized the user can input in the text box for the numeric number of items up three digits (XXX).
9) Equipment Operational Condition. User determines based on their operational need and mission parameters if the equipment is combat ready/mission capable. User selects the clear text operational condition which will be populated with the corresponding single-digit alpha code for data exchange. Operational (can perform all combat functions): A, Minor (can perform most combat functions); B, Degraded (can perform some combat functions; C, Deadline (cannot perform any combat functions – i.e., cannot shoot, move or communicate).
10) Date Maintenance Support is Required. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.
11) Location of Equipment. The physical location the equipment will be or is currently located. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical locations can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

12) Type of Maintenance Support Required. User selects the clear text maintenance support requested which will be populated with the corresponding two-digit alpha numeric code for data exchange. Repair: R1, Recovery – Retrieve: R2, Recovery – Free from Immobility: R3, Contact Team: R4. Defaults to None: XX

13) Type of Repair. User selects the clear text maintenance repair type requested which will be populated with the corresponding two-digit alpha numeric code for data exchange. This list activates only when the type of maintenance support required is “Repair – R1.” Modification: M1, Servicing - PM: S1, Servicing - Tune/Adjust: S2, Calibration: C1, Repair Defect: D1.

14) Repair Major Defect. List activates when the type of maintenance support required is “Repair: R1” or “Contact Team: R4” only. Based on the item serial number or NIIN the list of available option will dynamically change. If the item is vehicular in nature, then only defects unique to the vehicle will be selectable with other defects becoming hidden (i.e., “transmission/output circuitry” would not be selectable as this pertains to electronics not vehicles). User selects the “best fit” clear text maintenance repair major defect which will be populated with the corresponding four-digit alpha numeric code for data exchange.

    a. MD01 - Engine: Issues with the starter, alternator, belts, rough start, or anything in this area. Can also be applied to generators, heavy equipment, and water/fuel pumping systems.

    b. MD02 - Fuel: Issues with fuel systems, carburation, fuel mixture, fuel leaks, or anything pertaining to the onboard storage, filtering, fuel quality, or internal movement of the fuel.

    c. MD03 - Exhaust: Issues with the output burned gasses routing, color, rhythm, or smell.

    d. MD04 - Cooling: Issues with retaining adequate levels, leaks, overheating, or running cold.

    e. MD05 - Lubrication: Issues with engine, transmission, or other internal friction components’ ability to internally lubricate due to leaking, low levels, etc.

    f. MD06 - Electrical/Fuse: Issues with the onboard electrical systems to maintain system voltage, operate lights or connected components, and/or fuses/cables overheating/tripping/short circuiting.

    g. MD07 - Transmission/Power Train: Issues where the equipment does not efficiently transfer engine generated power to the system and/or does not operate at expected parameters under load.

    h. MD08 - Chassis/Body/Frame/Armor: Issues with the structural mounting of the system and/or ballistic protective layer(s).

    i. MD09 - Suspension/Axle/Differential/Shocks: Issues with the ability of the equipment to absorb ground impacts and/or maintain ground clearance and/or transfer power train power efficiently through the axle(s).

    j. MD10 - Brakes/Tires/Wheels/Hub/Track: Issues with tire inflation, internal run flat systems, central tire inflation system, tire spindles, mounting hubs, calibers, track, track shoes, and braking system.

k. MD11 - Lift/Winch/Boom/Turret: Issues with mounted special purpose equipment pertaining to the ability to lift, pull, rotate, etc.

l. MD12 - Hydraulics: Issues with the internal system to operate hydraulically controlled systems due to low pressure, leaks, maintaining fluid levels, and/or electromechanical control of hydraulically controlled systems.

m. MD13 - Receiver/Input circuitry: Issues with the ability to receive or input data from external sources.

n. MD14 - Transmitter/Output circuitry: Issues with the ability to transfer or output data to external sources.

o. MD15 - Display/Screen/Video: Issues with the graphical user interfaces or displays required to operate equipment.

p. MD16 - Weapon System: Issues with the ability operate, load, unload, aim, target, or execute firing of a weapon system (projectile, missile, and/or munition based).

q. NMAJ - Routine/No Major Defect: Used for required preventive maintenance and/or minor/degrading issues.

15) Attachment. Expressed as an embedded object of a video, picture, or voice recording of the maintenance condition. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

16) Narrative. Free text for additional information up to 100 characters.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Segment Number | 1 | No | 1/1 | Must Use | Pass |
| 2. Request Type | 1 | No | 2/2 | Must Use | Pass |
| 3. Request Priority | 1 | No | 2/2 | Must Use | Pass |
| 4. Serial Number | 1 | Yes | 1/15 | Optional | Pass |
| 5. National Item Identification Number | 1 | Yes | 9/9 | Must Use | Pass |
| 6. Model of Equipment | 1 | Yes | 1/20 | Optional | Pass |
| 7. Item Nomenclature | 1 | Yes | 1/20 | Optional | Pass |
| 8. Number of Pieces | 1 | Yes | 1/5 | Must Use | Pass |
| 9. Equipment Operational Condition | 1 | Yes | 1/1 | Must Use | Pass |
| 10. Date of Maintenance Support | 1 | Yes | 9/9 | Must Use | Pass |
| 11. Location of Equipment | 1 | Yes | 14/14 | Must Use | Pass |
| 12. Type of Maintenance Support | 1 | Yes | 2/2 | Must Use | Pass |
| 13. Type of Repair | 1 | Yes | 2/2 | Must Use | Pass |
| 14. Repair Major Defect | 1 | Yes | 4/4 | Must Use | Pass |
| 15. Attachment | 1 | Yes | 1/1 | Must Use | Pass |
| 16. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 18: Maintenance Data Exchange Requirements**

## 3.7 General Engineering Data Requirements

Unlike the other logistics functions, General Engineering does not have a distinct functional cycle. This function provides both critical services and materiel, however, it is the least supported function by information technology. Every service provided by this function involves collecting, recording, transcribing, and reporting service data in a pre-formatted, analog, paper-

based form which is NOT converted or transliterated into a logistics information system of any kind. Engineering data is therefore not readily accessible in digital environments.

Notable changes within this function:

1) Bulk water and bulk fuel are combined into a single sub-function of ‘Bulk Liquids’ as the production, storage, and dispensing are all similar functions.
2) Assault Bridging and other standard bridging systems are no longer organic capabilities of the Marine Corps and are therefore omitted from this analysis. However, Bridging remains a doctrinal sub-function of General Engineering.
3) Power Generation and Power Distribution is an operational/strategic-level planning event. The tactical user at the lowest echelon would not request services with regards to generating or distributing power and is reserved for a higher-level data planning.
4) Explosive Ordnance Disposal (EOD) functions as Support Forces, Supporting Establishment, Homeland Defense, and Special Operations Forces, by locating, accessing, identifying, rendering safe, neutralizing, and disposing of hazards from foreign and domestic Chemical Biological Radiological Nuclear and High-Yield Explosives (CBRNE), Unexploded Explosive Ordnance (UXO), Improvised Explosive Devices (IED), and Weapons of Mass Destruction (WMD) that present a threat to operations, installations, personnel, or materiel. These basic data exchange requirements are presented here for tactical users to request EOD support.
5) Facilities maintenance is an operational/strategic-level planned event, as required. Tactical users would not request this service, although they would identify significant landmarks for planning considerations at higher echelons.
6) Vertical construction is an operational/strategic-level event used to strengthen and rehabilitate existing infrastructure. Vertical infrastructure is a sustainment consideration and consumes human and mechanical resources.
7) Horizontal construction includes the excavation and construction of roads, railways, bridges, cantonment sites, and gap crossings. Construction at this level is an operational/strategic-level planning event as it involves the construction of airfields (not repair or utilization of existing airfields), constructing supply routes or improving existing routes (to include repair, but not tactical movement repair), ordnance structures that include bunkers of various explosive weights for long storage (not an expeditious/field ammo storage point), and site prep/bed down that includes hardened structures, that can use existing facilities for rehabilitation when requested as a vertical construction requirement.
8) Enemy munitions/ordnance exploitation is dependent on the planning outputs of the Marine Corps Planning Process (MCPP). These planning considerations are not pertinent to the tactical user. However, organizational planning would include the reporting of enemy munitions, ammunition, and ordnance—typically retrieved from information contained within the SALUTE report. Decisions for exploitation are made at the operational/strategic-level. There are occasions when data is required to “exploit” known enemy infrastructure, positions, and/or equipment/supplies. These types of reports are already contained within the Area and Zone Reconnaissance Reports with the option of “Attachment/Media Object” and “Narrative” to support exploitation decisions.

### 3.7.1 Functional Interdependencies

#### 3.7.1.1 Mobility

Initially, when movement is required beyond the engineer platoon's organic capability, a movement request is generated. When the request is generated, it is either as a ground, littoral, or aerial or a combination of any of those modes. Follow-on requests will be made once the reconnaissance report has been submitted and can be fulfilled in any manner.

Engineer equipment, especially heavy equipment, requires mobility to move from home location to destination, or from home location to port of embarkation.

#### 3.7.1.2 Supply

Supply procurement is generated from requirements for multiple shipping locations and variations in classes of supply. Each request for service can contain like-matter items such as Class III and IV or be singular in nature such as Class V. Often, the requirement for procurement comes from locally-sourced materials and require a contracting officer.

It is important to consider the possibility that a "make" or "buy" decision is needed as an interdependent requirement for maintenance services or in the performance of maintenance activities.

#### 3.7.1.3 Maintenance

Organizational repair is done organically, and, if needed, a request for evacuation of the item will be made. A request for higher echelon maintenance does not mean the equipment in need should or will be evacuated; it may simply need additional maintenance expertise that is not organic to the local engineer team.

There are occasions when an item of equipment cannot be repaired organically and must be evacuated back to the organizational-level, or, in most cases, intermediate-level maintenance facility. When this occurs, additional equipment is required to continue the operation. Forecasting at this point is vital for the mission to continue seamlessly.

### 3.7.2 Processing Cycle at the Functional Level

General Engineering does not have a functional-level cycle of events that takes place for a product or service. There are services that general engineering provides that are part of the planning; however, the resulting service output products do not exist in any digital realm or system.

### 3.7.3 Data Exchange Requirements

During the analysis it was noted that much of the data contained within Engineer Reconnaissance reports were duplicate entries and used outdated doctrine as a standard. When an engineer reconnaissance team (ERT) conducts a survey, media (i.e., images, geospatial tags, and remarks) is typically added to a journal entry. Equally, when estimations are performed for requesting or supplying bulk liquids, antiquated and inaccurate forecasting techniques are applied. There exists no official digital reference for accurately forecasting platform-specific bulk liquid consumption rates.

While the tactical user may not have specifically requested general engineering services (i.e., route reconnaissance), these services are essential in the performance the unit's assigned mission. As such, there exists a fundamental data interdependency on other functional areas. The data captured from the ERT should reflect the same geospatial area as a point of reference and seamlessly transfer to other teams and users to review and append as necessary.

Likewise, if an additional ERT is subsequently employed, real-time geospatial atmospherics should be captured to enable modification to estimations and mobility requirements.

At present, ERTs manually produce “sketches” using stand-alone, disconnected handheld electronic measuring devices. To enable “digital truth”, application tools that allow the user to remain covered/concealed must be incorporated into the information system, without the need to manually convert scale/distance of objects. Automating this process will greatly reduce—if not eliminate—the potential for quantitative human error factors when these calculations are conducted manually. Providing digital pictures, videos, and voice recorded descriptions are now an essential data requirement to provide precise information on areas, roads, and routes.

Once this data becomes available in the digital environment, recommendations for mission-specific EDLs may be automated.

#### 3.7.3.1 Engineer Reconnaissance – Area Report

This is a general report to detail information concerning threat, terrain, society, and infrastructure of an area a unit must move through or occupy.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.
2) Element Leader. References the reconnaissance/engineer leader performing the reported data. Expressed as either 8 to 9-digit ZAP or 10-digit EDI-PI. ZAP data element is a preformatted and system assigned abbreviation for USMC personnel. EDI-PI is DoD assigned for all DoD personnel (active, reserve, GS, and contractor). User should not necessarily need to type in these identifiers but select from a preloaded list on the local machine or scan a government DoD issued identification card if the individual is not in the list.
3) Area Location. Used to indicate the general area of the reconnaissance. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.
4) Items Report. General Engineering report embedded within the overall item instance of the mission report. This report consists of individual references of interest. Expressed as a 14 position MGRS locator and free text block for type/label up to 50 text characters.
5) Water Source. If free-flowing water features are found within the area, a report of the flow rate/velocity expressed as a fixed four position numeric value. Quantity (kiloliters) fixed 4 position numeric value.
6) Attachment/Media objects. Expressed as an embedded video, picture, or voice-recorded file at the time of request. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).
7) Narrative. Free text for additional information up to 100 characters in length to describe the area in general not contained in the items report.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Element Leader | 1 | No | 8/10 | Must Use | Pass |
| 3. Area Location | 1 | No | 14/14 | Must Use | Pass |
| 4. Water Source | 1 | No | 8/8 | Optional | Pass |
| 5. Items Report | 1 | Yes | 19/64 | Must Use | Pass |
| 6. Attachment/Media | 1 | No | 1/1 | Optional | Pass |
| 7. Narrative | 1 | No | 5/100 | Optional | Pass |

**Table 19: Engineer Reconnaissance - Area - Data Exchange**

#### 3.7.3.2 Engineer Reconnaissance – Zone Report

A zone reconnaissance is assigned when the threat situation is vague or when information concerning cross-country trafficability is desired.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.
2) Element Leader. Expressed as either 8 to 9-digit ZAP or 10-digit EDI-PI. ZAP data element is a preformatted and system assigned abbreviation for USMC personnel. EDI-PI is DoD assigned for all DoD personnel (active, reserve, GS, and contractor). User should not necessarily need to type in these identifiers but select from a preloaded list on the local machine or scan a government DoD issued identification card if the individual is not in the list.
3) Amphibious Crossing Classification. If a water feature can be crossed. This is not the same as a ford report as this is mainly used for amphibious vehicles (i.e., AAV/ACV, LAV, LCAC, etc.). Expressed in alpha numeric context in three data blocks, for example “EWHITE400” (Type = E; Classification=WHITE; Frontage in Meters=440)
4) Enemy Report. This report consists of individual references to enemy positions, emplacements, facilities, supplies, and/or equipment. 14 position MGRS locator and free text block for type/label up to 50 text characters.
5) Zone Location. Used to indicate the zone of the reconnaissance. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Attached media/object/free is needed for specific approach/cover and concealment/obstacles/Observations.
6) Water Source. If free flowing water features are found within the area a report of the flow rate/velocity expressed as a fixed four position numeric value. Quantity (kiloliters) fixed 4 position numeric value.
7) Items Report. General Engineering report embedded within the overall item instance of the mission report. This report consists of individual references of interest. 14 position MGRS locator and free text block for type/label up to 50 text characters.
8) Attachment/Media Objects. Expressed as an embedded object of a video, picture, or voice recording of the of the item(s) being requested. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

9) Narrative. Free text for additional information up to 100 characters to describe the zone not presented in the data.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Element leader | 1 | No | 2/2 | Must Use | Pass |
| 3. Amphibious Crossing Classification | 1 | No | 4/12 | Optional | Pass |
| 4. Enemy Report | 1 | Yes | 19/64 | Must Use | Pass |
| 5. Zone Location | 1 | No | 14/14 | Must Use | Pass |
| 6. Water Source | 1 | No | 8/8 | Optional | Pass |
| 7. Items Report | 1 | Yes | 19/64 | Must Use | Pass |
| 8. Attachment/Media | 1 | No | 1/1 | Optional | Pass |
| 9. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 20: Engineer Reconnaissance - Zone - Data Exchange**

#### 3.7.3.3 Engineer Reconnaissance – Route Report

Route reconnaissance is the directed effort to obtain information (focused on the commanders' reconnaissance objective(s)) along a specified route and on adjacent terrain (including lateral routes) over which movement may be restricted.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Element Leader. Expressed as either 8 to 9-digit ZAP or 10-digit EDI-PI. ZAP data element is a preformatted and system assigned abbreviation for USMC personnel. EDI-PI is DoD assigned for all DoD personnel (active, reserve, government service [GS], and contractor). User should not necessarily need to type in these identifiers but select from a preloaded list on the local machine or scan a government DoD issued identification card if the individual is not in the list.

3) Enemy Report. This report consists of individual references to enemy positions, emplacements, facilities, supplies, and/or equipment. 14 position MGRS locator and free text block for type/label up to 50 text characters or by drop down list.

4) Route Information. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Attached media/object/free is needed for specific approach/cover and concealment/obstacles/observations. May use 'way points' to determine the route reviewed.

5) Water Source. If free flowing water features are found within the area a report of the flow rate/velocity expressed as a fixed four position numeric value. Quantity (kiloliters) fixed 4 position numeric value.

6) Items Report. General Engineering report embedded within the overall item instance of the mission report. This report consists of individual references of interest. 14 position MGRS locator and free text block for type/label up to 50 text characters.

7) Attachment/Media Objects. Expressed as an embedded object of a video, picture, or voice recording of the of the item(s) being requested. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

8) Narrative. Free text for additional information up to 100 characters to describe the route not revealed in the data elements.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Element leader | 1 | No | 8/10 | Must Use | Pass |
| 3. Enemy | 1 | No | 14/14 | Must Use | Pass |
| 4. Route Location (way points) | 1 | Yes | 14/14 | Must Use | Pass |
| 5. Water Source | 1 | No | 8/8 | Optional | Pass |
| 6. Items Report | 1 | No | 19/64 | Must Use | Pass |
| 7. Attachment/Media | 1 | No | 1/1 | Optional | Pass |
| 8. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 21: Engineer Reconnaissance - Route - Data Exchange**

#### 3.7.3.4 Engineer Reconnaissance – Road Report

The road report evaluates the condition and trafficability of established roads. The road report is different from a route report as it focuses only on the road conditions along a predetermined start/end point and does not include cross-country routes. If a road encounters a tunnel or bridge, a separate report is submitted for each instance of such features.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Start Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

3) End Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

4) Road Classification. Expressed as a single digit code when the user selects the clear text option A: Flat Gradients/ Easy Curves, B: Steep Gradient (>7%), C: Sharp Curves (Less than 30M), D: Steep Gradients and Sharp Curves

5) Drainage. Expressed as a single digit code when the user selects the clear text option A: Adequate, Crown/Camber with adequate culverts in good condition, B: Inadequate, Poor crown/ Camber, culverts or ditches are blocked or otherwise in poor condition.

6) Foundation. Expressed as a single digit code when the user selects the clear text option A: Stabilized and compacted. B: Unstable, loose

7) Surface Type. Expressed as a single digit code when the user selects the clear text option A: Concrete, B: Bituminous, C: Brick, D: Stone, E: Crushed Rock, F: Water bound Macadum, G: Gravel, H: Lightly Metaled, I: Natural or Stabilized Soil, J: Other (activates text box for 50 characters)

8) Obstructions. A: Serial Number Assigned, B: Particulars (Obstacles, Utilities, Power lines, Intersections), C: Grid Reference, D: Remarks

9) Attachment. Expressed as an embedded object of a video, picture, or voice recording of the maintenance condition. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

10) Narrative. Free text for additional information up to 100 characters to describe the materiel demand or additional coordination.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Start Point Location | 1 | No | 14/14 | Must Use | Pass |
| 3. End Point Location | 1 | No | 14/14 | Must Use | Pass |
| 4. Road Classification | 1 | No | 1/1 | Must Use | Pass |
| 5. Drainage | 1 | No | 1/1 | Must Use | Pass |
| 6. Foundation | 1 | No | 1/1 | Must Use | Pass |
| 7. Surface Type | 1 | No | 1/1 | Must Use | Pass |
| 8. Obstructions | 1 | No | 1/1 | Must Use | Pass |
| 9. Attachment | 1 | No | 1/1 | Optional | Pass |
| 10. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 22: Engineer Reconnaissance - Road - Data Exchange**

#### 3.7.3.5 Engineer Reconnaissance – Tunnel Report

The tunnel report is supplemental to the road report or a rail network evaluation.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Start Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

3) End Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

4) Tunnel Name. Clear text box entry up to 50 characters if the tunnel has a known name.

5) Tunnel Direction. Expressed as a two-digit code. First digit is S: Single direction or B: Bidirectional and second digit equates to actual cardinal direction for single direction: E: East, W: West, N: North, S: South and 1: East/West or 2: North/South for bidirectional tunnels.

6) Tunnel Number. If the tunnel has a corresponding number expressed as a seven-digit alpha-numeric value with special characters (XXXXXX).

7) Tunnel Media Through Type. Expressed as a single digit code when the user selects the clear text option; A: Subaqueous, B: Rock, C: Soil
8) Tunnel Through Feature. Expressed as a single digit code when the user selects the clear text option; M: Mountain, W: Water
9) Tunnel Length. Expressed as a five-digit numeric position value in feet as whole numbers (XXXXX).
10) Tunnel Type. Expressed as a single digit code when the user selects the clear text option; 1: Road, 2: Rail, 3: Waterway, 4: Pedestrian
11) Rail Tunnel Tracks. Number of tracks expressed as two-digit numeric value as whole numbers (XX).
12) Roadway Tunnel Lanes. Number of lanes expressed as a two-digit numeric value as whole numbers (XX).
13) Roadway Tunnel Width. Expressed as a five-digit numeric value in feet as whole numbers (XXXXX).
14) Tunnel Vertical Clearance. Expressed as a five-digit numeric and single digit alpha value in feet as whole numbers (XXXXX).
15) Tunnel Horizonal Clearance. Expressed as a five-digit numeric and single digit alpha value in feet as whole numbers (XXXXX).
16) Tunnel Curve Grade. Expressed as a single digit code when the user selects the clear text option A: Flat Gradients/ Easy Curves, B: Steep Gradient (>7%), C: Sharp Curves (Less than 30M), D: Steep Gradients and Sharp Curves
17) Tunnel Grade. Expressed as a two-digit numeric value in percentage (XX%).
18) Tunnel Alignment. Expressed as a single digit code when the user selects the clear text option; S: Straight, C: Curve (with radius) activates a box to input the numeric value.
19) Tunnel Lining Surface. Expressed as a single digit code when the user selects the clear text option A: Concrete, C: Brick, D: Stone, E: Crushed Rock, F: Water bound Macadum, G: Gravel, H: Lightly Metaled, I: Natural or Stabilized Soil, J: Other (activates text box for 50 characters) (option B omitted from the list).
20) Tunnel Ventilation. Expressed as a single digit code when the user selects the clear text option, (1: Yes, 0: N).
21) Tunnel Ventilation Type. Expressed as a single digit code when the user selects the clear text option; W: Wind Turbine, P: Emergency Access Ports (input interval between ports – text box appears when this selection is made)
22) Tunnel Drainage. Expressed as a single digit code when the user selects the clear text option: Y or (1: Yes, 0: N).
23) Tunnel Drainage Media. Expressed as an embedded object of a video, picture, or voice recording of the drainage system. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).
24) Tunnel Demolition. Expressed as a single digit code when the user selects the clear text option; Is the tunnel chambered for demolition: (1: Yes, 0: N).
25) Tunnel Completion Date. Expressed as date CCYYMMMDD.

26) Tunnel Overall Condition. Expressed as a single digit code when the user selects the clear text option; A: Excellent, B: Good, C: Fair, D: Poor

27) Tunnel Bypass. If the tunnel can be reasonably bypassed (1: Yes, 0: N).

28) Bypass Grid. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

29) Alternate Crossing Grid. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

30) In Tunnel Restrictions. Expressed as a single digit code when the user selects the clear text option; A: Radius of curves, B: Steep grades, C: Obstacles, D: Roadway width, E: Combination of restrictions.

31) Portal to Portal Length. Expressed as a five-digit numeric value in feet as whole numbers (XXXXX).

32) Centerline Distance of Tunnel. Expressed as a five-digit numeric value in feet as whole numbers (XXXXX).

33) Rise of Tunnel Arch. Expressed as a five-digit numeric and single digit alpha value in feet as whole numbers (XXXXX).

34) Portal View. Expressed as a single digit code when the user selects the clear text option; A: Square with arched ceiling, B: Elliptical, C: Horseshoe, D: Semicircular.

35) Narrative. Free text for additional information up to 100 characters to describe the tunnel outside of data provided.

36) Attachment. Expressed as an embedded object of a video, picture, or voice recording of the maintenance condition. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Start Point Location | 1 | No | 14/14 | Must Use | Pass |
| 3. End Point Location | 1 | No | 14/14 | Must Use | Pass |
| 4. Tunnel Name | 1 | No | 5/50 | Must Use | Pass |
| 5. Tunnel Direction | 1 | No | 2/2 | Must Use | Pass |
| 6. Tunnel Number | 1 | No | 1/50 | Must Use | Pass |
| 7. Tunnel Media Through Type | 1 | No | 1/1 | Must Use | Pass |
| 8. Tunnel Feature | 1 | No | 1/1 | Must Use | Pass |
| 9. Tunnel Length | 1 | No | 2/11 | Must Use | Pass |
| 10. Tunnel Type | 1 | No | 1/1 | Must Use | Pass |
| 11. Rail Tunnel Tracks | 1 | No | 1/2 | Must Use | Pass |
| 12. Roadway Tunnel Lanes | 1 | No | 1/2 | Must Use | Pass |
| 13. Roadway Tunnel Width | 1 | No | 2/11 | Must Use | Pass |
| 14. Tunnel Vertical Clearance | 1 | No | 2/11 | Must Use | Pass |
| 15. Tunnel Horizonal Clearance | 1 | No | 2/11 | Must Use | Pass |
| 16. Tunnel Curve Grade | 1 | No | 1/1 | Must Use | Pass |
| 17. Tunnel Grade | 1 | No | 2/2 | Must Use | Pass |
| 18. Tunnel Alignment | 1 | No | 1/1 | Must Use | Pass |
| 19. Tunnel Lining Surface | 1 | No | 1/1 | Must Use | Pass |
| 20. Tunnel Ventilation | 1 | No | 1/1 | Must Use | Pass |
| 21. Tunnel Ventilation Type | 1 | No | 1/1 | Must Use | Pass |
| 22. Tunnel Drainage | 1 | No | 1/1 | Must Use | Pass |
| 23. Tunnel Drainage Media | 1 | No | 1/1 | Must Use | Pass |
| 24. Tunnel Demolition | 1 | No | 1/1 | Must Use | Pass |
| 25. Tunnel Completion Date | 1 | No | 8/8 | Must Use | Pass |
| 26. Tunnel Overall Condition | 1 | No | 1/1 | Must Use | Pass |
| 27. Tunnel Bypass | 1 | No | 1/1 | Must Use | Pass |
| 28. Bypass Grid | 1 | No | 14/14 | Must Use | Pass |
| 29. Alternate Crossing Grid | 1 | No | 14/14 | Must Use | Pass |
| 30. In Tunnel Restrictions | 1 | No | 1/1 | Must Use | Pass |
| 31. Portal to Portal Length | 1 | No | 2/11 | Must Use | Pass |
| 32. Centerline Distance of Tunnel | 1 | No | 2/11 | Must Use | Pass |
| 33. Rise of Tunnel Arch | 1 | No | 2/11 | Must Use | Pass |
| 34. Portal View | 1 | No | 1/1 | Must Use | Pass |
| 35. Narrative | 1 | Yes | 5/100 | Optional | Pass |
| 36. Attachment | 1 | No | 1/1 | Optional | Pass |

**Table 23: Engineer Reconnaissance - Tunnel - Data Exchange**

#### 3.7.3.6 Engineer Reconnaissance – Bridge Report (Site/Structure)

The bridge report is supplemental to the road report or a rail network evaluation.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Location/Position Information. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting)

and position 10-14: Numeric (Northing). Attached media/object/free is needed for specific approach/cover and concealment/obstacles/Observations.

3) Start Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

4) End Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

5) Gap Width. Expressed as a five-digit numeric and single digit alpha value in feet as whole numbers (XXXXX).

6) Clearance. Expressed as a five-digit numeric and single digit alpha value in feet as whole numbers (XXXXX).

7) Bridge Name. Clear text box entry up to 50 characters if the bridge has a known name.

8) Bridge Direction. Expressed as a two-digit code. First digit is S: Single direction or B: Bidirectional and second digit equates to actual cardinal direction for single direction: E: East, W: West, N: North, S: South and 1: East/West or 2: North/South for bidirectional bridges.

9) Bridge Serial/Number. If the bridge has a corresponding number expressed as a seven-digit alpha-numeric value with special characters (XXXXXX).

10) Bridge/Tunnel Connector. If the bridge it connected to a tunnel (1: Yes, 0: N).

11) Bridge Length. Expressed as a five-digit numeric value in feet as whole numbers (XXXXX).

12) Bridge Type. Expressed as a single digit code when the user selects the clear text option; 1: Road, 2: Rail, 3: Pedestrian

13) Bridge Bypass. If the bridge can be reasonably bypassed (1: Yes, 0: N).

14) Bypass Grid. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

15) Bridge Tracks. Number of tracks expressed as two-digit numeric value as whole numbers (XX).

16) Roadway Bridge Lanes. Number of lanes expressed as a two-digit numeric value as whole numbers (XX).

17) Roadway Bridge Width. Expressed as a five-digit numeric value in feet as whole numbers (XXXXX).

18) Under Bridge Clearance. Expressed as a five-digit numeric value in feet as whole numbers (XXXXX).

19) Bridge Vertical Clearance. Expressed as a five-digit numeric value in feet as whole numbers (XXXXX).

20) Bridge Horizontal Clearance. Expressed as a five-digit numeric value in feet as whole numbers (XXXXX).

21) Total Number of Bridge Spans. Expressed as a three-digit numeric value as whole numbers (XXX).

22) Bridge Span Construction. Expressed as a single digit code when the user selects the clear text option; A: Truss, B: Girder, C: Beam, D: Slab, E: Arch – Closed Spandrel, F: Arch – Open Spandrel, G: Suspension, H: Floating, J: Trunnion Swing, K: Single Leaf Trunnion Bascule, L: Rolling Lift Bascule, M: Double Trunnion Bascule, N: Vertical Life, P: Other (opens text box).

23) Bridge Span Material. Expressed as a single digit code when the user selects the clear text option A: Steel, B: Concrete, C: Reinforced Concrete, D: Prestressed Concrete, F: Stone, G: Brick, H: Wood, J: Other (activates text box for 50 characters).

24) Attachment. Expressed as an embedded object of a video, picture, or voice recording of the maintenance condition. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

25) Narrative. Free text for additional information up to 100 characters to describe the bridge outside of data provided.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Location/Position | 1 | No | 14/14 | Must Use | Pass |
| 3. Start Point Location | 1 | No | 14/14 | Must Use | Pass |
| 4. End Point Location | 1 | No | 14/14 | Must Use | Pass |
| 5. Gap Width | 1 | No | 1/5 | Must Use | Pass |
| 6. Clearance | 1 | No | 1/5 | Must Use | Pass |
| 7. Bridge Name | 1 | No | 5/50 | Must Use | Pass |
| 8. Bridge Direction | 1 | No | 2/2 | Must Use | Pass |
| 9. Bridge Serial/Number | 1 | No | 1/7 | Must Use | Pass |
| 10. Bridge/Tunnel Connector | 1 | No | 1/1 | Must Use | Pass |
| 11. Bridge Length | 1 | No | 1/5 | Must Use | Pass |
| 12. Bridge Type | 1 | No | 1/1 | Must Use | Pass |
| 13. Bridge Bypass | 1 | No | 1/1 | Must Use | Pass |
| 14. Bridge Bypass Grid | 1 | No | 14/14 | Must Use | Pass |
| 15. Bridge Tracks | 1 | No | 1/2 | Must Use | Pass |
| 16. Roadway Bridge Lanes | 1 | No | 1/2 | Must Use | Pass |
| 17. Roadway Bridge Width | 1 | No | 1/5 | Must Use | Pass |
| 18. Under Bridge Clearance | 1 | No | 1/5 | Must Use | Pass |
| 19. Bridge Vertical Clearance | 1 | No | 1/5 | Must Use | Pass |
| 20. Bridge Horizontal Clearance | 1 | No | 1/5 | Must Use | Pass |
| 21. Total Number of Bridge Span | 1 | No | 1/3 | Must Use | Pass |
| 22. Bridge Span Construction | 1 | No | 1/1 | Must Use | Pass |
| 23.Bridge Span Material | 1 | No | 1/1 | Must Use | Pass |
| 24. Attachment | 1 | No | 1/1 | Optional | Pass |
| 25. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 24: Engineer Reconnaissance - Bridge - Data Exchange**

#### 3.7.3.7 Engineer Reconnaissance – Ford Report

Usually part of a route report, the ford report presents best possible areas of crossing a water feature of some kind without a gap crossing solution in place.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Location/Position Information. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Attached media/object/free is needed for specific approach/cover and concealment/obstacles/Observations.

3) Start Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

4) End Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.
5) Minimum Width. Expressed as a five-digit numeric user inputted value in feet (XXXXX).
6) Minimum Depth. Expressed as a two-digit numeric user inputted value in feet (XX).
7) Ford Bottom Type. Expressed as a single-digit alpha value once the user selects the clear text value M: Mud, C: Clay, S: Sand, G: Gravel, R: Rock, P: Artificial Paving.
8) Route Number. If the ford is connected to an established route expressed as a three-digit numeric entry (XXX).
9) Ford Type. Expressed single-digit numeric value when the user selects the clear text option P: Pedestrian, V: Vehicular.
10) Stream Velocity. Expressed as a double data input with first single-digit alpha value when the user selects the clear text option S: Swift, M: Moderate, L: Slow, and second data input of measured flow rate as whole numbers in meters in a numeric two-digit value (X|XX).
11) Name of River/Stream/Body of Water. Clear text box entry up to 50 characters if the body of water has a known name.
12) Low Water Measurements. Multiple data input Width: four-digit numeric expressed in feet (XXXX), Depth: four-digit numeric expressed in feet (XXXX), Velocity: three-digit numeric measured speed (XXX), Date of Measurement expressed as local date CCYYMMMDD.
13) Median Water Measurements. Multiple data input Width: four-digit numeric expressed in feet (XXXX), Depth: four-digit numeric expressed in feet (XXXX), Velocity: three-digit numeric measured speed (XXX), Date of Measurement expressed as local date CCYYMMMDD.
14) High Water Measurements. Multiple data input Width: four-digit numeric expressed in feet (XXXX), Depth: four-digit numeric expressed in feet (XXXX), Velocity: three-digit numeric measured speed (XXX), Date of Measurement expressed as local date CCYYMMMDD.
15) Attachment/Media. Expressed as an embedded object of a video, picture, or voice recording of the ford area. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).
16) Narrative. Free text for additional information up to 100 characters to describe the ford outside of data provided.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Location/Position | 1 | No | 14/14 | Must Use | Pass |
| 3. Start Point Location | 1 | No | 14/14 | Must Use | Pass |
| 4. End Point Location | 1 | No | 14/14 | Must Use | Pass |
| 5. Minimum Width | 1 | No | 1/2 | Must Use | Pass |
| 6. Minimum Depth | 1 | No | 1/2 | Must Use | Pass |
| 7. Ford Bottom Type | 1 | No | 1/1 | Must Use | Pass |
| 8. Route Number | 1 | No | 1/3 | Must Use | Pass |
| 9. Ford Type | 1 | No | 1/1 | Must Use | Pass |
| 10. Stream Velocity | 1 | No | 3/3 | Must Use | Pass |
| 11. Name of River/Stream/Body of Water | 1 | No | 1/50 | Must Use | Pass |
| 12. Low Water Measurements | 1 | No | 12/20 | Must Use | Pass |
| 13. Median Water Measurements | 1 | No | 12/20 | Must Use | Pass |
| 14. High Water Measurements | 1 | No | 12/20 | Must Use | Pass |
| 15. Attachment | 1 | No | 1/1 | Optional | Pass |
| 16. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 25: Engineer Reconnaissance - Ford - Data Exchange**

#### 3.7.3.8 Engineer Reconnaissance – Ferry Report

Usually part of a route report, the ferry report presents best possible options to cross large water features efficiently when organic lift is unsuitable or unavailable.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Ferry Route Start Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

3) Ferry Route End Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

4) Waterway Velocity. Expressed as a double data input with first single-digit alpha value when the user selects the clear text option S: Swift, M: Moderate, L: Slow, and second data input of measured flow rate as whole numbers in meters in a numeric two-digit value (X|XX).

5) Crossing Distance. Expressed as a three-digit numeric value in miles as whole numbers (XXX).

6) Ferry Crossing Time. Total time to cross in one direction expressed in 24-hour clock time as follows: DDHHMM, where D = days (0-99), H = hours (00-23), M = minutes (00-59). If

the ferry crossing is greater than 24 hours, the DD entry is used however, the default is zero.

7) Ferry Terminal Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

8) Route Number. If the ferry is connected to an established route expressed as a three-digit numeric entry (XXX).

9) Ferry Type. Expressed single-digit numeric value when the user selects the clear text option P: Pedestrian, V: Vehicular.

10) Name of River/Stream/Body of Water. Clear text box entry up to 50 characters if the body of water has a known name.

11) Vessel Condition. Expressed single-digit numeric value when the user selects the clear text option E: Excellent, F: Fair, P: Poor.

12) Vessel Information. Registration Number, Operating Length, Operating Width, Operating Tonnage, Vehicle Deadweight Capacity, Railcar Deadweight Capacity

13) Terminal Condition. Expressed single-digit numeric value when the user selects the clear text option E: Excellent, F: Fair, P: Poor.

14) Terminal Entrance Condition. Expressed single-digit numeric value when the user selects the clear text option E: Excellent, F: Fair, P: Poor.

15) Terminal Entrance Width. Expressed as a three-digit numeric value in feet as whole numbers (XXX).

16) Terminal Entrance Depth. Expressed as a three-digit numeric value in feet as whole numbers (XXX).

17) Terminal Approach Surface Type. Expressed as a single digit code when the user selects the clear text option A: Concrete, B: Bituminous, C: Brick, D: Stone, E: Crushed Rock, F: Water bound Macadum, G: Gravel, H: Lightly Metaled, J: Natural or Stabilized Soil, K: Other (activates text box for 50 characters)

18) Flooding Risk. If ferry operations are at risk by the condition, expressed single-digit alpha value when the user selects the clear text option H: Excellent, F: Fair, L: Low.

19) Freezing Risk. If ferry operations are at risk by the condition, expressed single-digit alpha value when the user selects the clear text option H: Excellent, F: Fair, L: Low.

20) Tidal Risk. If ferry operations are at risk by the condition, expressed single-digit alpha value when the user selects the clear text option H: Excellent, F: Fair, L: Low.

21) Water Trafficability. Expressed in a fixed 6 position alpha configuration “go” or “slow-go” or “no-go”.

22) Attachment/Media. Expressed as an embedded object of a video, picture, or voice recording of the ford area. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

23) Narrative. Free text for additional information up to 100 characters to describe the ford outside of data provided.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Ferry Route Start Point Location | 1 | No | 14/14 | Must Use | Pass |
| 3. Ferry Route End Point Location | 1 | No | 14/14 | Must Use | Pass |
| 4. Stream Velocity | 1 | No | 3/3 | Must Use | Pass |
| 5. Crossing Distance | 1 | No | 1/3 | Must Use | Pass |
| 6. Ferry Crossing Time | 1 | No | 6/6 | Must Use | Pass |
| 7. Ferry Terminal Locations | 1 | No | 14/14 | Must Use | Pass |
| 8. Route Number | 1 | No | 1/3 | Must Use | Pass |
| 9. Ferry Type | 1 | No | 1/1 | Must Use | Pass |
| 10. Name of River/Stream/Body of Water | 1 | No | 3/50 | Must Use | Pass |
| 11. Vessel Condition | 1 | No | 1/1 | Must Use | Pass |
| 12. Vessel Information | 1 | No | 5/100 | Must Use | Pass |
| 13. Terminal Condition | 1 | No | 1/1 | Must Use | Pass |
| 14. Terminal Entrance Condition | 1 | No | 1/1 | Must Use | Pass |
| 15. Terminal Entrance Width | 1 | No | 1/3 | Must Use | Pass |
| 16. Terminal Entrance Depth | 1 | No | 1/3 | Must Use | Pass |
| 17. Terminal Approach Surface Type | 1 | No | 1/1 | Must Use | Pass |
| 18. Flooding Risk | 1 | No | 1/1 | Must Use | Pass |
| 19. Freezing Risk | 1 | No | 1/1 | Must Use | Pass |
| 20. Tidal Risk | 1 | No | 1/1 | Must Use | Pass |
| 21. Water Trafficability | 1 | No | 2/6 | Must Use | Pass |
| 22. Attachment | 1 | No | 1/1 | Optional | Pass |
| 23. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 26: Engineer Reconnaissance - Ferry - Data Exchange**

#### 3.7.3.9 Engineer Reconnaissance – River Report

Usually part of a route report, the river/stream report presents best possible emplacement of gap crossing options when required or requested by higher-echelon command and control elements. Additionally, the river report can also serve as a more detailed report when considering water sources for bulk water purposes or for potential bivouac sites. This report can also be used in conjunction with the ford report.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Initial Start Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

3) Route End Point Location. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting)

and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

4) Route Number. If the river is connected to an established route expressed as a three-digit numeric entry (XXX).
5) Name of River/Stream. Clear text box entry up to 50 characters if the body of water has a known name.
6) General River/Stream Flow Direction. 1: East to West, 2: West to East, 3: North to South, 4: South to North.
7) Water Trafficability. Expressed in a fixed sing position alpha configuration A: “go”, B: “slow-go”, or C: “no-go”.
8) Waterway Velocity. Expressed as a double data input with first single-digit alpha value when the user selects the clear text option S: Swift, M: Moderate, L: Slow, and second data input of measured flow rate as whole numbers in meters in a numeric two-digit value (X|XX).
9) Near shore Bank Height. Expressed as a two-digit numeric value in feet (XX).
10) Near shore Bank Slope. Expressed as a two-digit numeric value in degrees (XX).
11) Near shore Bank Stability. Expressed as a single-digit value after the user selects the clear text option F: Firm, S: Soft, U: Unstable.
12) Nearshore Bank Soil Type. Expressed as a single-digit alpha value once the user selects the clear text value M: Mud, C: Clay, S: Sand, G: Gravel, R: Rock, P: Artificial Paving.
13) Far Shore Bank Height. Expressed as a two-digit numeric value in feet (XX).
14) Far Shore Bank Slope. Expressed as a two-digit numeric value in degrees (XX).
15) Far Shore Bank Stability. Expressed as a single-digit alpha value when the user selects the clear text option F: Firm, S: Soft, U: Unstable.
16) Far Shore Bank Soil Type. Expressed as a single-digit alpha value once the user selects the clear text value M: Mud, C: Clay, S: Sand, G: Gravel, R: Rock, P: Artificial Paving.
17) Attachment/Media. Expressed as an embedded object of a video, picture, or voice recording of the ford area. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).
18) Narrative. Free text for additional information up to 100 characters to describe the ford outside of data provided.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Initial Start Point Location | 1 | No | 14/14 | Must Use | Pass |
| 3. Route End Point Location | 1 | No | 14/14 | Must Use | Pass |
| 4. Route Number | 1 | No | 1/3 | Must Use | Pass |
| 5. Name of River/Stream | 1 | No | 3/50 | Must Use | Pass |
| 6. River/Stream Flow Direction | 1 | No | 1/1 | Must Use | Pass |
| 7. Water Trafficability | 1 | No | 2/6 | Must Use | Pass |
| 8. Water way velocity | 1 | No | 3/3 | Must Use | Pass |
| 9. Near shore Bank Height | 1 | No | 2/2 | Must Use | Pass |
| 10. Near shore Bank Slope | 1 | No | 2/2 | Must Use | Pass |
| 11. Near shore Bank Stability | 1 | No | 1/1 | Must Use | Pass |
| 12. Near Bank Soil Type | 1 | No | 1/1 | Must Use | Pass |
| 13. Far shore Bank Height | 1 | No | 2/2 | Must Use | Pass |
| 14. Far shore Bank Slope | 1 | No | 2/2 | Must Use | Pass |
| 15. Far shore Bank Stability | 1 | No | 1/1 | Must Use | Pass |
| 16. Far shore Bank Soil Type | 1 | No | 1/1 | Must Use | Pass |
| 17. Attachment | 1 | No | 1/1 | Optional | Pass |
| 18. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 27: Engineer Reconnaissance - River - Data Exchange**

### 3.7.3.10 Engineer Reconnaissance – Landing Zone Report

The landing zone report is an evaluation of a section of usable terrain for the use of aircraft to land and take off. The landing zone report is not intended to evaluate existing airfields and facilities for aircraft staging, maintenance, etc.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.
2) Location/Position Information. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Attached media/object/free is needed for specific approach/cover and concealment/obstacles/Observations.
3) Estimate. Expressed as a single digit code (1: Yes, 0: No)
4) Layout Designation. Expressed as a two-digit alpha value. (Landing Zone: LZ, Landing Site: LS, Landing Point: LP)
5) Landing Point Capacity. Expressed as a one-digit numeric value (value = number of aircraft parking spaces) (1) Single airframe
6) Landing Zone Capacity. Expressed as a one-digit numeric value (value = number of aircraft parking spaces) (1) Single airframe, (2-3) multiple airframes.
7) Landing Site Capacity. Expressed as a one-digit numeric value (value = number of aircraft parking spaces) (1) Single airframe, (2-4) multiple airframes.
8) Landing Zone Width. Expressed as a three-digit numeric value in feet as whole numbers (XXX).

9) Landing Zone Length. Expressed as a four-digit numeric value in feet as whole numbers (XXX).

10) Aircraft Supportability. Automated calculation based on Landing Zone width and length. Expressed as a single-digit alpha value indicating rotor diameter. A/B = Size 4 (38'-78' Aircraft) Aircraft, C/D = Size 2 (44'-59' Aircraft), E/F Vertical Lift (Requires VTOL).

11) Landing Zone Approach. Expressed single-digit alpha value when the user selects the clear text option S: South, E: East, N: North, W: West.

12) Landing Zone Departure. Expressed single-digit alpha value when the user selects the clear text option S: South, E: East, N: North, W: West.

13) Landing Zone Surface Slope. Expressed as a single digit code when the user selects the clear text option A: <7º up slope landing B: >7º side slope landing.

14) Obstacle. Expressed in single digit code when the user selects the clear text option 1: remove, 2: reduce, 3: emplace.

15) Attachment/Media. Expressed as an embedded object of a video, picture, or voice recording of the ford area. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

16) Narrative. Free text for additional information up to 100 characters to describe the ford outside of data provided.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Location | 1 | No | 14/14 | Must Use | Pass |
| 3. Estimate | 1 | No | 1/1 | Must Use | Pass |
| 4. Layout Designation | 1 | No | 2/2 | Must Use | Pass |
| 5. Landing Point Capacity | 1 | No | 1/1 | Must Use | Pass |
| 6. Landing Zone Capacity | 1 | No | 1/1 | Must Use | Pass |
| 7. Landing Site Capacity | 1 | No | 1/1 | Must Use | Pass |
| 8. Landing Zone Width | 1 | No | 1/3 | Must Use | Pass |
| 9. Landing Zone Length | 1 | No | 1/3 | Must Use | Pass |
| 10 Aircraft Supportability | 1 | No | 1/1 | Must Use | Pass |
| 11. Landing Zone Approach | 1 | No | 1/1 | Must Use | Pass |
| 12. Landing Zone Departure | 1 | No | 1/1 | Must Use | Pass |
| 13. Landing Zone Surface Slope | 1 | No | 1/1 | Must Use | Pass |
| 14. Obstacle | 1 | No | 1/1 | Must Use | Pass |
| 15. Attachment | 1 | No | 1/1 | Optional | Pass |
| 16. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 28: Engineer Reconnaissance - Landing Zone - Data Exchange**

### 3.7.3.11 General Engineering – Obstacle Removal

Obstacles refer to any physical characteristics of the terrain that impede the mobility of a force along a route, road, or pathway. All obstacles are existing or reinforcing. Existing obstacles are inherent aspects of the terrain and can be natural, man-made, or a combination of both.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Location/Position Information. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Attached media/object/free is needed for specific approach/cover and concealment/obstacles/Observations.
3) Obstacle. Expressed in single digit code when the user selects the clear text option 1: Remove, 2: Reduce, 3: Emplace.
4) Minimum/Maximum Length. Expressed as a double data input with first five-digit numeric minimum value and the second five-digit numeric maximum value if feet (XXXXX/XXXXX).
5) Minimum/Maximum Width. Expressed as a double data input with first five-digit numeric minimum value and the second five-digit numeric maximum value in feet (XXXXX/XXXXX).
6) Minimum/Maximum Depth. Expressed as a double data input with first five-digit numeric minimum value and the second five-digit numeric maximum value in feet (XXXXX/XXXXX).
7) Route Number. If the obstacle close to a connected or established route expressed as a three-digit numeric entry (XXX).
8) Determination of Action: Engineer recommendation of action based on feasibility, risk, and materials available (used to inform commander's decision). Expressed in one-digit numerical value 1: Use bypass, 2: Neutralize Obstacle, 3: Breach, 4: Continue Mission
9) Bypass. If the obstacle can be reasonably bypassed (1: Yes, 0: N).
10) Bypass Grid. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.
11) Attachment/Media. Expressed as an embedded object of a video, picture, or voice recording of the ford area. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).
12) Narrative. Free text for additional information up to 100 characters to describe the ford outside of data provided.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation/Reconnaissance | 1 | No | 9/9 | Must Use | Pass |
| 2. Location | 1 | No | 14/14 | Must Use | Pass |
| 3. Obstacle | 1 | No | 1/1 | Must Use | Pass |
| 4. Minimum/Maximum Length | 1 | No | 2/10 | Must Use | Pass |
| 5. Minimum/Maximum Width | 1 | No | 2/10 | Must Use | Pass |
| 6. Minimum/Maximum Depth | 1 | No | 2/10 | Must Use | Pass |
| 7. Route Number | 1 | No | 1/3 | Must Use | Pass |
| 8. Determination of action | 1 | No | 1/1 | Must Use | Pass |
| 9. Bypass | 1 | No | 1/1 | Must Use | Pass |
| 10. Bypass Grid | 1 | No | 14/14 | Must Use | Pass |
| 11. Attachment | 1 | No | 1/1 | Optional | Pass |
| 12. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 29: General Engineering - Obstacle Removal - Data Exchange**

### 3.7.3.12 Explosive Ordnance Disposal (EOD) Clearing/Rendering Safe

1) Date of UXO Discovery. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.
2) Requested Date of EOD Action. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.
3) Location of UXO. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Attached media/object/free is needed for specific approach/cover and concealment/obstacles/Observations.
4) Type of CBRN Agent. Expressed in a single digit code when the user selects the clear text option 1: Chemical, 2: Biological, 3: Radiological, 4: Nuclear, 5: None (default)
5) Physical Property of CBRN Agent. Option appears if ‘Type of CBRN Agent’ is select. Expressed in a single digit code when the user selects the clear text option 1: Gas, 2: Liquid, 3: Aerosols, 4: Ionizing Radiation
6) Contamination Value of CBRN Agent. Option appears if ‘Type of CBRN Agent’ is select. Expressed in a single digit code when the user selects the clear text option E: External, I: Internal, W: Wound, C: Contagious Casualties.
7) Munition Color. Expressed as a double data input with the first being an alpha code and the second being three-digit alpha code expressing color. (X|XXX)
8) Munition Markings. Free text for additional information up to 100 characters to describe markings.
9) Munition Purpose. Expressed as a double-digit alpha code when the user selects the clear text option AA: Anti-Armor, AP: Anti-Personnel, FL: Flare, SM: Smoke, IM: Improvised
10) Munition Type. Expressed as a single digit alpha code when the user selects the clear text option E: Emplace, D: Drop, T: Throw, P: Project

11) Attachment/Media. Expressed as an embedded object of a video, picture, or voice recording of the ford area. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

12) Narrative. Free text for additional information up to 100 characters to describe the ford outside of data provided.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of UXO Discovery | 1 | No | 9/9 | Must Use | Pass |
| 2. Requested Date of EOD Action | 1 | No | 9/9 | Must Use | Pass |
| 3. Location of UXO | 1 | No | 14/14 | Must Use | Pass |
| 4. Type of CBRN agent | 1 | No | 1/1 | Must Use | Pass |
| 5. Physical Property of CBRN agent | 1 | No | 1/1 | Optional | Pass |
| 6. Contamination Value of CBRN agent | 1 | No | 1/1 | Optional | Pass |
| 7. Munition Color | 1 | No | 4/4 | Must Use | Pass |
| 8. Munition Marking | 1 | No | 5/100 | Must Use | Pass |
| 9. Munition Purpose | 1 | No | 2/2 | Must Use | Pass |
| 10. Munition Type | 1 | No | 1/1 | Must Use | Pass |
| 11. Attachment | 1 | No | 1/1 | Optional | Pass |
| 12. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 30: Explosive Ordnance Disposal – Clearing/Rendering Safe - Data Exchange**

### 3.7.3.13 General Engineering - Bulk Liquid Support

This report is based on fuels and liquids available at a local or commercial sites. Types of fuels as are referenced by DLA Energy as a category. [14]

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Location/Position Information. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Attached media/object/free is needed for specific approach/cover and concealment/obstacles/Observations.

3) Estimate. Expressed as a single digit code the data presented is or is not an estimate. If the data in the report is not precisely measured (estimate) or was measured (actual). (1: Yes, 0: No)

4) Fuel. Single position alpha character “F” with a fixed 4 position alpha/numeric “type” (i.e., JP8 = JP81, Diesel = DESL, Kerosene = KERO, etc.) followed by a 6-position numeric quantity and a 3-position alpha unit of measure designator (i.e., “FJP81000GAL” or “FDESL500GAL” or “FKERO0003QTS”).

5) Water Source. Single position alpha character “W” followed by a 6-position numeric quantity and a 3-position alpha unit of measure designator followed by a 4-position alpha water type designation. “W5000GALBULK” or “W0010LBSPURC”

6) Attachment/Media. Expressed as an embedded object of a video, picture, or voice recording of the ford area. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

7) Narrative. Free text for additional information up to 100 characters to describe the ford outside of data provided.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation | 1 | No | 9/9 | Must Use | Pass |
| 2. Location of Bulk Liquid | 1 | No | 14/14 | Must Use | Pass |
| 3. Estimate | 1 | No | 1/1 | Must Use | Pass |
| 4. Fuel | 1 | No | 13/13 | Optional | Pass |
| 5. Water | 1 | No | 14/14 | Optional | Pass |
| 6. Attachment | 1 | No | 1/1 | Optional | Pass |
| 7. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 31: General Engineering - Bulk Liquid Support**

### 3.7.3.14 General Engineering - Demolition

Demolition requests are based on the need of the tactical unit when an object presents an obstacle to mobility or area use. Types of explosives and demolitions per the US Army FM 3-34.214 (FM 5-250) Explosives and Demolitions Manual.

1) Date of Evaluation/Reconnaissance. Expressed as local date CCYYMMMDD assuming midnight to 23:59 as a full day.

2) Location/Position Information Object to be Demolished. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Attached media/object/free is needed for specific approach/cover and concealment/obstacles/Observations.

3) Type of Demolition. Expressed in a single digit alpha code and the second being single digit alpha code when the user selects the clear text option B: Bridge, D: Ditches, F: Facilities, O: Obstacle, V: Vehicle, T: Tunnel, V: Overpass

4) Route Number. If the demolition is close to a connected or established route expressed as a three-digit numeric entry (XXX).

5) Determination of Action. Engineer recommendation of action based on feasibility, risk, and materials available (used to inform commander's decision). Expressed in one-digit numerical value 1: Use Bypass, 2: Neutralize, 3: Breach, 4: Continue Mission

6) Bypass. If the obstacle can be reasonably bypassed (1: Yes, 0: N).

7) Bypass Grid. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., White Beach, Pier 3, etc.) for ease of use.

8) Attachment/Media. Expressed as an embedded object of a video, picture, or voice recording of the ford area. Flag indicator if there is an attachment with a single digit code (1: Yes, 0: N).

9) Narrative. Free text for additional information up to 100 characters to describe the ford outside of data provided.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Date of Evaluation | 1 | No | 9/9 | Must Use | Pass |
| 2. Location | 1 | No | 14/14 | Must Use | Pass |
| 3.Type of Demolition | 1 | No | 1/1 | Must Use | Pass |
| 4. Route Number | 1 | No | 3/3 | Must Use | Pass |
| 5. Determination of Action | 1 | No | 1/1 | Must Use | Pass |
| 6. Bypass | 1 | No | 1/1 | Must Use | Pass |
| 7. Bypass Grid | 1 | No | 14/14 | Must Use | Pass |
| 8. Attachment | 1 | No | 1/1 | Optional | Pass |
| 9. Narrative | 1 | Yes | 5/100 | Optional | Pass |

**Table 32: General Engineering - Demolition**

## 3.8 Health Services Data Requirements

Health services is one of the more data-driven logistics functions requiring accurate and real-time to near real-time information about patients, especially in forward operating locations. Unfortunately, the health services function is the least understood function of logistics in the Marine Corps due to a dependence on U.S. Navy (USN) capabilities and lack of education thereof. This has created a twofold problem in functional understanding:

1) The USMC does not understand its own current service-aligned health service support (HSS) requirements and has made USMC programs of record largely inaccessible to naval personnel. Naval personnel are inadequately integrated into the Marine Corps digital environment. For example, Training and Readiness (T&R) Standards for Hospital Corpsman are listed in MCTIMS, but commands are not ensuring their sailors meet those standards, or that their proficiency is properly documented.

2) The USN does not adequately communicate organizational structure, training, and equipment shortfalls to their USMC commanders.

Therefore, battlefield data collection in the realm of health services is severely lacking, with no digital tools available to first responders, who primarily rely on analog, paper solutions. Additionally, the understanding of the flow of care, the actions of Medical C2, and the supporting data exchange requirements are not integrated into operational impacts and capabilities.
The below graphic depicts the functional layout of health services within the tactical operating environment and the relationship within the sub-functions. This helps to illustrate the operating cycle within the health services function and how the sub-functions repeat until resolution.

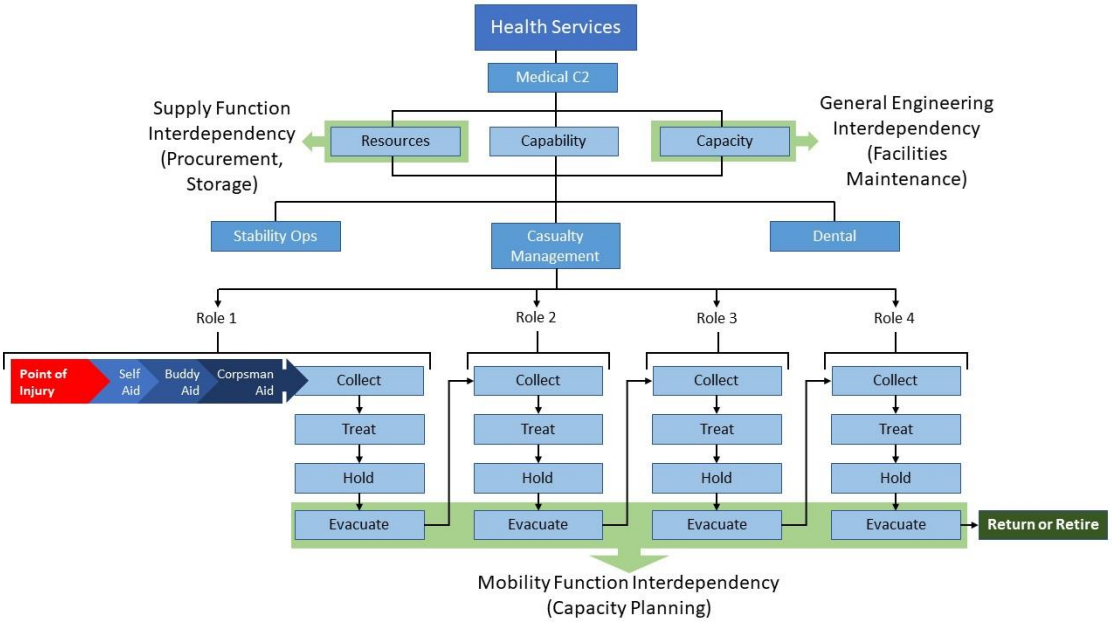


**Figure 17: Health Services Cycle**

## 3.8.1 Functional Interdependencies

### 3.8.1.1 Mobility

The movement and evacuation of casualties to/from/in between medical care areas/facilities are completely interdependent on the function of mobility and the ability to plan the fastest and most efficient mode of transportation. While forward units may have access to ground designated transportation (i.e., ambulances), all other modes of transportation must be planned, coordinated, and assigned by those working within the capacity planning sub-function.

### 3.8.1.2 Supply

Medical resources, specifically Class VIII (such as specialized equipment and supplies), are usually sourced through the Medical Logistics Company of the supporting logistics combat element (LCE). Resourcing is a critical Medical C2 function to maintain and sustain the continuum of care from the leading tactical edge throughout the operating environment. Therefore, the procurement, storage, and distribution of materiel resources within the medical care network is interdependent on the supply function to establish the required supply chains and data exchanges. Additionally, Medical C2 needs the visibility of relevant supply information to make strategic decisions as well as the need for rapid resupply of items to the tactical edge.

### 3.8.1.3 General Engineering

Beyond Role 1, the Health Services requires facilities, either permanent; semi-permanent; or temporary to perform medical care safely and efficiently nearest to the point of need. Health Services is interdependent on the ability of General Engineering for the identification, rehabilitation, and/or construction of appropriate medical care facilities (other than tents) as well as the maintenance of such facilities.

## 3.8.2 Processing Cycle at the Functional Level

The function of health services is multilayered in escalation of medical care under medical C2. To understand the data exchange requirements for health services requires understanding the roles of care. The following is the definition and responsibilities of each of the four roles:

#### 3.8.2.1 Role 1

Role 1 (also referred to as unit-level medical care) includes:

1) Immediate lifesaving measures.
2) Disease and nonbattle injury prevention and care.
3) Combat and operational stress preventive measures.
4) Patient location and acquisition (collection).

Treatment is provided by designated unit Corpsman. Major emphasis is placed on those measures necessary for the patient to return to duty or to stabilize them and allow for evacuation to the next role of care. These measures include maintaining the airway, stopping the bleeding, preventing shock, protecting wounds, immobilizing fractures, and other emergency measures. Escalation of care within this role starts at the point of injury and onward.

1) Self-Aid. All military personnel are trained in a variety of basic first-aid procedures. These procedures include aid for chemical casualties with particular emphasis on lifesaving tasks. This training enables the military personnel to apply first aid to alleviate potential life-threatening situations.
2) Buddy Aid/Combat Lifesaver Aid.
    a. Buddy aid is basic first-aid procedures applied on another individual using the casualty's Individual First-Aid Kit (IFAK) or other available medical supplies (issued and/or improvised).
    b. Combat lifesavers are non-medical military personnel selected by their unit commander for additional training beyond basic first-aid procedures. The primary duty of this individual does not change. The additional duty of the combat lifesaver is to provide enhanced first aid for injuries, based on Tactical Combat Casualty Care (TCCC) protocols and unit issued Combat Life Saver (CLS) medical kits, prior to the arrival of unit Corpsmen or other medical personnel.
3) Medical Personnel (Corpsman) Aid. Role 1 provides primary healthcare, specialized first aid, triage, resuscitation, and stabilization. Normally included within the basic Role 1 capabilities are routine sick call and the management of minor sick and injured personnel who can immediately return to duty. Role 1 also includes casualty collection and preparation of casualty's evacuation. Expanded medical treatment is provided by medical personnel with enhanced medical skills (i.e., Independent Duty Corpsman (IDC), Independent Duty Medical Technician (IDMT), or physician).

#### 3.8.2.2 Role 2

Role 2 provides advanced trauma management and emergency medical treatment, including continuation of resuscitation started in Role 1. Role 2 provides a greater capability to resuscitate trauma patients than is available at Role 1. If necessary, additional emergency measures are initiated, but they do not go beyond the measures dictated by immediate necessities. Role 2 care has the capability to provide packed blood products, limited X-ray, laboratory, dental support, combat and operational stress control, preventative medicine (PVTMED), and resuscitative surgical support. Role 2 has a limited hold capability necessitating the onward evacuation of casualties.

#### 3.8.2.3 Role 3

In Role 3, the patient is treated in medical treatment facility (MTF) staffed and equipped to provide care to all categories of patients, to include resuscitation, initial wound surgery, specialty surgery (general, orthopedic, urogenital, thoracic, ENT, neurosurgical) and post-operative treatment. This may include definitive surgery for local nationals depending on the current rules of engagement. This role of care expands the support provided at Role 2. Patients who are unable to tolerate and survive movement over long distances receive surgical care in a hospital as close to the supported unit as the tactical situation allows. This role includes provisions for:

1) Evacuating patients from supported units.
2) Providing care for all categories of patients in a MTF with the proper staff and equipment.
3) Providing area medical support to units without organic medical assets.

#### 3.8.2.4 Role 4

Role 4 medical care is found in U.S. hospitals and robust overseas facilities. These hospitals represent the most comprehensive medical care available to restore the individual to operational duty, or medically retire the individual for continuing, long term care.

#### 3.8.2.5 Role Data Similarities

Each of the roles are executing similar functions in collection, treatment, holding, and then finally evacuation. The initial evacuation from Role 1 to Role 2 would be a casualty evacuation (CASEVAC), however, once in the medical care system each subsequent evacuation is classified as a medical evacuation (MEDEVAC). If the patient is ambulatory, this is simply a PAX request. Therefore, no matter the role level, each data set is building upon the last role's data set in identifying the casualty, the injury, resulting treatment at each role, and subsequent movement. While casualty management is the function of health services, casualty tracking is the function of command services. This is currently a data layer gap between the Health Services function and Services function of logistics.

#### 3.8.2.6 Medical Data Limitations

DoD has mandated use of paper forms for patient/casualty data capture, to include DD Form 1380 and DA Form 7656, and TCCC Cards. The DoD's intent is as follows:

1) All data from these cards are expected to be manually-transcribed into the servicemember's electronic health record (EHR). While well intentioned, the expectation of analog data collected in the field to be then input into a digital record is unrealistic and prone to human error.
2) Services are to review and update current service-specific doctrine and training plans to incorporate use of the physical TCCC cards in pre-hospital and organizational first aid provided by both medical and non-medical first responders. However, non-medical first responders need continuous training to orient on and understand the complexities of medical abbreviations and definitions to accurately collect the relevant data.
3) Services are to coordinate with the Director, Defense Health Agency (DHA), to facilitate the development and deployment of an electronic version of the TCCC card. Again, these efforts have centered around and are limited to electromechanical variations of the paper card, appearing as fillable portable document formats (PDF). PDFs are not truly digital records. Additionally, it is highly unlikely that medical responders will have mobile devices with the PDF version of the TCCC card while forward deployed, and how such

data would be transmitted from the device to the next MTF and then transcribed into the casualty's EHR.

4) Since the TCCC card is a physical item, services are to ensure a physical TCCC card is stocked in organizational medical equipment sets, kits, and outfits to include CLS bags, IFAKs/JFAKs, and Unit First Aid Kits located on all operational platforms. Additionally, DHA requires the stocking of TCCC cards with expediency to support the appropriate documentation of care in the pre-hospital environment during all contingencies. The requirement for physical data capture cards comes with real cost to produce, space to store, and forecasting for re-stocking. This creates additional and unnecessary burden on the supply chain.

### 3.8.3 Data Exchange Requirements at Role 1 (pre-MTF)

Every role level exhibits the same process of collection, treatment, hold, and evacuation. Within the scope of the analysis, the actions at Role 1 are the building blocks for the data continuum throughout the patient's continuum of care, originating at the most forward tactical edge. Through analysis, the actions performed at Role 1 of collection, treatment, hold, and evacuation are all occurring, however, each stage in the process may be compressed or reduced due to situational urgency. Therefore, the data exchange requirement at each stage in the process are independent actions building on itself as a digital record. While recognizing each stage may be compressed to some point, the data flow appears as one seamless fluid action.

The collected data must be easily and seamlessly passed between care providers via local machine to local machine and/or to the next level of care. The next level of care should receive the casualty data prior to arrival of the patient, vice the data following in trace. The data originating in digital record at the point of injury serves as the basis of decision in continuing health care conclusions and enables effective/efficient Medical C2.

The initial buildout of available medications to first responders must be an automated process, completed prior to the execution of an exercise, operation, deployment. All data comprising a health services request must automatically update the servicemembers' electronic health record and be accessible and available at every level of care. If care is provided to an individual that is not within the service care network, or on an unregistered casualty (EPW, NON-US, US-CIV, etc.), a skeleton record is built based on the input data for later merging, updating, and/or archiving.

#### 3.8.3.1 Collection

Casualties are assembled at a predetermined location, or at a location of opportunity to protect from further injury. Data recorded at the point of collection are:

1) Segment Number. System assigned as the request is created under the header. A single digit indicator: up to nine independent request types can be added under the header. The segment number does not restrict the number of patients/casualties being reported or CASEVAC requested.

2) Request Type. Expressed as a two-digit background code when the user selects "Casualty Report". In this case CR is the code for an injury/casualty/illness report.

3) Request Priority. Expressed as two-digit code based on the urgency aligned to the element/unit's F/AD. Auto populated based on Triage Precedence: A and/or B = Critical, C = Urgent, D and/or E = Routine

4) ZAP/EDI-PI. Expressed as either 8 to 9-digit ZAP or 10-digit EDI-PI. ZAP data element is a preformatted and system assigned abbreviation for USMC personnel. EDI-PI is DoD

assigned for all DoD personnel (active, reserve, government service [GS], and contractor). User should not necessarily need to type in these identifiers but select from a preloaded list on the local machine or scan a government (U.S. Federal, Local, and/or foreign Passport/ID) issued identification card if the individual is not in the list.

5) Last Name. Auto populated from the local machine list if the ZAP/EDI-PI is resident. If not, input text box up to 15 characters for non-military U.S. or other. This information is passed only if the input text box is activated.

6) First Name. Auto populated from the local machine list if the ZAP/EDI-PI is resident. If not, input text box up to 15 characters for non-military U.S. or other. This information is passed only if the input text box is activated.

7) Service. Casualty's branch of service: Auto populated from the local machine list if the ZAP/EDI-PI is resident. If not, input text box with the following codes: USA, USSF, USAF, USCG, USN, USMC. For U.S. civilians (GS, contractor, etc.): US CIV. For non-U.S. Coalition and NATO personnel (military and civilian): NON-US. Enemy Prisoner of War: EPW. This information is passed only if the input text box is activated.

8) Element/Unit Identification/Call Sign. Injured/casualty's element/unit/call sign (not the reporting unit/element) if U.S military is auto populated from the local machine list if the ZAP/EDI-PI is resident. Expressed as a four-digit alpha-numeric reference key which aligns to the clear text naming convention. The clear text naming convention does not need to be passed, only the unique index key which was randomly assigned by system. The origination of callsigns/brevity codes for units/elements is with the operations function. This stresses the importance of operation planning and execution occurring in a digital environment. If not, input text box up to 15 characters. This information is passed only if the input text box is activated.

9) Allergies/Alerts. Ideally auto populated from source personnel medical records to the local machine prior to deployment. If transcribing from red medical tag, this becomes a text box for manual input up to 30 characters. No need to identify as "allergy" as the field is labeled and identified as the type of information when synchronized to the enterprise system. Example: if a red tag is found on the casualty indicating an allergy to penicillin, simply input "penicillin". This information is passed only if the input text box is activated. Auto populates with "NKDA" (no known drug allergies).

10) Date of Injury. Local: Date expressed as CCYYMMMDD. Defaults to the current date of the entry but can be overridden to a previous day.

11) Time of Injury. Local: System time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99). SS and DD are inconsequential for this data input and default to "00:00". Defaults to the current time of the entry but can be overridden to a previous time.

12) Location Injury Occurred. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., Hill 588, etc.) for ease of use, estimated location is acceptable as well as "pin drop" current location.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 1. Segment Number | 1 | No | 1/1 | Must Use | Pass |
| 2. Request Type | 1 | No | 2/2 | Must Use | Pass |
| 3. Request Priority | 1 | No | 2/2 | Must Use | Pass |
| 4. ZAP/EDI-PI | 1 | No | 8/10 | Must Use | Pass |
| 5. Last Name | 1 | No | 1/15 | Must Use | Pass |
| 6. First Name | 1 | No | 1/15 | Must Use | Pass |
| 7. Service | 1 | No | 3/6 | Optional | Not Passed |
| 8. Element/Unit Identification/Call Sign | 1 | No | 4/4 | Optional | Not Passed |
| 9. Allergies | 1 | No | 4/10 | Must Use | Pass |
| 10. Date of Injury | 1 | No | 9/9 | Must Use | Pass |
| 11. Time of Injury | 1 | No | 8/8 | Must Use | Pass |
| 12. Location Injury Occurred | 1 | No | 14/14 | Must Use | Pass |

**Table 33: Health Services - Collection Data**

#### 3.8.3.2 Treatment

Treatment includes the triage and application of care from self-aid through Role 4 capabilities. Triage is the process of casualty sorting and is basic to the successful operation of a patient movement system. Rapid evaluations must be made to identify which patients need immediate resuscitation and which patients can tolerate delay in treatment. After triage, treatment is applied to sustain the casualty until the next level of care. The following data elements are two separate actions within the same function and will be presented as “triage” and then “treatment.” The intent is to continue the record of data initiated at the point of collection. As the data is collected it can be easily categorized, arranged, and sorted as the user interface requires. For instance, the input data fields can be arranged to meet the MARCH-PAWS algorithm of care.

13) Primary Mechanism of Injury: Expressed as a main category and then a secondary category:

    a. Main Category 1. Injury by explosive. Indicates a rapid expansion of heated gases, solids, and/or liquids. Explosive effects on the body carry the possibility of shrapnel, concussive injuries, hemorrhaging, burns, and/or Chemical, Biological or Radiological vectors. Clear text selection is a two-digit code: E1: artillery/Indirect fire, grenade, E2: IED (improvised explosive device), E3: landmine, E4: RPG (rocket propelled grenade), and E5: other type of explosion.

    b. Main Category 2. Injury by a physical force. Effects of gravitational, velocity, and/or centrifugal forces. Effects of the body are lacerations, hemorrhaging, compression injuries, bruising, and/or fractures/breakage. Clear text selection is a two-digit code: P1: blunt force trauma, P2: fall, P3: velocity impact (e.g., vehicular crash/accident), P4: GSW (gunshot wound), and P5: knife/bayonet/sharp object.

    c. Main Category 3. Injury by illness/disease. Effects of abnormalities in the function and/or structure of body organs and systems. Effects on the body are lack of bodily control, extreme discomfort, choking, blistering, disorientation, and/or mental soundness. Clear text selection is a two-digit code: D1: pathogen (viral, bacterial, fungi, protozoa), D2: allergic reaction, D3: chemical/biological hazard exposure D4: pathological.

14) CBRN Related/Exposure. Expressed as a single code entry if casualty’s injury has been caused by CBRN vectors and/or the casualty has been or is suspected to have been

exposed to a CBRN environment. C: Chemical, B: Biological, R: Radiological, N: Nuclear, X: None. System defaults to X: None

15) Major Signs/Symptoms. Expressed as a clear text selection with a single digit code: B: bleeding, R: restricted breathing, X: burned, C: incapacitation. If “burn” is selected this will provide an overall calculated surface area estimate body percentage based on “Injury Location(s).”

16) Injury Location(s). User selects clear text entry which is converted into a three-digit data code for transmission. Only one area can be selected (one front head, one back head, one front chest, one back chest, etc.). Additionally, the selected location provides a burn estimate for calculation in parenthesis [XXX (XX)]: FH0 (4.5): Front Head (4.5), FH1: Front Head Ears, Nose, Throat, FH2 (4.5): Front Head Eyes, FH3 (4.5): Front Head Dental, BH0 (4.5): Back of the Head, FN0 (4.5): Front Neck, BN0 (4.5): Back Neck, C00 (18): Chest, RC0 (09): Right Chest, LC0 (09): Left Chest (09), CC0 (09): Center Chest, B00 (18): Back, UB0 (09): Upper Back, LB0 (09): Lower Back, LA0 (4.5): Left Arm, RA0 (4.5): Right Arm, LL0 (09): Left Leg, RL0 (09): Right Leg, G00 (01): Groin, P00 (01): Posterior, A00 (09): Abdominal

17) Triage/Treatment Vitals Check Date – Local. Date expressed as CCYYMMMDD. Defaults to the current date of the entry. This serves as a sub-heading for the six vitals checked at the time of triage/treatment.

18) Triage/Treatment Vitals Check Time – Local. System time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99). SS and DD are inconsequential for this data input and default to “00:00”. Defaults to the current time of the entry.

19) Body Temperature. Expressed as a three-digit whole number in Fahrenheit. System convertible to Celsius.

20) Pulse Rate/Location. Comprised of two data points. Pulse Rate: Expressed as a three-digit whole number. Pulse Location: Expressed as a single alpha code corresponding to the clear text selection W: Wrist (radial artery), N: Neck (carotid artery). XXX|X

21) Blood Pressure. Expressed as two sets of three-digit numbers (systolic and diastolic) with a pipe separator in between: XXX|XXX.

22) Respiratory Rate. Count of chest rises in a one minute while the casualty is at rest. Expressed as a two-digit numeric entry: XX.

23) Pulse Oximetry % $O^2$ Saturation. Expressed as two-digit whole numbers in a percentage usually from 90-99%: XX.

24) Responsiveness. Expressed with a clear text selection with a background single alpha code A: alert, V: responds to voice, P: responds to pain, U: unresponsive.

25) Pain Scale. If the casualty is alert or voice responsive a pain scale can be employed. The pain scale is a double numeric digit expressed as follows: 0: No pain, 1: Pain is very mild, barely noticeable. Most of the time the casualty doesn't think about it. 2: Minor pain. It's annoying. The casualty has a sharp pain now and then. 3: Noticeable pain. Distracting but the casualty gets used to it. 4: Moderate pain. If the casualty is involved in an activity, they can ignore the pain for a while, but it is still distracting. 5: Moderately strong pain. The casualty can't ignore it for more than a few minutes. But, with effort, the

casualty can still work or do some moderate activities. 6: Moderately stronger pain. The casualty avoids some normal daily activities and has trouble concentrating. 7: Strong pain. Casualty cannot perform normal activities. 8: Very strong pain. The casualty cannot perform any work or activities at all. 9: Pain that is very hard to tolerate. The casualty cannot or has difficulty carrying on a conversation.10: Worst pain possible.

26) Triage Precedence. Expressed with a clear text selection with a single alpha code. A: Urgent (surgical) - requires surgery while enroute to the MTF. B: Urgent (non-surgical) - patient can be stabilized until arrival to MTF. C: Priority - injuries that are not immediately life-threatening but could become life-threatening if medical care is not provided. D: Routine - non-life-threatening medical care beyond unit's ability, or deceased patient. E: Convenience - non-life-threatening care provided to personnel in austere conditions but requires professional medical evaluation. Note: If a tourniquet has been selected as a treatment type, the triage precedence defaults to B – Urgent (Non-Surgical), as a lower precedence cannot be applied. The precedence can be overridden to increase to A – Urgent (Surgical) by user intervention.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 13. Primary Mechanism of Injury | 1 | No | 2/2 | Must Use | Pass |
| 14. CBRN Related/Exposure | 1 | No | 1/1 | Must Use | Pass |
| 15. Major Signs/Symptoms | 1 | No | 1/1 | Must Use | Pass |
| 16. Injury Location(s) | 10 | Yes | 2/2 | Must Use | Pass |
| 17. Triage/Treatment Vitals Date - Local | 10 | Yes | 9/9 | Optional | Pass |
| 18. Triage/Treatment Vitals Time - Local | 10 | Yes | 8/8 | Optional | Pass |
| 19. Body Temperature | 10 | Yes | 3/3 | Optional | Pass |
| 20. Pulse Rate/Location | 10 | Yes | 4/4 | Optional | Pass |
| 21. Blood Pressure | 10 | Yes | 6/6 | Optional | Pass |
| 22. Respiratory Rate | 10 | Yes | 2/2 | Optional | Pass |
| 23. Pulse Oximetry | 10 | Yes | 2/2 | Optional | Pass |
| 24. Responsiveness | 10 | Yes | 1/1 | Optional | Pass |
| 25. Pain Scale | 10 | Yes | 1/2 | Optional | Pass |
| 26. Triage Precedence | 1 | No | 1/1 | Must Use | Pass |

**Table 34: Health Services - Treatment Data (Triage Segment)**

27) Extremity Bleeding Treatment (Tourniquets). Data set is a four-part series for placement, type, date, and time. All four parts must be recorded for a complete entry. The placement is the base or header data element with the type, date, and time being the subordinate and required data elements.

a. Placement. Clear text selection correlating to a four-digit alpha data code for transmission. No tourniquets: TQXX, Right Arm: TQRA, Left Arm: TQLA, Right Leg: TQRL, and Left Leg: TQLL

b. Type. Type of tourniquet applied is a single digit alpha code in response to the user's clear text input. The data element is the fifth data element in the string. E: extremity, J: junctional, and T: truncal

c. Date – Local. Date expressed as CCYYMMMDD. Defaults to the current date of the entry but can be overridden to a previous day.

d. Time – Local. System time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes

(00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99). SS and DD are inconsequential for this data input and default to “00:00”. Defaults to the current time of the entry but can be overridden to an earlier time.

28) Wounds Treatment: (Bandaging/Dressing). User selected clear text input provides background the corresponding code for date exchange Hemostatic Dressing: T1, Pressure Bandage: T2, Sling: T3, Splint: T3, Eye Shield (Left): T4, Eye Shield Right: T5, Eye Shield Both: T6, Hypothermia Prevention: T7

29) Airway Treatment. Information concerning the airway and treatments are expressed as two-digit alpha-numeric codes for data exchange when the user selects the clear text input. Airway Intact: A0, NPA (nasopharyngeal airway): A1, CRIC (cricothyroidotomy): A2, ET-Tube (endotracheal tube): A3, SGA (supraglottic airway): A4

30) Breathing Treatment. Information concerning the breathing treatments are expressed as two-digit alpha-numeric codes for data exchange when the user selects the clear text input. None: B0, O2: B1, Needle-D: B3, Chest Tube: B4, Chest Seal: B5

31) Fluid Circulation Resuscitation Interventions/Treatments. Expressed as a five-part data series.

    a. Fluid Name. Saline: S, Ringer’s Lactate: R, Hextend: H, Other (input clear text name up to 20 characters)

    b. Volume/Dose. Expressed as a four-digit numeric whole number for milliliters or cubic centimeters.

    c. Route. IV: Intravenous, IO: Interosseous

    d. Date – Local. Date expressed as CCYYMMMDD. Defaults to the current date of the entry but can be overridden to a previous day.

    e. Time – Local. System time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99). SS and DD are inconsequential for this data input and default to “00:00”. Defaults to the current time of the entry but can be overridden to an earlier time.

32) Blood Circulation Resuscitation Interventions/Treatments. Expressed as a five-part data series.

    a. Blood Product Name. whole blood: WBD, red blood cells: RBC, fresh frozen plasma: FFP, lyophilized (freeze-dried) plasma: FDP.

    b. Volume/Dose. Expressed as a four-digit numeric whole number for milliliters or cubic centimeters.

    c. Route. IV: Intravenous, IO: Interosseous

    d. Date – Local. Date expressed as CCYYMMMDD. Defaults to the current date of the entry but can be overridden to a previous day.

    e. Time – Local. System time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99). SS and DD

are inconsequential for this data input and default to “00:00”. Defaults to the current time of the entry but can be overridden to an earlier time.

33) Analgesic Medication Treatments. Due to the myriad of available medications, it is best if the medical personnel setup the local machine with the medications immediately carried and available to the first responder as list defaults before departing friendly lines. Expressed as a five-part data series.

    a. Medication Name. Ketamine: K, Fentanyl: F, Morphine: M, Other (input clear text name up to 20 characters)

    b. Volume/Dose. Expressed as a two-digit numeric whole number for milliliters or cubic centimeters.

    c. Route. R1: intrathecal (into the spine), R2: subcutaneous (under the skin), R3: intravenous (into a vein), R4: intramuscular (into a muscle), R5: oral, R6: inhale, R7: rectal

    d. Date – Local. Date expressed as CCYYMMMDD. Defaults to the current date of the entry but can be overridden to a previous day.

    e. Time – Local. System time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99). SS and DD are inconsequential for this data input and default to “00:00”. Defaults to the current time of the entry but can be overridden to an earlier time.

34) Antibiotic Medication Treatments. Due to the myriad of available medications, it is best if the medical personnel setup the local machine with the medications immediately carried and available to the first responder as list defaults before departing friendly lines. Expressed as a five-part data series.

    a. Medication Name. M: moxifloxacin, E: ertapenem, P: penicillin, A: azithromycin, Other (input clear text name up to 20 characters)

    b. Volume/Dose. Expressed as a two-digit numeric whole number for milliliters or cubic centimeters or milligrams.

    c. Route. R1: intrathecal (into the spine), R2: subcutaneous (under the skin), R3: intravenous (into a vein), R4: intramuscular (into a muscle), R5: oral, R6: inhale, R7: rectal

    d. Date – Local. Date expressed as CCYYMMMDD. Defaults to the current date of the entry but can be overridden to a previous day.

    e. Time – Local. System time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99). SS and DD are inconsequential for this data input and default to “00:00”. Defaults to the current time of the entry but can be overridden to an earlier time.

35) Other Medication Treatments. Due to the myriad of available medications, it is best if the medical personnel setup the local machine with the medications immediately carried and available to the first responder as list defaults before departing friendly lines.

a. Medication Name. I: Ibuprofen, T: tranexamic acid (TXA), Other (input clear text name up to 20 characters)

b. Volume/Dose. Expressed as a two-digit numeric whole number for milliliters or cubic centimeters or milligrams.

c. Route. R1: intrathecal (into the spine), R2: subcutaneous (under the skin), R3: intravenous (into a vein), R4: intramuscular (into a muscle), R5: oral, R6: inhale, R7: rectal

d. Date – Local. Date expressed as CCYYMMMDD. Defaults to the current date of the entry but can be overridden to a previous day.

e. Time – Local. System time expressed in 24-hour clock time as follows: HHMM, or HHMMSS, or HHMMSSD, or HHMMSSDD, where H = hours (00-23), M = minutes (00-59), S = integer seconds (00-59) and DD = decimal seconds; decimal seconds are expressed as follows: D = tenths (0-9) and DD = hundredths (00-99). SS and DD are inconsequential for this data input and default to “00:00”. Defaults to the current time of the entry but can be overridden to an earlier time.

36) Casualty Type. Indicate if the casualty is mobile or immobile. Expressed as a single alpha code when user selects clear text option. A: litter (cannot walk on their own), B: ambulatory (able to self-move)

37) First Responder ZAP/EDI-PI. Expressed as either 8 to 9-digit ZAP or 10-digit EDI-PI. ZAP data element is a preformatted and system assigned abbreviation for USMC/Navy personnel. Defaults to the local machine’s logged in user EDI-PI.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 27. Extremity Bleeding Treatment (TQ) | 4 | Yes | 22/22 | Optional | Pass |
| 28. Wounds Treatment | 7 | No | 2/2 | Must Use | Pass |
| 29. Airway Treatment | 1 | No | 2/2 | Must Use | Pass |
| 30. Breathing Treatment | 1 | No | 2/2 | Must Use | Pass |
| 31. Fluid Circulation Treatments | 5 | Yes | 24/34 | Optional | Pass |
| 32. Blood Circulation Treatments | 5 | Yes | 24/34 | Optional | Pass |
| 33. Analgesic Medication Treatments | 5 | Yes | 24/34 | Optional | Pass |
| 34. Antibiotic Medication Treatments | 5 | Yes | 24/34 | Optional | Pass |
| 35. Other Medication Treatments | 5 | Yes | 24/34 | Optional | Pass |
| 36. Casualty Type | 1 | Yes | 1/1 | Must Use | Pass |
| 37. First Responder ZAP/EDI-PI | 1 | No | 8/10 | Must Use | Pass |

**Table 35: Health Services - Treatment Data (Intervention Segment)**

#### 3.8.3.3 Hold

The holding of casualties for monitoring post-treatment or post-operation for onward care and/or movement. Holding of a casualty requires monitoring of vitals and administering continuing treatment methods as required. Additional data requirements are appended to the original record in terms of vitals and/or administering additional medications/treatments. All corresponding data elements can have additional date/time entries. No data is overwritten or deleted but is appended with additional date/time driven data entries.

#### 3.8.3.4 Evacuate (CASEVAC)

The movement of casualties to and/or between medical treatment facilities. Users select casualties from an auto generated list produced by the collection process to evacuate on the local machine. Once all casualties are selected (casualties can be selected from the list by filter i.e., by precedence, all, etc.) which will initiate the request via data form. The evacuation or CASEVAC is essential a specialized PAX movement request. When the evacuate request is initiated, key data elements recorded during the collection, treatment, and hold processes are brought forward into the request with minimal user intervention. The evacuate request data form prioritizes data 1) evacuation request (aka legacy 9-Line), 2) priority medical information (aka legacy Z-MIST), and 3) medical record/casualty report (aka legacy TCCC data collection card).

38) Request Priority. Auto populates based on the highest casualty triage precedence in the data set. A: Urgent (surgical) - i.e. requires surgeon to perform surgery while enroute to the MTF, B: Urgent (non-surgical) - i.e. patient can be stabilized until arrival to MTF, C: Priority - i.e. injuries that are not immediately life threatening but could become life threatening if medical care is not provided, D: Routine - i.e. non-life threatening medical care beyond unit's ability, E: Convenience - i.e. non life-threatening care provided to personnel in an austere condition but requires professional medical evaluation.

39) Location of Pickup. If predetermined, this can be loaded as a default setting. Geospatial reference expressed in Military Grid Reference System (MGRS). Position 1-2: Alpha (Grid Zone Designator, GZD), Position 3-4: Alpha (100K-meter Square Identifier), Position 5-14: Numeric (Numerical Location) depending on accuracy level within 1-, 10-, and 100-meter intervals: position 5-9: Numeric (Easting) and position 10-14: Numeric (Northing). Remarks: Physical can be pre-loaded as a predetermined clear text descriptor (i.e., Hill 588, etc.) for ease of use or use "pin drop" location.

40) Location Marking. User determines if location needs or will be marked. A: Panels, B: Pyrotechnic (i.e., flare, red star cluster), C: Smoke (provide smoke color), D: IR flash or signal beacon, E: None. (Optional, defaults to E: None)

41) Location Contamination. A: Nuclear/Radiological, B: Biological, C: Chemical, D: None (Optional, defaults to: D: None)

42) Contact Settings. Frequency and call-sign at pickup site. If predetermined, this can be loaded as a default setting. Defaults to user/element call sign and frequency. Four-digit callsign and 4-digit frequency setting XXXX|XX.XX

43) Count of Casualties – Precedence. Auto populated, calculated, and sorted by number of patients by precedence when casualties are selected. Two data elements: single digit priority and two-digit numeric number. X|XX

44) Count of Casualty – Types. Auto populated, calculated, and sorted by number of patients by type when casualties are selected. Two data elements: single digit type (A: ambulatory/L: Litter and two-digit numeric count. X|XX)

45) Requested Equipment. User indicates by clear text if special equipment is required, A: none, B: hoist, C: extraction equipment, D: ventilation (Optional, defaults to A: None)

46) Security. User determines if location maybe compromised by enemy action. N: No enemy troops in area, P: Possible enemy troops in area (approach with caution), E: Enemy troops in area (approach with caution), X: Enemy troops in area (armed escort required) (Optional, defaults to N: None)

47) ZAP/EDI-PI of Casualties/Treatments. Entire data set is auto populated when the casualties are selected for evacuation. The ZAP/EDI-PI serves a header for the treatment information. Expressed as either 8 to 9-digit ZAP or 10-digit EDI-PI.

   a. Mechanism of injuries auto populated with two-digit code when the casualties are selected for evacuation.

   b. Major injury auto populated with single digit code when the casualties are selected for evacuation.

   c. Signs last recorded readings data set auto populated when the casualties are selected for evacuation. Body Temperature: three-digit reading, Pulse Rate/Location: four-digit reading, Blood Pressure: six-digit reading, Respiratory Rate: two-digit reading.

   d. Treatment given (morphine, tourniquet etc.) last recorded readings auto populated when the casualties are selected for evacuation. Extremity Bleeding Treatment (TQ): 22-digit response X 1-4 occurrences (maximum of four, if applied), Wounds Treatment: 2 X 1-7 occurrences, Airway Treatment: two-digit code, Breathing Treatment: two-digit code, Fluid Circulation Treatments: 24 to 34 digit code (last application only), Blood Circulation Treatments: 24 to 34 digit code (last application only), Analgesic Medication Treatments: 24 to 34 digit code (last dose only), Antibiotic Medication Treatments: 24 to 34 digit code (last dose only), Other Medication Treatments: 24 to 34 digit code (last dose only)

   e. Casualty CBRN auto populated single digit code when the casualties are selected for evacuation.

| Segment Name | Max Use | Repeat | Min/Max | Usage | Exchange |
|---|---|---|---|---|---|
| 38. Request Priority | 1 | No | 1/1 | Must Use | Pass |
| 39. Location of Pickup | 1 | No | 1/1 | Must Use | Pass |
| 40. Location Marking | 1 | No | 1/1 | Optional | Pass |
| 41. Location Contamination | 1 | No | 1/1 | Optional | Pass |
| 42. Contact Settings | 1 | No | 8/8 | Must Use | Pass |
| 43. Count of Casualties – Precedence | 5 | No | 3/3 | Must Use | Pass |
| 44. Count of Casualty – Types | 2 | No | 3/3 | Must Use | Pass |
| 45. Requested Equipment | 1 | No | 1/1 | Optional | Pass |
| 46. Security | 1 | No | 1/1 | Optional | Pass |
| 47. ZAP/EDI-PI of Casualties | 1 | Yes | 29/305 | Must Use | Pass |

**Table 36: Health Services - Evacuation Segment**

# Section 4: Critical Capability Gaps and Recommended Doctrinal Changes

## 4.1 Digital Link Between Operational Planning and Logistics Gap

### 4.1.1 Problem Overview

As logistics is the pacing function for operations, the analysis has identified a critical capability gap in the warfighting function's data flow, organization, and data development/synthesis process to support the commander's command and control efforts. This gap is specifically and especially resident in the realm of operational logistics (OPLOG). Through the analysis of data exchange requirements, it was observed that operational requirements do not align well to logistical capabilities or were completely devoid of operational cohesion or assessment within the realm of logistics competencies.

The link between Logistics with Operations and Intelligence to provide true command & control awareness and capability to the Combatant Commander/Joint Force Commander is inherently flawed, inefficient, or not fully understood, negatively impacting a forces ability to forecast/anticipate future logistical requirements to meet the operational demand. The result is the commander exercising and making command and control decisions based on assessments which may not be fully informed despite best efforts, creating a commander's decision paradox.

Therefore, the analysis team identified a key exasperating condition where planners are developing courses of action within the confines of PowerPoints, whiteboards, and analog methodologies without the ability to connect digitally to logistics. Planners are constrained in relating and considering the relevance of after-action reports, expenditures, and lessons learned due to accessibility. Due to accessibility, the systematic and consistent consideration of enterprise-wide historical information is rarely deliberated in the planning process of similar operations. Due to the analog nature and the missing component of the lessons of the past, logistics is not fully aware of the requirements being formed and conversely planners are making assumptions on the logistics capabilities they are applying requirements on which may not be in the boundaries of logistical reach.

As such, planners currently do not have a common enterprise-wide mission planning and support tool that provides/displays a Common Operational Picture (COP) that efficiently interfaces, informs, and digitally feeds logistics planning, forecasting, and execution information technology tools. Additionally, key operational data on unit locations, callsigns, and other pertinent information directly impacting the ability to logistically sustain operations is not contained in any logistics information system and is not directly tied to an existing COP. C2PC is inefficient and ineffective in providing/capturing mission requirements, historical data, task organizing, and associated reports (POA&M, TEEP, FRAGO, WARNO, OPORD, Cmdr Guidance, mission reports, etc.) as expected or required by the Marine Corps Planning Process (MCPP).

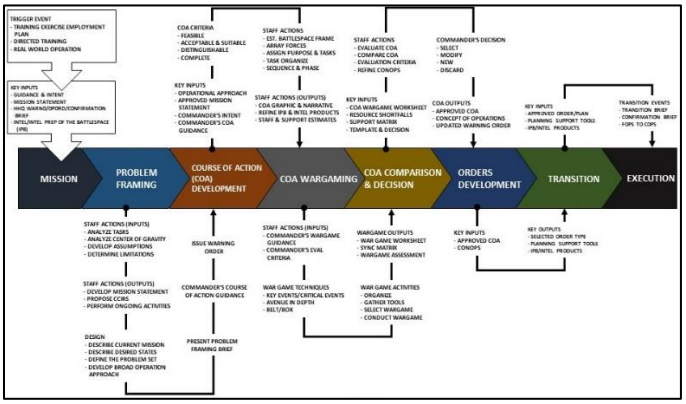


**Figure 18: Marine Corps Planning Process**

### 4.1.2 Impacts to Command and Control

Mission parameters/objectives and/or commander's intent establish within the functions the types of variables that need study to support the command-and-control decision. The staff then collect data from the appropriate sources, organizes the data, visualizes the data, and then analyzes the collected data to reach a set of conclusions, courses of actions, and options.

**Figure 19: Command and Control (C2)**

The best conclusions, courses of actions, and options are those that are realistic, manageable, and concise which paint the picture for the commander in realism, is manageable, and precise with an understanding of the influences on the other functions as to what is required from the interdependencies. All of these aspects culminate in the commander's decision advantage. Decision advantage is the ability of the commander to make clear strategically sound decisions faster than the competitor or adversary. As such, the current conditions of C2 does not lend itself to decision advantage especially where logistics is concerned.

### 4.1.3 Impacts to Operational Reach

Understanding operational reach is bound by the capabilities of logistics to provide what is required and where logistics is bound by time to provide the requirement. The function of logistics seeks to extend the boundaries of the operation. Hence, logistics is *the* pacing function for operations.

Operations is the focus of the Marine Corps specifically, its ability to allow Marine Corps forces to seamlessly deploy, project power, and fight from naval vessels or austere expeditionary bases, or any combination thereof.

The enabling aspect of operations is logistics. Logistics enables operational reach. Operational reach is the requirements to achieve an objective.

**Figure 20: Operational Boundaries (1 of 3)**

Operational reach boundaries are dictated by the ability of logistics (i.e., you can't go an extra mile on an empty tank of gas). Therefore, operations are bounded by logistics and can only operate within that boundary.

**Figure 21: Operational Boundaries (2 of 3)**

Even so, logistics itself is not boundless. Logistics boundaries are dictated by time and distance (space) which can or cannot enable speed, agility, and resilience in the context of the operational goal.

**Figure 22: Operational Boundaries (3 of 3)**

Operations are influenced by the ability of logistics to function effectively and efficiently within the parameters of time and distance. Logistics information systems are not adequately enabling the functions of logistics to overcome and anticipate operational requirements. In many cases the Marines are working for the system vice the system working for the Marines.

### 4.1.4 Operations Disconnect from Logistics

Planners are executing MCPP and planning functions manually and transferring these manual functions into analog nonstandard formats of PowerPoints, and Word Documents. Logisticians extract pertinent operational information from these formats into intermediatory logistics planning tools usually in Excel Spreadsheets. Once logistics planning is complete, this information is transferred into the various logistics systems for execution.

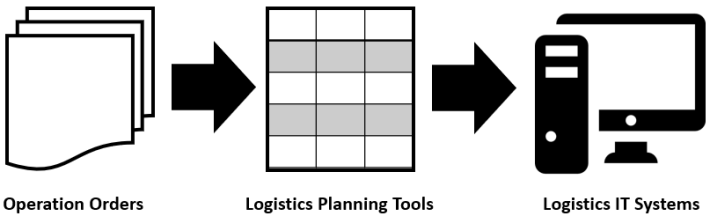


**Figure 23: Operational Informational Flow**

Through this process, the data becomes more fragmented and disassociated from its origination. Additionally, the manual nature of planning does not lend a direct correlation to lessons learned or informs the supporting logistics functions. Historical planning documents are archived in a variety of methods which are not standard throughout the enterprise.

### 4.1.5 Desired End State

A planner's workbench enables posting of a current, future, and historical operations in a single application, informing logistics planners through a common operating picture that is viewed jointly amongst operations, logistics, and intelligence communities on a cross domain solution (SIPR, NIPR, and CENTRIX).

Consistent with the OPLOG model proposal, implement an operations/log planning tool with the ability to:

1) Create mission/exercise portfolio's that assist planners identifying the mission requirement that drive logistics when applying supplies, equipment, material, and personnel readiness.
2) Present and produce products consistent with the MCPP process.
3) Assist planners in identifying, assigning forces, and track tasks associated to the execution of any mission. Once the unit is assigned the unit is responsible for dedicating the appropriate personal and equipment within the work bench, upon association this allows the unit and Higher headquarters a means to track the units' readiness to complete assigned mission.
4) Brief a commander from a mission/exercise dashboard within the program without necessarily building briefing products outside the program.
5) Waterfall or make available and tie logistics into the operation thus, supporting a common operating picture.

The desired end state provides a system for operations to plan. The system supports command and control over where operations, logistics, and intelligence all intersect. When the basic staff sections of operations, logistics, and intelligence all meet and touch in commonality, are mutually supporting, and indirectly/directly influence the other; this is where command and control exists and is exercised.

## 4.2 Doctrinal Shift in Defining a 'Supply Chain'

Traditionally, we think of a supply chain in terms of goods or material. However, anything tangible can be both in demand and be supplied. Even in this context, "inventory" has a meaning of whether or not what is demanded is available or unavailable which is not necessarily confined to materiel supplies or equipment. Therefore, within our military context we would extend the meaning of the term "supply chain" to include information and services as well as materiel as these are tangible products which can be forecast, accounted, inventoried, and quantified.

## 4.3 Doctrinal Shift in Defining a 'Request'

Our understanding of a request is placing a demand for something. Usually, we think of this demand primarily in terms of materiel, occasionally services, and very rarely applying it to information. However, agnostic of traditional views and focusing on the function of the request itself the analysis proposes a request would apply to all three of the tangible areas of information, service, and materiel regardless of logistics function or even warfighting function.

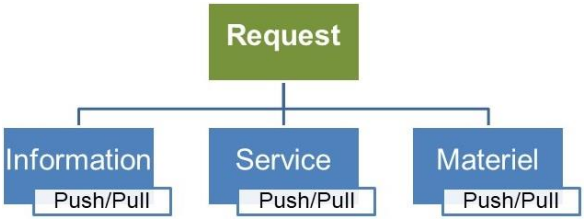


**Figure 24: Three Request Types**

### 4.3.1 Request for Information

As a tangible product, information represents anything the customer, observer, or consumer requires to accomplish their mission or conversely sends to another so they can accomplish their mission. The Marine Corps has established information and information supply chains deliberately but unknowingly.

For instance, a SALUTE report is a pre-formatted request for information on enemy activity. Planners want to know around the battlespace where the enemy is active and if a target of opportunity has presented itself. Additionally, such a report and the data elements can be analyzed to provide a pattern of activity or reveal enemy intentions. Thus, planners are demanding (requesting) any and all units provide information on such activities through the pre-formatted information form.

### 4.3.2 Request for Service

Service is the action of assisting or doing work for someone else. This type of request is an end action that is desired or demanded requiring a skill or application of skill plus equipment to accomplish the desired outcome. While the servicing organization or individual may require materials or information to conduct the work, the requesting or demanding organization is not necessarily responsible in knowing and/or requesting materials directly for a service action. Types of actions commonly categorized as “service” is a transportation request (i.e., CASEVAC, MEDEVAC, maintenance contact team, recovery, EOD UXO removal/render safe, route reconnaissance, etc.).

### 4.3.3 Request for Materiel

The most traditional view of a request as it applies to a tangible asset. Materiel is the finished supplies and equipment required to be employed for the mission and support people.

### 4.3.4 Push versus Pull

Push and pull request system refers to two different approaches to managing the flow of information, service, and material in supply chain management.

#### 4.3.4.1 Push Strategy

A push strategy is positioned to be proactive and provide what the customer needs before the customer requests them. This strategy requires an understanding of consumption rates and forecasts as well as accounting for delays due to time/space between the customer and point of delivery.

#### 4.3.4.2 Pull Strategy

A pull strategy reactively provides what the customer needs in accordance with the demand of the customer. Using this strategy conserves “inventory” and allocates resources only when demanded.

#### 4.3.4.3 Application of Push/Pull Strategies

Technically, every supply chain strategy is a hybrid between push and pull. A supply chain, no matter if is supplying information, service, or material is designed to flip between push and pull somewhere in the process. In all cases information, service, and material can be pushed or pulled anywhere along the supply chain.

## 4.4 Logistics Planning, Forecasting, and Estimation Tools Gap

A comprehensive review of the logistics information technology systems reveals a significant lack of planning, forecasting, and estimation tools within these existing programs. Most of these logistics applications stop data development at “information” by displaying or providing reports for the user to interpret as to the next step or action. Few applications had onboard dashboards displaying trends over a period of time however, even these dashboards did not lend themselves to a decision-making process and left interpretation to the user.

Users are forced to create their own intermediary planning, forecasting, and estimation tools which are more than likely a form of Excel with complex formulas. Results from these tools are then inputted into the various logistics IT programs for execution however, further tracking and updates remained in these intermediary tools requiring constant user interaction and sharing via email.

## 4.5 Recommended Doctrinal Changes to Transportation

### 4.5.1 Functional Change from ‘Transportation’ to ‘Mobility’

During the breakout sessions, the function of transportation under the current doctrinal structure is limited in its ability to incorporate capabilities and technologies in the wider scope of the competition continuum. How transportation is structured within the scope of Marine Corps doctrine is extremely narrow in its ability to apply and guide logisticians in emerging technologies as well as applying the appropriate level of data collection.

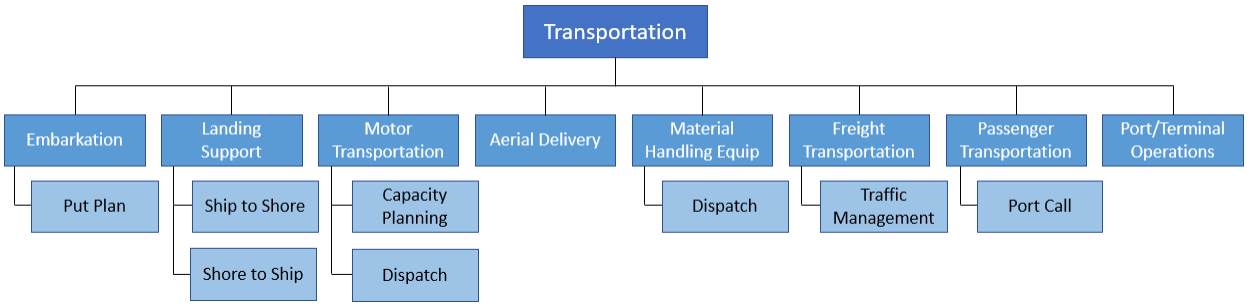


**Figure 25: Legacy Transportation Sub-functions**

#### 4.5.1.1 Mobility versus Transportation

The rationale for the differentiation between the terms “transportation” and “mobility” is in the context of doctrinal and policy development. Mobility is defined as the potential for movement and the ability to move something or someone from one place to another using one or more modes of transport (intermodal) to meet the needs of the organization. Mobility focuses on satisfactorily meeting demands, while transport (including mode, infrastructure, and accessibility) is the instrument which is required to carry out or realize mobility. Due to spatial separation of military activities, a demand for transport services arises. The type of transport services chosen to meet this need for mobility is the result of the platforms on hand, availability,

and accessibility coupled with the decision process of capacity planning of the available modes in relation to the threat/environment the requested action must take place.

#### 4.5.1.2 Intermodal Nature of Transportation

Through the process development of the Logistics IT OAG, it was noted transportation of both people and things are interdependent. When using more than one transportation mode in the conveyance of people or things is the definition of “intermodal.” Therefore, the doctrinal position of the Marine Corps is to group transportation methods together to communicate the modes available but also within the context of interdependencies of each mode in its conveyance. The deployment of Marines usually starts with vehicles, then to aircraft or ships, then to vehicles which illustrates intermodal connections in this scenario.

### 4.5.2 Littoral Capability Gap

The adversary threat and operating environment has given rise to the importance of littoral regions. By the way of definition, “littoral” refers to an area of or relating to a coastal or shore region to include a lake, sea, ocean or river, or any body of water (natural or man-made) with a shore region.

Under the littoral definition and competition continuum comes the requirement for littoral connectors that are smaller than current inventory is providing. The current inventory is built for major beach head operations (i.e., LCU or LCAC) that are owned by the Department of the Navy and operated by Naval personnel. The requirement for smaller more agile vessels that may be autonomous or operated by Marine Corps personnel and wholly owned by the Marine Corps is becoming more of an operational reality. Therefore, this necessitates a doctrinal change to account for both emerging technologies, current/future requirements, and anticipated employment by the organization. Therefore, three operating capabilities make up the littoral mode:

#### 4.5.2.1 Surface

The term “surface” means a vessel or watercraft operating on the surface of the water, including both manned and unmanned watercraft but is not capable of transitioning to a submersible or semisubmersible state.

#### 4.5.2.2 Submersible

The term “submersible” means a vessel or watercraft capable of operating completely below the surface of the water, including both manned and unmanned watercraft. While such craft can, at times, operate on the surface the central capability determines its classification.

#### 4.5.2.3 Semisubmersible

The term “semi-submersible” means any watercraft or vessel constructed or adapted to be capable of operating with most of its hull and bulk under the surface of the water, including both manned and unmanned watercraft. Operating with most of the hull under the surface of the water reduces the silhouette and cross-sectional profile from surveillance, detection, and targeting.

### 4.5.3 Unmanned and Small Aerial Craft Capabilities Gap

Emerging technologies in Unmanned Aircraft Systems (UAS), Unmanned Aerial Vehicles (UAV), and small aerial craft are becoming more prominent. As these technologies become more common place what is becoming apparent is owning and operating aerial craft is no longer the sole responsibility of the Marine Corps’ Air Component. Aerial craft have the ability to lift

small cargo packages, deliver ordnance, and transport people and are becoming more integrated into transportation mode selection.

#### 4.5.3.1 Unmanned Aircraft Systems (UAS)

UAS encompasses the entire system that supports and controls the UAV, including ground control stations, software, and communication systems. These components are ultimately necessary for the safe operation of the UAV within the operating environment. Such components can be as simple as a radio controlling device to more elaborate components such as satellite positioning, high frequency radio, and optical infrared guidance systems.

#### 4.5.3.2 Unmanned Aerial Vehicles (UAV)

The UAV is what is commonly known as the “drone.” It is the actual inflight mechanism to provide surveillance, information, ordnance, or delivery of material. UAVs range in size from as small as a couple of inches with a range of just 100 yards to as large as small airplane with a range of 100,000 kilometers.

#### 4.5.3.3 Small Aerial Craft

Small aerial craft are manned, autonomous or piloted aircraft which are capable of carrying one or two persons but are not as complex as traditional aircraft as they fly in a limited range and lower altitudes. Since these small aerial aircraft operate slower and at lower altitudes that do not require special training to operate, they do not rely on or require traditional air traffic control methods and can achieve autonomous flight. Commonly known as “personal drones” or “personal flying vehicles” these capabilities open up a new space for what was commonly considered “flying cars.” Technological advances in this area have created a new mode of transportation which is more accessible and capable for short or medium haul requirements (i.e., MEDIVAC, CASEVAC, Contact Teams, emergency services, etc.) that are faster and cheaper compared to mass or bulk ground transportation options and require less support than traditional fixed/rotor wing assets.

### 4.5.4 Logical Sub-functional Sequence

From an execution perspective, the sub-functions follow a “build on” data flow. Each of the four proposed sub-functions add to the successive data layer in the process of mobility as a whole to accomplish the assigned mission.

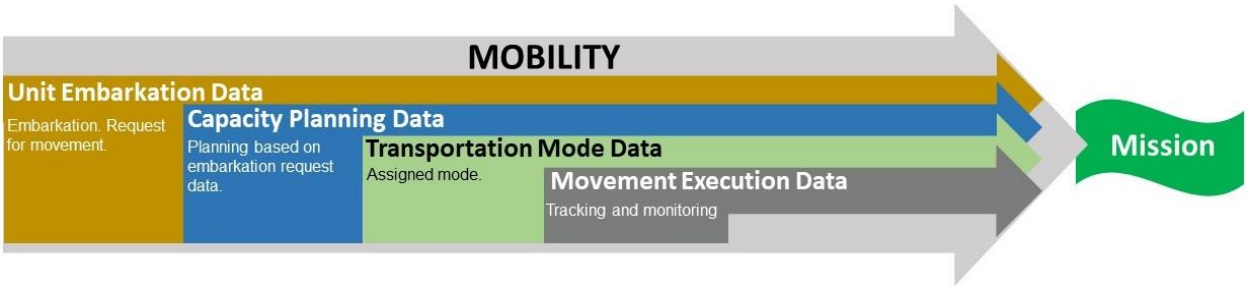


**Figure 26: Mobility Data Continuum**

#### 4.5.4.1 Embarkation

While we understand embarkation as the process of putting personnel and/or equipment into ships, vehicles, and/or aircraft, this is not usually decided at the tactical unit unless those assets are organically owned and operated. Embarkation at the tactical unit is the action of identifying and planning for the people and things to be embarked. It is more akin for a request for service. Using manning documents for people and equipment density lists for things, embarkation

personnel collect weights, measurements, and associated other data points to the next sequence of capacity planning. From the tactical unit perspective, embarkation is the preparatory work for the request for lift/movement by providing the requisite data.

#### 4.5.4.2 Capacity Planning

At the higher-level capacity is matched with the provided embarkation data. Even in terms of the tactical unit, capacity planning is occurring. Capacity planning is the process of determining the demands of the mission with available and accessible assets. This action occurs at all levels from the unit's motor transport section to TRANSCOM, formally and informally, agnostic to scale and scope regardless of transportation mode. Therefore, the elevation as a subfunction adjacent to embarkation ensures deliberate planning is considered and builds upon the previous embarkation request data.

#### 4.5.4.3 Transportation Mode

The output of the capacity planning process is transportation mode selection. The assumption should be made that the complete capacity plan will be intermodal by nature, meaning more than one mode is in support of the overall mission demand. Within this subfunction are the internal actions of assigning, allocating, dispatching, etc. The data drivers are common to all levels in scale and scope as someone owns, operates, and directs the assets to support the assigned mission.

#### 4.5.4.4 Movement Execution

Movement execution becomes working action under the assigned plan. Tracking and monitoring of the movement is a general requirement action which occurs at all levels. The end of the action is the confirmation the people and assets arrived at the destination and the assigned lift resources/transportation modes are returned. After action data is critical to the capacity planning cycle for consumption, resource allocation, and time expended and must not be discarded.

### 4.5.5 Summary of Changes from Transportation to Mobility

The structural changes to this function of logistics elevates the enabling subfunctions and is sequenced in a common occurring logical order.

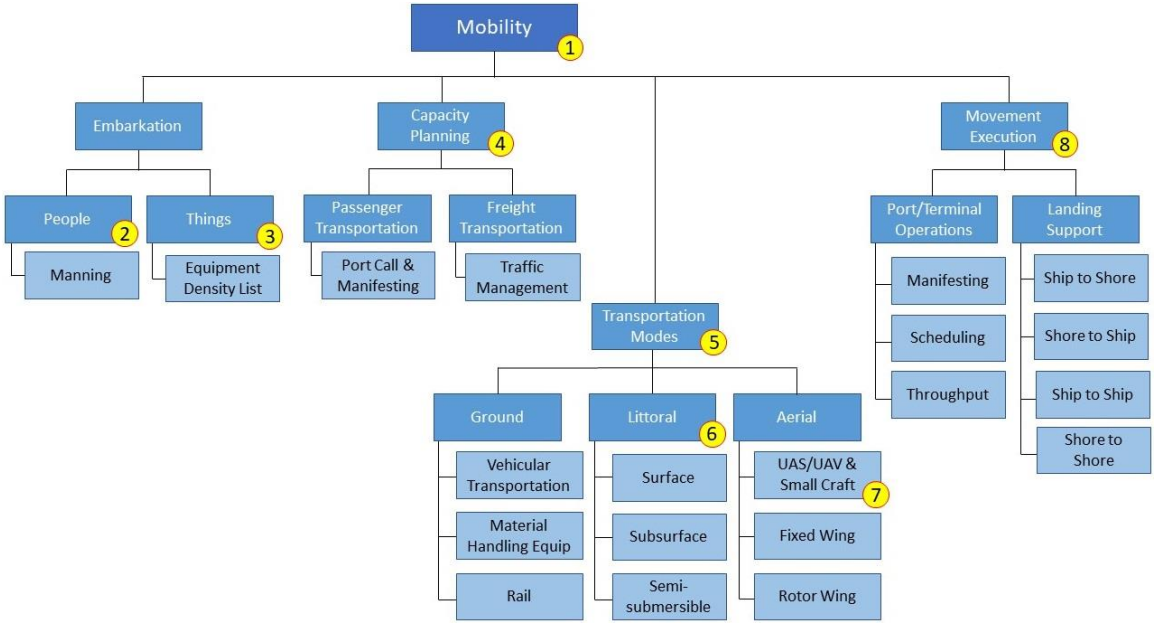


**Figure 27: Proposed Mobility Sub-functions**

The summary of changes corresponds to the numbers represented in the graphic.

1) Mobility. The ability, accessibility, and requirement to move people or things according to the organizational goals and requirements.

2) People (PAX). Within embarkation the movement of people has drifted into the realm of administration through the collection of manning or staffing documents. Calling out people as a planning factor within embarkation drives the unification of planning back to the embarkation function.

3) Things (Cargo). Within embarkation the things category is a separate category but on par with people.

4) Capacity Planning. The next phase within the mobility function from embarkation data collection to matching the need to move people and things to the available transportation modes. Passenger transportation is focused on the moving people either individually or as a group. Like passenger transportation, freight transportation is focused on the material things.

5) Transportation Modes. Capacity planning allocates and aligns to the best transportation modes accessible and available within the timeframe allocated. Within this subfunction, each mode is agnostic if the assets are organic, commercial, chartered, or laterally supported.

6) Littoral Mode. This mode, while a topic of greater conversation and is a wider influence in the competition continuum, is not accounted for as a specific mode. Traditional view of this mode is this is the Navy's realm. However, with the need for smaller, versatile, cheaper, and easily piloted systems the traditional view is being challenged. Waterborne craft are becoming autonomous and in varying sizes indicating the Marine Corps will own and operate these systems.

7) UAS/UAV & Small Aerial Craft. Typically, we see these as "drones" however, the capability is widening as technologies mature. These systems will become more and more common place transporting small and medium lift packages in and around the

battlespace as well as a low cost/low risk option in shuttling personnel between locations without using traditional aircraft for administrative use. Technology allows the capability to be employed without a large infrastructure and without extensive training.

8) Movement Execution. Establishing this subfunction encapsulates the action and decision execution after the mode of transportation has been selected/assigned. Port/terminal operations ensures the people/things are manifested, the transportation mode is scheduled, and the throughput monitored for arrivals and departures. Landing Support provides tactical support for tactical offload/onload of people/things without establishing a permanent presence.

## 4.6 Recommended Doctrinal Changes to Supply

### 4.6.1 Redundant/Multiple Supply Chains

The analysis observed the redundant and multiple supply chains for each of the Class of Supplies and in some cases for each sub-function. This is most pronounced and evident in Class IX where the supply chain is divided between “consumable” and “secondary reparable”.

The secondary reparable parallel supply chain is a legacy process born out of cost reduction and accountability from the late 1970’s. In order to reduce cost in the rebuild cycle, customers are required to provide the necessary consumable parts (call a ‘configuration’) to the turn-in facility along with non-working core (called a ‘carcass’). However, to ensure authorized issues are made to authorized persons and the configurations are correct leads to an extensive administrative burden to execute. For example:

1) Commanding Officers are required to generate letter correspondence for delayed turn-ins, issues without a corresponding turn-in, and turn-ins without a corresponding issue. These administrative letters are required even though the same commanding officer acting as an Approving Official has appointed representatives as a Financial Ordering Official on the DD 577 form for the unit. This calls into question if the CO/AO has assigned a financial representative on the DD 577 to obligate the command, why would additional authorization/notification be required for the reparable exchange activity to execute such issues/turn-ins? When faced with logistics administrative requirements, the enterprise must call into question the process value and its relevance. Without challenging these administrative requirements, the underlying processes become needlessly redundant and create a slow/unresponsive/inefficient logistics apparatus.

2) The customer is required to travel to the reparable exchange activity to make an exchange. Additionally, if the carcass doesn't meet 'configuration criteria,' it is rejected, sending the customer back to correct it. The organization is expending man-hours and material in the thousands to save a few hundred of dollars in easily obtained consumable parts. The maintenance section expends significant resources in time and distance to perform the SECREP exchange as it is a walk through, in person process.

Notably, these garrison administrative procedures were not followed in contingency operations. At no time in OIF/OEF were commanders denied an issue because of a delayed turn-in letter. Additionally, the SECREP exchange was not an in person, configuration check. SECREPs were ordered as a Class IX consumable part and the carcass returned in the FASTPAC box the new SECREP arrived.

The Marine Corps absolutely needs to collapse Class IX consumable and secondary reparable supply chains into a single supply chain to remove the antiquated administrative requirements

and provide a more efficient and effective supply chain. In this concept the following changes would apply:

1) All condition code “A” secondary reparable parts would fall under the intermediate supply activity. All orders, backorders, fulfilment, and follow on shipping for secondary reparable materiel is the responsibility of the intermediate supply activity.
2) The secondary reparable activity becomes the single supplier to the intermediate supply activity.
3) Customers order secondary reparable parts at they would any Class IX. This includes financially obligating the initial full price upon requisitioning.
4) The intermediate supply activity ships the secondary reparable to the customer. The customer receives the notification of the refund limitations and core refund price list.
5) The customer is responsible for putting a clean and complete carcass with the shipping paperwork back into the shipping box and is collected by the intermediate supply delivery truck.
6) The delivery truck will bring all collected carcasses to the secondary reparable facility for processing.
7) The secondary reparable facility will issue a financial refund for the core to the customer based on the carcass condition:
    a. Clean, complete, and paperwork is correct = 100% refund
    b. Clean, incomplete, and paperwork is correct = 75% refund
    c. Not clean, incomplete, and paperwork is correct = 65% refund
    d. Not clean, incomplete, and paperwork is incorrect = 50% refund
    e. No carcass returned after 30 days of new issue = 0% refund

### 4.6.2 Production/Manufacturing Capabilities Gap

The Marine Corps is transforming its supply chain to support internal production/manufacturing of parts/products to overcome time and distance in the competition continuum as well as supplement the peacetime challenges. Production/manufacturing is established in a “make or buy” decision performed by the supply chain management apparatus through available data (cost, time, location, transportability, etc.). The Marine Corps has yet to insert production/manufacturing within the context of supply chain doctrine on the ground side.
The condition has been present and applicable with traditional subtractive manufacturing forms and organic capabilities but has become most apparent with the rise and use of additive manufacturing. Therefore, the Marine Corps has pervasive problems linking the supply chain “make or buy” decision with existing and emerging production/manufacturing capabilities:

1) The requisite research and business case analysis to make or buy a part has not been conducted to focus the available resources against critical parts for critical assets.
2) Production/manufacturing capabilities are not systematically available throughout the supply chain either laterally or in a hierarchy nor are the capabilities centrally managed and standardized.
3) The ability to select a part to be bought or made is not a deliberate supply chain decision making process supported by data in cost, time, availability, and capability.

4) Logistics information systems are not enabled or capable to pass production requests electronically to the production facility or to escalate to another production facility (commercial or government).

Currently, the customer, usually the mechanic/technician, is required to be proactive in searching for, activating, and making use of organic production/manufacturing capabilities to satisfy demands. The current state ends up making the system reactionary, laborious, administratively intensive, and creates data gaps essential to supply chain planning.

#### 4.6.2.1 Make or Buy Decisions

Before a part or product is produced, a decision must be made based on the demand and what is available. Decisions made regarding the external purchase or internal manufacture of a product regardless of the method (traditional subtractive versus additive manufacturing) is a supply chain management decision. This type of decision is known as a “make or buy” decision. However, for this is supported by the analysis of data. To decide to make or buy requires a make or buy analysis.

#### 4.6.2.2 Make or Buy Analysis

The make or buy analysis is a process of gathering and organizing data about product requirements, available supply chain data, and against available alternatives. This analysis then determines the buy strategy and the make strategy for each part/product. Essential data elements include:

1) Product buy information: cost to buy, lead time from order to receipt, product criticality, end item information/criticality, average demand data over the course of three years, and seasonality alpha adjustments.

2) Product make information: cost to make, make time (includes administration time and post product finishing), available design, available material, make method, meets/exceeds specification data, design package (for the combination of multiple production methods i.e., plastic additive manufacturing and metal machine lathing to produce an end product), and any approvals/conditional requirements.

The results of the make or buy analysis then informs the make or buy strategy.

#### 4.6.2.3 Make or Buy Strategy

Make or buy strategy is applying the analysis to the overall impact and mission of the organization as it pertains to the criticality of the part/product to the mission. The objective of the make or buy strategy is to ensure the appropriate level of resources are applied to enable part/product availability across the enterprise. The supply chain decision may not be traditionally held at the organizational supply activity and the actual production/manufacturing also may not be performed at the organizational supply activity either. However, the science and data analysis are a supply function regardless of who or where it is performed.

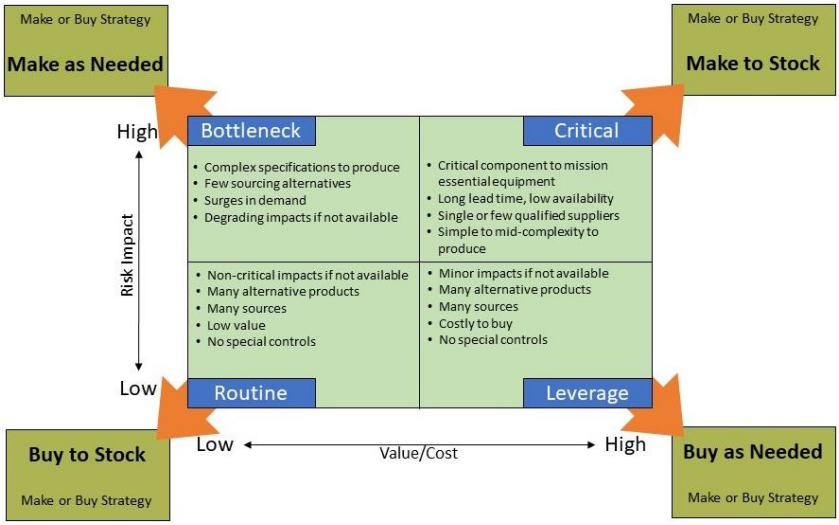


**Figure 28: Make or Buy Strategy Matrix**

The above graphic presents a make or buy strategy matrix example. The quadrant is indexed by value/cost on the Y axis and the risk impact to mission on the X axis.

The four labeled quadrants represent the general impacts between cost/value and risk impacts to the mission. An item best described by the quadrant then leads to the preferred make or buy strategy for that item:

1) Routine (low cost/value; low risk impact). Item is low cost to buy and is not mission critical if it is not available or becomes subject of a normal supply chain disruption. Therefore, the strategy for this item is to buy and stock.
2) Leverage (high cost/value; low risk impact). High-cost items which are not typically mission critical items. Therefore, the strategy for this item is to buy only when demanded from a customer without stocking the item.
3) Bottleneck (low cost/value; high risk impact). This item is not a cost driver but has a direct impact on the mission. Additionally, due to the limited number of suppliers or an increased susceptibility to supply chain disruptions it is best to be prepared to make this item when the traditional supply chain starts issuing backorders.
4) Critical (high cost/value; high risk impact). High dollar value coupled with the mission critical nature of this part means operations will come to halt if not available. Therefore, this item should be made to put on the shelf even if demands are infrequent.

#### 4.6.2.4 Advanced Manufacturing as a Production/Manufacturing Enabler

Advanced manufacturing describes the use of technologies and techniques to enhance production quality, efficiency, and process of manufacturing. Advanced Manufacturing encompasses the following: [15]

1) Efficient Production. The technologies adopted are related to design, followed by simulation. It addresses physical and computer modeling, all advanced production technologies, and related control techniques that a manufacturing unit may need to adopt. The focus is on simultaneous engineering rather than sequence based. Production technologies include rapid prototype creation and near net shape manufacturing as well as the techniques of precision casting, machining, and joining.

2) Intelligent Production. Intelligent production utilizes information communication technology across manufacturing and logistics systems. This system focuses on creating and implementing systems to optimize manufacturing facilities with effective monitoring coupled with regularly updated maintenance and repair plans (i.e., linked to predictive maintenance analysis or notifications).

3) Effective Organization. Effective organization relates to the development of effective creation and coordination of manufacturing resources. It comprises material resources as well as related knowledge and works in areas where technology can be used to enhance the participation and abilities of a SME or a lateral organization. Effective organization looks into related aspects of manufacturing such as sharing of facilities and resources, incubation units for newer ideas and technologies, knowledge management, and all aspects of product availability effectively becoming a cross-functional, multi-discipline organization.

4) Additive Manufacturing. Manufacturers use methods like 3-D printing, fused deposition modeling, or powder-bed laser printing systems to create complex models or replacement parts from a single material. Industries that use additive manufacturing are the aerospace sector, the medical field, prototyping, automotive, and consumer goods among many others. These technologies continue to cost less to implement.

5) Advanced and Composite Materials. Advanced materials enable extremely precise composite blends for particular applications like blends of metals, plastics, glass, and ceramics. Materials that require precision in its chemical and physical properties are manipulated to help enable performance breakthroughs while reducing material trade-offs. Composite materials used often include recyclable plastics, high-strength alloys, and advanced ceramics, among others.

6) Robotics and Automation. Robotics fit in naturally among advanced manufacturing techniques and are designed to automate processes that involve precision movements, lifting heavy objects, and joining of elements--consistent across production units. Additionally, robotics reduces the risk involved in hazardous jobs like the automotive and aerospace sectors.

7) Laser Machining and Welding. The laser process enables precision in machining and welding. Laser technology processes high precision parts rapidly as the integrity of the component is maintained with lower and steady amounts of heat. Such processes are used to make pressure vessels and proximity sensor welding. It is also seen in battery welding and sensitive electronics.

8) Nanotechnology. Devices are becoming smaller, and naturally, the need for smaller parts and components is on the rise. Nanotechnology has increasingly gained traction among designers who are looking to increase functionality and decrease the size of the devices. Nanotechnology has helped advanced manufacturing systems minimize their overall footprint while maximizing functionality across production facilities.

9) Network and IT Integration. With the internet, connectivity is unparalleled, and advancements in internet technology have enabled connectivity to and from systems and machines as well. Communication between sectors can be electronic and need not be done manually. With internet access to all aspects of manufacturing, there is an instant notification on any issues that may arise, enabling preemptive repairs and saving time and cost.

#### 4.6.2.5 Advanced Manufacturing Implementation

Advanced manufacturing technologies are not plug-and-play and are dependent on skilled people and formal processes. The Marine Corps needs to take four key steps for the implementation of advanced manufacturing technologies: [15]

1) Syncing Manufacturing Strategy to Mission Strategy. The Marine Corps must have a well thought out manufacturing strategy that ties in strongly with its mission/employment strategy. This can be achieved by quantifying aspects such as lead times, quality of the product, the flexibility of manufacturing, and costs. Once quantified, the data from these two strategies can be used to create manufacturing goals. In turn, data insights will help with capital expenditure and investment decisions for the adoption of newer technologies and their implementation. These strategies can be difficult if leadership does not have access to the right kind of data and the various cost options to achieve mission/employment goals. Another issue is the disconnect between departments and their strategies. The first issue can be handled by investing in strong data acquisition and analytical capabilities.
2) Tapping Networks for Technology Identification. The Marine Corps needs a central core production/manufacturing business development department to build strong internal and external production/manufacturing networks to identify new technologies relevant to the mission strategy as well as to build the key performance indicators (KPIs) to monitor performance. The production/manufacturing network needs the right framework with clear goals. Network performance must regularly be measured against a set of quantified input and output metrics to establish overall effectiveness, mission impacts, and value.
3) Technology Prioritization and Evaluation. Establish a strong set of processes in place to quantify the cost and merits of each production/manufacturing technology option when compared to the mission strategy. This process should be the central core production/manufacturing business development department who are staffed by risk takers and can pilot innovations in the right direction. Additionally, the department must also be ready to pull the plug on failing technologies when needed. The department needs the support of specialized analysts who can provide the right data and can forecast the impact of technology across a transition period.
4) Rapid Yet Strategic Implementation of Technologies. The Marine Corps is having the challenge of rapid piloting and rolling out of new technologies because it remains apprehensive about implementing new technologies and tend to hold out until it is too late to be effective or becomes immediately obsolete. It is difficult to balance the everyday functioning of the enterprise while doing R&D with new technologies. The way to handle this is to create a small, specialized team of uniformed “makers” paired with and supported by civilian contractors that can independently work on the R&D aspect of new technologies. This way, bringing in advanced technologies into the enterprise is easier and operationalized for immediate use.

#### 4.6.2.6 Challenges to Implementing Advanced Manufacturing

There are two main challenges to adopting advanced manufacturing: human resources and technology. [15]

1) Human Resource Issues. There is an increasingly unmet demand for professionals with IT and design skills at the management level who are risk takers to try something new.

There is a need for "makers"[15] to possess advanced design skills for shorter, customized production runs. Essential but not always available, is IT literacy and skill sets such as an understanding of CAD/CAM and CNC machine-based software.

2) Investment Issues. Advanced manufacturing technologies naturally require large investments in research and development. There is also a relatively high-risk factor enterprises take on, which at times affects decision-making, ultimately innovation. In several cases, organizations may find it difficult to access specific materials, or find it challenging to understand the characteristics of the material. Evaluation techniques and the processing of designs can also be complex to create and apply. Often simulation capability adopted by an organization does not keep up with technological development, causing delays. More cybersecurity investments are needed to keep up with technology innovations.

#### 4.6.2.7 Additive Manufacturing

Additive manufacturing is a subset of Advanced Manufacturing and is of particular interest to the Marine Corps because of the flexibility it provides relative to its physically small footprint making it ideal for deploying this capability forward.

By definition, additive manufacturing is the computer-controlled process of building a 3D object by compiling layer upon layer of material, typically ceramic or metal powder, on a build platform until the final product is finished. The layers are hardened using heat, a curing agent, or lasers.

Since this method is additive, there's less waste than with subtractive manufacturing, and therefore, lower costs. [16]

#### 4.6.2.8 Eight Types or Methods of Additive Manufacturing

The following is currently the types/methods of additive manufacturing each having unique differences. [16]

1) Binder Jetting. Binder jetting, also known as material jetting or inkjet powder printing, is among the most common additive manufacturing types. This method works similarly to an office printer, except it prints three-dimensional objects. Instead of jetting ink onto a page, binder jetting jets adhesive into a powder material. The print head moves horizontally and vertically, putting down a new layer of build material with every pass. This method uses a variety of materials: gypsum powder (chalk), plastics (usually polypropylene), ceramics, metal (usually alloys), glass, and sand. Binder jetting is an affordable additive manufacturing process due to its relatively low cost of entry and the inexpensive nature of its materials. It can also create objects faster than most other additive manufacturing processes in full color. Although it comes with many benefits, binder jetting products tend to be fragile and require post-processing, including cleaning and hardening, which can be tedious and time-consuming.

2) Directed Energy Deposition (DED). Directed energy deposition (DED) uses welding principles to create three-dimensional objects. The material - typically metal wire or powder - is melted by a focused energy source (laser or electron beam). The liquid material is then precisely poured onto the build platform, where it quickly hardens, forming a layer. This process repeats until the object is finished printing. One of DED's significant benefits is that it can be used for more than just item creation; it can also

[15] "Maker" in this sense means someone who creates or invents things, either using traditional crafts/methods or technology or the combination of both

repair and add material to existing parts or preforms. Additionally, it can utilize multiple materials in a single printing process.

3) Material Extrusion. Material extrusion works similarly to a hot glue gun. The material feeds into the printer from a coil. The tip of the nozzle heats and melts the material. The liquid material is then placed layer by layer on the build platform, where it can cool and solidify, forming the object. While this method is the most inexpensive method of additive manufacturing, material extrusion isn't without its limitations. Since the heating elements aren't powerful enough to melt high-density materials like metal, it is used primarily for plastic polymers - which may not be durable enough for some applications, such as tooling and fixturing.

4) Powder Bed Fusion (PBF). Powder bed fusion, otherwise known as electron beam melting (EBM), starts with a large bed of powdered material, typically plastic, metal, sand, or ceramic powders mixed with sand. The powder is selectively fused together using a laser or electron beam. Once a layer of material is fused, the working area moves down, and the new layer is built on top using the same process. With PBF, there's a high complexity level to the objects, making them stronger than objects created by some other types of additive manufacturing. Since it requires a bed of powder, it is difficult to maintain a clean working environment as it produces airborne dust. This method is not the best option for tight spaces which do not have dust abatement.

5) Sheet Lamination. Sheet lamination, otherwise known as ultrasonic additive manufacturing (UAM) or laminated object manufacturing (LOM) - is an additive manufacturing process that stacks thin sheets of material and bonds them together through ultrasonic welding, bonding, or brazing. As the layers stack up, the object takes shape. Once all the layers are stacked and laminated, a CNC machine or laser cutter removes excess material, creating the object's final form. Since it works in multiple layers, sheet lamination allows manufacturers to layer various materials, print in full color, and at a much lower cost than some other additive manufacturing processes. While sheet lamination is relatively inexpensive, it does require some post-processing, which can be tedious and time-consuming. This process is limited to materials that can come in sheets, such as paper, plastic, and metal. Since objects need to be trimmed once lamination is complete, more waste is involved in this additive manufacturing process compared to others.

6) Vat Polymerization. Vat polymerization is similar to powder bed fusion, except instead of a bed of powder, it uses a vat of photopolymer resin, which is hardened in layers by an ultraviolet laser. Once a layer is complete, more resin is added, and the next layer gets built. This process continues until the object is complete. Although resin can be expensive, vat polymerization can construct objects quickly while providing a high level of accuracy and a good finish. Despite its benefits, vat polymerization manufacturers are limited to using photo-resin materials and are required to perform a substantial level of post-processing. The printed objects need to be removed from the resin and cleaned before use.

7) Material Jetting. Similar to binder jetting, material jetting layers material to construct an object. However, instead of layering adhesive on a bed of powder, material jetting melts wax-like materials and precisely deposits droplets onto the build platform. As the layers build up, the object takes shape. Many manufacturers utilize material jetting because it's relatively inexpensive and provides outstanding accuracy with high-quality surface finishes. Unfortunately, material jetting is limited to using only wax-like materials, which can be fragile as well as a longer build time.

8) Cold Spray. Cold spray additive manufacturing (CSAM) (also called cold spray 3D printing) is a particular application of cold spraying, able to fabricate freestanding parts or to build features on existing parts or components. During the process, fine powder particles are accelerated in a high velocity compressed gas stream, and upon the impact on a substrate or backing plate, deform and bond together creating a layer. Moving the nozzle over a substrate repeatedly, a deposit is building up layer-by-layer, to form a part or component. A robot or computer-controlled manipulator controls the spray gun movements which complex shapes can be created. To achieve 3D shape, there are two different approaches. First, fix the substrate and move the cold spray gun/nozzle on a robotic arm or second, move the substrate with a robotic arm and keep the spray-gun nozzle fixed. There is also a possibility to combine these two approaches either using two robotic arms or other manipulators. The process always requires a substrate and uses only powder as raw material. This technique is distinct from selective laser melting or electron-beam additive manufacturing or other additive manufacturing process using laser or electron beam for melting the feedstock materials.

#### 4.6.2.9 Emerging Technologies

Advance manufacturing is rapidly driving major performance improvements in the industry sector. Newer model machines are allowing for unique and obscure product development with less waste and more stability. Additionally, emerging technologies are becoming faster and more accurate stream-lining the journey from product decision to product making to product delivery. Because AM depends on optimizing problems many users approach concepts with a "best guess" scenario, design the product, calculate the failure, redesign, produce and then redesign which is time consuming and at best suboptimal for limited resource AM manufacturing. Artificial intelligence (AI) is one emerging technology that takes the guess work out of calculations and adjusts to the design to ensure the best product is manufactured.

1) Contour Crafting. Uses a computer-controlled crane or gantry to build large structures fast using quick-setting concrete "like" material. With this technology, NASA is starting to refine the process to build on the moon and eventually Mars. Contour crafting machines have been developed and are being tested to utilize other materials such as thermoplastics, thermosets, and various ceramics.

2) Generative Design. Digital simulation and analysis can be evaluated in seconds. Algorithms can automatically adjust the geometry of a part between simulations, with no manual refinement required. Using artificial-intelligence techniques, these new generative design systems can compare the results of thousands of simulations to close in on a design that delivers the most favorable combination of attributes,

3) Ultrasonic Additive Manufacturing (UAM). A technique which employs high-frequency vibrations to fuse thin sheets of metal together without alterations to the microstructure of the metal. Currently in use by NASA.

4) Meta Material. A synthetic composite material with structures that are not found in natural materials.

#### 4.6.2.10 Required Data Capture Points

Through the LOG IT OAG, it was apparent the Marine Corps was not capturing key data points which would enable the make or buy analysis and decisions. Additionally, existing data is not curated to provide the necessary insights in the use, cost, and criticality of each part in context of the item to which it belongs or is used. The following key data points are required for capture:

1) Item Reference. The original manufactured part reference number (i.e., NSN). This is not the production version item reference which could be different.
2) Production Flag. The flag indicates the item can currently be produced (Y or N).
3) Production Mode. Production mode this applies (i.e., subtractive manufacturing, additive manufacturing, hybrid, etc.)
4) Production Cost. The total average cost, to include waste, to produce the item.
5) Production Time. The total average time, to include finishing, to produce the item.
6) Design Approved. The approved design is on file and is currently in an approved status (Y or N). Approved designs generally indicates the design and identified material meets or exceeds the original item specifications in fit, function, and performance.
7) Quality Control Flag. If the item requires a formal quality control inspection postproduction (Y or N).
8) Producer(s). Listing of producers who are capable of providing the finished product and serves as the point of shipment.

#### 4.6.2.11 Production/Manufacturing Summary

Organic and emerging production/manufacturing capabilities are part of the “make or buy” supply chain management decision process. However, these capabilities are not currently nested into the doctrinal function of supply and have been essentially put into the “maintenance” function.

Additionally, the Marine Corps is not poised to formally align these functional capabilities deliberately into the supply chain as it does collect/possess the data necessary to support the make or buy analysis for each potential part it could/should make and does not have the logistics IT solutions enabled to capitalize on integrating, tracking, and costing requests for produced part/products.

### 4.6.3 Distribution

Through the analysis, there seems to be an over emphasis of the function of distribution as a function of transportation to the point it seems these two functions have essentially merged. However, this should not be the case as at the functional level.

The Joint Publication 4-09, Distribution Operations, 05 February 2010 defines the aspect of distribution: “It is the operational process of synchronizing all elements of the logistic system to deliver the “right things” to the “right place” at the “right time” to support the geographic combatant commander (GCC).”

Additionally, the Marine Corps echoes distribution as, “an official delivery of anything, such as orders or supplies and that functional phase of military logistics that embraces the act of dispensing materiel, facilities, and services.”

The Marine Corps should refocus its efforts away from emphasizing distribution as a function of mobility. Functionally, distribution is bringing demand and capacity together in order to make supportability/sustainment decisions, which includes transportation modes but is not exclusively mutual.

### 4.6.4 Digital Tracking of Military Equipment

Through the analysis of determining asset/equipment data reliability, the enterprise does not possess “digital truth” of military equipment and assets anywhere at any time. There are two basic questions the enterprise struggles with:

1) Do I have all the equipment assigned to me in my possession (does it exist)?
2) Where is this equipment physically located (where is it or when was it last “seen”)?

To the first question, many assume they have the equipment in their possession and are required to check quarterly, annually, and/or as occurring by reconciling the digital report and manually matching physical assets. Currently, this method expends hundreds of man-hours reconciling the physical world with the digital world and usually, with limited success. Additionally, the report is a snapshot in time and only addresses the first question and not the second question.

To the second question, many correlate physicality of the equipment to the DoDAAC. For instance, if the equipment is owned by 3d Maintenance Battalion and the TAC 2 address is "Camp Kinser, Okinawa" it is assumed, for the most part, the equipment is physically on Okinawa and on Camp Kinser. However, Camp Kinser, Okinawa is the owner's address not the physical location of the equipment. 3d Maintenance Battalion has its equipment spread from Camp Kinser to Camp Hansen and supports contact teams in Korea, Philippines, Guam, and Hawaii.

To verify both existence and physical location takes a human being to go out to the lots, bays, rooms, warehouses, etc. to physically check and verify. If the equipment is not where it is supposed to be or thought to be leads to a process of research to determine when and where the equipment was last seen and its current location. Therefore, the enterprise as a whole does not possess “digital truth” of its equipment and assets as it fully relies on to ask "someone" (a human being) to go out and physically verify that X item with Y serial number is in location Z and if not, to research to figure disparities and discrepancies.

Due to the analog nature and lag between digital reporting and physical reality, it is impossible to fully perform force level equipment sourcing and formulate equipment density list on hands.

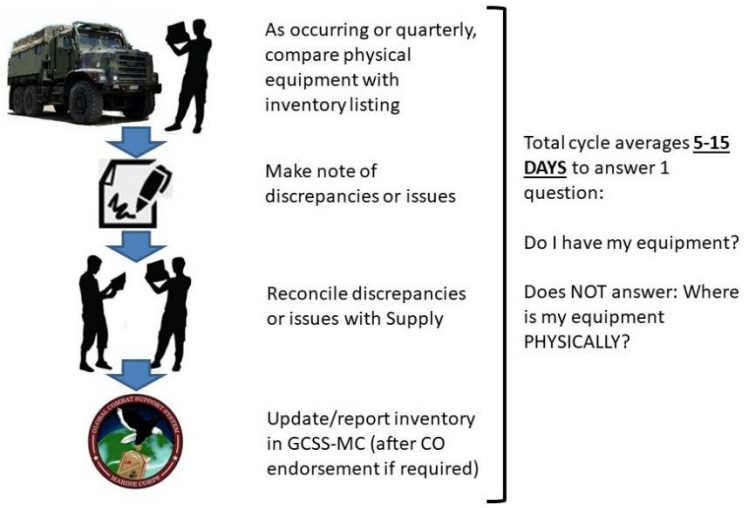


**Figure 29: Current USMC Asset Management**

The best solution for equipment/asset management on an enterprise scale in a global distributed environment is to employ pRFID technology. This technology has been the industry

standard for equipment/asset management since the late 1990's and much more superior than barcode scan tags.

| Specification | pRFID | Barcode |
|---|---|---|
| Line of Site | Not required (in most cases) | Required |
| Read Range | Passive UHF RFID:<br>- Up to 40 feet (fixed readers)<br>- Up to 20 feet (handheld readers) | Several inches up to several feet |
| Read Rate | 10's, 100's or 1000's simultaneously | Only one barcode at a time per reader |
| Identification | Can uniquely identify each item/asset tagged. | Most barcodes only identify the type of item (UPC Code) but not uniquely. |
| Read/Write | Many RFID tags are Read/Write | Read only |
| Technology | RF (Radio Frequency) | Optical (Laser) |
| Interference | Some RFID frequencies are degraded when used on Metal and Liquids. They can interfere with some RF Frequencies. | Obstructed barcodes cannot be read (dirt covering barcode, torn barcode, etc.) |
| Automation | Most "fixed" readers don't require human involvement to collect data (automated) | Most barcode scanners require a human to operate (labor intensive) |

**Table 37: pRFID and Barcode Capability Comparison**

Employing a pRFID solution achieves velocity, agility, and resilience in logistics by optimizing processes, removing process redundancy, and designing out human error. The technology has been a proven industry standard to achieve compliancy as the 'normative' state as well as avoid new system proliferation.

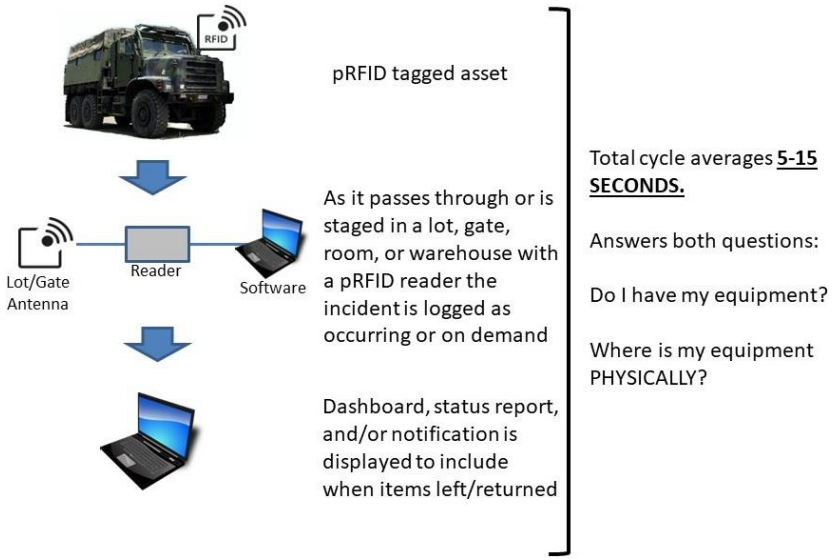


**Figure 30: pRFID Enabled Asset Management**

The current state of this technology is quite mature as implementation costs are dropping and reliability rising. The realization of return on investment for automated information technology

(AIT) and total asset visibility (TAV) is the increased reliability and “digital truth” of assets. Therefore, employment of this technology fits within the USMC vision for total asset visibility.

Using the current inventory management system, GCSS-MC, as the base of information the pRFID information confirms existence as well as physical movement of all assets by interval, as occurring, and/or on demand.

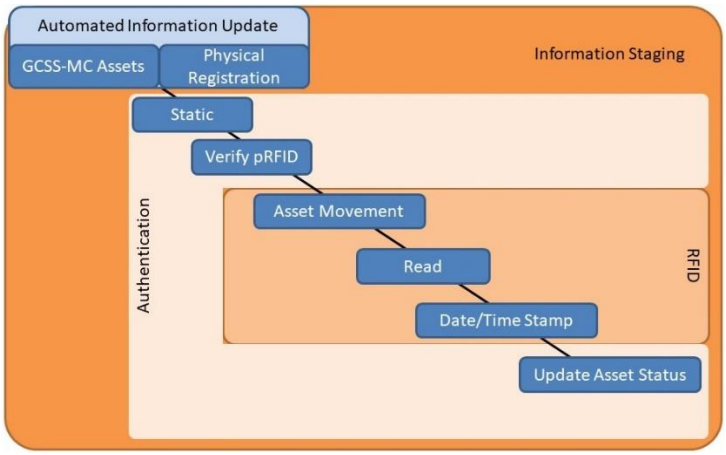


**Figure 31: pRFID Data Staging**

The pRFID technology presents an integrated solution for the automatic identification of equipment as it captures, stores, and correlates detailed information about all system transactions. Asset existence is the digital truth by pRFID logging, verifies/authenticates physical location, and tracks physical movement in and out of areas eliminating administrative challenges and manual processes. As an enterprise technology solution, pRFID can be realized in the short term with minimal upfront cost and is available anywhere a connected computer exists.

## 4.7 Health Services Capability Gap

Pertaining to the INDOPACOM operational scenario, the medical planners are forecasting requirements based on the assumption units will need an initial 15-day medical supply, after which will require some kind of resupply. This operating concept—as envisioned—will have small teams operating out of austere outposts over vast geographic distances with extremely limited reach back support. That scenario demands our forces be as self-sufficient as possible, necessitating that Marines and Corpsman leverage resources available in their locales and foraging for various classes of supply.

One of the common After-Action Report (AARs) narratives from the Marine Special Operations Companies (MSOC)/Marine Special Operations Teams (MSOT) returning from deployment to the Philippines was, “have more IV bags on-hand”, at least initially. Basilan, Philippines is hot and humid, and during sustained operating conditions, people dehydrate very quickly—especially when they’re not acclimated properly. If the scenario for which we are preparing ever manifests, the first medical supply to be depleted will be saline intravenous fluid (IV) bags, as is always the case. Once those get used up, then what? A single IV bag weighs 2 pounds and is sensitive to the elements. Additionally, it cannot be assumed the local populace and/or location will have enough IV bags—or any medical resources—to share with the miliary, especially given the contested transportation and supply chains. This situation poses a very real, tangible risk to

human life. That risk is also unnecessary, as we have the capacity to avoid this scenario before crisis emerges.

To avoid the logistically-contested aspect of medical supply availability, movement, storage, and tracking, the capability to produce things like saline IV bags locally and on-demand is essential. This capability has two developments:

1) NASA made a saline IV bag machine for the International Space Station (ISS)—arguably the most austere environment known to man. Therefore, NASA is not paying money and sacrificing limited space and weight on already-constrained rockets to transport intravenous fluids into orbit.
2) The Army has begun research and has gone through some testing for a similar capability: https://www.thedefensepost.com/2021/01/29/device-iv-fluid-wounded-soldiers/

Sustainment of forces requires the development and procurement of self-generating systems to provide resuscitation fluids like 0.9 percent saline, and an additional requirement to increase that concentration to 3% for Damage Control Resuscitation Treatments (DCRT) in treatment of Traumatic Brain Injuries (TBI).

# Appendix A
# Supply Function Data Points

| Sub-Function 1 | Sub-Function 2 | Data Type | Line Data Level |
|---|---|---|---|
| A. Requirements | 1. Planning | TO&E | Billet Alpha Grade |
| A. Requirements | 1. Planning | TO&E | Billet Branch of Service |
| A. Requirements | 1. Planning | TO&E | Billet Description |
| A. Requirements | 1. Planning | TO&E | Billet Identification Code |
| A. Requirements | 1. Planning | TO&E | Billet Military Occupational Specialty |
| A. Requirements | 1. Planning | TO&E | Billet Security Clearance |
| A. Requirements | 1. Planning | TO&E | Billet Weapon |
| A. Requirements | 1. Planning | TO&E | Table of Authorized Material Control Number |
| A. Requirements | 1. Planning | TO&E | Table of Equipment Authorized Quantity |
| A. Requirements | 1. Planning | TO&E | Table of Equipment Nomenclature Name and Model |
| A. Requirements | 1. Planning | TO&E | Table of Equipment Unit of Issue |
| A. Requirements | 1. Planning | TO&E | Table of Equipment Association Consists of Nomenclature Name and Model |
| A. Requirements | 1. Planning | TO&E | Table of Equipment Association Consists of Table of Authorized Material Control Number |
| A. Requirements | 1. Planning | TO&E | Unit Identifier Code |
| A. Requirements | 1. Planning | TO&E | Unit Identifier Code |
| A. Requirements | 1. Planning | OPLAN | Mission ID |
| A. Requirements | 1. Planning | OPLAN | Mission Name |
| A. Requirements | 1. Planning | OPLAN | Operation ID |
| A. Requirements | 1. Planning | OPLAN | Operation Name |
| A. Requirements | 1. Planning | OPLAN | Operational Plan Identification Number |
| A. Requirements | 1. Planning | OPLAN | Operation Start/Commencement Date |
| A. Requirements | 1. Planning | OPLAN | Operation End/Termination Date |
| A. Requirements | 1. Planning | Activity | Combatant Command |
| A. Requirements | 1. Planning | Activity | CONUS/OCONUS Indicator |
| A. Requirements | 1. Planning | Activity | Department of Defense Activity Address Code |
| A. Requirements | 1. Planning | Activity | DoDAAC Authority Code |
| A. Requirements | 1. Planning | Activity | Force/Activity Designator Code |
| A. Requirements | 1. Planning | Activity | Major Command Code |
| A. Requirements | 1. Planning | Activity | Table of Authorized Material Control Number |
| A. Requirements | 1. Planning | OPLAN | Country Entity Name |
| A. Requirements | 2. Allowancing | EDL | Table of Authorized Material Control Number |
| A. Requirements | 2. Allowancing | EDL | Table of Equipment Association Consists of Nomenclature Name and Model |
| A. Requirements | 2. Allowancing | EDL | Table of Equipment Association Consists of Table of Authorized Material Control Number |
| A. Requirements | 2. Allowancing | EDL | Unit of Issue |
| A. Requirements | 2. Allowancing | EDL | Authorized Quantity |
| A. Requirements | 2. Allowancing | Staffing | Unit Identifier Code |
| A. Requirements | 2. Allowancing | Staffing | Billet Alpha Grade |
| A. Requirements | 2. Allowancing | Staffing | Billet Branch of Service |
| A. Requirements | 2. Allowancing | Staffing | Billet Description |
| A. Requirements | 2. Allowancing | Staffing | Billet Military Occupational Specialty |
| A. Requirements | 2. Allowancing | Staffing | Billet Security Clearance |
| A. Requirements | 2. Allowancing | Staffing | Billet Weapon |
| A. Requirements | 2. Allowancing | Activity | Combatant Command |
| A. Requirements | 2. Allowancing | Activity | CONUS/OCONUS Indicator |
| A. Requirements | 2. Allowancing | Activity | Department of Defense Activity Address Code |
| A. Requirements | 2. Allowancing | Activity | DoDAAC Authority Code |
| A. Requirements | 2. Allowancing | Activity | Force/Activity Designator Code |
| A. Requirements | 2. Allowancing | Activity | Major Command Code |
| A. Requirements | 3. Forecasting | EDL | On Hand Quantity |
| A. Requirements | 3. Forecasting | Staffing | Total Manpower |
| A. Requirements | 3. Forecasting | OPLAN | Operation Start/Commencement Date |
| A. Requirements | 3. Forecasting | OPLAN | Operation End/Termination Date |
| B. Distribution | 1. Assigning | Item Catalog Information | Table of Authorized Material Control Number |
| B. Distribution | 1. Assigning | Item Catalog Information | National Stock Number |
| B. Distribution | 1. Assigning | Item Catalog Information | Unit of Issue |
| B. Distribution | 1. Assigning | Item Catalog Information | Unit Price |

| | | | |
|---|---|---|---|
| B. Distribution | 1. Assigning | Item Catalog Information | Nomenclature |
| B. Distribution | 1. Assigning | Item Catalog Information | Stores Account Code |
| B. Distribution | 1. Assigning | Item Catalog Information | Class of Supply |
| B. Distribution | 1. Assigning | Item Catalog Information | Sub-Class of Supply |
| B. Distribution | 1. Assigning | Item Identification | Item Unique Identification |
| B. Distribution | 1. Assigning | Item Identification | Lot Number |
| B. Distribution | 1. Assigning | Item Identification | Serial Number |
| B. Distribution | 1. Assigning | Item Attributes | Quantity |
| B. Distribution | 1. Assigning | Item Attributes | Assigned Activity |
| B. Distribution | 1. Assigning | Item Attributes | Assigned Subordinate Activity |
| B. Distribution | 1. Assigning | Item Attributes | Responsible Individual |
| B. Distribution | 1. Assigning | Item Attributes | Responsible Officer |
| B. Distribution | 1. Assigning | Item Attributes | Last Inventoried Date |
| B. Distribution | 3. Dispensing | Volume (Imperial) | Gallon |
| B. Distribution | 3. Dispensing | Volume (Imperial) | Quart |
| B. Distribution | 3. Dispensing | Item Identification | Lot Number |
| B. Distribution | 3. Dispensing | Requisitioning | Priority Designator |
| B. Distribution | 3. Dispensing | Requisitioning | Request ID/Tracking Number |
| B. Distribution | 3. Dispensing | Requisitioning | Request Type |
| B. Distribution | 3. Dispensing | Requisitioning | Requesting Element/Unit |
| B. Distribution | 3. Dispensing | Requisitioning | Requesting Element/Unit Location |
| B. Distribution | 3. Dispensing | Requisitioning | Request Quantity |
| B. Distribution | 3. Dispensing | Item Catalog Information | National Stock Number |
| B. Distribution | 3. Dispensing | Activity | Department of Defense Activity Address Code |
| B. Distribution | 3. Dispensing | Item Attributes | Last Inventoried Date |
| B. Distribution | 3. Dispensing | Item Attributes | Location ID |
| B. Distribution | 3. Dispensing | Item Attributes | Location Name |
| B. Distribution | 3. Dispensing | Issuing | Document Identifier Code |
| B. Distribution | 3. Dispensing | Issuing | Issued Quantity |
| B. Distribution | 2. Issuing | Material Release Order | Issue to Activity |
| B. Distribution | 2. Issuing | Material Release Order | Issue from Activity |
| B. Distribution | 2. Issuing | Material Release Order | Issue NSN |
| B. Distribution | 2. Issuing | Requisitioning | Priority Designator |
| B. Distribution | 2. Issuing | Requisitioning | Request ID/Tracking Number |
| B. Distribution | 2. Issuing | Requisitioning | Request Type |
| B. Distribution | 2. Issuing | Requisitioning | Requesting Element/Unit |
| B. Distribution | 2. Issuing | Requisitioning | Requesting Element/Unit Location |
| B. Distribution | 2. Issuing | Requisitioning | Request Quantity |
| B. Distribution | 2. Issuing | Procurement | Standard Document Number |
| B. Distribution | 2. Issuing | Procurement | Quantity |
| B. Distribution | 2. Issuing | Procurement | Priority Designator |
| B. Distribution | 2. Issuing | Procurement | Suffix Code |
| B. Distribution | 2. Issuing | Procurement | Supplementary Address Code |
| B. Distribution | 2. Issuing | Item Catalog Information | Table of Authorized Material Control Number |
| B. Distribution | 2. Issuing | Item Catalog Information | National Stock Number |
| B. Distribution | 2. Issuing | Item Catalog Information | Unit of Issue |
| B. Distribution | 2. Issuing | Item Catalog Information | Unit Price |
| B. Distribution | 2. Issuing | Item Catalog Information | Nomenclature |
| B. Distribution | 2. Issuing | Item Catalog Information | Stores Account Code |
| B. Distribution | 2. Issuing | Item Catalog Information | Class of Supply |
| B. Distribution | 2. Issuing | Item Catalog Information | Sub-Class of Supply |
| B. Distribution | 2. Issuing | Item Identification | Item Unique Identification |
| B. Distribution | 2. Issuing | Item Identification | Lot Number |
| B. Distribution | 2. Issuing | Item Identification | Serial Number |
| B. Distribution | 2. Issuing | Item Attributes | Quantity |
| B. Distribution | 2. Issuing | Item Attributes | Assigned Activity |
| B. Distribution | 2. Issuing | Item Attributes | Assigned Subordinate Activity |
| B. Distribution | 2. Issuing | Issuing | Document Identifier Code |
| B. Distribution | 2. Issuing | Issuing | Issued Quantity |
| B. Distribution | 4. Requisitioning | Requisitioning | Priority Designator |
| B. Distribution | 4. Requisitioning | Requisitioning | Request ID/Tracking Number |
| B. Distribution | 4. Requisitioning | Requisitioning | Request Type |
| B. Distribution | 4. Requisitioning | Requisitioning | Requesting Element/Unit |
| B. Distribution | 4. Requisitioning | Requisitioning | Requesting Element/Unit Location |
| B. Distribution | 4. Requisitioning | Requisitioning | Request Quantity |

| | | | |
|---|---|---|---|
| B. Distribution | 4. Requisitioning | Item Catalog Information | National Stock Number |
| B. Distribution | 4. Requisitioning | Activity | Department of Defense Activity Address Code |
| B. Distribution | 4. Requisitioning | Item Catalog Information | Nomenclature |
| B. Distribution | 4. Requisitioning | Item Catalog Information | Unit of Issue |
| C. Procurement | Class I (Subsistence) | Requisitioning | Priority Designator |
| C. Procurement | Class I (Subsistence) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class I (Subsistence) | Requisitioning | Request Type |
| C. Procurement | Class I (Subsistence) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class I (Subsistence) | Requisitioning | Priority Designator |
| C. Procurement | Class I (Subsistence) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class I (Subsistence) | Requisitioning | Request Quantity |
| C. Procurement | Class I (Subsistence) | Item Catalog Information | National Stock Number |
| C. Procurement | Class I (Subsistence) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class I (Subsistence) | Item Catalog Information | Nomenclature |
| C. Procurement | Class I (Subsistence) | Item Catalog Information | Unit of Issue |
| C. Procurement | Class I (Subsistence) | Procurement Action | Advice Code |
| C. Procurement | Class I (Subsistence) | Procurement Action | Demand Code |
| C. Procurement | Class I (Subsistence) | Procurement Action | Document Identifier Code |
| C. Procurement | Class I (Subsistence) | Procurement Action | Media and Status Code |
| C. Procurement | Class I (Subsistence) | Procurement Action | Non-Mission Capable Supply |
| C. Procurement | Class I (Subsistence) | Procurement Action | Priority Designator |
| C. Procurement | Class I (Subsistence) | Procurement Action | Project Code |
| C. Procurement | Class I (Subsistence) | Procurement Action | Quantity |
| C. Procurement | Class I (Subsistence) | Procurement Action | Required Delivery Date |
| C. Procurement | Class I (Subsistence) | Procurement Action | Signal Code |
| C. Procurement | Class I (Subsistence) | Procurement Action | Standard Document Number |
| C. Procurement | Class I (Subsistence) | Procurement Status | Suffix Code |
| C. Procurement | Class I (Subsistence) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class I (Subsistence) | Procurement Status | Supply Status Code |
| C. Procurement | Class I (Subsistence) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class I (Subsistence) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class I (Subsistence) | Item Catalog Information | Phrase Code |
| C. Procurement | Class I (Subsistence) | Item Catalog Information | Unit Price |
| C. Procurement | Class II (Minor) | Item Catalog Information | Table of Authorized Material Control Number |
| C. Procurement | Class II (Minor) | Requisitioning | Priority Designator |
| C. Procurement | Class II (Minor) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class II (Minor) | Requisitioning | Request Type |
| C. Procurement | Class II (Minor) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class II (Minor) | Requisitioning | Priority Designator |
| C. Procurement | Class II (Minor) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class II (Minor) | Requisitioning | Request Type |
| C. Procurement | Class II (Minor) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class II (Minor) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class II (Minor) | Requisitioning | Request Quantity |
| C. Procurement | Class II (Minor) | Item Catalog Information | National Stock Number |
| C. Procurement | Class II (Minor) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class II (Minor) | Item Catalog Information | Nomenclature |
| C. Procurement | Class II (Minor) | Item Catalog Information | Unit of Issue |
| C. Procurement | Class II (Minor) | Procurement Action | Advice Code |
| C. Procurement | Class II (Minor) | Procurement Action | Demand Code |
| C. Procurement | Class II (Minor) | Procurement Action | Document Identifier Code |
| C. Procurement | Class II (Minor) | Procurement Action | Media and Status Code |
| C. Procurement | Class II (Minor) | Procurement Action | Non-Mission Capable Supply |
| C. Procurement | Class II (Minor) | Procurement Action | Priority Designator |
| C. Procurement | Class II (Minor) | Procurement Action | Project Code |
| C. Procurement | Class II (Minor) | Procurement Action | Quantity |
| C. Procurement | Class II (Minor) | Procurement Action | Required Delivery Date |
| C. Procurement | Class II (Minor) | Procurement Action | Signal Code |
| C. Procurement | Class II (Minor) | Procurement Action | Standard Document Number |
| C. Procurement | Class II (Minor) | Procurement Status | Suffix Code |
| C. Procurement | Class II (Minor) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class II (Minor) | Procurement Status | Supply Status Code |
| C. Procurement | Class II (Minor) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class II (Minor) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class II (Minor) | Item Catalog Information | Phrase Code |

| | | | |
|---|---|---|---|
| C. Procurement | Class II (Minor) | Item Catalog Information | Unit Price |
| C. Procurement | Class III (POL) | Requisitioning | Priority Designator |
| C. Procurement | Class III (POL) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class III (POL) | Requisitioning | Request Type |
| C. Procurement | Class III (POL) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class III (POL) | Requisitioning | Priority Designator |
| C. Procurement | Class III (POL) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class III (POL) | Requisitioning | Request Quantity |
| C. Procurement | Class III (POL) | Item Catalog Information | National Stock Number |
| C. Procurement | Class III (POL) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class III (POL) | Item Catalog Information | Nomenclature |
| C. Procurement | Class III (POL) | Item Catalog Information | Unit of Issue |
| C. Procurement | Class III (POL) | Procurement Action | Advice Code |
| C. Procurement | Class III (POL) | Procurement Action | Demand Code |
| C. Procurement | Class III (POL) | Procurement Action | Document Identifier Code |
| C. Procurement | Class III (POL) | Procurement Action | Media and Status Code |
| C. Procurement | Class III (POL) | Procurement Action | Non-Mission Capable Supply |
| C. Procurement | Class III (POL) | Procurement Action | Priority Designator |
| C. Procurement | Class III (POL) | Procurement Action | Project Code |
| C. Procurement | Class III (POL) | Procurement Action | Quantity |
| C. Procurement | Class III (POL) | Procurement Action | Required Delivery Date |
| C. Procurement | Class III (POL) | Procurement Action | Signal Code |
| C. Procurement | Class III (POL) | Procurement Action | Standard Document Number |
| C. Procurement | Class III (POL) | Procurement Status | Suffix Code |
| C. Procurement | Class III (POL) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class III (POL) | Procurement Status | Supply Status Code |
| C. Procurement | Class III (POL) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class III (POL) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class III (POL) | Item Catalog Information | Phrase Code |
| C. Procurement | Class III (POL) | Item Catalog Information | Unit Price |
| C. Procurement | Class IV (Engineer/Barrier) | Requisitioning | Priority Designator |
| C. Procurement | Class IV (Engineer/Barrier) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class IV (Engineer/Barrier) | Requisitioning | Request Type |
| C. Procurement | Class IV (Engineer/Barrier) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class IV (Engineer/Barrier) | Requisitioning | Priority Designator |
| C. Procurement | Class IV (Engineer/Barrier) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class IV (Engineer/Barrier) | Requisitioning | Request Quantity |
| C. Procurement | Class IV (Engineer/Barrier) | Item Catalog Information | National Stock Number |
| C. Procurement | Class IV (Engineer/Barrier) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class IV (Engineer/Barrier) | Item Catalog Information | Nomenclature |
| C. Procurement | Class IV (Engineer/Barrier) | Item Catalog Information | Unit of Issue |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Advice Code |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Demand Code |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Document Identifier Code |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Media and Status Code |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Non-Mission Capable Supply |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Priority Designator |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Project Code |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Quantity |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Required Delivery Date |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Signal Code |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Standard Document Number |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Status | Suffix Code |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class IV (Engineer/Barrier) | Procurement Status | Supply Status Code |
| C. Procurement | Class IV (Engineer/Barrier) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class IV (Engineer/Barrier) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class IV (Engineer/Barrier) | Item Catalog Information | Phrase Code |
| C. Procurement | Class IV (Engineer/Barrier) | Item Catalog Information | Unit Price |
| C. Procurement | Class V (Ammunition) | Item Catalog Information | Department of Defense Identification Code |
| C. Procurement | Class V (Ammunition) | Requisitioning | Priority Designator |
| C. Procurement | Class V (Ammunition) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class V (Ammunition) | Requisitioning | Request Type |
| C. Procurement | Class V (Ammunition) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class V (Ammunition) | Requisitioning | Priority Designator |

| | | | |
|---|---|---|---|
| C. Procurement | Class V (Ammunition) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class V (Ammunition) | Requisitioning | Request Quantity |
| C. Procurement | Class V (Ammunition) | Item Catalog Information | National Stock Number |
| C. Procurement | Class V (Ammunition) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class V (Ammunition) | Item Catalog Information | Nomenclature |
| C. Procurement | Class V (Ammunition) | Item Catalog Information | Unit of Issue |
| C. Procurement | Class V (Ammunition) | Procurement Action | Advice Code |
| C. Procurement | Class V (Ammunition) | Procurement Action | Demand Code |
| C. Procurement | Class V (Ammunition) | Procurement Action | Document Identifier Code |
| C. Procurement | Class V (Ammunition) | Procurement Action | Media and Status Code |
| C. Procurement | Class V (Ammunition) | Procurement Action | Non-Mission Capable Supply |
| C. Procurement | Class V (Ammunition) | Procurement Action | Priority Designator |
| C. Procurement | Class V (Ammunition) | Procurement Action | Project Code |
| C. Procurement | Class V (Ammunition) | Procurement Action | Quantity |
| C. Procurement | Class V (Ammunition) | Procurement Action | Required Delivery Date |
| C. Procurement | Class V (Ammunition) | Procurement Action | Signal Code |
| C. Procurement | Class V (Ammunition) | Procurement Action | Standard Document Number |
| C. Procurement | Class V (Ammunition) | Procurement Status | Suffix Code |
| C. Procurement | Class V (Ammunition) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class V (Ammunition) | Procurement Status | Supply Status Code |
| C. Procurement | Class V (Ammunition) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class V (Ammunition) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class V (Ammunition) | Item Catalog Information | Phrase Code |
| C. Procurement | Class V (Ammunition) | Item Catalog Information | Unit Price |
| C. Procurement | Class VI (Personal Demand) | Requisitioning | Priority Designator |
| C. Procurement | Class VI (Personal Demand) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class VI (Personal Demand) | Requisitioning | Request Type |
| C. Procurement | Class VI (Personal Demand) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class VI (Personal Demand) | Requisitioning | Priority Designator |
| C. Procurement | Class VI (Personal Demand) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class VI (Personal Demand) | Requisitioning | Request Quantity |
| C. Procurement | Class VI (Personal Demand) | Item Catalog Information | National Stock Number |
| C. Procurement | Class VI (Personal Demand) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class VI (Personal Demand) | Item Catalog Information | Nomenclature |
| C. Procurement | Class VI (Personal Demand) | Item Catalog Information | Unit of Issue |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Advice Code |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Demand Code |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Document Identifier Code |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Media and Status Code |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Non-Mission Capable Supply |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Priority Designator |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Project Code |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Quantity |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Required Delivery Date |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Signal Code |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Standard Document Number |
| C. Procurement | Class VI (Personal Demand) | Procurement Status | Suffix Code |
| C. Procurement | Class VI (Personal Demand) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class VI (Personal Demand) | Procurement Status | Supply Status Code |
| C. Procurement | Class VI (Personal Demand) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class VI (Personal Demand) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class VI (Personal Demand) | Item Catalog Information | Phrase Code |
| C. Procurement | Class VI (Personal Demand) | Item Catalog Information | Unit Price |
| C. Procurement | Class VII (Major End Items) | Item Catalog Information | Table of Authorized Material Control Number |
| C. Procurement | Class VII (Major End Items) | Requisitioning | Priority Designator |
| C. Procurement | Class VII (Major End Items) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class VII (Major End Items) | Requisitioning | Request Type |
| C. Procurement | Class VII (Major End Items) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class VII (Major End Items) | Requisitioning | Priority Designator |
| C. Procurement | Class VII (Major End Items) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class VII (Major End Items) | Requisitioning | Request Quantity |
| C. Procurement | Class VII (Major End Items) | Item Catalog Information | National Stock Number |
| C. Procurement | Class VII (Major End Items) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class VII (Major End Items) | Item Catalog Information | Nomenclature |
| C. Procurement | Class VII (Major End Items) | Item Catalog Information | Unit of Issue |

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| C. Procurement | Class VII (Major End Items) | Procurement Action | Advice Code |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Demand Code |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Document Identifier Code |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Media and Status Code |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Non-Mission Capable Supply |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Priority Designator |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Project Code |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Quantity |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Required Delivery Date |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Signal Code |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Standard Document Number |
| C. Procurement | Class VII (Major End Items) | Procurement Status | Suffix Code |
| C. Procurement | Class VII (Major End Items) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class VII (Major End Items) | Procurement Status | Supply Status Code |
| C. Procurement | Class VII (Major End Items) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class VII (Major End Items) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class VII (Major End Items) | Item Catalog Information | Phrase Code |
| C. Procurement | Class VII (Major End Items) | Item Catalog Information | Unit Price |
| C. Procurement | Class VIII (Medical) | Item Catalog Information | Medical Name |
| C. Procurement | Class VIII (Medical) | Requisitioning | Priority Designator |
| C. Procurement | Class VIII (Medical) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class VIII (Medical) | Requisitioning | Request Type |
| C. Procurement | Class VIII (Medical) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class VIII (Medical) | Requisitioning | Priority Designator |
| C. Procurement | Class VIII (Medical) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class VIII (Medical) | Requisitioning | Request Quantity |
| C. Procurement | Class VIII (Medical) | Item Catalog Information | National Stock Number |
| C. Procurement | Class VIII (Medical) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class VIII (Medical) | Item Catalog Information | Nomenclature |
| C. Procurement | Class VIII (Medical) | Item Catalog Information | Unit of Issue |
| C. Procurement | Class VIII (Medical) | Procurement Action | Advice Code |
| C. Procurement | Class VIII (Medical) | Procurement Action | Demand Code |
| C. Procurement | Class VIII (Medical) | Procurement Action | Document Identifier Code |
| C. Procurement | Class VIII (Medical) | Procurement Action | Media and Status Code |
| C. Procurement | Class VIII (Medical) | Procurement Action | Non-Mission Capable Supply |
| C. Procurement | Class VIII (Medical) | Procurement Action | Priority Designator |
| C. Procurement | Class VIII (Medical) | Procurement Action | Project Code |
| C. Procurement | Class VIII (Medical) | Procurement Action | Quantity |
| C. Procurement | Class VIII (Medical) | Procurement Action | Required Delivery Date |
| C. Procurement | Class VIII (Medical) | Procurement Action | Signal Code |
| C. Procurement | Class VIII (Medical) | Procurement Action | Standard Document Number |
| C. Procurement | Class VIII (Medical) | Procurement Status | Suffix Code |
| C. Procurement | Class VIII (Medical) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class VIII (Medical) | Procurement Status | Supply Status Code |
| C. Procurement | Class VIII (Medical) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class VIII (Medical) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class VIII (Medical) | Item Catalog Information | Phrase Code |
| C. Procurement | Class VIII (Medical) | Item Catalog Information | Unit Price |
| C. Procurement | Class VIII (Medical) | Item Catalog Information | Department of Defense Identification Code |
| C. Procurement | Class IX (Repair Parts) | Requisitioning | Priority Designator |
| C. Procurement | Class IX (Repair Parts) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class IX (Repair Parts) | Requisitioning | Request Type |
| C. Procurement | Class IX (Repair Parts) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class IX (Repair Parts) | Requisitioning | Priority Designator |
| C. Procurement | Class IX (Repair Parts) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class IX (Repair Parts) | Requisitioning | Request Quantity |
| C. Procurement | Class IX (Repair Parts) | Item Catalog Information | National Stock Number |
| C. Procurement | Class IX (Repair Parts) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class IX (Repair Parts) | Item Catalog Information | Nomenclature |
| C. Procurement | Class IX (Repair Parts) | Item Catalog Information | Unit of Issue |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Advice Code |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Demand Code |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Document Identifier Code |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Media and Status Code |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Non-Mission Capable Supply |

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| C. Procurement | Class IX (Repair Parts) | Procurement Action | Priority Designator |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Project Code |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Quantity |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Required Delivery Date |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Signal Code |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Standard Document Number |
| C. Procurement | Class IX (Repair Parts) | Procurement Status | Suffix Code |
| C. Procurement | Class IX (Repair Parts) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class IX (Repair Parts) | Procurement Status | Supply Status Code |
| C. Procurement | Class IX (Repair Parts) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class IX (Repair Parts) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class IX (Repair Parts) | Item Catalog Information | Phrase Code |
| C. Procurement | Class IX (Repair Parts) | Item Catalog Information | Unit Price |
| C. Procurement | Class X (Nonmilitary Programs) | Requisitioning | Priority Designator |
| C. Procurement | Class X (Nonmilitary Programs) | Requisitioning | Request ID/Tracking Number |
| C. Procurement | Class X (Nonmilitary Programs) | Requisitioning | Request Type |
| C. Procurement | Class X (Nonmilitary Programs) | Requisitioning | Requesting Element/Unit |
| C. Procurement | Class X (Nonmilitary Programs) | Requisitioning | Priority Designator |
| C. Procurement | Class X (Nonmilitary Programs) | Requisitioning | Requesting Element/Unit Location |
| C. Procurement | Class X (Nonmilitary Programs) | Requisitioning | Request Quantity |
| C. Procurement | Class X (Nonmilitary Programs) | Item Catalog Information | National Stock Number |
| C. Procurement | Class X (Nonmilitary Programs) | Activity | Department of Defense Activity Address Code |
| C. Procurement | Class X (Nonmilitary Programs) | Item Catalog Information | Nomenclature |
| C. Procurement | Class X (Nonmilitary Programs) | Item Catalog Information | Unit of Issue |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Advice Code |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Demand Code |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Document Identifier Code |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Media and Status Code |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Non-Mission Capable Supply |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Priority Designator |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Project Code |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Quantity |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Required Delivery Date |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Signal Code |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Standard Document Number |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Status | Suffix Code |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Action | Supplementary Address Code |
| C. Procurement | Class X (Nonmilitary Programs) | Procurement Status | Supply Status Code |
| C. Procurement | Class X (Nonmilitary Programs) | Financials | Project, Organization, Expenditure, Type |
| C. Procurement | Class X (Nonmilitary Programs) | Item Catalog Information | Acquisition Advice Code |
| C. Procurement | Class X (Nonmilitary Programs) | Item Catalog Information | Phrase Code |
| C. Procurement | Class X (Nonmilitary Programs) | Item Catalog Information | Unit Price |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | Table of Authorized Material Control Number |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | National Stock Number |
| D. Storage | 3. Care-in-Storage | Item Attributes | Quantity |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | Unit of Issue |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | Unit Price |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | Nomenclature |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | Stores Account Code |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | Class of Supply |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | Sub-Class of Supply |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | Criticality Code |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | Environmental Attribute Code |
| D. Storage | 3. Care-in-Storage | Item Catalog Information | HazMat Indicator Code |
| D. Storage | 3. Care-in-Storage | Item Characteristics | Gross Weight |
| D. Storage | 3. Care-in-Storage | Item Characteristics | Shipping Height |
| D. Storage | 3. Care-in-Storage | Item Characteristics | Shipping Length |
| D. Storage | 3. Care-in-Storage | Item Characteristics | Shipping Width |
| D. Storage | 3. Care-in-Storage | Item Identification | Date of Pack |
| D. Storage | 3. Care-in-Storage | Item Identification | Expiration Date |
| D. Storage | 3. Care-in-Storage | Item Identification | Item Unique Identification |
| D. Storage | 3. Care-in-Storage | Item Identification | Lot Number |
| D. Storage | 3. Care-in-Storage | Item Identification | Serial Number |
| D. Storage | 3. Care-in-Storage | Item Attributes | Assigned Activity |
| D. Storage | 3. Care-in-Storage | Activity | Department of Defense Activity Address Code |

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| D. Storage | 3. Care-in-Storage | Item Attributes | Last Inventoried Date |
| D. Storage | 3. Care-in-Storage | Item Attributes | Location ID |
| D. Storage | 3. Care-in-Storage | Item Attributes | Location Name |
| D. Storage | 2. Inventory Control | Item Catalog Information | Table of Authorized Material Control Number |
| D. Storage | 2. Inventory Control | Item Catalog Information | National Stock Number |
| D. Storage | 2. Inventory Control | Inventory | Document Identifier Code |
| D. Storage | 2. Inventory Control | Inventory | Supply Condition Code |
| D. Storage | 2. Inventory Control | Item Attributes | Quantity |
| D. Storage | 2. Inventory Control | Inventory | On Hand Quantity |
| D. Storage | 2. Inventory Control | Item Catalog Information | Unit of Issue |
| D. Storage | 2. Inventory Control | Item Catalog Information | Unit Price |
| D. Storage | 2. Inventory Control | Item Catalog Information | Nomenclature |
| D. Storage | 2. Inventory Control | Item Catalog Information | Stores Account Code |
| D. Storage | 2. Inventory Control | Item Catalog Information | Class of Supply |
| D. Storage | 2. Inventory Control | Item Catalog Information | Sub-Class of Supply |
| D. Storage | 2. Inventory Control | Item Catalog Information | Controlled Inventory Item Code |
| D. Storage | 2. Inventory Control | Item Catalog Information | Controlled Item Code |
| D. Storage | 2. Inventory Control | Item Catalog Information | Criticality Code |
| D. Storage | 2. Inventory Control | EDL | Authorized Quantity |
| D. Storage | 2. Inventory Control | Inventory | Provisioning Type |
| D. Storage | 2. Inventory Control | Inventory | Provisioning Quantity |
| D. Storage | 2. Inventory Control | Inventory | Provisioning Date |
| D. Storage | 1. Receipting | Procurement | Standard Document Number |
| D. Storage | 1. Receipting | Procurement | Quantity |
| D. Storage | 1. Receipting | Procurement | Priority Designator |
| D. Storage | 1. Receipting | Procurement | Suffix Code |
| D. Storage | 1. Receipting | Procurement | Supplementary Address Code |
| D. Storage | 1. Receipting | Receipt Discrepancy | Discrepancy Indicator Code |
| D. Storage | 1. Receipting | Receipt Transaction | Document Identifier Code |
| D. Storage | 1. Receipting | Receipt Discrepancy | Supply Action Code |
| D. Storage | 1. Receipting | Receipt Discrepancy | Supply Discrepancy Code |
| D. Storage | 1. Receipting | Receipt Discrepancy | Supply Discrepancy Code Document Type Codes |
| D. Storage | 1. Receipting | Receipt Discrepancy | Supply Discrepancy Response Codes |
| D. Storage | 1. Receipting | Requisitioning | Priority Designator |
| D. Storage | 1. Receipting | Requisitioning | Request ID/Tracking Number |
| D. Storage | 1. Receipting | Requisitioning | Request Type |
| D. Storage | 1. Receipting | Requisitioning | Requesting Element/Unit |
| D. Storage | 1. Receipting | Requisitioning | Requesting Element/Unit Location |
| D. Storage | 1. Receipting | Requisitioning | Request Quantity |
| D. Storage | 1. Receipting | Receipting | Document Identifier Code |
| D. Storage | 1. Receipting | Receipting | Receipt Quantity |
| D. Storage | 4. Safekeeping | Personal Effects | Member Case Number |
| D. Storage | 4. Safekeeping | Personal Effects | Member Status |
| D. Storage | 4. Safekeeping | Personal Effects | Container Baggage Number |
| D. Storage | 4. Safekeeping | Personal Effects | Container Inventory |
| D. Storage | 4. Safekeeping | Personal Effects | Container Seal |
| D. Storage | 4. Safekeeping | Individual Member | Military/Service Occupation |
| D. Storage | 4. Safekeeping | Individual Member | Name, First |
| D. Storage | 4. Safekeeping | Individual Member | Name, Last |
| D. Storage | 4. Safekeeping | Individual Member | Name, Middle |
| D. Storage | 4. Safekeeping | Individual Member | Name, Suffix |
| D. Storage | 4. Safekeeping | Individual Member | Electronic Data Interchange Personal Identifier |
| D. Storage | 4. Safekeeping | Individual Member | Rank |
| D. Storage | 4. Safekeeping | Individual Member | Service Status |
| E. Salvage | 2. Recycle | Item Catalog Information | Controlled Inventory Item Code |
| E. Salvage | 2. Recycle | Item Catalog Information | Controlled Item Code |
| E. Salvage | 2. Recycle | Item Catalog Information | Criticality Code |
| E. Salvage | 2. Recycle | Item Catalog Information | Demilitarization Code |
| E. Salvage | 2. Recycle | Item Catalog Information | Nomenclature |
| E. Salvage | 2. Recycle | Item Catalog Information | Unit of Issue |
| E. Salvage | 2. Recycle | Item Catalog Information | National Stock Number |
| E. Salvage | 2. Recycle | Inventory | Supply Condition Code |
| E. Salvage | 2. Recycle | Item Attributes | Quantity |
| E. Salvage | 2. Recycle | Item Catalog Information | Unit of Issue |
| E. Salvage | 2. Recycle | Item Catalog Information | Unit Price |

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| E. Salvage | 2. Recycle | Item Catalog Information | Nomenclature |
| E. Salvage | 2. Recycle | Activity | Department of Defense Activity Address Code |
| E. Salvage | 2. Recycle | Inventory | Document Identifier Code |
| E. Salvage | 2. Recycle | Inventory | Supply Condition Code |
| E. Salvage | 2. Recycle | Item Catalog Information | Table of Authorized Material Control Number |
| E. Salvage | 2. Recycle | Item Identification | Item Unique Identification |
| E. Salvage | 2. Recycle | Item Identification | Lot Number |
| E. Salvage | 2. Recycle | Item Identification | Serial Number |
| E. Salvage | 2. Recycle | Activity | Department of Defense Activity Address Code |
| E. Salvage | 1. Reuse | Item Catalog Information | Controlled Inventory Item Code |
| E. Salvage | 1. Reuse | Item Catalog Information | Controlled Item Code |
| E. Salvage | 1. Reuse | Item Catalog Information | Criticality Code |
| E. Salvage | 1. Reuse | Item Catalog Information | Demilitarization Code |
| E. Salvage | 1. Reuse | Item Catalog Information | Nomenclature |
| E. Salvage | 1. Reuse | Item Catalog Information | Unit of Issue |
| E. Salvage | 1. Reuse | Item Catalog Information | National Stock Number |
| E. Salvage | 1. Reuse | Inventory | Supply Condition Code |
| E. Salvage | 1. Reuse | Item Attributes | Quantity |
| E. Salvage | 1. Reuse | Item Catalog Information | Unit of Issue |
| E. Salvage | 1. Reuse | Item Catalog Information | Unit Price |
| E. Salvage | 1. Reuse | Item Catalog Information | Nomenclature |
| E. Salvage | 1. Reuse | Activity | Department of Defense Activity Address Code |
| E. Salvage | 1. Reuse | Inventory | Document Identifier Code |
| E. Salvage | 1. Reuse | Inventory | Supply Condition Code |
| E. Salvage | 1. Reuse | Item Catalog Information | Combat Essentiality Code |
| E. Salvage | 1. Reuse | Item Catalog Information | Table of Authorized Material Control Number |
| E. Salvage | 1. Reuse | Item Identification | Item Unique Identification |
| E. Salvage | 1. Reuse | Item Identification | Lot Number |
| E. Salvage | 1. Reuse | Item Identification | Serial Number |
| E. Salvage | 1. Reuse | Activity | Department of Defense Activity Address Code |
| E. Salvage | 3. Scrap | Item Catalog Information | Controlled Inventory Item Code |
| E. Salvage | 3. Scrap | Item Catalog Information | Controlled Item Code |
| E. Salvage | 3. Scrap | Item Catalog Information | Criticality Code |
| E. Salvage | 3. Scrap | Item Catalog Information | Demilitarization Code |
| E. Salvage | 3. Scrap | Item Catalog Information | Nomenclature |
| E. Salvage | 3. Scrap | Item Catalog Information | Unit of Issue |
| E. Salvage | 3. Scrap | Item Catalog Information | National Stock Number |
| E. Salvage | 3. Scrap | Inventory | Supply Condition Code |
| E. Salvage | 3. Scrap | Item Attributes | Quantity |
| E. Salvage | 3. Scrap | Item Catalog Information | Unit of Issue |
| E. Salvage | 3. Scrap | Item Catalog Information | Unit Price |
| E. Salvage | 3. Scrap | Item Catalog Information | Nomenclature |
| E. Salvage | 3. Scrap | Activity | Department of Defense Activity Address Code |
| E. Salvage | 3. Scrap | Inventory | Document Identifier Code |
| E. Salvage | 3. Scrap | Inventory | Supply Condition Code |
| E. Salvage | 3. Scrap | Item Catalog Information | Table of Authorized Material Control Number |
| E. Salvage | 3. Scrap | Item Identification | Item Unique Identification |
| E. Salvage | 3. Scrap | Item Identification | Lot Number |
| E. Salvage | 3. Scrap | Item Identification | Serial Number |
| F. Disposal | 3. Abandonment | Location, Physical | Military Grid Reference System |
| F. Disposal | 3. Abandonment | Item Catalog Information | Demilitarization Code |
| F. Disposal | 3. Abandonment | Item Catalog Information | Controlled Item Code |
| F. Disposal | 3. Abandonment | Item Catalog Information | Precious Metals Indicator |
| F. Disposal | 3. Abandonment | Item Catalog Information | National Stock Number |
| F. Disposal | 3. Abandonment | Item Catalog Information | Nomenclature |
| F. Disposal | 3. Abandonment | Item Catalog Information | Unit of Issue |
| F. Disposal | 3. Abandonment | Item Attributes | Quantity |
| F. Disposal | 3. Abandonment | Item Catalog Information | Combat Essentiality Code |
| F. Disposal | 3. Abandonment | Item Catalog Information | Controlled Inventory Item Code |
| F. Disposal | 3. Abandonment | Item Catalog Information | Controlled Item Code |
| F. Disposal | 3. Abandonment | Item Catalog Information | Criticality Code |
| F. Disposal | 3. Abandonment | Item Catalog Information | Table of Authorized Material Control Number |
| F. Disposal | 3. Abandonment | Item Identification | Item Unique Identification |
| F. Disposal | 3. Abandonment | Item Identification | Lot Number |
| F. Disposal | 3. Abandonment | Item Identification | Serial Number |

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| F. Disposal | 3. Abandonment | Activity | Department of Defense Activity Address Code |
| F. Disposal | 3. Abandonment | Item Attributes | Operational Status |
| F. Disposal | 3. Abandonment | Item Attributes | Assigned Activity |
| F. Disposal | 3. Abandonment | Item Attributes | Assigned Subordinate Activity |
| F. Disposal | 3. Abandonment | Inventory | Document Identifier Code |
| F. Disposal | 3. Abandonment | Inventory | Supply Condition Code |
| F. Disposal | 2. Sale | Item Attributes | Re-sale Unit Price |
| F. Disposal | 2. Sale | Item Attributes | Re-sale Type (Foreign/Domestic) |
| F. Disposal | 2. Sale | Item Attributes | Re-sale Country/Entity |
| F. Disposal | 2. Sale | Item Catalog Information | Demilitarization Code |
| F. Disposal | 2. Sale | Item Catalog Information | Controlled Item Code |
| F. Disposal | 2. Sale | Item Catalog Information | Precious Metals Indicator |
| F. Disposal | 2. Sale | Item Catalog Information | National Stock Number |
| F. Disposal | 2. Sale | Item Catalog Information | Nomenclature |
| F. Disposal | 2. Sale | Item Catalog Information | Unit of Issue |
| F. Disposal | 2. Sale | Item Attributes | Quantity |
| F. Disposal | 2. Sale | Item Catalog Information | Combat Essentiality Code |
| F. Disposal | 2. Sale | Item Catalog Information | Controlled Inventory Item Code |
| F. Disposal | 2. Sale | Item Catalog Information | Controlled Item Code |
| F. Disposal | 2. Sale | Item Catalog Information | Criticality Code |
| F. Disposal | 2. Sale | Item Catalog Information | Table of Authorized Material Control Number |
| F. Disposal | 2. Sale | Item Identification | Item Unique Identification |
| F. Disposal | 2. Sale | Item Identification | Lot Number |
| F. Disposal | 2. Sale | Item Identification | Serial Number |
| F. Disposal | 2. Sale | Activity | Department of Defense Activity Address Code |
| F. Disposal | 2. Sale | Item Attributes | Operational Status |
| F. Disposal | 2. Sale | Inventory | Document Identifier Code |
| F. Disposal | 2. Sale | Inventory | Supply Condition Code |
| F. Disposal | 1. Transfer | Item Catalog Information | Demilitarization Code |
| F. Disposal | 1. Transfer | Item Catalog Information | Controlled Item Code |
| F. Disposal | 1. Transfer | Item Catalog Information | Precious Metals Indicator |
| F. Disposal | 1. Transfer | Item Catalog Information | National Stock Number |
| F. Disposal | 1. Transfer | Item Catalog Information | Nomenclature |
| F. Disposal | 1. Transfer | Item Catalog Information | Unit of Issue |
| F. Disposal | 1. Transfer | Item Attributes | Quantity |
| F. Disposal | 1. Transfer | Item Catalog Information | Combat Essentiality Code |
| F. Disposal | 1. Transfer | Item Catalog Information | Controlled Inventory Item Code |
| F. Disposal | 1. Transfer | Item Catalog Information | Controlled Item Code |
| F. Disposal | 1. Transfer | Item Catalog Information | Criticality Code |
| F. Disposal | 1. Transfer | Item Catalog Information | Table of Authorized Material Control Number |
| F. Disposal | 1. Transfer | Item Identification | Item Unique Identification |
| F. Disposal | 1. Transfer | Item Identification | Lot Number |
| F. Disposal | 1. Transfer | Item Identification | Serial Number |
| F. Disposal | 1. Transfer | Activity | Department of Defense Activity Address Code |
| F. Disposal | 1. Transfer | Item Attributes | Operational Status |
| F. Disposal | 1. Transfer | Inventory | Document Identifier Code |
| F. Disposal | 1. Transfer | Inventory | Supply Condition Code |

# Appendix B
# Maintenance Function Data Points

| Sub-Function 1 | Sub-Function 2 | Data Type | Line Data Level |
|---|---|---|---|
| A. Inspection | 1. Check | Predictive Indicator | Fault/Failure Detection |
| A. Inspection | 1. Check | Predictive Indicator | Condition Forecast |
| A. Inspection | 1. Check | Predictive Indicator | Out of Tolerance/Baseline |
| A. Inspection | 1. Check | Item Identification | Item Unique Identification |
| A. Inspection | 1. Check | Item Identification | Lot Number |
| A. Inspection | 1. Check | Item Identification | Serial Number |
| A. Inspection | 1. Check | Item Attributes | Assigned Activity |
| A. Inspection | 1. Check | Activity | Department of Defense Activity Address Code |
| A. Inspection | 1. Check | Item Catalog Information | Table of Authorized Material Control Number |
| A. Inspection | 1. Check | Item Catalog Information | National Stock Number |
| A. Inspection | 1. Check | Item Catalog Information | Identification Number |
| A. Inspection | 1. Check | Item Attributes | Assigned Activity |
| A. Inspection | 1. Check | Item Attributes | Assigned Subordinate Activity |
| A. Inspection | 1. Check | Requisitioning | Priority Designator |
| A. Inspection | 1. Check | Requisitioning | Request ID/Tracking Number |
| A. Inspection | 1. Check | Requisitioning | Request Type |
| A. Inspection | 1. Check | Requisitioning | Requesting Element/Unit |
| A. Inspection | 1. Check | Requisitioning | Requesting Element/Unit Location |
| A. Inspection | 1. Check | Requisitioning | Request Quantity |
| A. Inspection | 1. Check | Item Catalog Information | Model |
| A. Inspection | 2. Test | Item Attributes | Calibrated Date |
| A. Inspection | 2. Test | Item Attributes | Calibration Due Date |
| A. Inspection | 2. Test | Item Attributes | Calibration Status Code |
| A. Inspection | 2. Test | Item Attributes | Calibration Type |
| A. Inspection | 2. Test | Item Attributes | Last Odometer Reading |
| A. Inspection | 2. Test | Item Attributes | Last Operating Hour Reading |
| A. Inspection | 2. Test | Item Attributes | Last Round Count |
| A. Inspection | 2. Test | Item Attributes | Equipment Operation Time Code |
| A. Inspection | 2. Test | Item Catalog Information | Operational Test Code |
| A. Inspection | 2. Test | Item Identification | Item Unique Identification |
| A. Inspection | 2. Test | Item Identification | Lot Number |
| A. Inspection | 2. Test | Item Identification | Serial Number |
| A. Inspection | 2. Test | Item Attributes | Assigned Activity |
| A. Inspection | 2. Test | Activity | Department of Defense Activity Address Code |
| A. Inspection | 2. Test | Item Catalog Information | Table of Authorized Material Control Number |
| A. Inspection | 2. Test | Item Catalog Information | National Stock Number |
| A. Inspection | 2. Test | Item Catalog Information | Identification Number |
| A. Inspection | 2. Test | Item Attributes | Assigned Activity |
| A. Inspection | 2. Test | Item Attributes | Assigned Subordinate Activity |
| A. Inspection | 2. Test | Requisitioning | Priority Designator |
| A. Inspection | 2. Test | Requisitioning | Request ID/Tracking Number |
| A. Inspection | 2. Test | Requisitioning | Request Type |
| A. Inspection | 2. Test | Requisitioning | Requesting Element/Unit |
| A. Inspection | 2. Test | Requisitioning | Requesting Element/Unit Location |
| A. Inspection | 2. Test | Requisitioning | Request Quantity |
| A. Inspection | 2. Test | Item Catalog Information | Model |
| B. Classification | Assign Maintenance Category | Item Attributes | Corrosion Classification |
| B. Classification | Assign Maintenance Category | Item Attributes | Corrosion Inspection Date |
| B. Classification | Assign Maintenance Category | Item Identification | Item Unique Identification |
| B. Classification | Assign Maintenance Category | Item Identification | Lot Number |
| B. Classification | Assign Maintenance Category | Item Identification | Serial Number |
| B. Classification | Assign Maintenance Category | Item Attributes | Assigned Activity |
| B. Classification | Assign Maintenance Category | Activity | Department of Defense Activity Address Code |
| B. Classification | Assign Maintenance Category | Item Catalog Information | Table of Authorized Material Control Number |
| B. Classification | Assign Maintenance Category | Item Catalog Information | National Stock Number |
| B. Classification | Assign Maintenance Category | Item Catalog Information | Identification Number |
| B. Classification | Assign Maintenance Category | Item Attributes | Assigned Activity |
| B. Classification | Assign Maintenance Category | Item Attributes | Assigned Subordinate Activity |
| B. Classification | Assign Maintenance Category | Requisitioning | Priority Designator |

| | | | |
|---|---|---|---|
| B. Classification | Assign Maintenance Category | Requisitioning | Request ID/Tracking Number |
| B. Classification | Assign Maintenance Category | Requisitioning | Request Type |
| B. Classification | Assign Maintenance Category | Requisitioning | Requesting Element/Unit |
| B. Classification | Assign Maintenance Category | Requisitioning | Requesting Element/Unit Location |
| B. Classification | Assign Maintenance Category | Requisitioning | Request Quantity |
| B. Classification | Assign Maintenance Category | Item Identification | Item Unique Identification |
| B. Classification | Assign Maintenance Category | Item Identification | Lot Number |
| B. Classification | Assign Maintenance Category | Item Identification | Serial Number |
| B. Classification | Assign Maintenance Category | Item Attributes | Assigned Activity |
| B. Classification | Assign Maintenance Category | Activity | Department of Defense Activity Address Code |
| B. Classification | Assign Maintenance Category | Item Catalog Information | Table of Authorized Material Control Number |
| B. Classification | Assign Maintenance Category | Item Catalog Information | National Stock Number |
| B. Classification | Assign Maintenance Category | Item Catalog Information | Identification Number |
| B. Classification | Assign Maintenance Category | Item Attributes | Assigned Activity |
| B. Classification | Assign Maintenance Category | Item Attributes | Assigned Subordinate Activity |
| B. Classification | Assign Maintenance Category | Item Attributes | Operational Status |
| B. Classification | Assign Maintenance Category | Item Attributes | Model |
| B. Classification | Assign Maintenance Category | Item Attributes | Major Problem Code, Primary |
| B. Classification | Assign Maintenance Category | Item Attributes | Major Problem Code, Secondary |
| E. Servicing | 1. Minor Corrections | Activity | Department of Defense Activity Address Code |
| E. Servicing | 1. Minor Corrections | Item Catalog Information | Table of Authorized Material Control Number |
| E. Servicing | 1. Minor Corrections | Item Catalog Information | National Stock Number |
| E. Servicing | 1. Minor Corrections | Item Catalog Information | Identification Number |
| E. Servicing | 1. Minor Corrections | Item Attributes | Assigned Activity |
| E. Servicing | 1. Minor Corrections | Item Attributes | Assigned Subordinate Activity |
| E. Servicing | 1. Minor Corrections | Requisitioning | Priority Designator |
| E. Servicing | 1. Minor Corrections | Requisitioning | Request ID/Tracking Number |
| E. Servicing | 1. Minor Corrections | Requisitioning | Request Type |
| E. Servicing | 1. Minor Corrections | Requisitioning | Requesting Element/Unit |
| E. Servicing | 1. Minor Corrections | Requisitioning | Requesting Element/Unit Location |
| E. Servicing | 1. Minor Corrections | Requisitioning | Request Quantity |
| E. Servicing | 1. Minor Corrections | Item Identification | Item Unique Identification |
| E. Servicing | 1. Minor Corrections | Item Identification | Lot Number |
| E. Servicing | 1. Minor Corrections | Item Identification | Serial Number |
| E. Servicing | 1. Minor Corrections | Item Attributes | Assigned Activity |
| E. Servicing | 1. Minor Corrections | Activity | Department of Defense Activity Address Code |
| E. Servicing | 1. Minor Corrections | Item Catalog Information | Table of Authorized Material Control Number |
| E. Servicing | 1. Minor Corrections | Item Catalog Information | National Stock Number |
| E. Servicing | 1. Minor Corrections | Item Catalog Information | Identification Number |
| E. Servicing | 1. Minor Corrections | Item Attributes | Assigned Activity |
| E. Servicing | 1. Minor Corrections | Item Attributes | Assigned Subordinate Activity |
| E. Servicing | 1. Minor Corrections | Item Attributes | Operational Status |
| E. Servicing | 1. Minor Corrections | Item Attributes | Model |
| E. Servicing | 1. Minor Corrections | Item Attributes | Corrosion Classification |
| E. Servicing | 1. Minor Corrections | Item Attributes | Corrosion Inspection Date |
| E. Servicing | 1. Minor Corrections | Item Identification | Item Unique Identification |
| E. Servicing | 1. Minor Corrections | Item Identification | Lot Number |
| E. Servicing | 1. Minor Corrections | Item Identification | Serial Number |
| E. Servicing | 1. Minor Corrections | Item Attributes | Assigned Activity |
| E. Servicing | 1. Minor Corrections | Predictive Indicator | Fault/Failure Detection |
| E. Servicing | 1. Minor Corrections | Predictive Indicator | Condition Forecast |
| E. Servicing | 1. Minor Corrections | Predictive Indicator | Out of Tolerance/Baseline |
| E. Servicing | 1. Minor Corrections | Item Attributes | Equipment Operation Time Code |
| E. Servicing | 1. Minor Corrections | Item Catalog Information | Operational Test Code |
| E. Servicing | 1. Minor Corrections | Item Attributes | Warranty Expiration Date |
| E. Servicing | 1. Minor Corrections | Item Attributes | Warranty Issuing Entity |
| E. Servicing | 1. Minor Corrections | Item Attributes | Warranty Manager Name |
| E. Servicing | 1. Minor Corrections | Item Attributes | Warranty Managing Entity |
| E. Servicing | 1. Minor Corrections | Item Attributes | Warranty Policy Number |
| E. Servicing | 1. Minor Corrections | Item Attributes | Major Problem Code, Primary |
| E. Servicing | 1. Minor Corrections | Item Attributes | Major Problem Code, Secondary |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Last Odometer Reading |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Last Operating Hour Reading |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Last Round Count |
| E. Servicing | 2. Preventative Maintenance | Activity | Department of Defense Activity Address Code |

| | | | |
|---|---|---|---|
| E. Servicing | 2. Preventative Maintenance | Item Catalog Information | Table of Authorized Material Control Number |
| E. Servicing | 2. Preventative Maintenance | Item Catalog Information | National Stock Number |
| E. Servicing | 2. Preventative Maintenance | Item Catalog Information | Identification Number |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Assigned Activity |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Assigned Subordinate Activity |
| E. Servicing | 2. Preventative Maintenance | Requisitioning | Priority Designator |
| E. Servicing | 2. Preventative Maintenance | Requisitioning | Request ID/Tracking Number |
| E. Servicing | 2. Preventative Maintenance | Requisitioning | Request Type |
| E. Servicing | 2. Preventative Maintenance | Requisitioning | Requesting Element/Unit |
| E. Servicing | 2. Preventative Maintenance | Requisitioning | Requesting Element/Unit Location |
| E. Servicing | 2. Preventative Maintenance | Requisitioning | Request Quantity |
| E. Servicing | 2. Preventative Maintenance | Item Identification | Item Unique Identification |
| E. Servicing | 2. Preventative Maintenance | Item Identification | Lot Number |
| E. Servicing | 2. Preventative Maintenance | Item Identification | Serial Number |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Assigned Activity |
| E. Servicing | 2. Preventative Maintenance | Activity | Department of Defense Activity Address Code |
| E. Servicing | 2. Preventative Maintenance | Item Catalog Information | Table of Authorized Material Control Number |
| E. Servicing | 2. Preventative Maintenance | Item Catalog Information | National Stock Number |
| E. Servicing | 2. Preventative Maintenance | Item Catalog Information | Identification Number |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Assigned Activity |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Assigned Subordinate Activity |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Operational Status |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Model |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Corrosion Classification |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Corrosion Inspection Date |
| E. Servicing | 2. Preventative Maintenance | Item Identification | Item Unique Identification |
| E. Servicing | 2. Preventative Maintenance | Item Identification | Lot Number |
| E. Servicing | 2. Preventative Maintenance | Item Identification | Serial Number |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Assigned Activity |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Warranty Expiration Date |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Warranty Issuing Entity |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Warranty Manager Name |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Warranty Managing Entity |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Warranty Policy Number |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Major Problem Code, Primary |
| E. Servicing | 2. Preventative Maintenance | Item Attributes | Major Problem Code, Secondary |
| F. Repair | Correct Specific Failure | Predictive Indicator | Fault/Failure Detection |
| F. Repair | Correct Specific Failure | Predictive Indicator | Condition Forecast |
| F. Repair | Correct Specific Failure | Predictive Indicator | Out of Tolerance/Baseline |
| F. Repair | Correct Specific Failure | Item Identification | Item Unique Identification |
| F. Repair | Correct Specific Failure | Item Identification | Lot Number |
| F. Repair | Correct Specific Failure | Item Identification | Serial Number |
| F. Repair | Correct Specific Failure | Item Attributes | Assigned Activity |
| F. Repair | Correct Specific Failure | Activity | Department of Defense Activity Address Code |
| F. Repair | Correct Specific Failure | Item Catalog Information | Table of Authorized Material Control Number |
| F. Repair | Correct Specific Failure | Item Catalog Information | National Stock Number |
| F. Repair | Correct Specific Failure | Item Catalog Information | Identification Number |
| F. Repair | Correct Specific Failure | Item Attributes | Assigned Activity |
| F. Repair | Correct Specific Failure | Item Attributes | Assigned Subordinate Activity |
| F. Repair | Correct Specific Failure | Requisitioning | Priority Designator |
| F. Repair | Correct Specific Failure | Requisitioning | Request ID/Tracking Number |
| F. Repair | Correct Specific Failure | Requisitioning | Request Type |
| F. Repair | Correct Specific Failure | Requisitioning | Requesting Element/Unit |
| F. Repair | Correct Specific Failure | Requisitioning | Requesting Element/Unit Location |
| F. Repair | Correct Specific Failure | Requisitioning | Request Quantity |
| F. Repair | Correct Specific Failure | Item Catalog Information | Model |
| F. Repair | Correct Specific Failure | Item Attributes | Warranty Expiration Date |
| F. Repair | Correct Specific Failure | Item Attributes | Warranty Issuing Entity |
| F. Repair | Correct Specific Failure | Item Attributes | Warranty Manager Name |
| F. Repair | Correct Specific Failure | Item Attributes | Warranty Managing Entity |
| F. Repair | Correct Specific Failure | Item Attributes | Warranty Policy Number |
| F. Repair | Correct Specific Failure | Item Attributes | Major Problem Code, Primary |
| F. Repair | Correct Specific Failure | Item Attributes | Major Problem Code, Secondary |
| G. Modification | Change Design, Function, or Form | Item Identification | Item Unique Identification |
| G. Modification | Change Design, Function, or Form | Item Identification | Lot Number |

| | | | |
|---|---|---|---|
| G. Modification | Change Design, Function, or Form | Item Identification | Serial Number |
| G. Modification | Change Design, Function, or Form | Item Attributes | Assigned Activity |
| G. Modification | Change Design, Function, or Form | Activity | Department of Defense Activity Address Code |
| G. Modification | Change Design, Function, or Form | Item Catalog Information | Table of Authorized Material Control Number |
| G. Modification | Change Design, Function, or Form | Item Catalog Information | National Stock Number |
| G. Modification | Change Design, Function, or Form | Item Catalog Information | Identification Number |
| G. Modification | Change Design, Function, or Form | Item Attributes | Assigned Activity |
| G. Modification | Change Design, Function, or Form | Item Attributes | Assigned Subordinate Activity |
| G. Modification | Change Design, Function, or Form | Requisitioning | Priority Designator |
| G. Modification | Change Design, Function, or Form | Requisitioning | Request ID/Tracking Number |
| G. Modification | Change Design, Function, or Form | Requisitioning | Request Type |
| G. Modification | Change Design, Function, or Form | Requisitioning | Requesting Element/Unit |
| G. Modification | Change Design, Function, or Form | Requisitioning | Requesting Element/Unit Location |
| G. Modification | Change Design, Function, or Form | Requisitioning | Request Quantity |
| G. Modification | Change Design, Function, or Form | Item Catalog Information | Model |
| G. Modification | Change Design, Function, or Form | Item Attributes | Warranty Expiration Date |
| G. Modification | Change Design, Function, or Form | Item Attributes | Warranty Issuing Entity |
| G. Modification | Change Design, Function, or Form | Item Attributes | Warranty Manager Name |
| G. Modification | Change Design, Function, or Form | Item Attributes | Warranty Managing Entity |
| G. Modification | Change Design, Function, or Form | Item Attributes | Warranty Policy Number |
| G. Modification | Change Design, Function, or Form | Item Attributes | Modification Status |
| G. Modification | Change Design, Function, or Form | Item Attributes | Major Problem Code, Primary |
| G. Modification | Change Design, Function, or Form | Item Attributes | Major Problem Code, Secondary |
| K. Rebuild | --Depot Level Non-Organic | None | None |
| J. Overhaul | --Depot Level Non-Organic | None | None |
| H. Reclamation | Repair to Return to Supply | Item Identification | Item Unique Identification |
| H. Reclamation | Repair to Return to Supply | Item Identification | Lot Number |
| H. Reclamation | Repair to Return to Supply | Item Identification | Serial Number |
| H. Reclamation | Repair to Return to Supply | Activity | Department of Defense Activity Address Code |
| H. Reclamation | Repair to Return to Supply | Item Catalog Information | Table of Authorized Material Control Number |
| H. Reclamation | Repair to Return to Supply | Item Catalog Information | National Stock Number |
| H. Reclamation | Repair to Return to Supply | Item Catalog Information | Identification Number |
| H. Reclamation | Repair to Return to Supply | Requisitioning | Priority Designator |
| H. Reclamation | Repair to Return to Supply | Requisitioning | Request ID/Tracking Number |
| H. Reclamation | Repair to Return to Supply | Requisitioning | Request Type |
| H. Reclamation | Repair to Return to Supply | Requisitioning | Requesting Element/Unit |
| H. Reclamation | Repair to Return to Supply | Requisitioning | Requesting Element/Unit Location |
| H. Reclamation | Repair to Return to Supply | Requisitioning | Request Quantity |
| H. Reclamation | Repair to Return to Supply | Item Catalog Information | Model |
| H. Reclamation | Repair to Return to Supply | Inventory | Supply Condition Code |
| H. Reclamation | Repair to Return to Supply | Item Attributes | Major Problem Code, Primary |
| H. Reclamation | Repair to Return to Supply | Item Attributes | Major Problem Code, Secondary |
| I. Recovery | 2. Retrieve | Requisitioning | Priority Designator |
| I. Recovery | 2. Retrieve | Requisitioning | Request ID/Tracking Number |
| I. Recovery | 2. Retrieve | Requisitioning | Request Type |
| I. Recovery | 2. Retrieve | Requisitioning | Requesting Element/Unit |
| I. Recovery | 2. Retrieve | Requisitioning | Requesting Element/Unit Location |
| I. Recovery | 2. Retrieve | Requisitioning | Request Quantity |
| I. Recovery | 2. Retrieve | Item Identification | Item Unique Identification |
| I. Recovery | 2. Retrieve | Item Identification | Lot Number |
| I. Recovery | 2. Retrieve | Item Identification | Serial Number |
| I. Recovery | 2. Retrieve | Activity | Department of Defense Activity Address Code |
| I. Recovery | 2. Retrieve | Item Catalog Information | Table of Authorized Material Control Number |
| I. Recovery | 2. Retrieve | Item Catalog Information | National Stock Number |
| I. Recovery | 2. Retrieve | Item Catalog Information | Identification Number |
| I. Recovery | 2. Retrieve | Item Attributes | Assigned Activity |
| I. Recovery | 2. Retrieve | Manifest | Mission Number |
| I. Recovery | 2. Retrieve | Manifest | Priority |
| I. Recovery | 2. Retrieve | Manifest | Series |
| I. Recovery | 2. Retrieve | Manifest | Personnel Listing |
| I. Recovery | 2. Retrieve | Manifest | Cargo Listing |
| I. Recovery | 2. Retrieve | Item Catalog Information | National Motor Freight Classification |
| I. Recovery | 2. Retrieve | Item Characteristics | Shipping Height |
| I. Recovery | 2. Retrieve | Item Characteristics | Shipping Length |
| I. Recovery | 2. Retrieve | Item Characteristics | Shipping Width |

| | | | |
|---|---|---|---|
| I. Recovery | 2. Retrieve | Item Characteristics | Net Weight |
| I. Recovery | 2. Retrieve | Location, Physical | Destination Grid |
| I. Recovery | 1. Free from Immobility | Requisitioning | Priority Designator |
| I. Recovery | 1. Free from Immobility | Requisitioning | Request ID/Tracking Number |
| I. Recovery | 1. Free from Immobility | Requisitioning | Request Type |
| I. Recovery | 1. Free from Immobility | Requisitioning | Requesting Element/Unit |
| I. Recovery | 1. Free from Immobility | Requisitioning | Requesting Element/Unit Location |
| I. Recovery | 1. Free from Immobility | Requisitioning | Request Quantity |
| I. Recovery | 1. Free from Immobility | Item Identification | Item Unique Identification |
| I. Recovery | 1. Free from Immobility | Item Identification | Lot Number |
| I. Recovery | 1. Free from Immobility | Item Identification | Serial Number |
| I. Recovery | 1. Free from Immobility | Activity | Department of Defense Activity Address Code |
| I. Recovery | 1. Free from Immobility | Item Catalog Information | Table of Authorized Material Control Number |
| I. Recovery | 1. Free from Immobility | Item Catalog Information | National Stock Number |
| I. Recovery | 1. Free from Immobility | Item Catalog Information | Identification Number |
| I. Recovery | 1. Free from Immobility | Item Attributes | Assigned Activity |
| I. Recovery | 1. Free from Immobility | Manifest | Mission Number |
| I. Recovery | 1. Free from Immobility | Manifest | Priority |
| I. Recovery | 1. Free from Immobility | Manifest | Series |
| I. Recovery | 1. Free from Immobility | Manifest | Personnel Listing |
| I. Recovery | 1. Free from Immobility | Manifest | Cargo Listing |
| C. Evacuation | Move to Higher | Requisitioning | Priority Designator |
| C. Evacuation | Move to Higher | Requisitioning | Request ID/Tracking Number |
| C. Evacuation | Move to Higher | Requisitioning | Request Type |
| C. Evacuation | Move to Higher | Requisitioning | Requesting Element/Unit |
| C. Evacuation | Move to Higher | Requisitioning | Requesting Element/Unit Location |
| C. Evacuation | Move to Higher | Requisitioning | Request Quantity |
| C. Evacuation | Move to Higher | Item Identification | Item Unique Identification |
| C. Evacuation | Move to Higher | Item Identification | Lot Number |
| C. Evacuation | Move to Higher | Item Identification | Serial Number |
| C. Evacuation | Move to Higher | Activity | Department of Defense Activity Address Code |
| C. Evacuation | Move to Higher | Item Catalog Information | Table of Authorized Material Control Number |
| C. Evacuation | Move to Higher | Item Catalog Information | National Stock Number |
| C. Evacuation | Move to Higher | Item Catalog Information | Identification Number |
| C. Evacuation | Move to Higher | Item Attributes | Assigned Activity |
| C. Evacuation | Move to Higher | Manifest | Mission Number |
| C. Evacuation | Move to Higher | Manifest | Priority |
| C. Evacuation | Move to Higher | Manifest | Series |
| C. Evacuation | Move to Higher | Manifest | Personnel Listing |
| C. Evacuation | Move to Higher | Manifest | Cargo Listing |
| C. Evacuation | Move to Higher | Item Catalog Information | National Motor Freight Classification |
| C. Evacuation | Move to Higher | Item Characteristics | Shipping Height |
| C. Evacuation | Move to Higher | Item Characteristics | Shipping Length |
| C. Evacuation | Move to Higher | Item Characteristics | Shipping Width |
| C. Evacuation | Move to Higher | Item Characteristics | Net Weight |
| C. Evacuation | Move to Higher | Location, Physical | Destination Grid |
| C. Evacuation | Move to Higher | Item Attributes | Operational Status |
| C. Evacuation | Move to Higher | Item Attributes | Major Problem Code, Primary |
| C. Evacuation | Move to Higher | Item Attributes | Major Problem Code, Secondary |
| D. Calibration | Calibration | Activity | Department of Defense Activity Address Code |
| D. Calibration | Calibration | Item Catalog Information | Table of Authorized Material Control Number |
| D. Calibration | Calibration | Item Catalog Information | National Stock Number |
| D. Calibration | Calibration | Item Catalog Information | Identification Number |
| D. Calibration | Calibration | Item Attributes | Assigned Activity |
| D. Calibration | Calibration | Item Attributes | Assigned Subordinate Activity |
| D. Calibration | Calibration | Requisitioning | Priority Designator |
| D. Calibration | Calibration | Requisitioning | Request ID/Tracking Number |
| D. Calibration | Calibration | Requisitioning | Request Type |
| D. Calibration | Calibration | Requisitioning | Requesting Element/Unit |
| D. Calibration | Calibration | Requisitioning | Requesting Element/Unit Location |
| D. Calibration | Calibration | Requisitioning | Request Quantity |
| D. Calibration | Calibration | Item Identification | Item Unique Identification |
| D. Calibration | Calibration | Item Identification | Lot Number |
| D. Calibration | Calibration | Item Identification | Serial Number |
| D. Calibration | Calibration | Item Attributes | Assigned Activity |

| | | | |
|---|---|---|---|
| D. Calibration | Calibration | Activity | Department of Defense Activity Address Code |
| D. Calibration | Calibration | Item Catalog Information | Table of Authorized Material Control Number |
| D. Calibration | Calibration | Item Catalog Information | National Stock Number |
| D. Calibration | Calibration | Item Catalog Information | Identification Number |
| D. Calibration | Calibration | Item Attributes | Assigned Activity |
| D. Calibration | Calibration | Item Attributes | Assigned Subordinate Activity |
| D. Calibration | Calibration | Item Attributes | Operational Status |
| D. Calibration | Calibration | Item Attributes | Model |
| D. Calibration | Calibration | Item Attributes | Corrosion Classification |
| D. Calibration | Calibration | Item Attributes | Corrosion Inspection Date |
| D. Calibration | Calibration | Item Identification | Item Unique Identification |
| D. Calibration | Calibration | Item Identification | Lot Number |
| D. Calibration | Calibration | Item Identification | Serial Number |
| D. Calibration | Calibration | Item Attributes | Assigned Activity |
| D. Calibration | Calibration | Predictive Indicator | Fault/Failure Detection |
| D. Calibration | Calibration | Predictive Indicator | Condition Forecast |
| D. Calibration | Calibration | Predictive Indicator | Out of Tolerance/Baseline |
| D. Calibration | Calibration | Item Attributes | Calibrated Date |
| D. Calibration | Calibration | Item Attributes | Calibration Due Date |
| D. Calibration | Calibration | Item Attributes | Calibration Status Code |
| D. Calibration | Calibration | Item Attributes | Calibration Type |
| D. Calibration | Calibration | Item Attributes | Equipment Operation Time Code |
| D. Calibration | Calibration | Item Catalog Information | Operational Test Code |
| D. Calibration | Calibration | Item Attributes | Warranty Expiration Date |
| D. Calibration | Calibration | Item Attributes | Warranty Issuing Entity |
| D. Calibration | Calibration | Item Attributes | Warranty Manager Name |
| D. Calibration | Calibration | Item Attributes | Warranty Managing Entity |
| D. Calibration | Calibration | Item Attributes | Warranty Policy Number |
| D. Calibration | Calibration | Item Attributes | Major Problem Code, Primary |
| D. Calibration | Calibration | Item Attributes | Major Problem Code, Secondary |

# Appendix C
# Mobility Function Data Points

| Sub-Function 1 | Sub-Function 2 | Data Type | Line Data Level |
|---|---|---|---|
| A. Embarkation | Put Plan | Activity | Department of Defense Activity Address Code |
| A. Embarkation | Put Plan | Item Catalog Information | Table of Authorized Material Control Number |
| A. Embarkation | Put Plan | Item Catalog Information | National Stock Number |
| A. Embarkation | Put Plan | Item Catalog Information | Unit of Issue |
| A. Embarkation | Put Plan | Item Catalog Information | Unit Price |
| A. Embarkation | Put Plan | Item Catalog Information | Nomenclature |
| A. Embarkation | Put Plan | Item Catalog Information | Class of Supply |
| A. Embarkation | Put Plan | Item Catalog Information | Sub-Class of Supply |
| A. Embarkation | Put Plan | Item Identification | Item Unique Identification |
| A. Embarkation | Put Plan | Item Identification | Lot Number |
| A. Embarkation | Put Plan | Item Identification | Serial Number |
| A. Embarkation | Put Plan | Item Attributes | Quantity |
| A. Embarkation | Put Plan | Item Attributes | Assigned Activity |
| A. Embarkation | Put Plan | Item Attributes | Assigned Subordinate Activity |
| A. Embarkation | Put Plan | Item Catalog Information | Model |
| A. Embarkation | Put Plan | Item Catalog Information | National Motor Freight Classification |
| A. Embarkation | Put Plan | Item Characteristics | Shipping Height |
| A. Embarkation | Put Plan | Item Characteristics | Shipping Length |
| A. Embarkation | Put Plan | Item Characteristics | Shipping Width |
| A. Embarkation | Put Plan | Item Characteristics | Net Weight |
| A. Embarkation | Put Plan | Item Characteristics | Gross Weight |
| A. Embarkation | Put Plan | Item Characteristics | Tare (Empty) Weight |
| A. Embarkation | Put Plan | Transportation | Cargo Container ID |
| A. Embarkation | Put Plan | Transportation | Cargo Container Type |
| B. Landing Support | 1. Ship to Shore | Activity | Department of Defense Activity Address Code |
| B. Landing Support | 1. Ship to Shore | Item Catalog Information | Table of Authorized Material Control Number |
| B. Landing Support | 1. Ship to Shore | Item Catalog Information | National Stock Number |
| B. Landing Support | 1. Ship to Shore | Item Catalog Information | Unit of Issue |
| B. Landing Support | 1. Ship to Shore | Item Catalog Information | Nomenclature |
| B. Landing Support | 1. Ship to Shore | Item Identification | Item Unique Identification |
| B. Landing Support | 1. Ship to Shore | Item Identification | Lot Number |
| B. Landing Support | 1. Ship to Shore | Item Identification | Serial Number |
| B. Landing Support | 1. Ship to Shore | Item Attributes | Quantity |
| B. Landing Support | 1. Ship to Shore | Item Attributes | Assigned Activity |
| B. Landing Support | 1. Ship to Shore | Item Attributes | Assigned Subordinate Activity |
| B. Landing Support | 1. Ship to Shore | Item Catalog Information | Model |
| B. Landing Support | 1. Ship to Shore | Item Catalog Information | National Motor Freight Classification |
| B. Landing Support | 1. Ship to Shore | Item Characteristics | Shipping Height |
| B. Landing Support | 1. Ship to Shore | Item Characteristics | Shipping Length |
| B. Landing Support | 1. Ship to Shore | Item Characteristics | Shipping Width |
| B. Landing Support | 1. Ship to Shore | Item Characteristics | Net Weight |
| B. Landing Support | 1. Ship to Shore | Item Characteristics | Gross Weight |
| B. Landing Support | 1. Ship to Shore | Item Characteristics | Tare (Empty) Weight |
| B. Landing Support | 1. Ship to Shore | Transportation | Cargo Container ID |
| B. Landing Support | 1. Ship to Shore | Transportation | Cargo Container Type |
| B. Landing Support | 1. Ship to Shore | Manifest | Bump Plan |
| B. Landing Support | 1. Ship to Shore | Manifest | Chalk |
| B. Landing Support | 1. Ship to Shore | Manifest | Mission Number |
| B. Landing Support | 1. Ship to Shore | Manifest | Platform |
| B. Landing Support | 1. Ship to Shore | Manifest | Platform ID |
| B. Landing Support | 1. Ship to Shore | Manifest | Platform Type |
| B. Landing Support | 1. Ship to Shore | Manifest | Priority |
| B. Landing Support | 1. Ship to Shore | Manifest | Series |
| B. Landing Support | 1. Ship to Shore | Manifest | Stick |
| B. Landing Support | 1. Ship to Shore | Individual Member | Zap Number |
| B. Landing Support | 1. Ship to Shore | Individual Member | Rank |
| B. Landing Support | 1. Ship to Shore | Individual Member | Name, First |
| B. Landing Support | 1. Ship to Shore | Individual Member | Name, Last |
| B. Landing Support | 1. Ship to Shore | Individual Member | Electronic Data Interchange Personal Identifier |

| | | | |
|---|---|---|---|
| B. Landing Support | 2. Shore to Ship | Activity | Department of Defense Activity Address Code |
| B. Landing Support | 2. Shore to Ship | Item Catalog Information | Table of Authorized Material Control Number |
| B. Landing Support | 2. Shore to Ship | Item Catalog Information | National Stock Number |
| B. Landing Support | 2. Shore to Ship | Item Catalog Information | Unit of Issue |
| B. Landing Support | 2. Shore to Ship | Item Catalog Information | Nomenclature |
| B. Landing Support | 2. Shore to Ship | Item Identification | Item Unique Identification |
| B. Landing Support | 2. Shore to Ship | Item Identification | Lot Number |
| B. Landing Support | 2. Shore to Ship | Item Identification | Serial Number |
| B. Landing Support | 2. Shore to Ship | Item Attributes | Quantity |
| B. Landing Support | 2. Shore to Ship | Item Attributes | Assigned Activity |
| B. Landing Support | 2. Shore to Ship | Item Attributes | Assigned Subordinate Activity |
| B. Landing Support | 2. Shore to Ship | Item Catalog Information | Model |
| B. Landing Support | 2. Shore to Ship | Item Catalog Information | National Motor Freight Classification |
| B. Landing Support | 2. Shore to Ship | Item Characteristics | Shipping Height |
| B. Landing Support | 2. Shore to Ship | Item Characteristics | Shipping Length |
| B. Landing Support | 2. Shore to Ship | Item Characteristics | Shipping Width |
| B. Landing Support | 2. Shore to Ship | Item Characteristics | Net Weight |
| B. Landing Support | 2. Shore to Ship | Item Characteristics | Gross Weight |
| B. Landing Support | 2. Shore to Ship | Item Characteristics | Tare (Empty) Weight |
| B. Landing Support | 2. Shore to Ship | Transportation | Cargo Container ID |
| B. Landing Support | 2. Shore to Ship | Transportation | Cargo Container Type |
| B. Landing Support | 2. Shore to Ship | Manifest | Bump Plan |
| B. Landing Support | 2. Shore to Ship | Manifest | Chalk |
| B. Landing Support | 2. Shore to Ship | Manifest | Mission Number |
| B. Landing Support | 2. Shore to Ship | Manifest | Platform |
| B. Landing Support | 2. Shore to Ship | Manifest | Platform ID |
| B. Landing Support | 2. Shore to Ship | Manifest | Platform Type |
| B. Landing Support | 2. Shore to Ship | Manifest | Priority |
| B. Landing Support | 2. Shore to Ship | Manifest | Series |
| B. Landing Support | 2. Shore to Ship | Manifest | Stick |
| B. Landing Support | 2. Shore to Ship | Individual Member | Zap Number |
| B. Landing Support | 2. Shore to Ship | Individual Member | Rank |
| B. Landing Support | 2. Shore to Ship | Individual Member | Name, First |
| B. Landing Support | 2. Shore to Ship | Individual Member | Name, Last |
| B. Landing Support | 2. Shore to Ship | Individual Member | Electronic Data Interchange Personal Identifier |
| C. Motor Transport | 2. Dispatch | Requisitioning | Priority Designator |
| C. Motor Transport | 2. Dispatch | Requisitioning | Request ID/Tracking Number |
| C. Motor Transport | 2. Dispatch | Requisitioning | Request Type |
| C. Motor Transport | 2. Dispatch | Requisitioning | Requesting Element/Unit |
| C. Motor Transport | 2. Dispatch | Requisitioning | Requesting Element/Unit Location |
| C. Motor Transport | 2. Dispatch | Requisitioning | Request Quantity |
| C. Motor Transport | 2. Dispatch | Individual Member | Zap Number |
| C. Motor Transport | 2. Dispatch | Individual Member | Rank |
| C. Motor Transport | 2. Dispatch | Individual Member | Name, First |
| C. Motor Transport | 2. Dispatch | Individual Member | Name, Last |
| C. Motor Transport | 2. Dispatch | Individual Member | Electronic Data Interchange Personal Identifier |
| C. Motor Transport | 2. Dispatch | Individual Member | Licensing |
| C. Motor Transport | 2. Dispatch | Individual Member | Certifications |
| C. Motor Transport | 2. Dispatch | Item Catalog Information | Model |
| C. Motor Transport | 2. Dispatch | Item Catalog Information | National Motor Freight Classification |
| C. Motor Transport | 2. Dispatch | Activity | Department of Defense Activity Address Code |
| C. Motor Transport | 2. Dispatch | Item Catalog Information | Table of Authorized Material Control Number |
| C. Motor Transport | 2. Dispatch | Item Catalog Information | National Stock Number |
| C. Motor Transport | 2. Dispatch | Item Catalog Information | Nomenclature |
| C. Motor Transport | 2. Dispatch | Item Identification | Item Unique Identification |
| C. Motor Transport | 2. Dispatch | Item Identification | Serial Number |
| C. Motor Transport | 2. Dispatch | Item Attributes | Operational Status |
| C. Motor Transport | 2. Dispatch | Item Catalog Information | Model |
| C. Motor Transport | 2. Dispatch | Item Characteristics | Fuel Type |
| C. Motor Transport | 2. Dispatch | Item Characteristics | On Board Fuel Tank Capacity (Volume) |
| C. Motor Transport | 2. Dispatch | Manifest | Mission Number |
| C. Motor Transport | 2. Dispatch | Manifest | Priority |
| C. Motor Transport | 2. Dispatch | Manifest | Series |
| C. Motor Transport | 2. Dispatch | Manifest | Personnel Listing |
| C. Motor Transport | 2. Dispatch | Manifest | Cargo Listing |

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|---|---|---|---|
| C. Motor Transport | 1. Capacity Planning | Requisitioning | Priority Designator |
| C. Motor Transport | 1. Capacity Planning | Requisitioning | Request ID/Tracking Number |
| C. Motor Transport | 1. Capacity Planning | Requisitioning | Request Type |
| C. Motor Transport | 1. Capacity Planning | Requisitioning | Requesting Element/Unit |
| C. Motor Transport | 1. Capacity Planning | Requisitioning | Requesting Element/Unit Location |
| C. Motor Transport | 1. Capacity Planning | Requisitioning | Request Quantity |
| C. Motor Transport | 1. Capacity Planning | Item Catalog Information | Table of Authorized Material Control Number |
| C. Motor Transport | 1. Capacity Planning | Item Catalog Information | National Stock Number |
| C. Motor Transport | 1. Capacity Planning | Item Catalog Information | Unit of Issue |
| C. Motor Transport | 1. Capacity Planning | Item Catalog Information | Nomenclature |
| C. Motor Transport | 1. Capacity Planning | Item Identification | Item Unique Identification |
| C. Motor Transport | 1. Capacity Planning | Item Identification | Lot Number |
| C. Motor Transport | 1. Capacity Planning | Item Identification | Serial Number |
| C. Motor Transport | 1. Capacity Planning | Item Attributes | Quantity |
| C. Motor Transport | 1. Capacity Planning | Item Attributes | Assigned Activity |
| C. Motor Transport | 1. Capacity Planning | Item Attributes | Assigned Subordinate Activity |
| C. Motor Transport | 1. Capacity Planning | Item Catalog Information | Model |
| C. Motor Transport | 1. Capacity Planning | Item Catalog Information | National Motor Freight Classification |
| C. Motor Transport | 1. Capacity Planning | Item Characteristics | Shipping Height |
| C. Motor Transport | 1. Capacity Planning | Item Characteristics | Shipping Length |
| C. Motor Transport | 1. Capacity Planning | Item Characteristics | Shipping Width |
| C. Motor Transport | 1. Capacity Planning | Item Characteristics | Net Weight |
| C. Motor Transport | 1. Capacity Planning | Item Characteristics | Gross Weight |
| C. Motor Transport | 1. Capacity Planning | Item Characteristics | Tare (Empty) Weight |
| C. Motor Transport | 1. Capacity Planning | Transportation | Cargo Container ID |
| C. Motor Transport | 1. Capacity Planning | Transportation | Cargo Container Type |
| C. Motor Transport | 1. Capacity Planning | Manifest | Mission Number |
| C. Motor Transport | 1. Capacity Planning | Manifest | Priority |
| C. Motor Transport | 1. Capacity Planning | Manifest | Series |
| C. Motor Transport | 1. Capacity Planning | Manifest | Personnel Listing |
| C. Motor Transport | 1. Capacity Planning | Manifest | Cargo Listing |
| C. Motor Transport | 1. Capacity Planning | Item Catalog Information | Environmental Attribute Code |
| C. Motor Transport | 1. Capacity Planning | Item Catalog Information | HazMat Indicator Code |
| D. Aerial Delivery | Aerial Delivery | Requisitioning | Priority Designator |
| D. Aerial Delivery | Aerial Delivery | Requisitioning | Request ID/Tracking Number |
| D. Aerial Delivery | Aerial Delivery | Requisitioning | Request Type |
| D. Aerial Delivery | Aerial Delivery | Requisitioning | Requesting Element/Unit |
| D. Aerial Delivery | Aerial Delivery | Requisitioning | Requesting Element/Unit Location |
| D. Aerial Delivery | Aerial Delivery | Requisitioning | Request Quantity |
| D. Aerial Delivery | Aerial Delivery | Item Catalog Information | Table of Authorized Material Control Number |
| D. Aerial Delivery | Aerial Delivery | Item Catalog Information | National Stock Number |
| D. Aerial Delivery | Aerial Delivery | Item Catalog Information | Unit of Issue |
| D. Aerial Delivery | Aerial Delivery | Item Catalog Information | Nomenclature |
| D. Aerial Delivery | Aerial Delivery | Item Identification | Item Unique Identification |
| D. Aerial Delivery | Aerial Delivery | Item Identification | Lot Number |
| D. Aerial Delivery | Aerial Delivery | Item Identification | Serial Number |
| D. Aerial Delivery | Aerial Delivery | Item Attributes | Quantity |
| D. Aerial Delivery | Aerial Delivery | Item Attributes | Assigned Activity |
| D. Aerial Delivery | Aerial Delivery | Item Attributes | Assigned Subordinate Activity |
| D. Aerial Delivery | Aerial Delivery | Item Catalog Information | Model |
| D. Aerial Delivery | Aerial Delivery | Item Catalog Information | National Motor Freight Classification |
| D. Aerial Delivery | Aerial Delivery | Item Characteristics | Shipping Height |
| D. Aerial Delivery | Aerial Delivery | Item Characteristics | Shipping Length |
| D. Aerial Delivery | Aerial Delivery | Item Characteristics | Shipping Width |
| D. Aerial Delivery | Aerial Delivery | Item Characteristics | Net Weight |
| D. Aerial Delivery | Aerial Delivery | Item Characteristics | Gross Weight |
| D. Aerial Delivery | Aerial Delivery | Item Characteristics | Tare (Empty) Weight |
| D. Aerial Delivery | Aerial Delivery | Transportation | Cargo Container ID |
| D. Aerial Delivery | Aerial Delivery | Transportation | Cargo Container Type |
| D. Aerial Delivery | Aerial Delivery | Manifest | Mission Number |
| D. Aerial Delivery | Aerial Delivery | Manifest | Priority |
| D. Aerial Delivery | Aerial Delivery | Manifest | Series |
| D. Aerial Delivery | Aerial Delivery | Manifest | Personnel Listing |
| D. Aerial Delivery | Aerial Delivery | Manifest | Cargo Listing |
| D. Aerial Delivery | Aerial Delivery | Item Catalog Information | Environmental Attribute Code |

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|---|---|---|---|
| D. Aerial Delivery | Aerial Delivery | Item Catalog Information | HazMat Indicator Code |
| F. Freight Transportation | Traffic Management | Requisitioning | Priority Designator |
| F. Freight Transportation | Traffic Management | Requisitioning | Request ID/Tracking Number |
| F. Freight Transportation | Traffic Management | Requisitioning | Request Type |
| F. Freight Transportation | Traffic Management | Requisitioning | Requesting Element/Unit |
| F. Freight Transportation | Traffic Management | Requisitioning | Requesting Element/Unit Location |
| F. Freight Transportation | Traffic Management | Requisitioning | Request Quantity |
| F. Freight Transportation | Traffic Management | Item Catalog Information | Table of Authorized Material Control Number |
| F. Freight Transportation | Traffic Management | Item Catalog Information | National Stock Number |
| F. Freight Transportation | Traffic Management | Item Catalog Information | Unit of Issue |
| F. Freight Transportation | Traffic Management | Item Catalog Information | Nomenclature |
| F. Freight Transportation | Traffic Management | Item Identification | Item Unique Identification |
| F. Freight Transportation | Traffic Management | Item Identification | Lot Number |
| F. Freight Transportation | Traffic Management | Item Identification | Serial Number |
| F. Freight Transportation | Traffic Management | Item Attributes | Quantity |
| F. Freight Transportation | Traffic Management | Item Attributes | Assigned Activity |
| F. Freight Transportation | Traffic Management | Item Attributes | Assigned Subordinate Activity |
| F. Freight Transportation | Traffic Management | Item Catalog Information | Model |
| F. Freight Transportation | Traffic Management | Item Catalog Information | National Motor Freight Classification |
| F. Freight Transportation | Traffic Management | Item Characteristics | Shipping Height |
| F. Freight Transportation | Traffic Management | Item Characteristics | Shipping Length |
| F. Freight Transportation | Traffic Management | Item Characteristics | Shipping Width |
| F. Freight Transportation | Traffic Management | Item Characteristics | Net Weight |
| F. Freight Transportation | Traffic Management | Item Characteristics | Gross Weight |
| F. Freight Transportation | Traffic Management | Item Characteristics | Tare (Empty) Weight |
| F. Freight Transportation | Traffic Management | Transportation | Cargo Container ID |
| F. Freight Transportation | Traffic Management | Transportation | Cargo Container Type |
| F. Freight Transportation | Traffic Management | Manifest | Mission Number |
| F. Freight Transportation | Traffic Management | Manifest | Priority |
| F. Freight Transportation | Traffic Management | Manifest | Series |
| F. Freight Transportation | Traffic Management | Manifest | Personnel Listing |
| F. Freight Transportation | Traffic Management | Manifest | Cargo Listing |
| F. Freight Transportation | Traffic Management | Item Catalog Information | Environmental Attribute Code |
| F. Freight Transportation | Traffic Management | Item Catalog Information | HazMat Indicator Code |
| F. Freight Transportation | Traffic Management | Financials | Transportation Account Code |
| F. Freight Transportation | Traffic Management | Activity | Assigned Aerial Port of Debarkation |
| F. Freight Transportation | Traffic Management | Activity | Assigned Water Port of Debarkation |
| F. Freight Transportation | Traffic Management | Transportation | Port of Debarkation |
| F. Freight Transportation | Traffic Management | Transportation | Port of Embarkation |
| G. Passenger Transportation | Port Call | Individual Member | Assigned Billet Identification Code |
| G. Passenger Transportation | Port Call | Individual Member | Assigned Monitored Command Code |
| G. Passenger Transportation | Port Call | Individual Member | Blood Group |
| G. Passenger Transportation | Port Call | Individual Member | Blood RhD |
| G. Passenger Transportation | Port Call | Individual Member | Citizenship |
| G. Passenger Transportation | Port Call | Individual Member | Digital Signature |
| G. Passenger Transportation | Port Call | Individual Member | Duty Location |
| G. Passenger Transportation | Port Call | Individual Member | Electronic Data Interchange Personal Identifier |
| G. Passenger Transportation | Port Call | Individual Member | Home of Record |
| G. Passenger Transportation | Port Call | Individual Member | Medical Readiness |
| G. Passenger Transportation | Port Call | Individual Member | Military/Service Occupation |
| G. Passenger Transportation | Port Call | Individual Member | Name, First |
| G. Passenger Transportation | Port Call | Individual Member | Name, Last |
| G. Passenger Transportation | Port Call | Individual Member | Name, Middle |
| G. Passenger Transportation | Port Call | Individual Member | Name, Suffix |
| G. Passenger Transportation | Port Call | Individual Member | Passport Expiration Date |
| G. Passenger Transportation | Port Call | Individual Member | Passport Issuing Government |
| G. Passenger Transportation | Port Call | Individual Member | Passport Number |
| G. Passenger Transportation | Port Call | Individual Member | Passport Type |
| G. Passenger Transportation | Port Call | Individual Member | Pay Grade |
| G. Passenger Transportation | Port Call | Individual Member | Rank |
| G. Passenger Transportation | Port Call | Individual Member | Security Clearance |
| G. Passenger Transportation | Port Call | Individual Member | Service Status |
| G. Passenger Transportation | Port Call | Individual Member | Zap Number |
| G. Passenger Transportation | Port Call | Financials | Transportation Account Code |
| G. Passenger Transportation | Port Call | Activity | Assigned Aerial Port of Debarkation |

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|---|---|---|---|
| G. Passenger Transportation | Port Call | Activity | Assigned Water Port of Debarkation |
| G. Passenger Transportation | Port Call | Transportation | Port of Debarkation |
| G. Passenger Transportation | Port Call | Transportation | Port of Embarkation |
| G. Passenger Transportation | Port Call | Requisitioning | Priority Designator |
| G. Passenger Transportation | Port Call | Requisitioning | Request ID/Tracking Number |
| G. Passenger Transportation | Port Call | Requisitioning | Request Type |
| G. Passenger Transportation | Port Call | Requisitioning | Requesting Element/Unit |
| G. Passenger Transportation | Port Call | Requisitioning | Requesting Element/Unit Location |
| G. Passenger Transportation | Port Call | Requisitioning | Request Quantity |
| E. Material Handling Equipment | Dispatch | Requisitioning | Priority Designator |
| E. Material Handling Equipment | Dispatch | Requisitioning | Request ID/Tracking Number |
| E. Material Handling Equipment | Dispatch | Requisitioning | Request Type |
| E. Material Handling Equipment | Dispatch | Requisitioning | Requesting Element/Unit |
| E. Material Handling Equipment | Dispatch | Requisitioning | Requesting Element/Unit Location |
| E. Material Handling Equipment | Dispatch | Requisitioning | Request Quantity |
| E. Material Handling Equipment | Dispatch | Individual Member | Zap Number |
| E. Material Handling Equipment | Dispatch | Individual Member | Rank |
| E. Material Handling Equipment | Dispatch | Individual Member | Name, First |
| E. Material Handling Equipment | Dispatch | Individual Member | Name, Last |
| E. Material Handling Equipment | Dispatch | Individual Member | Electronic Data Interchange Personal Identifier |
| E. Material Handling Equipment | Dispatch | Individual Member | Licensing |
| E. Material Handling Equipment | Dispatch | Individual Member | Certifications |
| E. Material Handling Equipment | Dispatch | Item Catalog Information | Model |
| E. Material Handling Equipment | Dispatch | Item Catalog Information | National Motor Freight Classification |
| E. Material Handling Equipment | Dispatch | Activity | Department of Defense Activity Address Code |
| E. Material Handling Equipment | Dispatch | Item Catalog Information | Table of Authorized Material Control Number |
| E. Material Handling Equipment | Dispatch | Item Catalog Information | National Stock Number |
| E. Material Handling Equipment | Dispatch | Item Catalog Information | Nomenclature |
| E. Material Handling Equipment | Dispatch | Item Identification | Item Unique Identification |
| E. Material Handling Equipment | Dispatch | Item Identification | Serial Number |
| E. Material Handling Equipment | Dispatch | Item Attributes | Operational Status |
| E. Material Handling Equipment | Dispatch | Item Catalog Information | Model |
| E. Material Handling Equipment | Dispatch | Item Characteristics | Fuel Type |
| E. Material Handling Equipment | Dispatch | Item Characteristics | On Board Fuel Tank Capacity (Volume) |
| E. Material Handling Equipment | Dispatch | Manifest | Mission Number |
| E. Material Handling Equipment | Dispatch | Manifest | Priority |
| E. Material Handling Equipment | Dispatch | Manifest | Series |
| E. Material Handling Equipment | Dispatch | Manifest | Cargo Listing |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Port of Debarkation |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Port of Embarkation |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Mode of Shipment |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Passenger Itinerary |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Cargo Container ID |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Cargo Container Manifest |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Cargo Container pRFID |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Cargo Container Type |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Cargo Itinerary |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Document Identifier Code |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | In Transit Visibility |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Mission Itinerary |
| H. Port and Terminal Operations | Port & Terminal Operations | Transportation | Mode of Shipment |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Personnel Listing |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Cargo Listing |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Bump Plan |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Chalk |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Mission Number |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Platform |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Platform ID |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Platform Type |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Priority |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Series |
| H. Port and Terminal Operations | Port & Terminal Operations | Manifest | Stick |
| H. Port and Terminal Operations | Port & Terminal Operations | Individual Member | Zap Number |
| H. Port and Terminal Operations | Port & Terminal Operations | Individual Member | Rank |
| H. Port and Terminal Operations | Port & Terminal Operations | Individual Member | Name, First |

| | | | |
|---|---|---|---|
| H. Port and Terminal Operations | Port & Terminal Operations | Individual Member | Name, Last |
| H. Port and Terminal Operations | Port & Terminal Operations | Individual Member | Electronic Data Interchange Personal Identifier |
| H. Port and Terminal Operations | Port & Terminal Operations | Activity | DoDAAC Authority Code |
| H. Port and Terminal Operations | Port & Terminal Operations | Activity | Unit Identifier Code |
| H. Port and Terminal Operations | Port & Terminal Operations | Requisitioning | Priority Designator |
| H. Port and Terminal Operations | Port & Terminal Operations | Requisitioning | Request ID/Tracking Number |
| H. Port and Terminal Operations | Port & Terminal Operations | Requisitioning | Request Type |
| H. Port and Terminal Operations | Port & Terminal Operations | Requisitioning | Requesting Element/Unit |
| H. Port and Terminal Operations | Port & Terminal Operations | Requisitioning | Requesting Element/Unit Location |
| H. Port and Terminal Operations | Port & Terminal Operations | Requisitioning | Request Quantity |

# Appendix D
# Engineer Function Data Points

| Sub-Function 1 | Sub-Function 2 | Data Type | Line Data Level |
|---|---|---|---|
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Air Mobility Command (AMC) Suitability |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Facilities: Aprons |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Facilities: General Information |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Facilities: Hardstands |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Facilities: Helipads |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Facilities: Taxiways |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Maximum Usage |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Runway Length and Width |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Schedule and Workload |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Airfield Type (ICAO or MILAIR) |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Seaport Facilities: Berth Information |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Seaport Facilities: Cargo Pier Depth |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Seaport Facilities: General Information |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Seaport Facilities: Maximum Size Vessel |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Seaports: Channel Depth |
| A. Engineer Reconnaissance (ER) | 1. Area | Transportation | Seaports: General Information |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Country Entity Name |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Military Grid Reference System |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Gridded Reference Graphic |
| A. Engineer Reconnaissance (ER) | 1. Area | Procurement | Acquisitions & Cross Servicing Agreements |
| A. Engineer Reconnaissance (ER) | 1. Area | Procurement | Bulk Fuel Contracts |
| A. Engineer Reconnaissance (ER) | 1. Area | Procurement | Local/Regional Water Sources |
| A. Engineer Reconnaissance (ER) | 1. Area | Report, General Engineering | File Number |
| A. Engineer Reconnaissance (ER) | 1. Area | Report, General Engineering | Report Number |
| A. Engineer Reconnaissance (ER) | 1. Area | Date | Date Time Group |
| A. Engineer Reconnaissance (ER) | 1. Area | Element Leader | Military/Service Occupation |
| A. Engineer Reconnaissance (ER) | 1. Area | Element Leader | Name, First |
| A. Engineer Reconnaissance (ER) | 1. Area | Element Leader | Name, Last |
| A. Engineer Reconnaissance (ER) | 1. Area | Element Leader | Pay Grade |
| A. Engineer Reconnaissance (ER) | 1. Area | Element Leader | Rank |
| A. Engineer Reconnaissance (ER) | 1. Area | Element Leader | Electronic Data Interchange Personal Identifier |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Work (Time/Capacity) |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Soil |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Geology |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Drainage |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Topography |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Earthwork |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Obstacle Crossing |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Bridge |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Accessibility to Materials |
| A. Engineer Reconnaissance (ER) | 1. Area | Estimate | Accessibility to Utilities |
| A. Engineer Reconnaissance (ER) | 1. Area | Requisitioning | Priority Designator |
| A. Engineer Reconnaissance (ER) | 1. Area | Requisitioning | Request ID/Tracking Number |
| A. Engineer Reconnaissance (ER) | 1. Area | Requisitioning | Request Type |
| A. Engineer Reconnaissance (ER) | 1. Area | Requisitioning | Requesting Element/Unit |
| A. Engineer Reconnaissance (ER) | 1. Area | Requisitioning | Requesting Element/Unit Location |
| A. Engineer Reconnaissance (ER) | 1. Area | Map Overlay Object | Colors |
| A. Engineer Reconnaissance (ER) | 1. Area | Map Overlay Object | Shapes |
| A. Engineer Reconnaissance (ER) | 1. Area | Map Overlay Object | Military Graphics and Symbols |
| A. Engineer Reconnaissance (ER) | 1. Area | Media Object | Sound |
| A. Engineer Reconnaissance (ER) | 1. Area | Media Object | Picture |
| A. Engineer Reconnaissance (ER) | 1. Area | Media Object | Video |
| A. Engineer Reconnaissance (ER) | 1. Area | Media Object | Uniform Resource Locator |
| A. Engineer Reconnaissance (ER) | 1. Area | Media Object | Office Productivity File |
| A. Engineer Reconnaissance (ER) | 1. Area | Inventory | Supply Condition Code |

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|---|---|---|---|
| A. Engineer Reconnaissance (ER) | 1. Area | Inventory | On Hand Quantity |
| A. Engineer Reconnaissance (ER) | 1. Area | Item Catalog Information | Class of Supply |
| A. Engineer Reconnaissance (ER) | 1. Area | Item Catalog Information | Nomenclature |
| A. Engineer Reconnaissance (ER) | 1. Area | Facility | Condition |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Postal/ZIP Code |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | State/Providence/Prefecture |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Street Name |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Street Number |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Uncommon Address: Area ID/Designation |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Uncommon Address: Range ID/Number |
| A. Engineer Reconnaissance (ER) | 1. Area | Location, Physical | Uncommon Address: Structure ID/Number |
| A. Engineer Reconnaissance (ER) | 1. Area | Facility | Intended Use Type |
| A. Engineer Reconnaissance (ER) | 1. Area | Water Source | Type |
| A. Engineer Reconnaissance (ER) | 1. Area | Water Source | Quantity |
| A. Engineer Reconnaissance (ER) | 1. Area | Water Source | Flow Rate |
| A. Engineer Reconnaissance (ER) | 1. Area | Land/Terrain | General Features/Type |
| A. Engineer Reconnaissance (ER) | 1. Area | Land/Terrain | Intended Use |
| A. Engineer Reconnaissance (ER) | 1. Area | Land/Terrain | Soil Type |
| A. Engineer Reconnaissance (ER) | 1. Area | Storage | Local/Regional Storage Facilities |
| A. Engineer Reconnaissance (ER) | 1. Area | Item Characteristics | Fuel Load Capacity (Volume) |
| A. Engineer Reconnaissance (ER) | 1. Area | Item Characteristics | Fuel Type |
| A. Engineer Reconnaissance (ER) | 1. Area | Report, General Engineering | Report Location |
| A. Engineer Reconnaissance (ER) | 1. Area | Report, General Engineering | Report Type/Purpose |
| A. Engineer Reconnaissance (ER) | 2. Zone | Report, General Engineering | Report Location |
| A. Engineer Reconnaissance (ER) | 2. Zone | Report, General Engineering | Report Type/Purpose |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Country Entity Name |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Military Grid Reference System |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Gridded Reference Graphic |
| A. Engineer Reconnaissance (ER) | 2. Zone | Report, General Engineering | File Number |
| A. Engineer Reconnaissance (ER) | 2. Zone | Report, General Engineering | Report Number |
| A. Engineer Reconnaissance (ER) | 2. Zone | Date | Date Time Group |
| A. Engineer Reconnaissance (ER) | 2. Zone | Element Leader | Military/Service Occupation |
| A. Engineer Reconnaissance (ER) | 2. Zone | Element Leader | Name, First |
| A. Engineer Reconnaissance (ER) | 2. Zone | Element Leader | Name, Last |
| A. Engineer Reconnaissance (ER) | 2. Zone | Element Leader | Pay Grade |
| A. Engineer Reconnaissance (ER) | 2. Zone | Element Leader | Rank |
| A. Engineer Reconnaissance (ER) | 2. Zone | Element Leader | Electronic Data Interchange Personal Identifier |
| A. Engineer Reconnaissance (ER) | 2. Zone | Estimate | Work (Time/Capacity) |
| A. Engineer Reconnaissance (ER) | 2. Zone | Estimate | Soil |
| A. Engineer Reconnaissance (ER) | 2. Zone | Estimate | Geology |
| A. Engineer Reconnaissance (ER) | 2. Zone | Estimate | Drainage |
| A. Engineer Reconnaissance (ER) | 2. Zone | Estimate | Topography |
| A. Engineer Reconnaissance (ER) | 2. Zone | Estimate | Earthwork |
| A. Engineer Reconnaissance (ER) | 2. Zone | Estimate | Obstacle Crossing |
| A. Engineer Reconnaissance (ER) | 2. Zone | Estimate | Bridge |
| A. Engineer Reconnaissance (ER) | 2. Zone | Requisitioning | Priority Designator |
| A. Engineer Reconnaissance (ER) | 2. Zone | Requisitioning | Request ID/Tracking Number |
| A. Engineer Reconnaissance (ER) | 2. Zone | Requisitioning | Request Type |
| A. Engineer Reconnaissance (ER) | 2. Zone | Requisitioning | Requesting Element/Unit |
| A. Engineer Reconnaissance (ER) | 2. Zone | Requisitioning | Requesting Element/Unit Location |
| A. Engineer Reconnaissance (ER) | 2. Zone | Map Overlay Object | Colors |
| A. Engineer Reconnaissance (ER) | 2. Zone | Map Overlay Object | Shapes |
| A. Engineer Reconnaissance (ER) | 2. Zone | Map Overlay Object | Military Graphics and Symbols |
| A. Engineer Reconnaissance (ER) | 2. Zone | Media Object | Sound |
| A. Engineer Reconnaissance (ER) | 2. Zone | Media Object | Picture |
| A. Engineer Reconnaissance (ER) | 2. Zone | Media Object | Video |
| A. Engineer Reconnaissance (ER) | 2. Zone | Media Object | Uniform Resource Locator |
| A. Engineer Reconnaissance (ER) | 2. Zone | Media Object | Office Productivity File |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Postal/ZIP Code |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | State/Providence/Prefecture |

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| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Street Name |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Street Number |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Uncommon Address: Area ID/Designation |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| A. Engineer Reconnaissance (ER) | 2. Zone | Location, Physical | Uncommon Address: Range ID/Number |
| A. Engineer Reconnaissance (ER) | 2. Zone | Water Source | Type |
| A. Engineer Reconnaissance (ER) | 2. Zone | Water Source | Flow Rate |
| A. Engineer Reconnaissance (ER) | 2. Zone | Storage | Local/Regional Storage Facilities |
| A. Engineer Reconnaissance (ER) | 2. Zone | Item Characteristics | Fuel Load Capacity (Volume) |
| A. Engineer Reconnaissance (ER) | 2. Zone | Item Characteristics | Fuel Type |
| A. Engineer Reconnaissance (ER) | 2. Zone | Amphibious Crossing | Type |
| A. Engineer Reconnaissance (ER) | 2. Zone | Amphibious Crossing | Frontage (Meters) |
| A. Engineer Reconnaissance (ER) | 2. Zone | Amphibious Crossing | Classification |
| A. Engineer Reconnaissance (ER) | 2. Zone | Enemy Information | Size |
| A. Engineer Reconnaissance (ER) | 2. Zone | Enemy Information | Equipment |
| A. Engineer Reconnaissance (ER) | 2. Zone | Enemy Information | Activity |
| A. Engineer Reconnaissance (ER) | 2. Zone | Enemy Information | Unit Markings |
| A. Engineer Reconnaissance (ER) | 2. Zone | Enemy Information | Location |
| A. Engineer Reconnaissance (ER) | 2. Zone | Enemy Information | DTG (of Observation) |
| A. Engineer Reconnaissance (ER) | 2. Zone | Weather | Atmospheric Obstruction |
| A. Engineer Reconnaissance (ER) | 2. Zone | Weather | Temperature (Ambient) |
| A. Engineer Reconnaissance (ER) | 2. Zone | Weather | Illumination (Daytime) |
| A. Engineer Reconnaissance (ER) | 2. Zone | Weather | Illumination (Nighttime) |
| A. Engineer Reconnaissance (ER) | 2. Zone | Weather | Precipitation Type |
| A. Engineer Reconnaissance (ER) | 2. Zone | Position Information | Location, Key terrain |
| A. Engineer Reconnaissance (ER) | 2. Zone | Position Information | Location, Observation/Fields of Fire |
| A. Engineer Reconnaissance (ER) | 2. Zone | Position Information | Location, Cover and Concealment |
| A. Engineer Reconnaissance (ER) | 2. Zone | Position Information | Location, Obstacles |
| A. Engineer Reconnaissance (ER) | 2. Zone | Position Information | Location, Avenues of Approach |
| A. Engineer Reconnaissance (ER) | 3. Route | Report, General Engineering | Report Location |
| A. Engineer Reconnaissance (ER) | 3. Route | Report, General Engineering | Report Type/Purpose |
| A. Engineer Reconnaissance (ER) | 3. Route | Location, Physical | Country Entity Name |
| A. Engineer Reconnaissance (ER) | 3. Route | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| A. Engineer Reconnaissance (ER) | 3. Route | Location, Physical | Military Grid Reference System |
| A. Engineer Reconnaissance (ER) | 3. Route | Location, Physical | Gridded Reference Graphic |
| A. Engineer Reconnaissance (ER) | 3. Route | Report, General Engineering | File Number |
| A. Engineer Reconnaissance (ER) | 3. Route | Report, General Engineering | Report Number |
| A. Engineer Reconnaissance (ER) | 3. Route | Date | Date Time Group |
| A. Engineer Reconnaissance (ER) | 3. Route | Element Leader | Military/Service Occupation |
| A. Engineer Reconnaissance (ER) | 3. Route | Element Leader | Name, First |
| A. Engineer Reconnaissance (ER) | 3. Route | Element Leader | Name, Last |
| A. Engineer Reconnaissance (ER) | 3. Route | Element Leader | Pay Grade |
| A. Engineer Reconnaissance (ER) | 3. Route | Element Leader | Rank |
| A. Engineer Reconnaissance (ER) | 3. Route | Element Leader | Electronic Data Interchange Personal Identifier |
| A. Engineer Reconnaissance (ER) | 3. Route | Estimate | Soil |
| A. Engineer Reconnaissance (ER) | 3. Route | Estimate | Drainage |
| A. Engineer Reconnaissance (ER) | 3. Route | Estimate | Obstacle Crossing |
| A. Engineer Reconnaissance (ER) | 3. Route | Estimate | Bridge |
| A. Engineer Reconnaissance (ER) | 3. Route | Requisitioning | Priority Designator |
| A. Engineer Reconnaissance (ER) | 3. Route | Requisitioning | Request ID/Tracking Number |
| A. Engineer Reconnaissance (ER) | 3. Route | Requisitioning | Request Type |
| A. Engineer Reconnaissance (ER) | 3. Route | Requisitioning | Requesting Element/Unit |
| A. Engineer Reconnaissance (ER) | 3. Route | Requisitioning | Requesting Element/Unit Location |
| A. Engineer Reconnaissance (ER) | 3. Route | Map Overlay Object | Colors |
| A. Engineer Reconnaissance (ER) | 3. Route | Map Overlay Object | Shapes |
| A. Engineer Reconnaissance (ER) | 3. Route | Map Overlay Object | Military Graphics and Symbols |
| A. Engineer Reconnaissance (ER) | 3. Route | Media Object | Sound |
| A. Engineer Reconnaissance (ER) | 3. Route | Media Object | Picture |
| A. Engineer Reconnaissance (ER) | 3. Route | Media Object | Video |
| A. Engineer Reconnaissance (ER) | 3. Route | Media Object | Uniform Resource Locator |
| A. Engineer Reconnaissance (ER) | 3. Route | Media Object | Office Productivity File |
| A. Engineer Reconnaissance (ER) | 3. Route | Location, Physical | Uncommon Address: Area ID/Designation |

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|---|---|---|---|
| A. Engineer Reconnaissance (ER) | 3. Route | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| A. Engineer Reconnaissance (ER) | 3. Route | Location, Physical | Uncommon Address: Range ID/Number |
| A. Engineer Reconnaissance (ER) | 3. Route | Water Source | Type |
| A. Engineer Reconnaissance (ER) | 3. Route | Water Source | Flow Rate |
| A. Engineer Reconnaissance (ER) | 3. Route | Enemy Information | Size |
| A. Engineer Reconnaissance (ER) | 3. Route | Enemy Information | Equipment |
| A. Engineer Reconnaissance (ER) | 3. Route | Enemy Information | Activity |
| A. Engineer Reconnaissance (ER) | 3. Route | Enemy Information | Unit Markings |
| A. Engineer Reconnaissance (ER) | 3. Route | Enemy Information | Location |
| A. Engineer Reconnaissance (ER) | 3. Route | Enemy Information | DTG (of Observation) |
| A. Engineer Reconnaissance (ER) | 3. Route | Position Information | Location, Key terrain |
| A. Engineer Reconnaissance (ER) | 3. Route | Position Information | Location, Observation/Fields of Fire |
| A. Engineer Reconnaissance (ER) | 3. Route | Position Information | Location, Cover and Concealment |
| A. Engineer Reconnaissance (ER) | 3. Route | Position Information | Location, Obstacles |
| A. Engineer Reconnaissance (ER) | 3. Route | Position Information | Location, Avenues of Approach |
| A. Engineer Reconnaissance (ER) | 3. Route | Water Source | Type |
| A. Engineer Reconnaissance (ER) | 3. Route | Water Source | Flow Rate/Current Velocity |
| A. Engineer Reconnaissance (ER) | 3. Route | Water Source | Type, Natural Bottom/Bed |
| A. Engineer Reconnaissance (ER) | 3. Route | Water Source | Water Line Height (High) |
| A. Engineer Reconnaissance (ER) | 3. Route | Water Source | Water Line Height (Average) |
| A. Engineer Reconnaissance (ER) | 3. Route | Water Source | Water Line Height (Low) |
| A. Engineer Reconnaissance (ER) | 3. Route | Transportation | Land Infrastructure |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Clearance, Horizontal |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Clearance, Vertical |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Clearance, Under/Beneath |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Type, Construction |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Type, Material (Primary) |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Type, Material (Secondary) |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Number of Spans |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Overall Length |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Overall Width |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Overall Condition |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Load Class |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Type, Roadway |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Structure | Bridge By-Pass Location |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Site | Gap Width, Far Bank |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Site | Gap Width, Near Bank |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Site | Gap Width, At Water Level |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Site | Gap Width, At Bottom |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Site | Embankment Height Above Water Line (Near) |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Site | Embankment Height Above Water Line (Far) |
| A. Engineer Reconnaissance (ER) | 3. Route | Bridge Site | Type, Turnout |
| J. Bridging | Deleted | Deleted | Deleted |
| H. Horizontal Construction | 2. MSR construction | Requisitioning | Priority Designator |
| H. Horizontal Construction | 2. MSR construction | Requisitioning | Request ID/Tracking Number |
| H. Horizontal Construction | 2. MSR construction | Requisitioning | Request Type |
| H. Horizontal Construction | 2. MSR construction | Requisitioning | Requesting Element/Unit |
| H. Horizontal Construction | 2. MSR construction | Requisitioning | Requesting Element/Unit Location |
| H. Horizontal Construction | 2. MSR construction | Location, Physical | Uncommon Address: Area ID/Designation |
| H. Horizontal Construction | 2. MSR construction | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| H. Horizontal Construction | 2. MSR construction | Location, Physical | Uncommon Address: Range ID/Number |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Key terrain |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Observation/Fields of Fire |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Cover and Concealment |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Obstacles |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Avenues of Approach |
| H. Horizontal Construction | 2. MSR construction | Map Overlay Object | Colors |
| H. Horizontal Construction | 2. MSR construction | Map Overlay Object | Shapes |
| H. Horizontal Construction | 2. MSR construction | Map Overlay Object | Military Graphics and Symbols |
| H. Horizontal Construction | 2. MSR construction | Media Object | Sound |
| H. Horizontal Construction | 2. MSR construction | Media Object | Picture |
| H. Horizontal Construction | 2. MSR construction | Media Object | Video |
| H. Horizontal Construction | 2. MSR construction | Media Object | Uniform Resource Locator |

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| H. Horizontal Construction | 2. MSR construction | Media Object | Office Productivity File |
| H. Horizontal Construction | 2. MSR construction | Estimate | Work (Time/Capacity) |
| H. Horizontal Construction | 2. MSR construction | Estimate | Materials |
| H. Horizontal Construction | 2. MSR construction | Estimate | Equipment |
| H. Horizontal Construction | 2. MSR construction | Estimate | Energy |
| H. Horizontal Construction | 2. MSR construction | Estimate | Soil |
| H. Horizontal Construction | 2. MSR construction | Estimate | Geology |
| H. Horizontal Construction | 2. MSR construction | Estimate | Drainage |
| H. Horizontal Construction | 2. MSR construction | Estimate | Topography |
| H. Horizontal Construction | 2. MSR construction | Estimate | Earthwork |
| H. Horizontal Construction | 2. MSR construction | Estimate | Obstacle Crossing |
| H. Horizontal Construction | 2. MSR construction | Estimate | Bridge |
| H. Horizontal Construction | 2. MSR construction | Estimate | Accessibility to Materials |
| H. Horizontal Construction | 2. MSR construction | Estimate | Accessibility to Utilities |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Key terrain |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Observation/Fields of Fire |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Cover and Concealment |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Obstacles |
| H. Horizontal Construction | 2. MSR construction | Position Information | Location, Avenues of Approach |
| H. Horizontal Construction | 2. MSR construction | Location, Physical | Country Entity Name |
| H. Horizontal Construction | 2. MSR construction | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| H. Horizontal Construction | 2. MSR construction | Location, Physical | Military Grid Reference System |
| H. Horizontal Construction | 2. MSR construction | Location, Physical | Gridded Reference Graphic |
| H. Horizontal Construction | 2. MSR construction | Transportation | Land Infrastructure |
| H. Horizontal Construction | 2. MSR construction | Element Leader | Military/Service Occupation |
| H. Horizontal Construction | 2. MSR construction | Element Leader | Name, First |
| H. Horizontal Construction | 2. MSR construction | Element Leader | Name, Last |
| H. Horizontal Construction | 2. MSR construction | Element Leader | Pay Grade |
| H. Horizontal Construction | 2. MSR construction | Element Leader | Rank |
| H. Horizontal Construction | 2. MSR construction | Element Leader | Electronic Data Interchange Personal Identifier |
| H. Horizontal Construction | 1. Expeditionary Airfields | Requisitioning | Priority Designator |
| H. Horizontal Construction | 1. Expeditionary Airfields | Requisitioning | Request ID/Tracking Number |
| H. Horizontal Construction | 1. Expeditionary Airfields | Requisitioning | Request Type |
| H. Horizontal Construction | 1. Expeditionary Airfields | Requisitioning | Requesting Element/Unit |
| H. Horizontal Construction | 1. Expeditionary Airfields | Requisitioning | Requesting Element/Unit Location |
| H. Horizontal Construction | 1. Expeditionary Airfields | Element Leader | Military/Service Occupation |
| H. Horizontal Construction | 1. Expeditionary Airfields | Element Leader | Name, First |
| H. Horizontal Construction | 1. Expeditionary Airfields | Element Leader | Name, Last |
| H. Horizontal Construction | 1. Expeditionary Airfields | Element Leader | Pay Grade |
| H. Horizontal Construction | 1. Expeditionary Airfields | Element Leader | Rank |
| H. Horizontal Construction | 1. Expeditionary Airfields | Element Leader | Electronic Data Interchange Personal Identifier |
| H. Horizontal Construction | 1. Expeditionary Airfields | Location, Physical | Uncommon Address: Area ID/Designation |
| H. Horizontal Construction | 1. Expeditionary Airfields | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| H. Horizontal Construction | 1. Expeditionary Airfields | Location, Physical | Uncommon Address: Range ID/Number |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Key terrain |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Observation/Fields of Fire |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Cover and Concealment |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Obstacles |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Avenues of Approach |
| H. Horizontal Construction | 1. Expeditionary Airfields | Map Overlay Object | Colors |
| H. Horizontal Construction | 1. Expeditionary Airfields | Map Overlay Object | Shapes |
| H. Horizontal Construction | 1. Expeditionary Airfields | Map Overlay Object | Military Graphics and Symbols |
| H. Horizontal Construction | 1. Expeditionary Airfields | Media Object | Sound |
| H. Horizontal Construction | 1. Expeditionary Airfields | Media Object | Picture |
| H. Horizontal Construction | 1. Expeditionary Airfields | Media Object | Video |
| H. Horizontal Construction | 1. Expeditionary Airfields | Media Object | Uniform Resource Locator |
| H. Horizontal Construction | 1. Expeditionary Airfields | Media Object | Office Productivity File |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Work (Time/Capacity) |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Materials |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Equipment |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Energy |

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| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Soil |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Geology |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Drainage |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Topography |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Earthwork |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Accessibility to Materials |
| H. Horizontal Construction | 1. Expeditionary Airfields | Estimate | Accessibility to Utilities |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Key terrain |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Observation/Fields of Fire |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Cover and Concealment |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Obstacles |
| H. Horizontal Construction | 1. Expeditionary Airfields | Position Information | Location, Avenues of Approach |
| H. Horizontal Construction | 1. Expeditionary Airfields | Location, Physical | Country Entity Name |
| H. Horizontal Construction | 1. Expeditionary Airfields | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| H. Horizontal Construction | 1. Expeditionary Airfields | Location, Physical | Military Grid Reference System |
| H. Horizontal Construction | 1. Expeditionary Airfields | Location, Physical | Gridded Reference Graphic |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Requisitioning | Priority Designator |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Requisitioning | Request ID/Tracking Number |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Requisitioning | Request Type |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Requisitioning | Requesting Element/Unit |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Requisitioning | Requesting Element/Unit Location |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Element Leader | Military/Service Occupation |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Element Leader | Name, First |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Element Leader | Name, Last |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Element Leader | Pay Grade |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Element Leader | Rank |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Element Leader | Electronic Data Interchange Personal Identifier |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Location, Physical | Uncommon Address: Area ID/Designation |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Location, Physical | Uncommon Address: Range ID/Number |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Key terrain |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Observation/Fields of Fire |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Cover and Concealment |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Obstacles |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Avenues of Approach |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Map Overlay Object | Colors |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Map Overlay Object | Shapes |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Map Overlay Object | Military Graphics and Symbols |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Media Object | Sound |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Media Object | Picture |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Media Object | Video |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Media Object | Uniform Resource Locator |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Media Object | Office Productivity File |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Work (Time/Capacity) |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Materials |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Equipment |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Energy |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Soil |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Geology |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Drainage |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Topography |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Earthwork |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Accessibility to Materials |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Estimate | Accessibility to Utilities |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Key terrain |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Observation/Fields of Fire |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Cover and Concealment |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Obstacles |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Position Information | Location, Avenues of Approach |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Location, Physical | Country Entity Name |

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| H. Horizontal Construction | 4. Site Prep/Bed Down | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Location, Physical | Military Grid Reference System |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Location, Physical | Gridded Reference Graphic |
| H. Horizontal Construction | 4. Site Prep/Bed Down | Facility | Intended Use Type |
| H. Horizontal Construction | 3. Ordnance Storage | Requisitioning | Priority Designator |
| H. Horizontal Construction | 3. Ordnance Storage | Requisitioning | Request ID/Tracking Number |
| H. Horizontal Construction | 3. Ordnance Storage | Requisitioning | Request Type |
| H. Horizontal Construction | 3. Ordnance Storage | Requisitioning | Requesting Element/Unit |
| H. Horizontal Construction | 3. Ordnance Storage | Requisitioning | Requesting Element/Unit Location |
| H. Horizontal Construction | 3. Ordnance Storage | Location, Physical | Uncommon Address: Area ID/Designation |
| H. Horizontal Construction | 3. Ordnance Storage | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| H. Horizontal Construction | 3. Ordnance Storage | Location, Physical | Uncommon Address: Range ID/Number |
| H. Horizontal Construction | 3. Ordnance Storage | Element Leader | Military/Service Occupation |
| H. Horizontal Construction | 3. Ordnance Storage | Element Leader | Name, First |
| H. Horizontal Construction | 3. Ordnance Storage | Element Leader | Name, Last |
| H. Horizontal Construction | 3. Ordnance Storage | Element Leader | Pay Grade |
| H. Horizontal Construction | 3. Ordnance Storage | Element Leader | Rank |
| H. Horizontal Construction | 3. Ordnance Storage | Element Leader | Electronic Data Interchange Personal Identifier |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Key terrain |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Observation/Fields of Fire |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Cover and Concealment |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Obstacles |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Avenues of Approach |
| H. Horizontal Construction | 3. Ordnance Storage | Map Overlay Object | Colors |
| H. Horizontal Construction | 3. Ordnance Storage | Map Overlay Object | Shapes |
| H. Horizontal Construction | 3. Ordnance Storage | Map Overlay Object | Military Graphics and Symbols |
| H. Horizontal Construction | 3. Ordnance Storage | Media Object | Sound |
| H. Horizontal Construction | 3. Ordnance Storage | Media Object | Picture |
| H. Horizontal Construction | 3. Ordnance Storage | Media Object | Video |
| H. Horizontal Construction | 3. Ordnance Storage | Media Object | Uniform Resource Locator |
| H. Horizontal Construction | 3. Ordnance Storage | Media Object | Office Productivity File |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Work (Time/Capacity) |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Materials |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Equipment |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Energy |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Soil |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Geology |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Drainage |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Topography |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Earthwork |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Accessibility to Materials |
| H. Horizontal Construction | 3. Ordnance Storage | Estimate | Accessibility to Utilities |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Key terrain |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Observation/Fields of Fire |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Cover and Concealment |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Obstacles |
| H. Horizontal Construction | 3. Ordnance Storage | Position Information | Location, Avenues of Approach |
| H. Horizontal Construction | 3. Ordnance Storage | Location, Physical | Country Entity Name |
| H. Horizontal Construction | 3. Ordnance Storage | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| H. Horizontal Construction | 3. Ordnance Storage | Location, Physical | Military Grid Reference System |
| H. Horizontal Construction | 3. Ordnance Storage | Location, Physical | Gridded Reference Graphic |
| H. Horizontal Construction | 3. Ordnance Storage | Item Characteristics | Gross Weight |
| H. Horizontal Construction | 3. Ordnance Storage | Item Catalog Information | National Stock Number |
| H. Horizontal Construction | 3. Ordnance Storage | Item Catalog Information | Nomenclature |
| H. Horizontal Construction | 3. Ordnance Storage | Item Catalog Information | Department of Defense Identification Code |
| I. Vertical Construction | 3. Wood/Masonry | Requisitioning | Priority Designator |
| I. Vertical Construction | 3. Wood/Masonry | Requisitioning | Request ID/Tracking Number |
| I. Vertical Construction | 3. Wood/Masonry | Requisitioning | Request Type |
| I. Vertical Construction | 3. Wood/Masonry | Requisitioning | Requesting Element/Unit |
| I. Vertical Construction | 3. Wood/Masonry | Requisitioning | Requesting Element/Unit Location |

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|---|---|---|---|
| I. Vertical Construction | 3. Wood/Masonry | Location, Physical | Uncommon Address: Area ID/Designation |
| I. Vertical Construction | 3. Wood/Masonry | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| I. Vertical Construction | 3. Wood/Masonry | Location, Physical | Uncommon Address: Range ID/Number |
| I. Vertical Construction | 3. Wood/Masonry | Map Overlay Object | Colors |
| I. Vertical Construction | 3. Wood/Masonry | Map Overlay Object | Shapes |
| I. Vertical Construction | 3. Wood/Masonry | Map Overlay Object | Military Graphics and Symbols |
| I. Vertical Construction | 3. Wood/Masonry | Element Leader | Military/Service Occupation |
| I. Vertical Construction | 3. Wood/Masonry | Element Leader | Name, First |
| I. Vertical Construction | 3. Wood/Masonry | Element Leader | Name, Last |
| I. Vertical Construction | 3. Wood/Masonry | Element Leader | Pay Grade |
| I. Vertical Construction | 3. Wood/Masonry | Element Leader | Rank |
| I. Vertical Construction | 3. Wood/Masonry | Element Leader | Electronic Data Interchange Personal Identifier |
| I. Vertical Construction | 3. Wood/Masonry | Media Object | Sound |
| I. Vertical Construction | 3. Wood/Masonry | Media Object | Picture |
| I. Vertical Construction | 3. Wood/Masonry | Media Object | Video |
| I. Vertical Construction | 3. Wood/Masonry | Media Object | Uniform Resource Locator |
| I. Vertical Construction | 3. Wood/Masonry | Media Object | Office Productivity File |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Work (Time/Capacity) |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Materials |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Equipment |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Energy |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Soil |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Geology |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Drainage |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Topography |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Earthwork |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Accessibility to Materials |
| I. Vertical Construction | 3. Wood/Masonry | Estimate | Accessibility to Utilities |
| I. Vertical Construction | 3. Wood/Masonry | Position Information | Location, Key terrain |
| I. Vertical Construction | 3. Wood/Masonry | Position Information | Location, Observation/Fields of Fire |
| I. Vertical Construction | 3. Wood/Masonry | Position Information | Location, Cover and Concealment |
| I. Vertical Construction | 3. Wood/Masonry | Position Information | Location, Obstacles |
| I. Vertical Construction | 3. Wood/Masonry | Position Information | Location, Avenues of Approach |
| I. Vertical Construction | 3. Wood/Masonry | Location, Physical | Country Entity Name |
| I. Vertical Construction | 3. Wood/Masonry | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| I. Vertical Construction | 3. Wood/Masonry | Location, Physical | Military Grid Reference System |
| I. Vertical Construction | 3. Wood/Masonry | Location, Physical | Gridded Reference Graphic |
| I. Vertical Construction | 3. Wood/Masonry | Facility | Intended Use Type |
| I. Vertical Construction | 2. Existing Facility Rehab | Facility | Intended Use Type |
| I. Vertical Construction | 2. Existing Facility Rehab | Facility | Condition |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Postal/ZIP Code |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | State/Providence/Prefecture |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Street Name |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Street Number |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Uncommon Address: Area ID/Designation |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Uncommon Address: Range ID/Number |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Uncommon Address: Structure ID/Number |
| I. Vertical Construction | 2. Existing Facility Rehab | Requisitioning | Priority Designator |
| I. Vertical Construction | 2. Existing Facility Rehab | Requisitioning | Request ID/Tracking Number |
| I. Vertical Construction | 2. Existing Facility Rehab | Requisitioning | Request Type |
| I. Vertical Construction | 2. Existing Facility Rehab | Requisitioning | Requesting Element/Unit |
| I. Vertical Construction | 2. Existing Facility Rehab | Requisitioning | Requesting Element/Unit Location |
| I. Vertical Construction | 2. Existing Facility Rehab | Element Leader | Military/Service Occupation |
| I. Vertical Construction | 2. Existing Facility Rehab | Element Leader | Name, First |
| I. Vertical Construction | 2. Existing Facility Rehab | Element Leader | Name, Last |
| I. Vertical Construction | 2. Existing Facility Rehab | Element Leader | Pay Grade |
| I. Vertical Construction | 2. Existing Facility Rehab | Element Leader | Rank |
| I. Vertical Construction | 2. Existing Facility Rehab | Element Leader | Electronic Data Interchange Personal Identifier |
| I. Vertical Construction | 2. Existing Facility Rehab | Map Overlay Object | Colors |

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|---|---|---|---|
| I. Vertical Construction | 2. Existing Facility Rehab | Map Overlay Object | Shapes |
| I. Vertical Construction | 2. Existing Facility Rehab | Map Overlay Object | Military Graphics and Symbols |
| I. Vertical Construction | 2. Existing Facility Rehab | Media Object | Sound |
| I. Vertical Construction | 2. Existing Facility Rehab | Media Object | Picture |
| I. Vertical Construction | 2. Existing Facility Rehab | Media Object | Video |
| I. Vertical Construction | 2. Existing Facility Rehab | Media Object | Uniform Resource Locator |
| I. Vertical Construction | 2. Existing Facility Rehab | Media Object | Office Productivity File |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Work (Time/Capacity) |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Materials |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Equipment |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Energy |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Soil |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Geology |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Drainage |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Topography |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Earthwork |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Accessibility to Materials |
| I. Vertical Construction | 2. Existing Facility Rehab | Estimate | Accessibility to Utilities |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Country Entity Name |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Military Grid Reference System |
| I. Vertical Construction | 2. Existing Facility Rehab | Location, Physical | Gridded Reference Graphic |
| I. Vertical Construction | 1. Structural Reinforcement | Facility | Intended Use Type |
| I. Vertical Construction | 1. Structural Reinforcement | Facility | Condition |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Postal/ZIP Code |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | State/Providence/Prefecture |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Street Name |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Street Number |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Uncommon Address: Area ID/Designation |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Uncommon Address: Range ID/Number |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Uncommon Address: Structure ID/Number |
| I. Vertical Construction | 1. Structural Reinforcement | Requisitioning | Priority Designator |
| I. Vertical Construction | 1. Structural Reinforcement | Requisitioning | Request ID/Tracking Number |
| I. Vertical Construction | 1. Structural Reinforcement | Requisitioning | Request Type |
| I. Vertical Construction | 1. Structural Reinforcement | Requisitioning | Requesting Element/Unit |
| I. Vertical Construction | 1. Structural Reinforcement | Requisitioning | Requesting Element/Unit Location |
| I. Vertical Construction | 1. Structural Reinforcement | Element Leader | Military/Service Occupation |
| I. Vertical Construction | 1. Structural Reinforcement | Element Leader | Name, First |
| I. Vertical Construction | 1. Structural Reinforcement | Element Leader | Name, Last |
| I. Vertical Construction | 1. Structural Reinforcement | Element Leader | Pay Grade |
| I. Vertical Construction | 1. Structural Reinforcement | Element Leader | Rank |
| I. Vertical Construction | 1. Structural Reinforcement | Element Leader | Electronic Data Interchange Personal Identifier |
| I. Vertical Construction | 1. Structural Reinforcement | Map Overlay Object | Colors |
| I. Vertical Construction | 1. Structural Reinforcement | Map Overlay Object | Shapes |
| I. Vertical Construction | 1. Structural Reinforcement | Map Overlay Object | Military Graphics and Symbols |
| I. Vertical Construction | 1. Structural Reinforcement | Media Object | Sound |
| I. Vertical Construction | 1. Structural Reinforcement | Media Object | Picture |
| I. Vertical Construction | 1. Structural Reinforcement | Media Object | Video |
| I. Vertical Construction | 1. Structural Reinforcement | Media Object | Uniform Resource Locator |
| I. Vertical Construction | 1. Structural Reinforcement | Media Object | Office Productivity File |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Work (Time/Capacity) |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Materials |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Equipment |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Energy |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Soil |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Geology |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Drainage |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Topography |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Earthwork |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Accessibility to Materials |
| I. Vertical Construction | 1. Structural Reinforcement | Estimate | Accessibility to Utilities |

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|---|---|---|---|
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Country Entity Name |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Military Grid Reference System |
| I. Vertical Construction | 1. Structural Reinforcement | Location, Physical | Gridded Reference Graphic |
| G. Facilities Maintenance | Facilities Maintenance | Facility | Intended Use Type |
| G. Facilities Maintenance | Facilities Maintenance | Facility | Condition |
| G. Facilities Maintenance | Facilities Maintenance | Element Leader | Military/Service Occupation |
| G. Facilities Maintenance | Facilities Maintenance | Element Leader | Name, First |
| G. Facilities Maintenance | Facilities Maintenance | Element Leader | Name, Last |
| G. Facilities Maintenance | Facilities Maintenance | Element Leader | Pay Grade |
| G. Facilities Maintenance | Facilities Maintenance | Element Leader | Rank |
| G. Facilities Maintenance | Facilities Maintenance | Element Leader | Electronic Data Interchange Personal Identifier |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Postal/ZIP Code |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | State/Providence/Prefecture |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Street Name |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Street Number |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Uncommon Address: Area ID/Designation |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Uncommon Address: Range ID/Number |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Uncommon Address: Structure ID/Number |
| G. Facilities Maintenance | Facilities Maintenance | Requisitioning | Priority Designator |
| G. Facilities Maintenance | Facilities Maintenance | Requisitioning | Request ID/Tracking Number |
| G. Facilities Maintenance | Facilities Maintenance | Requisitioning | Request Type |
| G. Facilities Maintenance | Facilities Maintenance | Requisitioning | Requesting Element/Unit |
| G. Facilities Maintenance | Facilities Maintenance | Requisitioning | Requesting Element/Unit Location |
| G. Facilities Maintenance | Facilities Maintenance | Map Overlay Object | Colors |
| G. Facilities Maintenance | Facilities Maintenance | Map Overlay Object | Shapes |
| G. Facilities Maintenance | Facilities Maintenance | Map Overlay Object | Military Graphics and Symbols |
| G. Facilities Maintenance | Facilities Maintenance | Media Object | Sound |
| G. Facilities Maintenance | Facilities Maintenance | Media Object | Picture |
| G. Facilities Maintenance | Facilities Maintenance | Media Object | Video |
| G. Facilities Maintenance | Facilities Maintenance | Media Object | Uniform Resource Locator |
| G. Facilities Maintenance | Facilities Maintenance | Media Object | Office Productivity File |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Work (Time/Capacity) |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Materials |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Equipment |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Energy |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Soil |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Geology |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Drainage |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Topography |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Earthwork |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Accessibility to Materials |
| G. Facilities Maintenance | Facilities Maintenance | Estimate | Accessibility to Utilities |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Country Entity Name |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Military Grid Reference System |
| G. Facilities Maintenance | Facilities Maintenance | Location, Physical | Gridded Reference Graphic |
| F. Demolition | Demolition | Demolition | Type, Demolition (Shape or Obliterate) |
| F. Demolition | Demolition | Demolition | Type, Breach |
| F. Demolition | Demolition | Requisitioning | Priority Designator |
| F. Demolition | Demolition | Requisitioning | Request ID/Tracking Number |
| F. Demolition | Demolition | Requisitioning | Request Type |
| F. Demolition | Demolition | Requisitioning | Requesting Element/Unit |
| F. Demolition | Demolition | Requisitioning | Requesting Element/Unit Location |
| F. Demolition | Demolition | Element Leader | Military/Service Occupation |
| F. Demolition | Demolition | Element Leader | Name, First |
| F. Demolition | Demolition | Element Leader | Name, Last |
| F. Demolition | Demolition | Element Leader | Pay Grade |
| F. Demolition | Demolition | Element Leader | Rank |

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|---|---|---|---|
| F. Demolition | Demolition | Element Leader | Electronic Data Interchange Personal Identifier |
| F. Demolition | Demolition | Item Catalog Information | National Stock Number |
| F. Demolition | Demolition | Item Catalog Information | Nomenclature |
| F. Demolition | Demolition | Item Catalog Information | Department of Defense Identification Code |
| F. Demolition | Demolition | Location, Physical | Country Entity Name |
| F. Demolition | Demolition | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| F. Demolition | Demolition | Location, Physical | Military Grid Reference System |
| F. Demolition | Demolition | Location, Physical | Gridded Reference Graphic |
| F. Demolition | Demolition | Estimate | Materials |
| F. Demolition | Demolition | Estimate | Equipment |
| F. Demolition | Demolition | Estimate | Work (Time/Capacity) |
| F. Demolition | Demolition | Map Overlay Object | Colors |
| F. Demolition | Demolition | Map Overlay Object | Shapes |
| F. Demolition | Demolition | Map Overlay Object | Military Graphics and Symbols |
| F. Demolition | Demolition | Media Object | Sound |
| F. Demolition | Demolition | Media Object | Picture |
| F. Demolition | Demolition | Media Object | Video |
| F. Demolition | Demolition | Media Object | Uniform Resource Locator |
| F. Demolition | Demolition | Media Object | Office Productivity File |
| F. Demolition | Demolition | Facility | Intended Use Type |
| F. Demolition | Demolition | Facility | Condition |
| F. Demolition | Demolition | Location, Physical | Postal/ZIP Code |
| F. Demolition | Demolition | Location, Physical | State/Providence/Prefecture |
| F. Demolition | Demolition | Location, Physical | Street Name |
| F. Demolition | Demolition | Location, Physical | Street Number |
| F. Demolition | Demolition | Location, Physical | Uncommon Address: Area ID/Designation |
| F. Demolition | Demolition | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| F. Demolition | Demolition | Location, Physical | Uncommon Address: Range ID/Number |
| F. Demolition | Demolition | Location, Physical | Uncommon Address: Structure ID/Number |
| B. Obstacle Removal | Obstacle Removal | Obstacle | Type, Obstacle |
| B. Obstacle Removal | Obstacle Removal | Obstacle | Type, Lane Marking |
| B. Obstacle Removal | Obstacle Removal | Obstacle | Quantity, Lanes |
| B. Obstacle Removal | Obstacle Removal | Element Leader | Military/Service Occupation |
| B. Obstacle Removal | Obstacle Removal | Element Leader | Name, First |
| B. Obstacle Removal | Obstacle Removal | Element Leader | Name, Last |
| B. Obstacle Removal | Obstacle Removal | Element Leader | Pay Grade |
| B. Obstacle Removal | Obstacle Removal | Element Leader | Rank |
| B. Obstacle Removal | Obstacle Removal | Element Leader | Electronic Data Interchange Personal Identifier |
| B. Obstacle Removal | Obstacle Removal | Requisitioning | Priority Designator |
| B. Obstacle Removal | Obstacle Removal | Requisitioning | Request ID/Tracking Number |
| B. Obstacle Removal | Obstacle Removal | Requisitioning | Request Type |
| B. Obstacle Removal | Obstacle Removal | Requisitioning | Requesting Element/Unit |
| B. Obstacle Removal | Obstacle Removal | Requisitioning | Requesting Element/Unit Location |
| B. Obstacle Removal | Obstacle Removal | Item Catalog Information | National Stock Number |
| B. Obstacle Removal | Obstacle Removal | Item Catalog Information | Nomenclature |
| B. Obstacle Removal | Obstacle Removal | Item Catalog Information | Department of Defense Identification Code |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Country Entity Name |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Military Grid Reference System |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Gridded Reference Graphic |
| B. Obstacle Removal | Obstacle Removal | Estimate | Materials |
| B. Obstacle Removal | Obstacle Removal | Estimate | Equipment |
| B. Obstacle Removal | Obstacle Removal | Estimate | Work (Time/Capacity) |
| B. Obstacle Removal | Obstacle Removal | Map Overlay Object | Colors |
| B. Obstacle Removal | Obstacle Removal | Map Overlay Object | Shapes |
| B. Obstacle Removal | Obstacle Removal | Map Overlay Object | Military Graphics and Symbols |
| B. Obstacle Removal | Obstacle Removal | Media Object | Sound |
| B. Obstacle Removal | Obstacle Removal | Media Object | Picture |
| B. Obstacle Removal | Obstacle Removal | Media Object | Video |

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| B. Obstacle Removal | Obstacle Removal | Media Object | Uniform Resource Locator |
| B. Obstacle Removal | Obstacle Removal | Media Object | Office Productivity File |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Postal/ZIP Code |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | State/Providence/Prefecture |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Street Name |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Street Number |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Uncommon Address: Area ID/Designation |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Uncommon Address: Range ID/Number |
| B. Obstacle Removal | Obstacle Removal | Location, Physical | Uncommon Address: Structure ID/Number |
| C. Explosive Ordnance Disposal | 1. Clearing | Requisitioning | Priority Designator |
| C. Explosive Ordnance Disposal | 1. Clearing | Requisitioning | Request ID/Tracking Number |
| C. Explosive Ordnance Disposal | 1. Clearing | Requisitioning | Request Type |
| C. Explosive Ordnance Disposal | 1. Clearing | Requisitioning | Requesting Element/Unit |
| C. Explosive Ordnance Disposal | 1. Clearing | Requisitioning | Requesting Element/Unit Location |
| C. Explosive Ordnance Disposal | 1. Clearing | Time | DTG (of Observation) |
| C. Explosive Ordnance Disposal | 1. Clearing | Communications | Phone Number, Landline |
| C. Explosive Ordnance Disposal | 1. Clearing | Communications | Phone Number, Satellite Phone |
| C. Explosive Ordnance Disposal | 1. Clearing | Communications | Phone Number, Mobile Cellular |
| C. Explosive Ordnance Disposal | 1. Clearing | Communications | Email, Official |
| C. Explosive Ordnance Disposal | 1. Clearing | Communications | Email, Personal |
| C. Explosive Ordnance Disposal | 1. Clearing | Communications | Type, Radio |
| C. Explosive Ordnance Disposal | 1. Clearing | Communications | Frequency, Radio |
| C. Explosive Ordnance Disposal | 1. Clearing | Munition Characteristics | Type, Munition (Emplace, Drop, Throw, Project) |
| C. Explosive Ordnance Disposal | 1. Clearing | Munition Characteristics | Munition Purpose (Anti-Personnel, Anti-Armor, Flare, Smoke, Improvised) |
| C. Explosive Ordnance Disposal | 1. Clearing | Munition Characteristics | Color, Primary |
| C. Explosive Ordnance Disposal | 1. Clearing | Munition Characteristics | Color, Secondary |
| C. Explosive Ordnance Disposal | 1. Clearing | Munition Characteristics | Markings |
| C. Explosive Ordnance Disposal | 1. Clearing | CBRN | Type, Contaminant (Vesicant - Blistering, Cyanogenic - Blood, Pulmonary - Choking, Riot Control/Tear) |
| C. Explosive Ordnance Disposal | 1. Clearing | CBRN | Type, Vector (Smoke, Mist, Droplets, Solid) |
| C. Explosive Ordnance Disposal | 1. Clearing | Media Object | Sound |
| C. Explosive Ordnance Disposal | 1. Clearing | Media Object | Picture |
| C. Explosive Ordnance Disposal | 1. Clearing | Media Object | Video |
| C. Explosive Ordnance Disposal | 1. Clearing | Media Object | Uniform Resource Locator |
| C. Explosive Ordnance Disposal | 1. Clearing | Report, General Engineering | File Number |
| C. Explosive Ordnance Disposal | 1. Clearing | Report, General Engineering | Report Number |
| C. Explosive Ordnance Disposal | 1. Clearing | Report, General Engineering | Report Location |
| C. Explosive Ordnance Disposal | 1. Clearing | Report, General Engineering | Report Type/Purpose |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Requisitioning | Priority Designator |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Requisitioning | Request ID/Tracking Number |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Requisitioning | Request Type |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Requisitioning | Requesting Element/Unit |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Requisitioning | Requesting Element/Unit Location |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Time | DTG (of Observation) |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Communications | Phone Number, Landline |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Communications | Phone Number, Satellite Phone |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Communications | Phone Number, Mobile Cellular |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Communications | Email, Official |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Communications | Email, Personal |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Communications | Type, Radio |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Communications | Frequency, Radio |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Munition Characteristics | Type, Munition (Emplace, Drop, Throw, Project) |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Munition Characteristics | Munition Purpose (Anti-Personnel, Anti-Armor, Flare, Smoke, Improvised) |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Munition Characteristics | Color, Primary |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Munition Characteristics | Color, Secondary |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Munition Characteristics | Markings |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | CBRN | Type, Contaminant (Vesicant - Blistering, Cyanogenic - Blood, Pulmonary - Choking, Riot Control/Tear) |

| | | | |
|---|---|---|---|
| C. Explosive Ordnance Disposal | 2. Rendering Safe | CBRN | Type, Vector (Smoke, Mist, Droplets, Solid) |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Media Object | Sound |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Media Object | Picture |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Media Object | Video |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Media Object | Uniform Resource Locator |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Report, General Engineering | File Number |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Report, General Engineering | Report Number |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Report, General Engineering | Report Location |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Report, General Engineering | Report Type/Purpose |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Report, General Engineering | File Number |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Report, General Engineering | Report Number |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Report, General Engineering | Report Location |
| C. Explosive Ordnance Disposal | 2. Rendering Safe | Report, General Engineering | Report Type/Purpose |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Requisitioning | Priority Designator |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Requisitioning | Request ID/Tracking Number |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Requisitioning | Request Type |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Requisitioning | Requesting Element/Unit |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Requisitioning | Requesting Element/Unit Location |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Time | DTG (of Observation) |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Communications | Phone Number, Landline |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Communications | Phone Number, Satellite Phone |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Communications | Phone Number, Mobile Cellular |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Communications | Email, Official |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Communications | Email, Personal |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Communications | Type, Radio |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Communications | Frequency, Radio |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Munition Characteristics | Type, Munition (Emplace, Drop, Throw, Project) |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Munition Characteristics | Munition Purpose (Anti-Personnel, Anti-Armor, Flare, Smoke, Improvised) |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Munition Characteristics | Color, Primary |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Munition Characteristics | Color, Secondary |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Munition Characteristics | Markings |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | CBRN | Type, Contaminant (Vesicant - Blistering, Cyanogenic - Blood, Pulmonary - Choking, Riot Control/Tear) |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | CBRN | Type, Vector (Smoke, Mist, Droplets, Solid) |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Media Object | Sound |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Media Object | Picture |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Media Object | Video |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Media Object | Uniform Resource Locator |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Report, General Engineering | File Number |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Report, General Engineering | Report Number |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Report, General Engineering | Report Location |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Report, General Engineering | Report Type/Purpose |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Report, General Engineering | File Number |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Report, General Engineering | Report Number |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Report, General Engineering | Report Location |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Report, General Engineering | Report Type/Purpose |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Munition Characteristics | Type, Triggering Device/Method |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Munition Characteristics | Type, Pre-Cursor |
| C. Explosive Ordnance Disposal | 3. Foreign/Enemy Exploitation | Munition Characteristics | Type, Accelerant |
| E. Bulk Liquid Support | 1. Fuel | Item Characteristics | Fuel Load Capacity (Volume) |
| E. Bulk Liquid Support | 1. Fuel | Item Characteristics | Fuel Type |
| E. Bulk Liquid Support | 1. Fuel | Item Catalog Information | Model |
| E. Bulk Liquid Support | 1. Fuel | Item Catalog Information | National Stock Number |
| E. Bulk Liquid Support | 1. Fuel | Item Catalog Information | Table of Authorized Material Control Number |
| E. Bulk Liquid Support | 1. Fuel | Item Identification | Serial Number |
| E. Bulk Liquid Support | 1. Fuel | Location, Physical | Gridded Reference Graphic |
| E. Bulk Liquid Support | 1. Fuel | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| E. Bulk Liquid Support | 1. Fuel | Material Release Order | Issue to Activity |
| E. Bulk Liquid Support | 1. Fuel | Material Release Order | Issue from Activity |
| E. Bulk Liquid Support | 1. Fuel | Material Release Order | Issue NSN |
| E. Bulk Liquid Support | 1. Fuel | Material Release Order | Issue Quantity |

| | | | |
|---|---|---|---|
| E. Bulk Liquid Support | 1. Fuel | Item Attributes | Dispensing Capacity per Hour |
| E. Bulk Liquid Support | 1. Fuel | Item Attributes | Dispensing Velocity |
| E. Bulk Liquid Support | 1. Fuel | Item Attributes | Dispensing Pressure |
| E. Bulk Liquid Support | 1. Fuel | Procurement | Bulk Fuel Contracts |
| E. Bulk Liquid Support | 1. Fuel | Procurement | Fuels |
| E. Bulk Liquid Support | 1. Fuel | Procurement | Seaborne Fuel, Ammunition, and Supplies |
| E. Bulk Liquid Support | 1. Fuel | Location, Physical | Military Grid Reference System |
| E. Bulk Liquid Support | 1. Fuel | Procurement | Acquisitions & Cross Servicing Agreements |
| E. Bulk Liquid Support | 1. Fuel | Requisitioning | Priority Designator |
| E. Bulk Liquid Support | 1. Fuel | Requisitioning | Request ID/Tracking Number |
| E. Bulk Liquid Support | 1. Fuel | Requisitioning | Request Type |
| E. Bulk Liquid Support | 1. Fuel | Requisitioning | Requesting Element/Unit |
| E. Bulk Liquid Support | 1. Fuel | Requisitioning | Requesting Element/Unit Location |
| E. Bulk Liquid Support | 2. Water | Item Characteristics | Water Load Capacity (Volume) |
| E. Bulk Liquid Support | 2. Water | Item Attributes | Water Generation Capability (per Hour) |
| E. Bulk Liquid Support | 2. Water | Item Catalog Information | Model |
| E. Bulk Liquid Support | 2. Water | Item Catalog Information | National Stock Number |
| E. Bulk Liquid Support | 2. Water | Item Catalog Information | Table of Authorized Material Control Number |
| E. Bulk Liquid Support | 2. Water | Item Identification | Serial Number |
| E. Bulk Liquid Support | 2. Water | Location, Physical | Gridded Reference Graphic |
| E. Bulk Liquid Support | 2. Water | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| E. Bulk Liquid Support | 2. Water | Material Release Order | Issue to Activity |
| E. Bulk Liquid Support | 2. Water | Material Release Order | Issue from Activity |
| E. Bulk Liquid Support | 2. Water | Material Release Order | Issue NSN |
| E. Bulk Liquid Support | 2. Water | Material Release Order | Issue Quantity |
| E. Bulk Liquid Support | 2. Water | Item Attributes | Dispensing Capacity per Hour |
| E. Bulk Liquid Support | 2. Water | Item Attributes | Dispensing Velocity |
| E. Bulk Liquid Support | 2. Water | Item Attributes | Dispensing Pressure |
| E. Bulk Liquid Support | 2. Water | Procurement | Local/Regional Water Sources |
| E. Bulk Liquid Support | 2. Water | Location, Physical | Military Grid Reference System |
| E. Bulk Liquid Support | 2. Water | Procurement | Acquisitions & Cross Servicing Agreements |
| E. Bulk Liquid Support | 2. Water | Water Source | Type |
| E. Bulk Liquid Support | 2. Water | Water Source | Quantity |
| E. Bulk Liquid Support | 2. Water | Requisitioning | Priority Designator |
| E. Bulk Liquid Support | 2. Water | Requisitioning | Request ID/Tracking Number |
| E. Bulk Liquid Support | 2. Water | Requisitioning | Request Type |
| E. Bulk Liquid Support | 2. Water | Requisitioning | Requesting Element/Unit |
| E. Bulk Liquid Support | 2. Water | Requisitioning | Requesting Element/Unit Location |
| K. Water Production/Storage | Put With Bulk Liquid Support | Put With Bulk Liquid Support | Put With Bulk Liquid Support |
| D. Power Generation/Distribution | Power Generation/Distribution | Estimate | Energy |
| D. Power Generation/Distribution | Power Generation/Distribution | General Engineering | Energy Sources |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Catalog Information | Model |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Catalog Information | National Stock Number |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Catalog Information | Table of Authorized Material Control Number |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Identification | Serial Number |
| D. Power Generation/Distribution | Power Generation/Distribution | Location, Physical | Gridded Reference Graphic |
| D. Power Generation/Distribution | Power Generation/Distribution | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Characteristics | Electrical Generation Capacity (Ampere) |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Characteristics | Electrical Generation Capacity (Volts) |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Characteristics | Electrical Generation Capacity (Watts) |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Characteristics | Electrical Phase Output |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Characteristics | Hourly Fuel Consumption Rate (High) |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Characteristics | Hourly Fuel Consumption Rate (Low) |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Characteristics | Fuel Load Capacity (Volume) |
| D. Power Generation/Distribution | Power Generation/Distribution | Item Characteristics | On Board Fuel Tank Capacity (Volume) |
| D. Power Generation/Distribution | Power Generation/Distribution | Requisitioning | Priority Designator |
| D. Power Generation/Distribution | Power Generation/Distribution | Requisitioning | Request ID/Tracking Number |
| D. Power Generation/Distribution | Power Generation/Distribution | Requisitioning | Request Type |
| D. Power Generation/Distribution | Power Generation/Distribution | Requisitioning | Requesting Element/Unit |
| D. Power Generation/Distribution | Power Generation/Distribution | Requisitioning | Requesting Element/Unit Location |

# Appendix E
# Health Services Function Data Points

| Sub-Function 1 | Sub-Function 2 | Data Type | Line Data Level |
|---|---|---|---|
| B. Casualty Management | 1. Collection | Location, Physical | Military Grid Reference System (Collection Area) |
| B. Casualty Management | 1. Collection | Communications | Frequency, Radio (Collection Area) |
| B. Casualty Management | 1. Collection | Activity | Call Sign (Collection Area) |
| B. Casualty Management | 1. Collection | Health Services | Medical Priority (Urgent, Urgent-Surgical, Priority, Routine, Convenience) |
| B. Casualty Management | 1. Collection | Health Services | Type, Patient (Litter, Ambulatory) |
| B. Casualty Management | 1. Collection | CBRN | Type, Contaminant (Vesicant - Blistering, Cyanogenic - Blood, Pulmonary - Choking, Riot Control/Tear) |
| B. Casualty Management | 1. Collection | CBRN | Type, Vector (Smoke, Mist, Droplets, Solid) |
| B. Casualty Management | 1. Collection | Individual Member | Status |
| B. Casualty Management | 1. Collection | Individual Member | Citizenship/Nationality |
| B. Casualty Management | 1. Collection | Individual Member | Component/Service |
| B. Casualty Management | 1. Collection | Individual Member | Zap Number |
| B. Casualty Management | 1. Collection | Health Services | Mechanism of Injury |
| B. Casualty Management | 1. Collection | Health Services | Type, Injury/Illness |
| B. Casualty Management | 1. Collection | Health Services | Symptoms |
| B. Casualty Management | 1. Collection | Health Services | Vitals, Airway |
| B. Casualty Management | 1. Collection | Health Services | Vitals, Breathing Rate |
| B. Casualty Management | 1. Collection | Health Services | Vitals, Pulse Rate |
| B. Casualty Management | 1. Collection | Health Services | Vitals, Conscious/Unconscious |
| B. Casualty Management | 1. Collection | Health Services | Vitals, Other |
| B. Casualty Management | 1. Collection | Health Services | Treatment (Field/Expedient), Performed/Given |
| B. Casualty Management | 1. Collection | Individual Member | Allergies |
| B. Casualty Management | 1. Collection | Health Services | Triage Priority |
| B. Casualty Management | 1. Collection | Health Services | Collected from Unit |
| B. Casualty Management | 1. Collection | Health Services | Collected from Location Grid |
| B. Casualty Management | 1. Collection | Location, Physical | Destination Grid |
| B. Casualty Management | 1. Collection | Activity | Destination Activity |
| B. Casualty Management | 1. Collection | Activity | Organizational Name |
| B. Casualty Management | 1. Collection | Transportation | Passenger Itinerary |
| B. Casualty Management | 1. Collection | Transportation | Date Time Group (Arrived) |
| B. Casualty Management | 1. Collection | Transportation | Date Time Group (Departed) |
| B. Casualty Management | 1. Collection | Health Services | Date Time Group (Vitals Taken) |
| B. Casualty Management | 3. Evacuation | Location, Physical | Military Grid Reference System |
| B. Casualty Management | 3. Evacuation | Communications | Frequency, Radio |
| B. Casualty Management | 3. Evacuation | Activity | Call Sign |
| B. Casualty Management | 3. Evacuation | Health Services | Medical Priority (Urgent, Urgent-Surgical, Priority, Routine, Convenience) |
| B. Casualty Management | 3. Evacuation | Health Services | Special Equipment (Hoist, Extraction Equipment, Ventilator) |
| B. Casualty Management | 3. Evacuation | Health Services | Type, Patients (Litter, Ambulatory) |
| B. Casualty Management | 3. Evacuation | Operations | Status, Enemy (None, Possible, Confirmed, Engaged) |
| B. Casualty Management | 3. Evacuation | Communications | Visual Signal (Panel, Pyrotechnic, Smoke) |
| B. Casualty Management | 3. Evacuation | CBRN | Type, Contaminant (Vesicant - Blistering, Cyanogenic - Blood, Pulmonary - Choking, Riot Control/Tear) |
| B. Casualty Management | 3. Evacuation | CBRN | Type, Vector (Smoke, Mist, Droplets, Solid) |
| B. Casualty Management | 3. Evacuation | Individual Member | Status |
| B. Casualty Management | 3. Evacuation | Individual Member | Citizenship/Nationality |
| B. Casualty Management | 3. Evacuation | Individual Member | Component/Service |
| B. Casualty Management | 3. Evacuation | Individual Member | Zap Number |
| B. Casualty Management | 3. Evacuation | Health Services | Mechanism of Injury |
| B. Casualty Management | 3. Evacuation | Health Services | Type, Injury/Illness |
| B. Casualty Management | 3. Evacuation | Health Services | Symptoms |
| B. Casualty Management | 3. Evacuation | Health Services | Vitals, Airway |
| B. Casualty Management | 3. Evacuation | Health Services | Vitals, Breathing Rate |
| B. Casualty Management | 3. Evacuation | Health Services | Vitals, Pulse Rate |
| B. Casualty Management | 3. Evacuation | Health Services | Vitals, Conscious/Unconscious |

| | | | |
|---|---|---|---|
| B. Casualty Management | 3. Evacuation | Health Services | Vitals, Other |
| B. Casualty Management | 3. Evacuation | Health Services | Treatment (Field/Expedient), Performed/Given |
| B. Casualty Management | 3. Evacuation | Health Services | Date Time Group (Vitals Taken) |
| B. Casualty Management | 3. Evacuation | Transportation | Date Time Group (Departed) |
| B. Casualty Management | 4. Treatment | Location, Physical | Military Grid Reference System (Treatment Facility) |
| B. Casualty Management | 4. Treatment | Activity | Organizational Name |
| B. Casualty Management | 4. Treatment | Health Services | Medical Priority (Urgent, Urgent-Surgical, Priority, Routine, Convenience) |
| B. Casualty Management | 4. Treatment | Health Services | Type, Patient (Litter, Ambulatory) |
| B. Casualty Management | 4. Treatment | CBRN | Type, Contaminant (Vesicant - Blistering, Cyanogenic - Blood, Pulmonary - Choking, Riot Control/Tear) |
| B. Casualty Management | 4. Treatment | CBRN | Type, Vector (Smoke, Mist, Droplets, Solid) |
| B. Casualty Management | 4. Treatment | Individual Member | Status |
| B. Casualty Management | 4. Treatment | Individual Member | Citizenship/Nationality |
| B. Casualty Management | 4. Treatment | Individual Member | Component/Service |
| B. Casualty Management | 4. Treatment | Individual Member | Zap Number |
| B. Casualty Management | 4. Treatment | Health Services | Mechanism of Injury |
| B. Casualty Management | 4. Treatment | Health Services | Type, Injury/Illness |
| B. Casualty Management | 4. Treatment | Health Services | Symptoms |
| B. Casualty Management | 4. Treatment | Health Services | Vitals, Airway |
| B. Casualty Management | 4. Treatment | Health Services | Vitals, Breathing Rate |
| B. Casualty Management | 4. Treatment | Health Services | Vitals, Pulse Rate |
| B. Casualty Management | 4. Treatment | Health Services | Vitals, Conscious/Unconscious |
| B. Casualty Management | 4. Treatment | Health Services | Vitals, Other |
| B. Casualty Management | 4. Treatment | Health Services | Treatment (Field/Expedient), Performed/Given |
| B. Casualty Management | 4. Treatment | Individual Member | Allergies |
| B. Casualty Management | 4. Treatment | Health Services | Triage Priority |
| B. Casualty Management | 4. Treatment | Health Services | Collected from Unit |
| B. Casualty Management | 4. Treatment | Health Services | Collected from Location Grid |
| B. Casualty Management | 4. Treatment | Location, Physical | Destination Grid |
| B. Casualty Management | 4. Treatment | Activity | Destination Activity |
| B. Casualty Management | 4. Treatment | Activity | Organizational Name |
| B. Casualty Management | 4. Treatment | Transportation | Passenger Itinerary |
| B. Casualty Management | 4. Treatment | Transportation | Date Time Group (Arrived) |
| B. Casualty Management | 4. Treatment | Transportation | Date Time Group (Departed) |
| B. Casualty Management | 4. Treatment | Health Services | Date Time Group (Vitals Taken) |
| B. Casualty Management | 4. Treatment | Health Services | Referral, Specialized Care Order |
| B. Casualty Management | 4. Treatment | Health Services | Referral, Date |
| B. Casualty Management | 4. Treatment | Health Services | Referral, Requesting Individual |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Type (Self, Buddy, Corpsman, etc.) |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Date |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Diagnosis |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Severity |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Results/Progress |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Prescription(s) Drug Name |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Prescription(s) Date Prescribed |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Prescription(s) Duration |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Prescription(s) Refill |
| B. Casualty Management | 4. Treatment | Health Services | Treatment Prescription(s) Prescribing Individual |
| A. Medical Command and Control | 2. Local Services | Health Services | Local/Regional Emergency Services |
| A. Medical Command and Control | 2. Local Services | Health Services | Local/Regional Medical Facilities/Capabilities |
| A. Medical Command and Control | 2. Local Services | Activity | Organizational Name |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Apartment/Suite/Room |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Burrough/Sub-division/District/Precinct |
| A. Medical Command and Control | 2. Local Services | Location, Physical | City/Town |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Country Entity Name |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Country Name Abbreviation (2 Character) |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Floor |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Gridded Reference Graphic |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Latitude, Longitude |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Military Grid Reference System |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Postal/ZIP Code |

| | | | |
|---|---|---|---|
| A. Medical Command and Control | 2. Local Services | Location, Physical | State/Providence/Prefecture |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Street Name |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Street Number |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Uncommon Address: Area ID/Designation |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| A. Medical Command and Control | 2. Local Services | Location, Physical | Uncommon Address: Structure ID/Number |
| A. Medical Command and Control | 2. Local Services | Health Services | Unit/Activity Medical Capacity Level (Bed Spaces) |
| A. Medical Command and Control | 2. Local Services | Health Services | Unit/Activity Medical Care Level (Echelon/Role 1-5) Capability |
| A. Medical Command and Control | 2. Local Services | Item Catalog Information | Medical Name |
| A. Medical Command and Control | 2. Local Services | Item Catalog Information | National Stock Number |
| A. Medical Command and Control | 2. Local Services | Item Catalog Information | Nomenclature |
| A. Medical Command and Control | 2. Local Services | Item Catalog Information | Class of Supply |
| A. Medical Command and Control | 2. Local Services | Inventory | On Hand Quantity |
| A. Medical Command and Control | 1. Unit Capabilities | Health Services | Unit/Activity Medical Capacity Level (Bed Spaces) |
| A. Medical Command and Control | 1. Unit Capabilities | Health Services | Unit/Activity Medical Care Level (Echelon/Role 1-5) Capability |
| A. Medical Command and Control | 1. Unit Capabilities | Activity | Organizational Name |
| A. Medical Command and Control | 1. Unit Capabilities | Location, Physical | Gridded Reference Graphic |
| A. Medical Command and Control | 1. Unit Capabilities | Location, Physical | Latitude, Longitude |
| A. Medical Command and Control | 1. Unit Capabilities | Location, Physical | Military Grid Reference System |
| A. Medical Command and Control | 1. Unit Capabilities | Health Services | Bed ID |
| A. Medical Command and Control | 1. Unit Capabilities | Health Services | Bed, Assigned Ward |
| A. Medical Command and Control | 1. Unit Capabilities | Health Services | Bed Occupied/Vacancy |
| A. Medical Command and Control | 1. Unit Capabilities | Health Services | Bed Overflow (Inactive) |
| A. Medical Command and Control | 1. Unit Capabilities | Health Services | Ward |
| A. Medical Command and Control | 1. Unit Capabilities | Element Leader | Electronic Data Interchange Personal Identifier |
| A. Medical Command and Control | 1. Unit Capabilities | Element Leader | Military/Service Occupation |
| A. Medical Command and Control | 1. Unit Capabilities | Element Leader | Name, First |
| A. Medical Command and Control | 1. Unit Capabilities | Element Leader | Name, Last |
| A. Medical Command and Control | 1. Unit Capabilities | Element Leader | Pay Grade |
| A. Medical Command and Control | 1. Unit Capabilities | Element Leader | Rank |
| A. Medical Command and Control | 1. Unit Capabilities | Item Catalog Information | Medical Name |
| A. Medical Command and Control | 1. Unit Capabilities | Item Catalog Information | National Stock Number |
| A. Medical Command and Control | 1. Unit Capabilities | Item Catalog Information | Nomenclature |
| A. Medical Command and Control | 1. Unit Capabilities | Inventory | On Hand Quantity |
| A. Medical Command and Control | 1. Unit Capabilities | Item Catalog Information | Class of Supply |
| A. Medical Command and Control | 1. Unit Capabilities | Individual Member | Status |
| A. Medical Command and Control | 1. Unit Capabilities | Individual Member | Citizenship/Nationality |
| A. Medical Command and Control | 1. Unit Capabilities | Individual Member | Component/Service |
| A. Medical Command and Control | 1. Unit Capabilities | Individual Member | Zap Number |
| D. Medical Stability Operations | 2. Balanced Score Card | Activity | Organizational Name |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Apartment/Suite/Room |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Burrough/Sub-division/District/Precinct |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | City/Town |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Country Entity Name |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Country Name Abbreviation (2 Character) |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Floor |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Gridded Reference Graphic |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Latitude, Longitude |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Military Grid Reference System |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Postal/ZIP Code |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | State/Providence/Prefecture |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Street Name |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Street Number |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Uncommon Address: Area ID/Designation |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| D. Medical Stability Operations | 2. Balanced Score Card | Location, Physical | Uncommon Address: Structure ID/Number |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Unit/Activity Medical Capacity Level (Bed Spaces) |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Unit/Activity Medical Care Level (Echelon/Role 1-5) Capability |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Bed, Assigned Ward |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Ward |

| | | | |
|---|---|---|---|
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Staff Coverage |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Functionality/Availability of Equipment |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Drug/Immunization Availability |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Guidelines for Standards of Care |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Training |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Supervision |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Quality of Care (Clinical Observation) |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Health Worker Satisfaction |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Patient Satisfaction |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, Patient Perception of Quality |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Corrective Goal/Plan/Condition |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Evaluation, DTG |
| D. Medical Stability Operations | 2. Balanced Score Card | Element Leader | Electronic Data Interchange Personal Identifier |
| D. Medical Stability Operations | 2. Balanced Score Card | Element Leader | Military/Service Occupation |
| D. Medical Stability Operations | 2. Balanced Score Card | Element Leader | Name, First |
| D. Medical Stability Operations | 2. Balanced Score Card | Element Leader | Name, Last |
| D. Medical Stability Operations | 2. Balanced Score Card | Element Leader | Pay Grade |
| D. Medical Stability Operations | 2. Balanced Score Card | Element Leader | Rank |
| D. Medical Stability Operations | 2. Balanced Score Card | Health Services | Responsibility Transfer DTG |
| D. Medical Stability Operations | 1. Tactical Conflict Assessment Planning Framework | TACAPF | Header |
| D. Medical Stability Operations | 1. Tactical Conflict Assessment Planning Framework | TACAPF | Livelihood Questionnaire |
| D. Medical Stability Operations | 1. Tactical Conflict Assessment Planning Framework | TACAPF | Basic Services Questionnaire |
| D. Medical Stability Operations | 1. Tactical Conflict Assessment Planning Framework | TACAPF | Rule of Law Questionnaire |
| D. Medical Stability Operations | 1. Tactical Conflict Assessment Planning Framework | TACAPF | Security Questionnaire |
| D. Medical Stability Operations | 1. Tactical Conflict Assessment Planning Framework | TACAPF | Governance Questionnaire |
| D. Medical Stability Operations | 1. Tactical Conflict Assessment Planning Framework | TACAPF | Basic Questionnaire |
| C. Dental | Dental | Health Services | Date Time Group (Vitals Taken) |
| C. Dental | Dental | Health Services | Referral, Specialized Care Order |
| C. Dental | Dental | Health Services | Referral, Date |
| C. Dental | Dental | Health Services | Referral, Requesting Individual |
| C. Dental | Dental | Health Services | Treatment Type (Self, Buddy, Corpsman, etc.) |
| C. Dental | Dental | Health Services | Treatment Date |
| C. Dental | Dental | Health Services | Treatment Diagnosis |
| C. Dental | Dental | Health Services | Treatment Severity |
| C. Dental | Dental | Health Services | Treatment Results/Progress |
| C. Dental | Dental | Health Services | Treatment Prescription(s) Drug Name |
| C. Dental | Dental | Health Services | Treatment Prescription(s) Date Prescribed |
| C. Dental | Dental | Health Services | Treatment Prescription(s) Duration |
| C. Dental | Dental | Health Services | Treatment Prescription(s) Refill |
| C. Dental | Dental | Health Services | Treatment Prescription(s) Prescribing Individual |
| C. Dental | Dental | Item Catalog Information | Medical Name |
| C. Dental | Dental | Item Catalog Information | National Stock Number |
| C. Dental | Dental | Item Catalog Information | Nomenclature |
| C. Dental | Dental | Item Catalog Information | Class of Supply |
| C. Dental | Dental | Inventory | On Hand Quantity |
| C. Dental | Dental | Activity | Organizational Name |
| C. Dental | Dental | Location, Physical | Apartment/Suite/Room |
| C. Dental | Dental | Location, Physical | Burrough/Sub-division/District/Precinct |
| C. Dental | Dental | Location, Physical | City/Town |
| C. Dental | Dental | Location, Physical | Country Entity Name |
| C. Dental | Dental | Location, Physical | Country Name Abbreviation (2 Character) |
| C. Dental | Dental | Location, Physical | Country Name Abbreviation (3 Character Geopolitical Entities, Names, and Codes (GENC) Standard) |
| C. Dental | Dental | Location, Physical | Floor |
| C. Dental | Dental | Location, Physical | Gridded Reference Graphic |
| C. Dental | Dental | Location, Physical | Latitude, Longitude |
| C. Dental | Dental | Location, Physical | Military Grid Reference System |
| C. Dental | Dental | Location, Physical | Postal/ZIP Code |

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|---|---|---|---|
| C. Dental | Dental | Location, Physical | State/Providence/Prefecture |
| C. Dental | Dental | Location, Physical | Street Name |
| C. Dental | Dental | Location, Physical | Street Number |
| C. Dental | Dental | Location, Physical | Uncommon Address: Area ID/Designation |
| C. Dental | Dental | Location, Physical | Uncommon Address: Code Name/Brevity Code |
| C. Dental | Dental | Location, Physical | Uncommon Address: Structure ID/Number |
| C. Dental | Dental | Health Services | Unit/Activity Medical Capacity Level (Bed Spaces) |
| C. Dental | Dental | Individual Member | Allergies |
| C. Dental | Dental | Health Services | Triage Priority |
| C. Dental | Dental | Individual Member | Dental Classification |
| C. Dental | Dental | Element Leader | Electronic Data Interchange Personal Identifier |
| C. Dental | Dental | Element Leader | Military/Service Occupation |
| C. Dental | Dental | Element Leader | Name, First |
| C. Dental | Dental | Element Leader | Name, Last |
| C. Dental | Dental | Element Leader | Pay Grade |
| C. Dental | Dental | Element Leader | Rank |
| C. Dental | Dental | Health Services | Mechanism of Injury |
| C. Dental | Dental | Health Services | Type, Injury/Illness |
| C. Dental | Dental | Health Services | Symptoms |
| C. Dental | Dental | Health Services | Vitals, Airway |
| C. Dental | Dental | Health Services | Vitals, Breathing Rate |
| C. Dental | Dental | Health Services | Vitals, Pulse Rate |
| C. Dental | Dental | Health Services | Vitals, Conscious/Unconscious |
| C. Dental | Dental | Health Services | Vitals, Other |
| B. Casualty Management | 2. Temporary Holding | Individual Member | Status |
| B. Casualty Management | 2. Temporary Holding | Individual Member | Citizenship/Nationality |
| B. Casualty Management | 2. Temporary Holding | Individual Member | Component/Service |
| B. Casualty Management | 2. Temporary Holding | Individual Member | Zap Number |
| B. Casualty Management | 2. Temporary Holding | Health Services | Mechanism of Injury |
| B. Casualty Management | 2. Temporary Holding | Health Services | Type, Injury/Illness |
| B. Casualty Management | 2. Temporary Holding | Health Services | Symptoms |
| B. Casualty Management | 2. Temporary Holding | Health Services | Vitals, Airway |
| B. Casualty Management | 2. Temporary Holding | Health Services | Vitals, Breathing Rate |
| B. Casualty Management | 2. Temporary Holding | Health Services | Vitals, Pulse Rate |
| B. Casualty Management | 2. Temporary Holding | Health Services | Vitals, Conscious/Unconscious |
| B. Casualty Management | 2. Temporary Holding | Health Services | Vitals, Other |
| B. Casualty Management | 2. Temporary Holding | Health Services | Treatment (Field/Expedient), Performed/Given |
| B. Casualty Management | 2. Temporary Holding | Individual Member | Allergies |
| B. Casualty Management | 2. Temporary Holding | Health Services | Triage Priority |
| B. Casualty Management | 2. Temporary Holding | Health Services | Collected from Unit |
| B. Casualty Management | 2. Temporary Holding | Health Services | Collected from Location Grid |
| B. Casualty Management | 2. Temporary Holding | Location, Physical | Destination Grid |
| B. Casualty Management | 2. Temporary Holding | Activity | Destination Activity |
| B. Casualty Management | 2. Temporary Holding | Activity | Organizational Name |
| B. Casualty Management | 2. Temporary Holding | Time | DTG Received |

# Appendix F
# Logistics Data Blocks

| Category | Data Block | Data Element | Data Field |
|---|---|---|---|
| Activity | Department of Defense Activity Address Directory | Header | 1. Department of Defense Activity Address Code |
| Activity | Department of Defense Activity Address Directory | Header | 2. Unit Identifier Code |
| Activity | Department of Defense Activity Address Directory | Header | 3. Service/Agency |
| Activity | Department of Defense Activity Address Directory | Header | 4. Organization Type Code |
| Activity | Department of Defense Activity Address Directory | Header | 5. Authority Code |
| Activity | Department of Defense Activity Address Directory | Header | 6. Major Command |
| Activity | Department of Defense Activity Address Directory | Header | 7. Sub Tier Code |
| Activity | Department of Defense Activity Address Directory | Header | 8. Deletion Date |
| Activity | Department of Defense Activity Address Directory | Header | 9. Contractor |
| Activity | Department of Defense Activity Address Directory | Header | 10. Contractor Administration Office |
| Activity | Department of Defense Activity Address Directory | Header | 11. Last Update |
| Activity | Department of Defense Activity Address Directory | Header | 12. Common Governmentwide Accounting Classification |
| Activity | Department of Defense Activity Address Directory | Header | 13. Procurement Authority |
| Activity | Department of Defense Activity Address Directory | Header | 14. Grant Authority |
| Activity | Department of Defense Activity Address Directory | Header | 15. Funding Office |
| Activity | Department of Defense Activity Address Directory | COMMRI | 1. Bill |
| Activity | Department of Defense Activity Address Directory | COMMRI | 2. Data Pattern |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 1. T1_Line 1 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 2. T1_Line 2 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 3. T1_Line 3 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 4. T1_Line 4 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 5. T1_Line 5 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 6. T1_City |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 7. T1_State/Province |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 8. T1_Country Name |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 9. T1_GENC CC |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 10. T1_Point of Contact (Name) |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 11. T1_Point of Contact Phone |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 12. T1_Zone Improvement Plan (ZIP) Code |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 13. T1_International Postal Code |

| | | | |
|---|---|---|---|
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 14. T1_State(A)/Province |
| Activity | Department of Defense Activity Address Directory | Type Address Code 1 (Postal/Mail) | 1. T1_Effective Date |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 2. T2_Line 1 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 3. T2_Line 2 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 4. T2_Line 3 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 5. T2_Line 4 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 6. T2_Line 5 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 7. T2_City |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 8. T2_State/Province |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 9. T2_Country Name |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 10. T2_GENC CC |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 11. T2_Combatant Command |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 12. T2_Standard Point Location Code |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 13. T2_Air Lines of Communication |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 14. T2_CONUS/OCONUS |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 15. T2_Point of Contact (Name) |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 16. T2_Point of Contact Phone |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 17. T2_Effective Date |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 18. T2_Deletion Date |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 19. T2_Zone Improvement Plan (ZIP) Code |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 20. T2_International Postal Code |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 21. T2_State(A)/Province |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 22. T2_Containerization Point |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 23. T2_Break Bulk Point |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 24. T2_Water Port of Debarkation |
| Activity | Department of Defense Activity Address Directory | Type Address Code 2 (Freight) | 25. T2_Aerial Port of Debarkation |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 1. T3_Line 1 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 2. T3_Line 2 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 3. T3_Line 3 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 4. T3_Line 4 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 5. T3_Line 5 |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 6. T3_City |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 7. T3_Country Name |

| | | | |
|---|---|---|---|
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 8. T3_GENC CC |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 9. T3_Effective Date |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 10. T3_Deletion Date |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 11. T3_Accounting Disbursing Station Number/Fiscal Station Number |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 12. T3_Zone Improvement Plan (ZIP) Code |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 13. T3_International Postal Code |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 14. T3_State(A)/Province |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 15. T3_Point of Contact (Name) |
| Activity | Department of Defense Activity Address Directory | Type Address Code 3 (Billing) | 16. T3_Point of Contact Phone |
| Activity | Department of Defense Activity Address Directory | Department of Defense Activity Address Code | 1. Service Type |
| Activity | Department of Defense Activity Address Directory | Department of Defense Activity Address Code | 2. Service Identifier |
| Activity | Total Force Management | Billet Identification Code | Position 1-6: Alpha/Numeric (UIC) |
| Activity | Total Force Management | Billet Identification Code | Position 7-11: Numeric (Line Number) |
| Activity | Unit ID | Unit Identifier Code | Position 1: Alpha (Indicates Service) |
| Activity | Unit ID | Unit Identifier Code | Position 2-6: Alpha/Numeric (Identifier) |
| Activity | Unit ID | Break Bulk Point | Position 1: Alpha (Indicates Service) |
| Activity | Unit ID | Break Bulk Point | Position 2-6: Alpha/Numeric (Identifier) |
| Communications | Telephonic | Phone Number, Landline | Position 1-3: Numeric (Phone Number Country Code) |
| Communications | Telephonic | Phone Number, Landline | Position 4-6: Numeric (Phone Number Area Code) |
| Communications | Telephonic | Phone Number, Landline | Position 7-10: Numeric (Phone Number Prefix/Central Office) |
| Communications | Telephonic | Phone Number, Landline | Position 11-15: Numeric (Phone Line Number) |
| Communications | Telephonic | Phone Number, Mobile Cellular | Position 1-3: Numeric (Phone Number Country Code) |
| Communications | Telephonic | Phone Number, Mobile Cellular | Position 4-6: Numeric (Phone Number Area Code) |
| Communications | Telephonic | Phone Number, Mobile Cellular | Position 7-10: Numeric (Phone Number Prefix/Central Office) |
| Communications | Telephonic | Phone Number, Mobile Cellular | Position 11-15: Numeric (Phone Line Number) |
| Communications | Telephonic | Phone Number, Satellite Phone | Position 1-3: Numeric (Phone Number Country Code) |
| Communications | Telephonic | Phone Number, Satellite Phone | Position 4-6: Numeric (Phone Number Area Code) |
| Communications | Telephonic | Phone Number, Satellite Phone | Position 7-10: Numeric (Phone Number Prefix/Central Office) |
| Communications | Telephonic | Phone Number, Satellite Phone | Position 11-15: Numeric (Phone Line Number) |
| Date | Military Time Reference | Date Time Group | Position 1-2: Numeric (2-digit calendar day) |
| Date | Military Time Reference | Date Time Group | Position 3-4: Numeric (2-digit military hour) |
| Date | Military Time Reference | Date Time Group | Position 5-6: Numeric (2-digit clock minute) |
| Date | Military Time Reference | Date Time Group | Position 7: Alpha (Time Zone Indicator) |
| Date | Military Time Reference | Date Time Group | Position 8-10: Alpha (Month Abbreviation) |
| Date | Military Time Reference | Date Time Group | Position 11-12: Numeric (Last 2 digits of calendar year) |
| Date | Military Time Reference | Julian Date | Position 1: Last digit of the calendar year |
| Date | Military Time Reference | Julian Date | Position 2-4: Number of day |
| Financials | Accounting Reference | Project | Position 1-6: Alpha/Numeric (Fiscal Station) |
| Financials | Accounting Reference | Project | Position 7-12: Alpha/Numeric (Project Owning Organization) |
| Financials | Accounting Reference | Project | Position 13-18: Alpha/Numeric (Project Number) |

| | | | |
|---|---|---|---|
| Financials | Accounting Reference | Expenditure | Position 1-2: Alpha/Numeric (Treasury Account Symbol) |
| Financials | Accounting Reference | Expenditure | Position 3-6: Numeric (Beginning Fiscal Year) |
| Financials | Accounting Reference | Expenditure | Position 7-10: Numeric (Ending Fiscal Year) |
| Financials | Accounting Reference | Expenditure | Position 11-14: Alpha/Numeric (Basic Symbol) |
| Financials | Accounting Reference | Expenditure | Position 15-17: Numeric (Sub Account) |
| Financials | Accounting Reference | Expenditure | Position 18: Alpha (Direct or Reimbursable) |
| Financials | Accounting Reference | Expenditure | Position 19-22: Alpha/Numeric (Budget Line Item) |
| Financials | Accounting Reference | Expenditure | Position 23-26: Numeric (Limit) |
| Financials | Accounting Reference | Expenditure | Position 27-47: Alpha/Numeric (Task Number) |
| Financials | Accounting Reference | Type | Position 1-3: Numeric (Object Class Code) |
| Financials | Accounting Reference | Task | Position 4-6: Numeric (Expenditure Type) |
| Financials | Accounting Reference | Task Number | Position 1-6: Alpha/Numeric (Expenditure Organization) |
| Financials | Accounting Reference | Task Number | Position 7-8: Numeric (Fiscal Year) |
| Financials | Accounting Reference | Task Number | Position 9-12: Numeric (Budget Line Item) |
| Financials | Accounting Reference | Task Number | Position 13-22: Alpha Numeric (Program) |
| Financials | Accounting Reference | Project, Organization, Expenditure, Type | Position 1-18: Alpha/Numeric (Project) |
| Financials | Accounting Reference | Project, Organization, Expenditure, Type | Position 19-24: Alpha/Numeric (Organization) |
| Financials | Accounting Reference | Project, Organization, Expenditure, Type | Position 25-72: Alpha/Numeric (Expenditure) |
| Financials | Accounting Reference | Project, Organization, Expenditure, Type | Position 73-78: Type |
| Individual Member | Activity | Assigned Billet Identification Code | Position 1-6: Alpha/Numeric (UIC) |
| Individual Member | Activity | Assigned Billet Identification Code | Position 7-11: Numeric (Line Number) |
| Individual Member | Identifier | Zap Number | Position 3-6: Numeric (Last 4 numbers of EDIPI) |
| Individual Member | Identifier | Zap Number | Position 7-8: Alpha (Blood Group) |
| Individual Member | Identifier | Zap Number | Position: 9: Numeric (Blood RhD) |
| Item | Item Catalog Information | Table of Authorized Material Control Number | Position 1: Alpha (Commodity Group) |
| Item | Item Catalog Information | Table of Authorized Material Control Number | Position 2-5: Numeric (Item Number) |
| Item | Item Catalog Information | Table of Authorized Material Control Number | Position 6: Numeric (Class of Supply) |
| Item | Item Catalog Information | Table of Authorized Material Control Number | Position 7: Alpha (Sub-Class of Supply) |
| Item | Item Catalog Information | Table of Authorized Material Control Number, Consists Of | Position 1: Alpha (Commodity Group) |
| Item | Item Catalog Information | Table of Authorized Material Control Number, Consists Of | Position 2-5: Numeric (Item Number) |
| Item | Item Catalog Information | Table of Authorized Material Control Number, Consists Of | Position 6: Numeric (Class of Supply) |
| Item | Item Catalog Information | Table of Authorized Material Control Number, Consists Of | Position 7: Alpha (Sub-Class of Supply) |
| Item | Item Catalog Information | Table of Authorized Material Control Number, Consists Of | Position 8: Alpha (Sub-Class of Supply of Parent) |
| Item Catalog Information | Item | Identification Number | Position 1-5: Numeric (Item Number) |
| Item Catalog Information | Item | Identification Number | Position 6: Alpha (Item Code) |
| Item Catalog Information | Item | National Stock Number | Position 1-4: Numeric (Fed Supply Class) |
| Item Catalog Information | Item | National Stock Number | Position 9-13: Alpha/Numeric (National Item ID Number) |
| Item Catalog Information | National Stock Number | Federal Supply Class | Position 1-2: Numeric (Fed Stock Group) |
| Item Catalog Information | National Stock Number | Federal Supply Class | Position 3-4: Numeric (Fed Stock Class) |
| Item Catalog Information | National Stock Number | National Item ID Number | Position 1-2: Alpha/Numeric (Country Code) |
| Item Catalog Information | National Stock Number | National Item ID Number | Position 3-5: Alpha/Numeric (Grouping Number) |
| Item Catalog Information | National Stock Number | National Item ID Number | Position 6-9: Alpha/Numeric (Item Number) |
| Location, Physical | Geospatial Reference | Military Grid Reference System | Position 1-2: Alpha (Grid Zone Designator, GZD) |

| | | | |
|---|---|---|---|
| Location, Physical | Geospatial Reference | Military Grid Reference System | Position 3-4: Alpha (100K-meter Square Identifier) |
| Location, Physical | Geospatial Reference | Military Grid Reference System | Position 5-14: Numeric (Numerical Location) |
| Location, Physical | Military Grid Reference System | Numerical Location | Position 5: Numeric (Easting) |
| Location, Physical | Military Grid Reference System | Numerical Location | Position 5: Numeric (Northing) |
| Supply | Procurement | Standard Document Number | Position 1-6: Alpha/Numeric (DoDAAC) |
| Supply | Procurement | Standard Document Number | Position 7-10: Numeric (Julian Date) |
| Supply | Procurement | Standard Document Number | Position 11-14: Alpha/Numeric (Serial) |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 1 | Have there been changes in the village population in the last year? |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 1A | What is the reason why for the change in population? |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 2 | What is the most important problem facing the village? |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 2A | Why is the most important problem? |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 3 | Who do you believe can solve this problem? |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 3A | Why do you believe it is their problem to solve? |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 4 | What should be done first to help the village? |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 4A | Why is this the first thing that should be done? |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 1 Response | Increase, decrease, no change, don't know, no comment |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 1A Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 2 Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 2A Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 3 Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 3A Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 4 Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Basic Questionnaire | Question 4A Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1 | Primary schools are available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1A | Secondary/vocational schools are available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1B | Healthcare is available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1C | Medicine is available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1D | Housing is available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1E | Power is available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1F | Potable water is available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1G | Sanitation is available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1H | Garbage collection is available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1I | Drivable roads are available and village residents have access to use these services |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1 Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1A Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1B Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |

| | | | |
|---|---|---|---|
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1C Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1D Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1E Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1F Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1G Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1H Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 1I Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2 | Primary schools are the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2A | Secondary/vocational schools is the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2B | Healthcare is the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2C | Medicine is the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2D | Housing is the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2E | Power is the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2F | Potable is the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2G | Sanitation is the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2H | Garbage is the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2I | Drivable roads are the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2J | Other: (Text) is the urgently needed service by the residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2H | No Comment |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2I | I do not know |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2 Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2A Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2B Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2C Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2D Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2E Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2F Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2G Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2H Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2I Response | Priority 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2J Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2H Response | 990 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 2I Response | 999 |

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| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 3 | Available services provided by the government are provided equally to all residents |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 3 Response | Yes - 2, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 4 | Is there a group provided least or no services by the government? |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 4 Response | Yes - 1, No - 2, I do not know - 9, No Comment - 99 |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 4 Response, Follow up | Other: (Text) is provided least or no services by the government |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 5 | Why do you believe this group is provided less or no services by the government? |
| Tactical Conflict Assessment Planning Framework | Basic Services Questionnaire | Question 5 Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 1 | Does the government have the capability and capacity to perform basic government functions? |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 1 Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2 | How much do residents of this village trust the National Government |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2A | How much do residents of this village trust the Provincial Government |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2B | How much do residents of this village trust the Local Government |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2C | How much do residents of this village trust the Civil Courts |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2D | How much do residents of this village trust the Criminal Courts |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2E | How much do residents of this village trust the Police |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2F | How much do residents of this village trust the Armed Forces |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2G | How much do residents of this village trust the Foreign/Allied/UN Forces/Peacekeepers |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2H | How much do residents of this village trust the Tribal Leaders |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2I | How much do residents of this village trust the Religious Leaders |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2J | How much do residents of this village trust the Local Social Service Organizations |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2K | How much do residents of this village trust the Radio Broadcasts |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2L | How much do residents of this village trust the Newspapers |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2M | How much do residents of this village trust the Television Reports |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2 Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2A Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2B Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2C Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2D Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2E Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2F Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2G Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2H Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |

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| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2I Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2J Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2K Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2L Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2M Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2 Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2A Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2B Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2C Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2D Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2E Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2F Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2G Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2H Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2I Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2J Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2K Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2L Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 2M Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 3 | How often do residents of the village express their grievances to local government authorities? |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 3 Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 4 | Does the government undertake regular community consultations? |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 4 Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 5 | Do you support the local government? |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 5 Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 6 | Does the local government conduct its affairs in a transparent and accountable manner? |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 6 Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 7 | Is the government more effective than it was one year ago? |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 7 Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Governance Questionnaire | Question 7 Response, Follow-up | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Header | Date | Date Time Group |
| Tactical Conflict Assessment Planning Framework | Header | Location, Physical | Burrough/Sub-division/District/Precinct |
| Tactical Conflict Assessment Planning Framework | Header | Location, Physical | City/Town/Village |

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| Tactical Conflict Assessment Planning Framework | Header | Location, Physical | Country 3 Digit Identifier |
| Tactical Conflict Assessment Planning Framework | Header | Location, Physical | Gridded Reference Graphic |
| Tactical Conflict Assessment Planning Framework | Header | Location, Physical | Latitude, Longitude |
| Tactical Conflict Assessment Planning Framework | Header | Location, Physical | Military Grid Reference System |
| Tactical Conflict Assessment Planning Framework | Header | Location, Physical | Population |
| Tactical Conflict Assessment Planning Framework | Header | Interviewer | Name, Last |
| Tactical Conflict Assessment Planning Framework | Header | Interviewer | Name, First |
| Tactical Conflict Assessment Planning Framework | Header | Interviewer | Component/Service |
| Tactical Conflict Assessment Planning Framework | Header | Interviewer | Zap Number |
| Tactical Conflict Assessment Planning Framework | Header | Interviewer | Electronic Data Interchange Personal Identifier |
| Tactical Conflict Assessment Planning Framework | Header | Interviewer | Pay Grade |
| Tactical Conflict Assessment Planning Framework | Header | Interviewer | Rank |
| Tactical Conflict Assessment Planning Framework | Header | Translator | Name, Last |
| Tactical Conflict Assessment Planning Framework | Header | Translator | Name, First |
| Tactical Conflict Assessment Planning Framework | Header | Translator | Component/Service |
| Tactical Conflict Assessment Planning Framework | Header | Translator | Zap Number |
| Tactical Conflict Assessment Planning Framework | Header | Translator | Electronic Data Interchange Personal Identifier |
| Tactical Conflict Assessment Planning Framework | Header | Translator | Pay Grade |
| Tactical Conflict Assessment Planning Framework | Header | Translator | Rank |
| Tactical Conflict Assessment Planning Framework | Header | Interviewee/Subject | Name, Last |
| Tactical Conflict Assessment Planning Framework | Header | Interviewee/Subject | Name, First |
| Tactical Conflict Assessment Planning Framework | Header | Interviewee/Subject | Gender |
| Tactical Conflict Assessment Planning Framework | Header | Interviewee/Subject | Tribe/Ethnicity |
| Tactical Conflict Assessment Planning Framework | Header | Interviewee/Subject | Profession |
| Tactical Conflict Assessment Planning Framework | Header | Interviewee/Subject | Title/Position |
| Tactical Conflict Assessment Planning Framework | Header | Interviewee/Subject | Male Age Category (Fighting/Old) |
| Tactical Conflict Assessment Planning Framework | Header | Interviewee/Subject | Female Age Category (Childbearing/Old) |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 1 | Residents have access to sufficient food for their families |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 1 Response | Always - 5, Mostly - 4, Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2 | Households pay for basic necessities by earning money through regular employment |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2A | Households pay for basic necessities by earning money through odd/handy man jobs |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2B | Households pay for basic necessities by receiving money from the government |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2C | Households pay for basic necessities by selling personal items from the home |

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| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2D | Households pay for basic necessities by receiving gifts from relatives |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2E | Households pay for basic necessities by borrowing money |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2F | Other: (Text) pays for basic necessities |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2G | No Comment |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2H | I do not know |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2A Response | Priority 1, 2, 3, 4, 5, 6 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2B Response | Priority 1, 2, 3, 4, 5, 6 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2C Response | Priority 1, 2, 3, 4, 5, 6 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2D Response | Priority 1, 2, 3, 4, 5, 6 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2E Response | Priority 1, 2, 3, 4, 5, 6 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2F Response | Priority 1, 2, 3, 4, 5, 6 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2G Response | 990 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 2H Response | 999 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 3 | What percentage of the village is unemployed that is able to work? |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 3 Response | 0-20% - 5, 21-40% - 4, 41-60% - 3, 61-80% - 2, 81-100% - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 4 | Are some groups financially better off than others? |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 4 Response | Yes - 2, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 5 | Is there a group that is financially better off than others? |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 5 Response | Yes - 1, No - 2, I do not know - 9, No Comment - 99 |
| Tactical Conflict Assessment Planning Framework | Livelihood Questionnaire | Question 5 Response, Follow-up | Other: (Text) is financially better off than others |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1 | If I were a victim of crime, I would report it to the National Government. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1A | If I were a victim of crime, I would report it to the Provincial Government. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1B | If I were a victim of crime, I would report it to the Local Government. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1C | If I were a victim of crime, I would report it to the Criminal Courts. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1D | If I were a victim of crime, I would report it to the Police. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1E | If I were a victim of crime, I would report it to the Armed Forces. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1F | If I were a victim of crime, I would report it to the Foreign/Allied/UN Forces/Peacekeepers. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1G | If I were a victim of crime, I would report it to the Tribal Leaders. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1H | If I were a victim of crime, I would report it to the Religious Leaders. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1I | If I were a victim of crime, I would report it to the Local Social Service Organizations. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1 Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1A Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1B Response | Yes - 0, No - 1, I do not know - 9 |

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| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1C Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1D Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1E Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1F Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1G Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1H Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 1I Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 2 | How effectively do you believe crime would be investigated and resolved? |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 2 Response | Effectively - 3, Neutral - 2, Ineffectively - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3 | If I had property/economic dispute, I would report it to the National Government. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3A | If I had property/economic dispute, I would report it to the Provincial Government. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3B | If I had property/economic dispute, I would report it to the Local Government. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3C | If I had property/economic dispute, I would report it to the Criminal Courts. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3D | If I had property/economic dispute, I would report it to the Police. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3E | If I had property/economic dispute, I would report it to the Armed Forces. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3F | If I had property/economic dispute, I would report it to the Foreign/Allied/UN Forces/Peacekeepers. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3G | If I had property/economic dispute, I would report it to the Tribal Leaders. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3H | If I had property/economic dispute, I would report it to the Religious Leaders. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3I | If I had property/economic dispute, I would report it to the Local Social Service Organizations. |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3 Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3A Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3B Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3C Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3D Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3E Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3F Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3G Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3H Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 3I Response | Yes - 0, No - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 4 | Is misconduct among government officials investigated and punished? |
| Tactical Conflict Assessment Planning Framework | Rule of Law Questionnaire | Question 4 Response | Always - 5, Mostly - 4, Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 1 | How is the security situation in this village? |

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| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 1 Response | Extremely Dangerous - 1, Dangerous - 2, Neutral - 3, Secure - 4, Extremely Secure - 5, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 2 | Are the Police responsible for security in the village? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 2A | Is the Armed Forces responsible for security in the village? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 2B | Is the Foreign/Allied/UN Forces/Peacekeepers responsible for security in the village? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 2 Response | Yes - 2, No - 1, I do not know - 9, Not Identified - 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 2A Response | Yes - 2, No - 1, I do not know - 9, Not Identified - 1000 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 2B Response | Yes - 2, No - 1, I do not know - 9, Not Identified - 1001 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 2C | Other: (Text) |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 2C Response | Yes - 2, No - 1, I do not know - 9, Not Identified - 1001 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3 | Random violence against individuals is the biggest fear to personal safety and security |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3A | Targeted violence against individuals is the biggest fear to personal safety and security |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3B | Robbery is the biggest fear to personal safety and security |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3C | Hijacking is the biggest fear to personal safety and security |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3D | Extortion is the biggest fear to personal safety and security |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3E | Kidnapping is the biggest fear to personal safety and security |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3F | Terrorist attacks is the biggest fear to personal safety and security |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3G | Other: (Text) is the biggest fear to personal safety and security |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3H | Nothing is the biggest fear to personal safety and security |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3I | No Comment |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3J | I do not know |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3 Response | Priority 1, 2, 3, 4, 5, 6, 7 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3A Response | Priority 1, 2, 3, 4, 5, 6, 7 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3B Response | Priority 1, 2, 3, 4, 5, 6, 7 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3C Response | Priority 1, 2, 3, 4, 5, 6, 7 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3D Response | Priority 1, 2, 3, 4, 5, 6, 7 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3E Response | Priority 1, 2, 3, 4, 5, 6, 7 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3F Response | Priority 1, 2, 3, 4, 5, 6, 7 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3G Response | Why: (Text Reply) |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3H Response | 0 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3I Response | 990 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 3J Response | 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4 | How often local security addresses random violence? |

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| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4A | How often local security addresses targeted violence? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4B | How often local security addresses robbery? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4C | How often local security addresses hijacking? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4D | How often local security addresses extortion? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4E | How often local security addresses kidnapping? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4F | How often local security addresses terrorist attacks? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4G | How often local security addresses Other: (Text)? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4 Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9, Not ID'd - 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4A Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9, Not ID'd - 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4B Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9, Not ID'd - 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4C Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9, Not ID'd - 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4D Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9, Not ID'd - 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4E Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9, Not ID'd - 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4F Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9, Not ID'd - 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 4G Response | Always - 5, Most of the Time - 4; Sometimes - 3, Rarely - 2, Never - 1, I do not know - 9, Not ID'd - 999 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5 | Do you think the National Government can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5A | Do you think the Provincial Government can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5B | Do you think the Local Government can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5C | Do you think the Criminal Courts can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5D | Do you think the Police can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5E | Do you think the Armed Forces can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5F | Do you think the Foreign/Allied/UN Forces/Peacekeepers can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5G | Do you think the Tribal Leaders can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5H | Do you think the Religious Leaders can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5I | Do you think the Local Social Service Organizations can effectively address these security threats? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5 Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5A Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5B Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |

| | | | |
|---|---|---|---|
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5C Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5D Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5E Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5F Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5G Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5H Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 5I Response | Not at all - 0, Just a little - 1, Somewhat - 2, A lot - 3, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 6 | Are the residents of this village more or less safe than they were a year ago? |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 6 Response | More - 3, Same - 2, Less - 1, I do not know - 9 |
| Tactical Conflict Assessment Planning Framework | Security Questionnaire | Question 6 Response, Follow-up | Why: (Text Reply) |
| Transportation | Traffic Management | Transportation Control Number | Position 1-14: Alpha/Numeric (SDN) |
| Transportation | Traffic Management | Transportation Control Number | Position 15: Alpha/Numeric (Suffix Code) |
| Transportation | Traffic Management | Transportation Control Number | Position 16: Alpha/Numeric (Partial Shipment Code) |
| Transportation | Traffic Management | Transportation Control Number | Position: 17: Alpha/Numeric (Split Shipment Code) |

# Appendix G
# Authoritative Data Sources and Managers

| Data Source | Source Manager |
|---|---|
| User input | User |
| Defense Automated Addressing System (DAAS) | DLA |
| Total Force Structure Management System (TFSMS) | USMC |
| MCO P1080.20 | USMC |
| Marine Corps Total Force System (MCTFS) | USMC |
| TDM-Catalyst | USMC |
| Federal Logistics Information System (FLIS) | DLA |
| Universal Transverse Mercator (UTM) or Universal Polar Stereographic (UPS) grid system | DoD |
| United States Naval Observatory | Department of Defense |
| Defense Enrollment Eligibility Reporting System (DEERS) | Defense Manpower Data Center |
| Modernized Integrated Database (MIDB) (SIPR only)<br>Homeland Security Infrastructure Program (HSIP) | DoD |
| Modernized Integrated Database (MIDB) (SIPR only)<br>Homeland Security Infrastructure Program (HSIP)<br>Real Property Assets Database (RPAD) | DoD |
| RAPIDS via Common Access Card | Defense Manpower Data Center |
| Medical Readiness Reporting System (MRRS) | USMC |
| Defense Logistics Services Center | DLA |
| Cesium | Open Source |
| ISO 3166 | International Standards Organization |
| Geographic Names Information System (GNIS) | US Board on Geographic Names |
| STANAG 1059 | Standardization Agreement (STANAG) North Atlantic Treaty Organization (NATO/OTAN) |
| System Generated | System |
| JOPES | DoD |
| CBM+ Enterprise Tool (yet to be developed) | USMC |
| ACSA Intellipedia | Under Secretary of Defense for Acquisition & Sustainment (OUSD A&S) - International Cooperation (IC) |
| DLA-Contracting Information System (DLA-CIS) | DLA |
| Reporting Emergency Petroleum Oils and Lubricants (REPOL)<br>Enterprise External Business Portal (EEBP) | DLA |
| National Level Ammunition Capability (NLAC) | Army |
| Global Ammunition Control-Point (GAC-P)<br>Ordnance Information System (OIS) | DoD |
| National Level Ammunition Capability (NLAC)<br>Office of Naval Intelligence (ONI)<br>Enterprise External Business Portal (EEBP) | Army<br>Navy<br>DLA |
| DLMS | DLA |
| IDE/GTN (Integrated Development Environment/Global Transportation Network) Convergence (IGC) | TRANSCOM |
| ISO 8601 | Information Standards Organization |
| Automated Air Facility Information File (AAFIF)<br>Global Decision Support System (GDSS) | National Geospatial-Intelligence Agency (NGA)<br>US Air Force |
| Defense Transportation Regulation | USTRANSCOM |
| World Port Index (WPI) | National Geospatial-Intelligence Agency (NGA) |
| Intelligent Road and Rail Information Server (IRRIS)<br>World Port Index (WPI) | Army<br>National Geospatial-Intelligence Agency (NGA) |

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